

Betriebsanleitung

Operating instructions

Kennwort
Code

NARAYANGONJ Bestell-Nr.: 45017726

Besteller
Customer

Lösche GmbH, Düsseldorf

Flender Auftrags-Nr.
Flender order no.

412 3221-

FLENDER

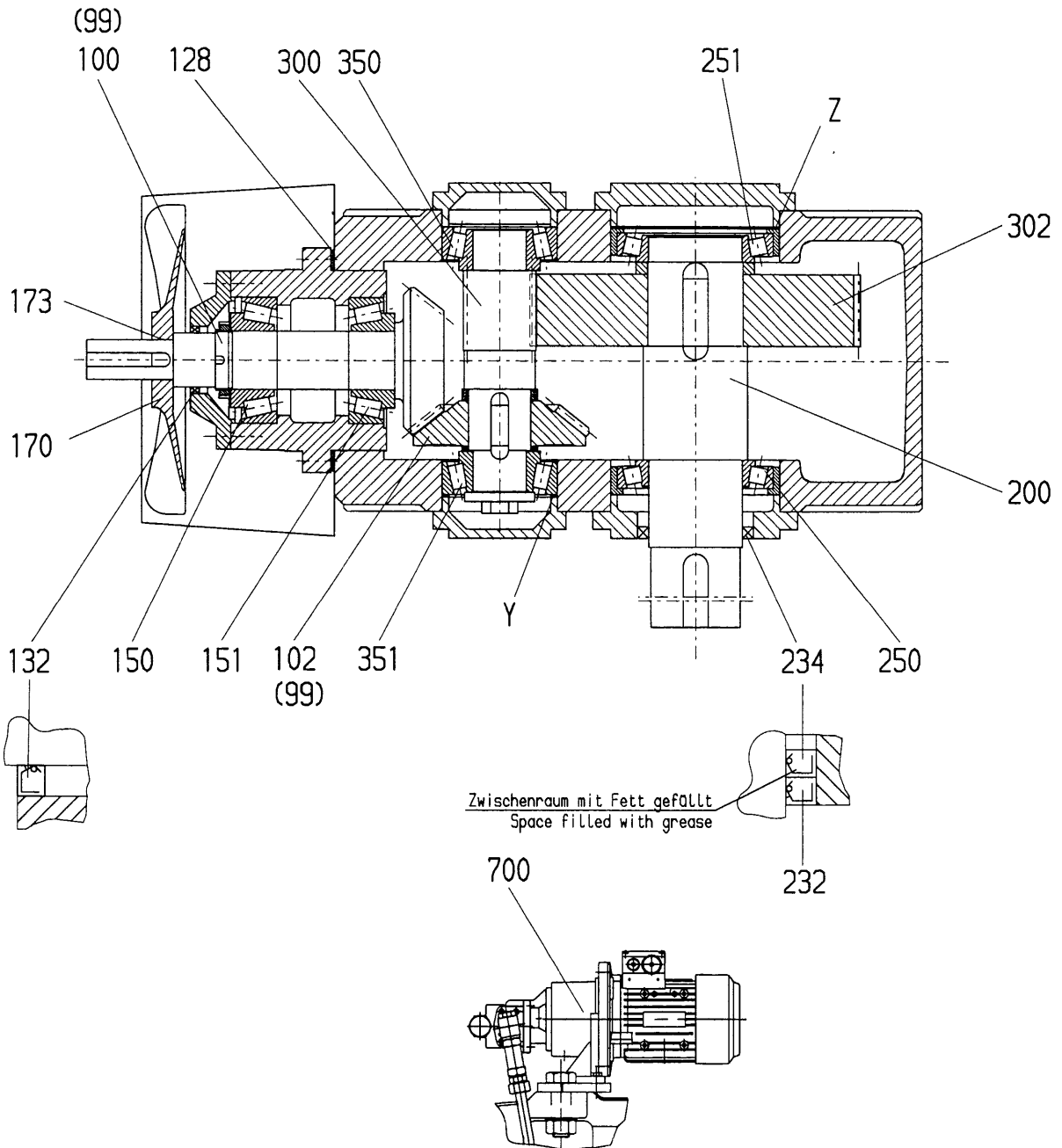
A. Friedr. Flender AG · D 46393 Bocholt · Tel. 02871/92-0 · Telefax 02871/92-2596 · <http://www.flender.com>

FLENDER**FLENDER - Zahnradgetriebe**
Gear unit

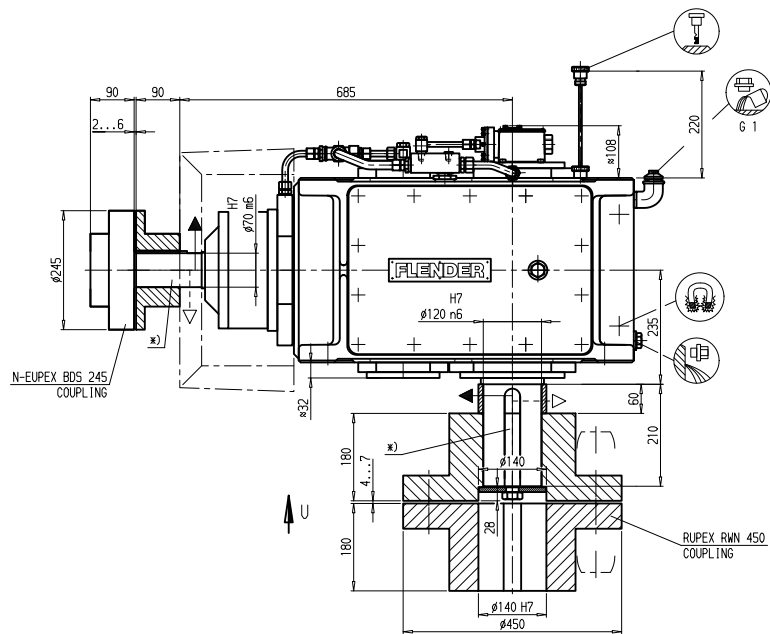
Ersatzteilzeichnung/ Spare part drawing

Bauart
Type **B2SV**Ausführung
DesignGröße
Size **7****C**

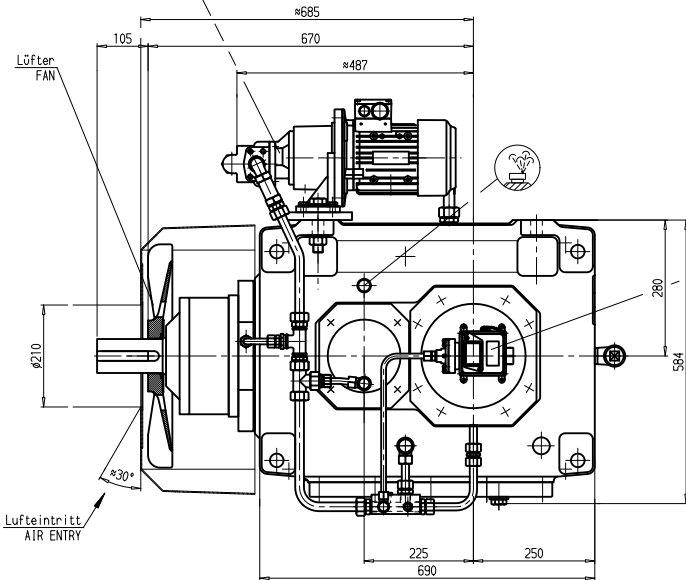
Bei Korrespondenz bitte angeben / Please quote in correspondence

EZ 4123221-040-1...1**Y, Z** = Paßscheibe / ShimFLENDER-Getriebewerk Penig GmbH, Thierbacher Str. 24,
09320 Penig, Tel. 037381/60, Telefax 0373816286Datum / Date
2004-06-14Name/Name Kästner
ez4123221_40ZG
.

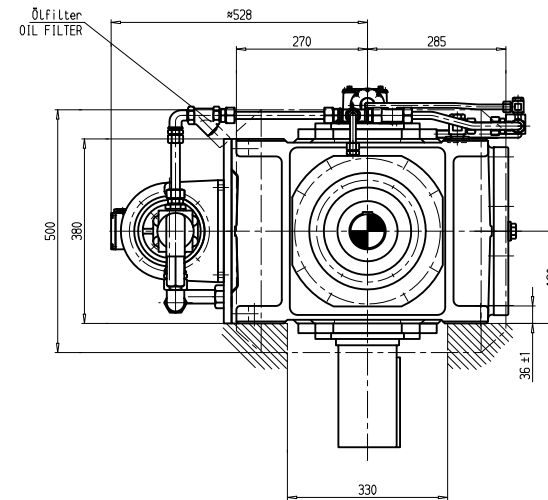
FLENDER		Ersatzteilliste (EL) Spare parts list		Bauart B2SV Type Größe 7,0 Size Übersetzung 8,569 ratio		Seite Page 1/1	
Bei Korrespondenz bitte angeben Please quote in correspondence				EL 4123221-040 DE/EN			
Hierzu gehört Zeichnungs-Nr. Please refer to DWG No.							
Teil-Nr.	Menge	Benennung		Zeichnungs-Nr.	Material-Nr.	Gw(kg)	
Part No.	No. off	Description		Drawings No.	Ident no.	Weight	
E 0100	1	ST	KEGELRADW R ZS 267/2,24 1.6587 BEVEL PINION SHAFT R ZS 267/2,24 1.6587	5881176/B	000.000.561.176	18,0	
E 0102	1	ST	KEGELRAD L ZS 267/2,24 1.6587 BEVEL GEAR L ZS 267/2,24 1.6587	5881341/B	000.000.561.341	17,9	
E 0128	1	ST	RING-PASS 241X336X06 F ST ADAPTER RING 241X336X06 F ST	5881755/D	000.000.561.755	2,2	
E 0132	1	ST	WDR 075X095X10 BAUMSL W 5710 75FPM585 SHAFT SEAL 075X095X10 BAUMSL W 5710		000.000.306.054	0,0	
E 0150	1	ST	WÄELZL 32316 B D 720 ST-KAEF ROLLING CONTACT BEARING 32316 B D 720		000.000.380.569	5,7	
E 0151	1	ST	WÄELZL 32316 B D 720 ST-KAEF ROLLING CONTACT BEARING 32316 B D 720		000.000.380.569	5,7	
E 0170	1	ST	LUEFTER 460/ 30,0/ 70 AC45000 FAN 460/ 30,0/ 70 AC45000	5882403/E	000.000.562.403	5,6	
E 0173	1	ST	RING-TOL BN0820-065-01 W 3372 FED-ST DISTANCE RING BN0820-065-01 W 3372		000.000.346.008	0,0	
E 0200	1	ST	WELLE 150X0621 16MnCrS5 SHAFT 150X0621 16MnCrS5	5876622/B	000.000.676.622	67,7	
E 0232	1	ST	WDR 130X160X12 BAUMSL W 5710 75FPM585 SHAFT SEAL 130X160X12 BAUMSL W 5710		000.000.306.001	0,1	
E 0234	1	ST	WDR 130X160X15 BAUM W 5710 75FPM585 SHAFT SEAL 130X160X15 BAUM W 5710		000.000.781.850	0,2	
E 0250	1	ST	WÄELZL 32026 X D 720 ST-KAEF ROLLING CONTACT BEARING 32026 X D 720		000.000.776.049	5,3	
E 0251	1	ST	WÄELZL 32026 X D 720 ST-KAEF ROLLING CONTACT BEARING 32026 X D 720		000.000.776.049	5,3	
E 0300	1	ST	STRDWELLE LP 225,0/3,83 1.6587 PINION SHAFT LP 225,0/3,83 1.6587	5877267/-	000.000.677.267	17,3	
E 0302	1	ST	STIRNRAD RP 225,0/3,83 1.6587 GEAR WHEEL RP 225,0/3,83 1.6587	5876711/-	000.000.676.711	65,6	
E 0350	1	ST	WÄELZL 32315 B D 720 ST-KAEF ROLLING CONTACT BEARING 32315 B D 720		000.000.380.568	5,5	
E 0351	1	ST	WÄELZL 32315 B D 720 ST-KAEF ROLLING CONTACT BEARING 32315 B D 720		000.000.380.568	5,5	
E 0700	1	ST	PUMPENGRUPPE GR.02 SF2/8 F5958 LOESCHE PUMP GROUPRUPPE GR.02 SF2/8 F5958 LOESCH		000.001.152.434	18,1	
Die mit * gekennzeichneten Teile gehören zu einer Baugruppe (G). Die Baugruppe ist nur komplett auszutauschen. The parts marked with * belong to a subassembly (G). The subassembly must be replaced complete.							
A. Friedr.Flender AG, D 46393 Bocholt, Tel. + 49(0)2871/92-0, Fax + 49(0)2871/922596, http://www.flender.com				Datum Date 15.06.2004	BOTHENDORF, RICO Rev.:		PE-403



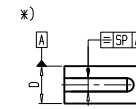
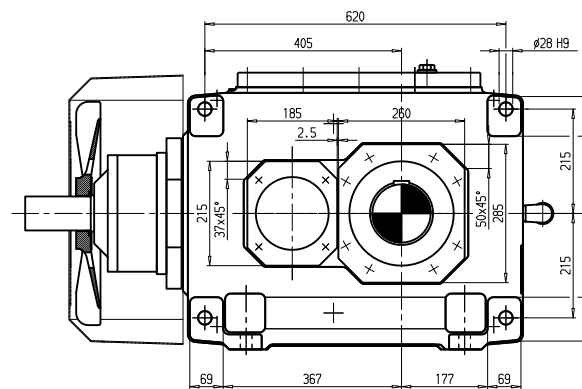
Motorpumpe SF 2/8
P= 0,75 kW, n=1395 1/min
400V, 50Hz, IP55
PUMP



CCS DUAL SNAP Druckwächter 0.5 bar
Modell 604 G1
0,4A/125V DC, 6A/28V DC
15 A/ 125 ~ 480 V AC 50 ~ 60 Hz
PRESSURE CONTROL CCS DUAL SNAP
SWITCHING CAPACITY
TYPE OF PROTECTION



U
Ansicht ohne Ölversorgung
VIEW WITHOUT PIPING PLAN



D	SP
8	<22 0,035
>22	<50 0,035
>50	<85 0,055
>85	<150 0,055
>150	<230 0,07
>230	<290 0,07
>290	<500 0,1

Angeschlossen an kundenseitiges Auswertegerät. Motor AUS bei Öl-druck < 0.5 bar.

Connected with switch board (not supplied by FLENDER). Main motor switch off by oil pressure <0.5 bar.

Vor Inbetriebnahme des Getriebes ist die Ölpumpe einzuschalten.

BEFORE START-UP GEAR UNIT START OIL PUMP

Kennwort: NARAYANGONJ

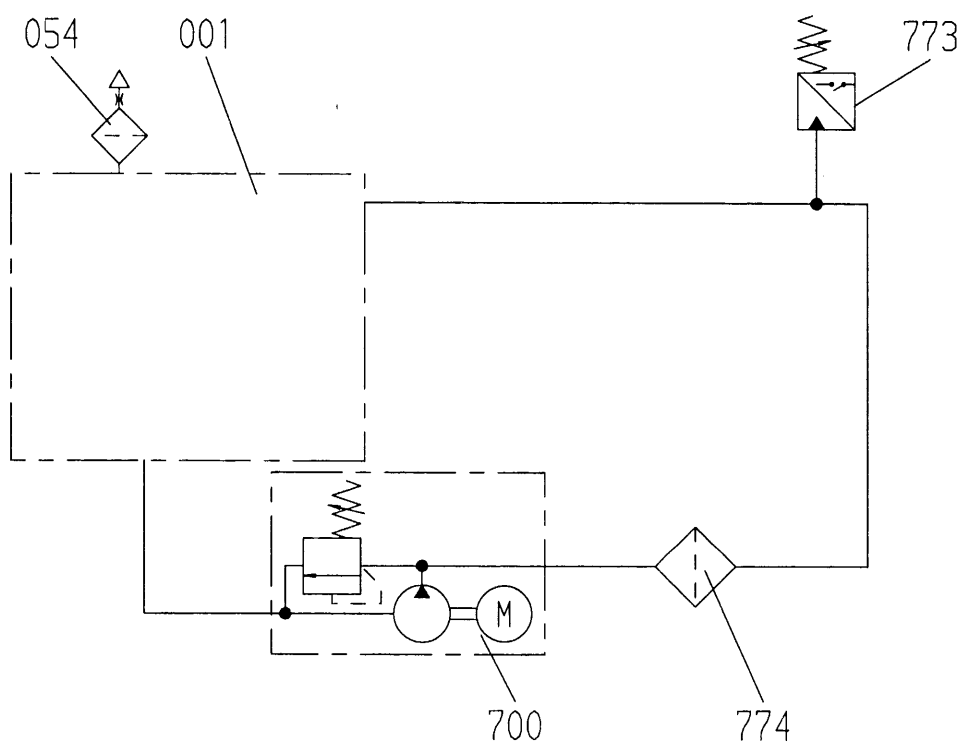
SERIAL NO.	FABRIK-NR.	412 3221-040-1...1
TYPE	BAUART	B2SV
SIZE	GRÖÖE	07
DESIGN	AUSFÖHRUNG	C
POWER RATING	LEISTUNG	P1/P2 200/170 kW
INPUT SPEED	ANTRIEBSDREHZAH	n1 510-1500 1/min
OUTPUT SPEED	ABTRIEBSDREHZAH	n2 59,5-175 1/min
EXACT RATIO	ÜBERSETZUNG	I ist 8,569
OIL QUANTITY	ÖLMENGE	W 35 L
OIL VISCOSITY	ÖL-VISKOSITÄT	ISO VG 320 PG-Öel (synth. Öl)

Tolerierung nach DIN ISO 8015		Allgemeintoleranzen <30 ±0,3 sonst nach DIN ISO 2768-mH	
FLENDER		Werkstoff /-Nr.	Benennung
Nicht gekennzeichnete Oberflächen:		Gez. ZG 2004-06-10 Kästner	Schl.-Nr.: 0290
Rauwerte Ra in µm		Gepr. ZG 2004-06-10 Bothendorf	MASSZCHNG B2SV 07
Abt. Datum		DIMENSION DRAWING	
Maßstab 1:5,5		Rauteil nach Zeichnung sonst nach Zeichnung	
Trägheitsmoment		Format Art Zeichnungs-Nr.	
Gewicht (kg) 615		6 M 6 063 183	
Ersatz für		Aus Zeichn.	
Schutzvermerk DIN 34 beachten			

PLOTT-DATUM: 02-SEP-2004 16:03:31 H. Dietze, Harald/PZ

The purchaser is responsible for the provision of safety guards.	Parallel keyway(s) to DIN 6885-1, Light seating and centering(s) to DIN 332, shape D5	Arrows show relative directions of rotation	Cylindricity tolerance for shafts and bores IT6 to DIN ISO 286
Umlaufende Teile müssen vom Käufer gegen unbeabsichtigtes Berühren gesichert werden! (Gesetz über technische Arbeitsmittel)	Passfedernut(en) nach DIN 6885-1, Leichter Sitz und Zentrierung(en) nach DIN 332, Form D5	Gleichartige Pfeile kennzeichnen die Abhängigkeit der Drehrichtungen!	Zylinderformtoleranz für Wellen und Bohrungen IT6 nach DIN ISO 286

001	Getriebe B2SV07 / GEAR UNIT
054	Entlüfter / BREATHER
700	Pumpengruppe SF2/8 / MOTORPUMP
724	Grobfilter / FILTER
773	Druckwächter / PRESSURE MONITOR



KENNWORT/CODE: NARAYANGONJ

2004-03-23

4123221/40

Tolerierung nach DIN ISO 8015

Allgemeintoleranzen $\leq 30 \pm 0,3$ sonst nach DIN ISO 2768-mH

FLENDER

1417

SCHEMA-HYDR-SCHMIER

Lubrication diagram

ZG 2004-06-15 Bothendorf

ZG 2004-06-15 Bothendorf

ZG . . .

1:1

9

6 048 277

4 Maße in mm

PLOTT-DATUM: 15-JUN-2004 08:15:38 H. Hoefler/PZ8

/PE

LOESCHE**MOTORDATENBLATT**

Kennwort: NARAYANGONJ

Zchg. Nr.:

Name:

Datum: 2004-06-15

Seite: 1 von 2

Dateiname: SF2-

Änderung	Datum	Name	Index

Umgebungstemperatur : °C
Aufstellungshöhe : m über NN

Mechanische Daten:

Hersteller : BIRKENBEUL
Typ : 5AP80-4
Baugröße : 80
Bauform : B5/200
Nennleistung : 0,75 kW
Nenn Drehzahl : 1395 min⁻¹
Nennmoment : 5,1 Nm
Anzugsmoment : 2,5 -faches Nennmoment
Kippmoment : 2,6 -faches Nennmoment
Trägheitsmoment : 0.0018 kgm²
Wirkungsgrad : 74 %

Einschaltung:

direkt ☒ / Stern-Dreieck ☐ / Anlasser ☐
zulässige Einschalthäufigkeit :
zulässige Hochlaufzeit :

Elektrische Daten:

Nennspannung : 380...420 V
Nennstrom : 2 A
Anzugsstrom : 5 -facher Nennstrom
Nennfrequenz : 50 Hz
Leistungsfaktor : 0,79
Schutzart : IP 55
Läufer :
Läuferstillstandsspannung : V
Läuferstrom : A
Isolationsklasse : Läufer: Ständer:F
Tropenfeste Isolierung : ja ☐ / nein ☒

Allgemeine Daten:

Kühlung : IC 411
Drehrichtung *) : links ☐ / rechts ☒ / beliebig ☐
Klemmenkasten *) : links ☐ / rechts ☐ / oben ☒
Maßbild :
Gewicht : 9,4 Kg *) Auf Welle gesehen

Überwachungen:

Wicklungstemperatur : Kaltleiter ☐ / PT 100 ☐ / nein ☒
Schalttemperatur Kaltleiter : °C °C
Lagertemperatur : ja ☐ / nein ☒
Drehzahlüberwachung : ja ☐ / nein ☒
Initiator -Hersteller :
-Typ :
Nockenzahl :

Lieferant: FLENDER

Datum: 15.06.04

Unterschrift:



MOTORDATENBLATT

Kennwort: NARAYANGONJ

Zchg. Nr.:

Name:

Datum: 2004-06-15

Seite: 2 von 2

Dateiname: SF2-

Änderung	Datum	Name	Index

Operating Instructions

BA 5011 EN 08.98

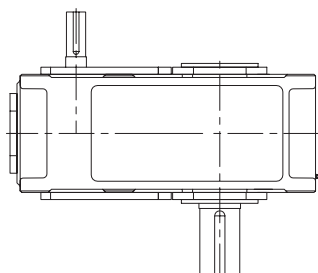
Gear Unit Types

H.SV, H.VV, H.HV, H.KV, H.DV, H.FV

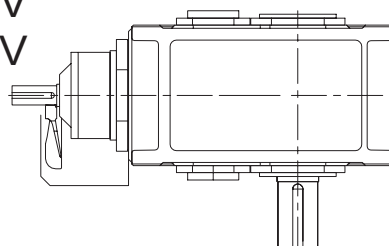
B.SV, B.VV, B.HV, B.KV, B.DV, B.FV

Sizes 1 to 26

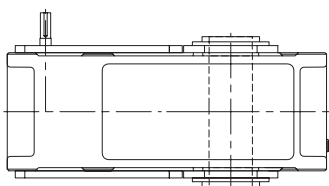
H.SV
H.VV



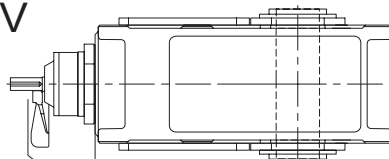
B.SV
B.VV



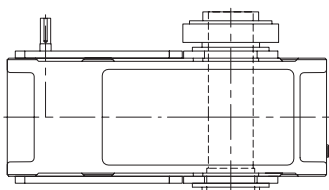
H.HV
H.KV



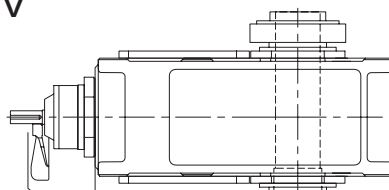
B.HV
B.KV



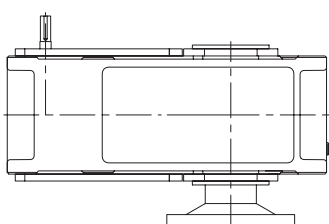
H.DV



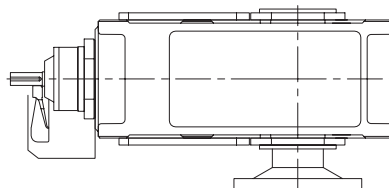
B.DV



H.FV



B.FV



FLENDER

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1. Technical data

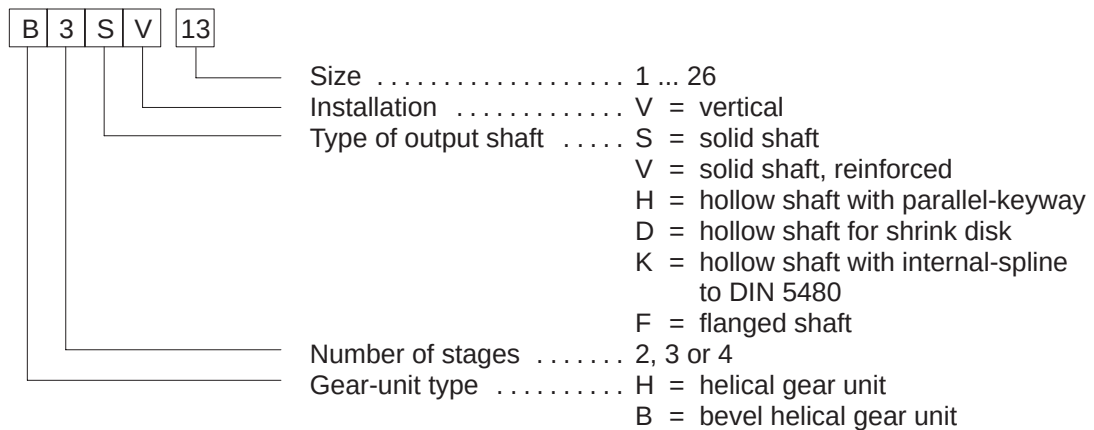
1.1 General technical data

The most important technical data are shown on the rating plate. These data along with the contractual agreements on the gear unit determine the limits of its proper use.

①		
②		
③		
④	⑤	
⑥	⑦	
⑧	⑨	⑩
⑪		
⑫		

- | | |
|---|---|
| <p>① Company logo and production location</p> <p>② Special information</p> <p>③ Order no. - item - serial no.</p> <p>④ Type / Size *)</p> <p>⑤ Power rating P in kW or T₂ in Nm</p> <p>⑥ Speed n₁</p> | <p>⑦ Speed n₂</p> <p>⑧ Type of oil</p> <p>⑨ Viscosity of oil in VG class</p> <p>⑩ Quantity of oil in litres for main gear housing</p> <p>⑪ Number of operating instruction(s)</p> <p>⑫ Special information</p> |
|---|---|

e.g. *)



Data on weights and measuring surface sound levels of the various gear types are given in sections 1.1.1 and 1.1.2.

For further technical details, refer to the drawings in the gear-unit documentation.

1.1.1 Weights

Type	Approx. weight (kg) for size											
	1	2	3	4	5	6	7	8	9	10	11	12
H2.V	–	–	115	190	300	355	505	590	830	960	1335	1615
H3.V	–	–	–	–	320	365	540	625	875	1020	1400	1675
H4.V	–	–	–	–	–	–	550	645	875	1010	1460	1725
B2.V	65	90	140	235	360	410	615	700	1000	1155	1640	1910
B3.V	–	–	130	210	325	380	550	635	890	1020	1455	1730
B4.V	–	–	–	–	335	385	555	655	890	1025	1485	1750

Type	Approx. weight (kg) for size													
	13	14	15	16	17	18	19	20	21	22	23	24	25	26
H2.V	1880	2430	3240	3465	4420	4870	5000	6150	6950	7550	10000	11350	12600	14400
H3.V	2155	2490	3260	3625	4250	4740	4750	6250	6550	7050	8400	8750	11500	13200
H4.V	2270	2600	3440	3740	4445	4915	5300	5950	7250	7750	9350	10450	12800	14700
B2.V	2350	2725	3795	4160	5320	5860	–	–	–	–	–	–	–	–
B3.V	2260	2615	3540	3765	4760	5240	6050	6710	8190	8950	10500	12000	14800	16900
B4.V	2280	2605	3435	3765	4460	4930	5400	6000	7350	7850	9400	10500	12900	14800

Table 1.1: Weights (approx. values only)

Note: All weights are for gear unit without oil charge or attachments. For the exact weights, refer to the drawings in the gear-unit documentation.

1.1.2 Measuring-surface sound level

The measuring-surface sound level for the gear unit at a distance of 1 m can be found in tables 1.1.2.1 to 1.1.2.4.

The measurement is carried out to DIN 45635 parts 1 and 23 using the sound-intensity method.

The workplace of the operating personnel is defined as the area on the measuring surface at a distance of 1 metre, in whose vicinity persons may be present.

The sound level applies for the warmed-up gear unit at input speed n_1 and input power P_1 stated on the rating plate. If several figures are given, the highest speed and power values apply.

The measuring-surface sound level includes attached lubrication units (if applicable). With outgoing and ingoing pipes, the interfaces are the flanges.

If repeat measurements on site do not produce conclusive results with regard to measuring technology, the measurement achieved on the FLENDER test bench will apply.

The sound level stated in the table was obtained by statistical calculation by our Quality Control Dept. The gear unit may be expected to comply with these sound levels with statistical probability.

1.1.2.1 Measuring-surface sound level for bevel helical-gear units (B..V) with fan

Measuring-surface sound level L _{pA} in dB(A) for bevel helical-gear units with fan																												
Type	i _N	n ₁ 1/min	Gear-unit size																									
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
B2	5	3000	79	83	85	89	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	.	1500	65	70	73	76	79	81	83	84	85	87	88	89	91	92	94	—	—	—	—	—	—	—	—	—	—	—
	.	1000	1)	1)	67	71	73	74	77	78	79	80	82	83	84	85	87	89	90	—	—	—	—	—	—	—	—	—
	8	750	1)	1)	61	64	66	67	70	71	72	73	75	76	77	78	81	82	83	85	—	—	—	—	—	—	—	—
	9	3000	79	81	83	87	89	90	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	65	67	70	73	75	76	78	81	82	83	84	85	86	87	88	90	—	—	—	—	—	—	—	—	—	
	.	1000	1)	61	63	67	68	70	73	74	75	77	79	80	81	82	83	84	86	87	—	—	—	—	—	—	—	
	14	750	1)	1)	1)	61	62	64	66	67	68	70	72	73	74	75	77	78	79	80	—	—	—	—	—	—	—	—
	16	3000	77	79	81	85	88	89	90	91	92	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	63	65	67	71	74	76	78	79	80	81	83	84	87	88	89	90	—	—	—	—	—	—	—	—	—	
	.	1000	1)	1)	60	64	67	68	70	72	73	74	78	79	80	81	82	83	84	84	—	—	—	—	—	—	—	—
	22.4	750	1)	1)	1)	1)	61	63	65	67	68	69	71	72	73	73	74	74	75	76	—	—	—	—	—	—	—	—
B3	12.5	3000	—	—	82	86	87	88	90	92	93	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	—	—	69	72	75	77	79	80	81	82	83	85	88	89	90	91	93	93	93	93	95	95	—	—	—	
	.	1000	—	—	62	65	68	69	71	72	73	74	77	78	80	82	83	83	84	85	86	86	88	88	89	90	91	
	31.5	750	—	—	1)	1)	63	64	66	68	69	70	71	73	74	75	76	77	78	78	79	79	81	81	82	84	85	
	35.5	3000	—	—	81	83	85	86	87	88	90	92	95	96	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	—	—	67	69	72	73	74	75	77	79	82	84	86	87	88	89	90	91	92	92	93	93	—	—	—	
	.	1000	—	—	1)	63	65	66	67	69	71	72	73	75	77	78	79	80	81	82	83	84	85	86	86	87	88	
	56	750	—	—	1)	1)	1)	62	64	65	67	69	70	71	72	73	74	75	76	77	78	79	79	80	81	82	82	
	63	3000	—	—	80	82	84	85	87	88	90	92	94	95	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	—	—	66	68	70	71	73	74	76	78	81	83	85	86	87	88	89	90	91	91	92	92	—	—	—	
	.	1000	—	—	1)	61	63	64	66	68	69	71	73	75	77	78	79	80	81	81	82	82	83	84	84	85	86	
	90	750	—	—	1)	1)	1)	61	63	64	66	67	68	70	71	72	73	74	75	75	76	77	77	78	79	79	80	

1) $L_{pA} < 60$ dB(A)

1.1.2.2 Measuring-surface sound level for bevel helical-gear units (B..V) without fan

Measuring-surface sound level L _{pA} in dB(A) for bevel helical-gear units without fan																												
Type	i _N	n ₁ 1/min	Gear-unit size																									
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
B2	5	3000	75	81	84	88	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	.	1500	63	69	72	75	78	80	82	83	84	86	87	88	89	90	93	—	—	—	—	—	—	—	—	—	—	—
	.	1000	1)	1)	66	70	72	73	76	77	78	79	81	82	83	84	86	88	89	—	—	—	—	—	—	—	—	—
	8	750	1)	1)	1)	63	65	66	69	71	72	73	74	75	77	78	80	82	83	84	—	—	—	—	—	—	—	—
	9	3000	73	77	80	83	86	87	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	61	64	67	71	74	75	77	79	80	81	83	84	85	86	87	89	—	—	—	—	—	—	—	—	—	
	.	1000	1)	1)	61	65	67	69	72	73	74	76	77	78	80	81	82	83	85	86	—	—	—	—	—	—	—	
	14	750	1)	1)	1)	1)	60	63	65	66	67	69	71	72	73	74	76	77	78	79	—	—	—	—	—	—	—	
	16	3000	69	72	76	79	81	83	85	86	87	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	1)	60	63	66	69	71	72	74	75	77	78	80	81	82	85	85	—	—	—	—	—	—	—	—	—	
	.	1000	1)	1)	1)	61	63	65	67	68	69	71	72	74	75	77	79	80	81	81	—	—	—	—	—	—	—	
	22.4	750	1)	1)	1)	1)	1)	1)	60	62	63	64	66	67	68	70	72	73	74	75	—	—	—	—	—	—	—	
B3	12.5	3000	—	—	77	81	84	86	87	88	90	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	—	—	65	68	71	74	75	76	77	79	81	83	84	85	86	87	87	88	89	90	91	92	—	—	—	
	.	1000	—	—	1)	63	66	68	69	70	72	73	75	77	78	80	80	81	82	82	84	85	86	86	88	89	90	—
	31.5	750	—	—	1)	1)	1)	61	62	64	65	66	68	71	71	73	73	74	75	75	77	78	79	79	81	82	83	84
	35.5	3000	—	—	72	77	80	82	83	84	84	86	89	92	—	—	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	—	—	60	65	67	70	71	71	72	74	77	79	80	81	82	83	83	84	86	86	88	88	—	—	—	
	.	1000	—	—	1)	1)	62	65	65	66	66	69	71	73	75	76	76	77	77	78	80	81	82	83	84	85	86	87
	56	750	—	—	1)	1)	1)	1)	1)	1)	1)	62	65	67	68	69	70	70	71	72	74	74	75	76	78	80	80	80
	63	3000	—	—	69	73	76	84	80	80	81	83	84	88	89	90	—	—	—	—	—	—	—	—	—	—	—	
	.	1500	—	—	1)	61	64	70	67	68	68	70	73	75	76	78	78	79	79	80	82	83	84	84	—	—	—	
	.	1000	—	—	1)	1)	1)	63	62	62	62	65	68	70	71	72	73	73	74	75	76	77	78	79	81	81	83	83
	90	750	—	—	1)	1)	1)	1)	1)	1)	1)	61	63	64	65	66	67	67	68	70	70	72	72	74	75	76	76	
B4	80	3000	—	—	—	—	76	77	79	81	82	85	87	89	90	91	92	—	—	—	—	—	—	—	—	—	—	
	.	1500	—	—	—	—	64	65	67	68	70	72	75	76	77	79	80	81	82	83	84	85	86	86	87	88	87	89
	.	1000	—	—	—	—	1)	1)	61	63	64	67	69	70	72	73	74	75	76	77	78	79	80	80	81	82	82	83
	125	750	—	—	—	—	1)	1)	1)	1)	1)	1)	62	64	65	66	68	68	69	71	71	72	73	74	75	75	76	
	140	3000	—	—	—	—	72	74	76	77	78	81	84	85	86	87	89	89	90	92	—	—	—	—	—	—	—	
	.	1500	—	—	—	—	60	61	63	65	66	68	71	72	73	75	76	77	78	79	80	81	82	82	83	84	85	86
	.	1000	—	—	—	—	1)	1)	1)	1)	61	63	65	67	68	69	71	71	72	74	75	75	76	77	78	78	79	80
	224	750	—	—	—	—	1)	1)	1)	1)	1)	1)	1)	1)	61	62	64	65	66	67	68	69	69	70	71	72	73	73
	250	3000	—	—	—	—	69	70	72	74	75	77	80	81	82	84	85	86	87	88	89	90	91	—	—	—	—	
	.	1500	—	—	—	—	1)	1)	1)	62	63	65	67	69	70	71	73	73	75	76	77	77	78	79	80	80	81	82
	.	1000	—	—	—	—	1)	1)	1)	1)	1)	1)	62	63	64	66	67	68	69	70	71	72	73	73	74	75	76	77
	400	750	—	—	—	—	1)	1)	1)	1)	1)	1)	1)	1)	1)	1)	1)	61	62	63	64	65	66	66	67	68	69	70

1) $L_{pA} < 60$ dB(A)

1.1.2.3 Measuring-surface sound level for helical-gear units (H..V) with fan

Measuring-surface sound level L _{pA} in dB(A) for helical-gear units with fan																												
Type	i _N	n ₁ 1/min	Gear-unit size																									
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
H2	6.3	3000	–	–	–	85	87	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
		1500	–	–	–	75	76	77	80	81	82	84	85	86	88	90	92	94	96	96	–	–	–	–	–	–	–	–
		1000	–	–	–	69	71	72	74	75	77	79	80	81	83	84	85	86	87	88	88	89	90	–	–	–	–	–
	10	750	–	–	–	66	68	69	70	72	73	75	76	77	79	80	81	82	83	83	84	84	85	85	86	–	–	–
		11.2	3000	–	–	–	84	86	87	91	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
			1500	–	–	–	73	75	77	79	80	81	82	85	88	90	91	92	93	95	95	–	–	–	–	–	–	–
	1000		–	–	–	68	69	70	72	73	75	77	79	80	82	83	84	85	85	86	86	87	87	87	88	88	–	–
	16	750	–	–	–	64	66	67	69	70	71	73	74	76	78	79	79	80	81	81	82	82	83	83	83	84	84	85
		18	3000	–	–	–	83	84	85	88	90	92	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
			1500	–	–	–	71	73	75	77	78	80	82	84	86	87	90	91	92	93	94	94	95	95	95	–	–	–
	1000		–	–	–	65	67	68	71	72	73	75	77	78	80	81	82	83	83	84	85	85	86	86	86	87	87	87
	28	750	–	–	–	62	64	65	67	68	69	71	73	74	75	77	78	79	79	80	80	81	81	81	82	82	82	83
H3		22.4	3000	–	–	–	–	84	84	87	87	90	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
			1500	–	–	–	–	71	72	75	75	77	77	80	80	81	81	84	84	84	85	–	–	–	–	–	–	–
	1000		–	–	–	–	65	66	69	70	71	72	74	75	75	75	78	78	78	79	–	–	–	–	–	–	–	
	35.5	750	–	–	–	–	62	62	66	67	67	68	70	70	71	72	74	74	75	76	–	–	–	–	–	–	–	
		40	3000	–	–	–	–	84	84	86	86	89	89	92	93	–	–	–	–	–	–	–	–	–	–	–	–	–
			1500	–	–	–	–	70	71	73	74	76	76	79	79	80	80	83	82	83	83	–	–	–	–	–	–	–
	1000		–	–	–	–	64	65	67	68	69	70	73	73	73	74	77	77	77	77	–	–	–	–	–	–	–	
	63	750	–	–	–	–	62	62	63	64	65	66	69	69	69	70	72	73	73	73	–	–	–	–	–	–	–	
		71	3000	–	–	–	–	83	83	85	85	89	89	92	92	92	92	–	–	–	–	–	–	–	–	–	–	–
			1500	–	–	–	–	70	70	72	72	75	75	78	78	78	78	82	82	82	82	–	–	–	–	–	–	–
	1000		–	–	–	–	64	64	65	66	68	69	71	72	72	72	75	75	75	76	–	–	–	–	–	–	–	
	112	750	–	–	–	–	61	61	62	62	64	65	67	67	68	68	71	71	71	72	–	–	–	–	–	–	–	

1) $L_{pA} < 60$ dB(A)

1.1.2.4 Measuring-surface sound level for helical-gear units (H..V) without fan

Measuring-surface sound level L _{pA} in dB(A) for helical-gear units without fan																												
Type	i _N	n ₁ 1/min	Gear-unit size																									
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
H2	6.3	3000	—	—	79	80	83	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		1500	—	—	70	71	74	75	76	77	79	79	80	81	81	82	84	85	85	86	—	—	—	—	—	—	—	—
		1000	—	—	65	66	69	70	71	72	74	74	75	76	76	77	80	80	80	81	83	83	84	—	—	—	—	—
		750	—	—	62	63	66	67	67	69	70	71	72	73	73	74	76	77	77	78	80	80	81	81	81	—	—	—
	11.2	3000	—	—	77	78	81	82	83	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		1500	—	—	68	69	72	73	74	75	77	77	78	79	79	80	82	83	83	84	—	—	—	—	—	—	—	—
		1000	—	—	63	64	67	68	69	70	72	72	73	74	74	75	77	78	78	79	81	81	82	82	83	83	—	—
		750	—	—	1)	61	64	65	66	67	69	69	70	71	71	72	74	75	75	76	77	78	79	79	79	80	81	81
	16	3000	—	—	75	75	78	79	80	81	83	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		1500	—	—	66	66	69	70	71	72	74	74	75	76	77	78	80	80	81	82	83	84	84	85	—	—	—	—
		1000	—	—	60	61	64	65	66	68	69	69	70	71	72	73	75	75	76	77	78	79	79	80	80	81	81	82
		750	—	—	1)	1)	61	62	63	64	66	66	77	68	69	70	72	72	73	73	75	75	76	76	77	77	78	78
18	3000	—	—	75	75	78	79	80	81	83	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
	1500	—	—	66	66	69	70	71	72	74	74	75	76	77	78	80	80	81	82	83	84	84	85	—	—	—	—	
	1000	—	—	60	61	64	65	66	68	69	69	70	71	72	73	75	75	76	77	78	79	79	80	80	81	81	82	
	750	—	—	1)	1)	61	62	63	64	66	66	77	68	69	70	72	72	73	73	75	75	76	76	77	77	78	78	
H3	22.4	3000	—	—	—	—	77	79	82	81	83	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	
		1500	—	—	—	—	68	69	73	74	74	75	77	77	78	79	81	81	82	83	83	84	85	86	86	87	—	—
		1000	—	—	—	—	63	65	68	69	69	71	72	73	73	74	76	77	77	78	79	79	81	81	81	82	83	83
		750	—	—	—	—	60	61	65	66	65	67	69	69	70	71	73	73	74	75	75	76	77	78	78	79	79	80
	35.5	3000	—	—	—	—	74	76	79	80	80	82	83	84	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		1500	—	—	—	—	65	67	70	71	71	73	74	75	76	76	78	79	79	80	81	81	83	83	84	84	—	—
		1000	—	—	—	—	1)	62	65	66	66	68	69	70	71	72	73	74	75	75	76	77	78	78	79	79	80	80
		750	—	—	—	—	1)	1)	62	63	63	65	66	67	67	68	70	71	71	72	73	73	75	75	76	76	77	77
	40	3000	—	—	—	—	74	76	79	80	80	82	83	84	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		1500	—	—	—	—	65	67	70	71	71	73	74	75	76	76	78	79	79	80	81	81	83	83	84	84	—	—
		1000	—	—	—	—	1)	62	65	66	66	68	69	70	71	72	73	74	75	75	76	77	78	78	79	79	80	80
		750	—	—	—	—	1)	1)	62	63	63	65	66	67	67	68	70	71	71	72	73	73	75	75	76	76	77	77
63	3000	—	—	—	—	71	73	76	77	77	79	80	81	82	83	—	—	—	—	—	—	—	—	—	—	—	—	
	1500	—	—	—	—	62	64	67	68	68	70	71	72	73	74	76	76	77	78	78	79	80	81	81	82	82	83	
	1000	—	—	—	—	1)	1)	62	63	63	65	66	67	68	69	71	71	72	73	73	74	75	76	76	77	77	78	
	750	—	—	—	—	1)	1)	1)	1)	1)	62	63	64	65	66	68	68	69	70	70	71	72	72	73	73	74	75	
H4	100	3000	—	—	—	—	—	75	76	77	78	79	80	81	82	84	84	—	—	—	—	—	—	—	—	—	—	
		1500	—	—	—	—	—	66	67	68	69	70	71	72	73	75	75	76	76	77	78	78	78	79	80	81	81	
		1000	—	—	—	—	—	62	63	63	64	65	66	67	68	70	70	71	72	72	73	73	74	74	76	76	77	
		750	—	—	—	—	—	1)	1)	1)	61	62	63	64	64	66	67	68	68	69	69	70	70	71	72	73	73	
	140	3000	—	—	—	—	—	73	74	75	75	77	78	78	79	81	82	82	83	—	—	—	—	—	—	—	—	—
		1500	—	—	—	—	—	64	65	66	66	68	68	69	70	72	73	73	74	74	75	75	76	76	78	78	79	
		1000	—	—	—	—	—	1)	60	61	62	63	64	64	65	67	68	68	69	70	70	71	71	72	73	74	74	
		750	—	—	—	—	—	1)	1)	1)	1)	60	61	61	62	64	64	65	66	66	67	67	68	68	70	70	71	
	160	3000	—	—	—	—	—	70	71	72	73	74	75	76	77	78	79	79	80	81	82	82	82	—	—	—	—	—
		1500	—	—	—	—	—	61	62	63	64	65	66	67	67	69	70	70	71	72	72	73	73	74	75	76	76	
		1000	—	—	—	—	—	1)	1)	1)	1)	60	61	62	63	64	65	66	66	67	68	68	68	69	70	71	72	
		750	—	—	—	—	—	1)	1)	1)	1)	1)	1)	1)	1)	61	62	62	63	64	64	55	65	65	67	68	68	
250	3000	—	—	—	—	—	70	71	72	73	74	75	76	77	78	79	79	80	81	82	82	82	—	—	—	—	—	
	1500	—	—	—	—	—	61	62	63	64	65	66	67	67	69	70	70	71	72	72	73	73	74	75	76	76		
	1000	—	—	—	—	—	1)	1)	1)	1)	60	61	62	63	64	65	66	66	67	68	68	68	69	70	71	72		
	750	—	—	—	—	—	1)	1)	1)	1)	1)	1)	1)	1)	61	62	62	63	64	64	55	65	65	67	68	68		
280	3000	—	—	—	—	—	70	71	72	73	74	75	76	77	78	79	79	80	81	82	82	82	—	—	—	—	—	
	1500	—	—	—	—	—	61	62	63	64	65	66	67	67	69	70	70	71	72	72	73	73	74	75	76	76		
	1000	—	—	—	—	—	1)	1)	1)	1)	60	61	62	63	64	65	66	66	67	68	68	68	69	70	71	72		
	750	—	—	—	—	—	1)	1)	1)	1)	1)	1)	1)	1)	61	62	62	63	64	64	55	65	65	67	68	68		
450	3000	—	—	—	—	—	70	71	72	73	74	75	76	77	78	79	79	80	81	82	82	82	—	—	—	—	—	
	1500	—	—	—	—	—	61	62	63	64	65	66	67	67	69	70	70	71	72	72	73	73	74	75	76	76		
	1000	—	—	—	—	—	1)	1)	1)	1)	60	61	62	63	64	65	66	66	67	68	68	68	69	70	71	72		
	750	—	—	—	—	—	1)	1)	1)	1)	1)	1)	1)	1)	61	62	62	63	64	64	55	65	65	67	68	68		

1) $L_{pA} < 60$ dB(A)

2. General notes

2.1 Introduction

These Operating Instructions (BA) are an integral part of the gear unit supplied and must be kept in its vicinity for reference at all times.

Caution!

All persons involved in the installation, operation, maintenance and repair of the gear unit must have read and understood these Operating Instructions and must comply with them at all times. We accept no responsibility for damage or disruption caused by disregard of these Instructions.

The “**FLENDER gear unit**” described in these Instructions has been developed for stationary use in general engineering applications. Possible areas of use for gear units of this type are (e.g.) sewage treatment, excavators, chemical industry, iron and steel industry, conveyor equipment, crane equipment, foodstuffs industry, paper machinery, cableways, cement industry etc.

The gear unit is designed only for the application described in section 1 “Technical data”. Possible use in any other area must be contractually agreed.

The gear unit described in these Instructions reflects the state of technical development at the time these Instructions went to print.

In the interest of technical progress, we reserve the right to make changes to the individual assemblies and accessories which we regard as necessary to preserve their essential characteristics and improve their efficiency and safety.

2.2 Copyright

The copyright to these Operating Instructions is held by **FLENDER AG**.

These Operating Instructions must not be wholly or partly reproduced for competitive purposes, used in any unauthorised way, or made available to third parties without our agreement.

Technical enquiries should be addressed to the following plants:

for gear sizes **1 ... 12**

A. FRIEDR. FLENDER AG
Getriebewerk Penig
Thierbacher Straße 24
D-09322 Penig

Tel.: 037381/60
Fax: 037381/80286

for gear sizes **13 ... 26**

A. FRIEDR. FLENDER AG

D-46393 Bocholt

Tel.: 02871/92-0
Fax: 02871/92-2596

or to our customer-service addresses. A list of our customer-service addresses is given in section 11 “Spare parts, customer-service addresses”.

3. Safety notes

3.1 Proper use

- The gear unit has been manufactured in accordance with the state of the art and is delivered in a condition for safe and reliable use. All changes to the gear unit on the part of the user which may affect its safety and reliability are prohibited. This applies equally to safety features designed to prevent accidental contact.
- The gear unit should be used and operated only within the context of the conditions laid down in the contract governing performance and supply.

3.2 Obligations of the user

- The user must ensure that all persons involved in the installation, operation, maintenance and repair of the gear unit have read and understood these Operating Instructions and comply with them at all times in order to:

- avoid injury or damage,
- ensure the safety and reliability of the unit,

and

- avoid disruptions and environmental damage through incorrect use.
- During transport, assembly, installation, dismantling, operation and maintenance of the unit, the relevant safety and environmental regulations must be complied with at all times.
- The gear unit should be operated, maintained or repaired by authorised, trained and qualified personnel.
- The gear unit must not be cleaned using high-pressure cleaning equipment.
- All work on the gear unit must be carried out with great care and with due regard to safety.
- All work on the gear unit must be carried out only when it is not in operation.
The drive assembly must be secured against being switched on accidentally (e.g. by locking the key switch or removing the fuses from the power supply). A notice should be attached to the ON switch stating clearly that work is in progress.
- No welding work should be done on the gear unit. The gear unit must not be used as an earthing point for welding operations. Delicate teeth and bearings may be irreparably damaged by welding.
- If any changes are noticed during operation (e.g. overheating or unusual noises), the drive assembly must be switched off immediately.
- Rotating drive components such as couplings, gearwheels, belt drives etc. must be protected against accidental contact.
- If the gear unit is intended for installation in plant or machinery, the manufacturer of such plant or machinery must ensure that the contents of these Instructions are incorporated in his own instructions.
- Notices attached to the gear unit, e.g. rating plate, direction arrows etc. must always be observed. They must be kept free from dirt and paint at all times.
- All spare parts must be obtained from FLENDER.

3.3 Environmental protection

- When changing oil, the used oil must be collected in suitable containers. Any spillage of oil must be removed immediately.
- Conservation agent should be stored separately from used oil.
- Used oil, conservation agent, oil-binding agents and oil-soaked cloths must be disposed of in accordance with environmental legislation.

3.4 Special dangers

- Depending on operating conditions, the surface of the gear unit may heat up considerably.
Danger of burns!
- When changing oil, take care to prevent scalding by hot oil.

3.5 Warnings and symbols used in these Instructions



This symbol indicates safety measures which must be observed to avoid **personal injury**.

Caution!

This symbol indicates safety measures which must be observed to avoid **damage to the gear unit**.

Note:

This symbol indicates general **operating procedures** which are of particular importance.

4. Handling and storage

Note: Observe the "Safety notes" in section 3.

4.1 Scope of supply

The products supplied are listed in the dispatch papers. Check immediately on receipt to ensure that all the products listed have actually been delivered. Damaged or missing parts must be notified in writing immediately.

4.2 Handling

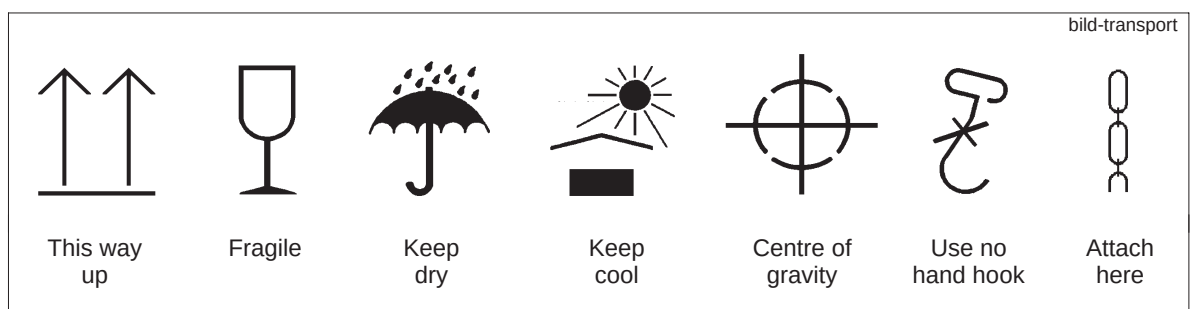


When handling FLENDER products use only lifting and handling equipment of sufficient load-carrying capacity!

The gear unit is delivered in the fully assembled state. Additional items (e.g. oil cooler, pipes and fittings) are delivered separately packaged.

Different forms of packaging may be used depending on the size of the unit and method of transport. Unless otherwise agreed, the packaging complies with the **HPE Packaging Guidelines**.

The symbols marked on the packaging must be observed at all times. These have the following meanings:



Caution!

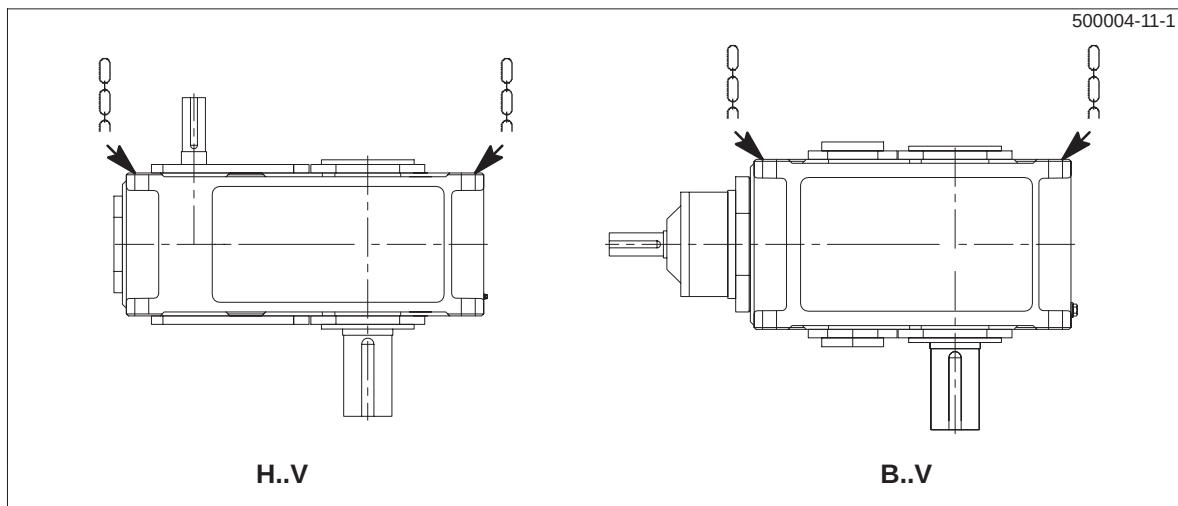
The gear unit must always be transported with due care to avoid danger to persons and property. Avoid bumping the ends of shafts as this may damage the gear unit.

Note:

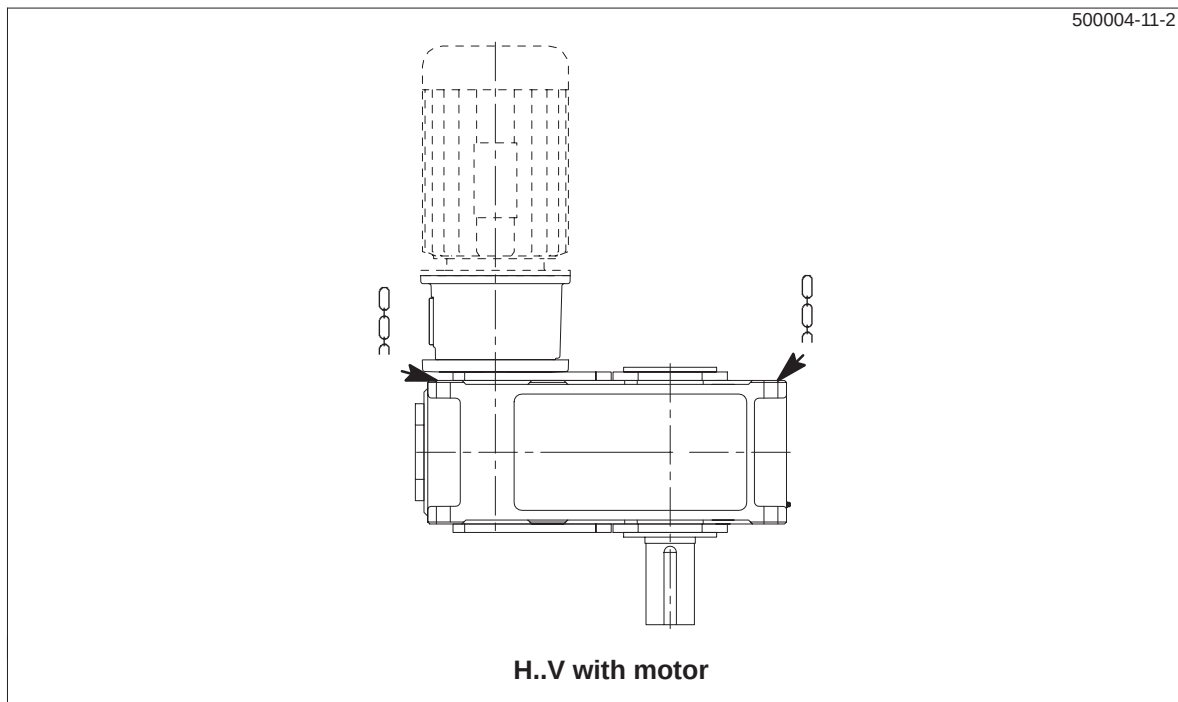
The gear unit must be transported using suitable equipment only. Never transport the gear unit with an oil charge.

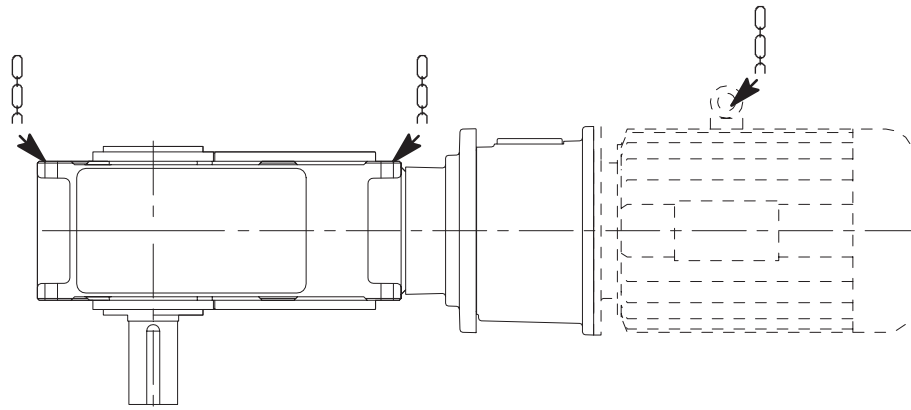
Caution!

Use only the 4 eyes provided for attaching lifting equipment to the unit. Do not use the front thread at the end of the shaft for attaching ring bolts for transport purposes.

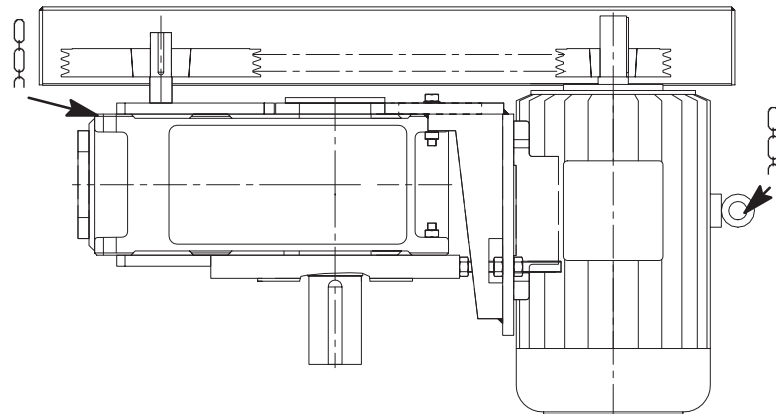


For drive units, where additional components (e.g. motor, coupling etc.) are mounted on the gear unit, an additional attachment point may be required because of the shift in the centre of gravity.





B..V with motor



B..with motor bedplate

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

The position of the attachment points is shown in the documentation for each specific order.

4.3 Storage

The gear unit must be stored in the position of use in a sheltered place. It must be placed on a vibration-free wooden base and covered over.



Do not stack gear units on top of one another.

Caution!

If the gear unit is being stored out of doors, it must be covered with particular care. It must be ensured that neither moisture nor foreign material can collect on the unit.

Note: Provision for special environmental conditions during transport (e.g. transport by ship) and storage (climate, termites etc.) must be contractually agreed.

4.4 Standard corrosion protection

The gear unit is filled with conservation agent. The free shaft ends are painted for protection.

The outer paint coating is resistant to dilute acids, alkalis, oils and solvents. It is also resistant to sea water, tropical-climate conditions and temperatures up to 140 °C.

Note: Ensure that the paint is not damaged!
Mechanical damage (scratches), chemical damage (acids, alkalis) and thermal damage (sparks, welding beads, heat) cause corrosion which may cause failure of the exterior protective coating.

Note: Unless otherwise agreed, we grant a guarantee of 24 months on the interior corrosion protection, and 12 months on the protective coating on the free shaft ends. The guarantee period commences on the date of delivery.

For longer periods of storage (> 24 months) we advise regular checking and, if necessary, renewal of the interior corrosion protection (see section 7 "Start-up").

5. Technical description

Note: Observe the "Safety notes" in section 3.

5.1 General description

The gear unit is supplied as a two-, three- or four-stage helical or bevel-helical gear unit. It is designed for installation in a vertical position. If necessary, it can also be designed for installation in a different position.

It can be operated in both directions of rotation.

A number of shaft configurations (types and rotation directions) are possible. These are shown in the following table as solid shafts:

Type	Configuration ¹⁾			
	A	B	C	D
H2SV H2HV H2DV H2KV H2FV				
H3SV H3HV H3DV H3KV H3FV				
H4SV H4HV H4DV H4KV H4FV				
B2SV B2HV B2DV B2KV B2FV				
B3SV B3HV B3DV B3KV B3FV				
B4SV B4HV B4DV B4KV B4FV				

1) If the input and output drive shafts are on both sides, refer to the drawings in the gear documentation for the type designation and the directions of rotation.

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

The gear units are characterised by a low noise level. This is achieved by helical and bevel-helical gears with a high contact ratio and special sound-damping housings.

The good temperature characteristics of the gear unit are achieved by its high degree of efficiency, large housing surface and performance-related cooling system.

Note: Unless otherwise agreed by contract, the gear units must not be exposed to harmful environmental factors such as chemical products, high-air pollution and temperatures outside the range 0 to +20 °C.

5.2 Housing

The housings are of cast iron. If required, it may also be of steel.

Housings up to size 12 are made in one part. The types 13 - 26 have a two-part gear housing. The housing is rigid in design and due to its form has excellent noise and temperature characteristics.

The housing is fitted with adequately dimensioned lifting eyes, inspection or assembly covers.

It is also fitted with a dipstick for checking the oil level, a draining screw for oil change and a venting screw.

Colour codes for venting, oil inlet, oil level and oil drainage:

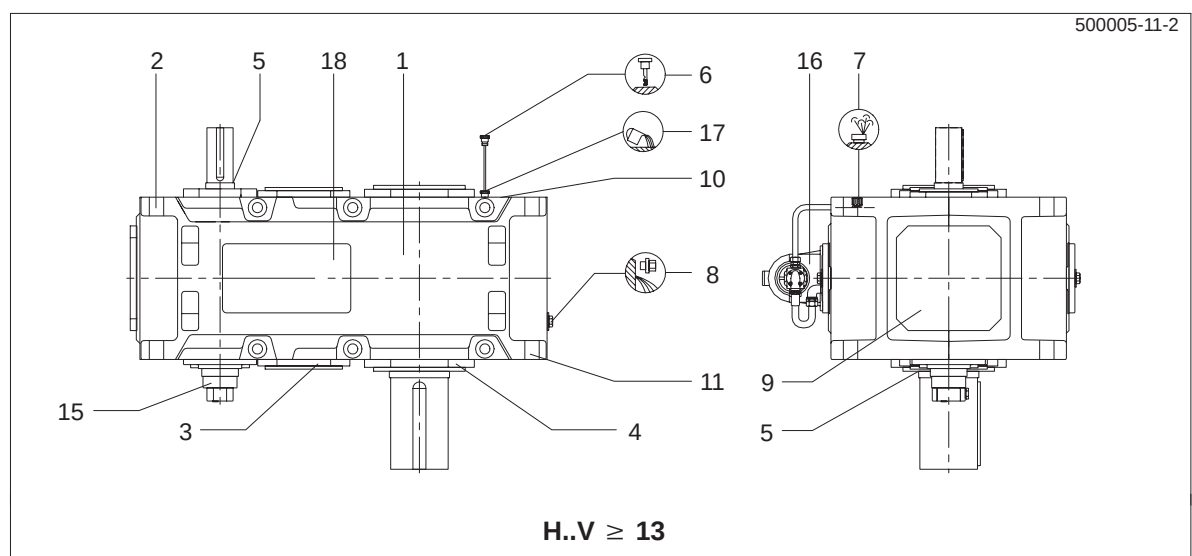
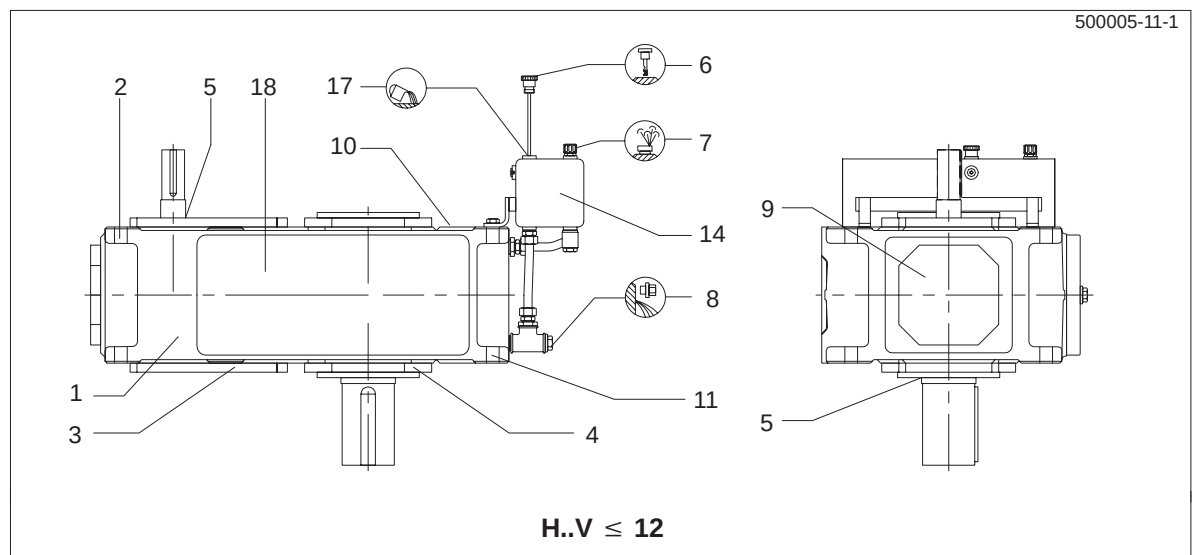
Venting: yellow 

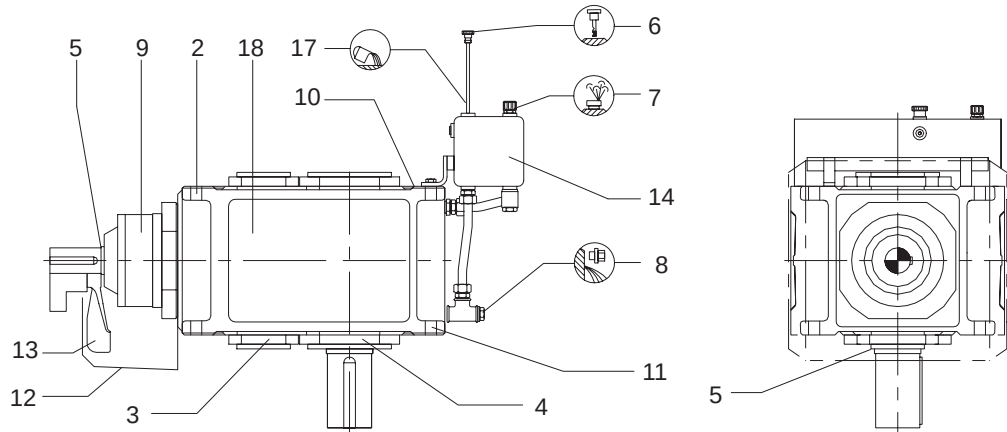
Oil inlet: yellow 

Oil level: red 

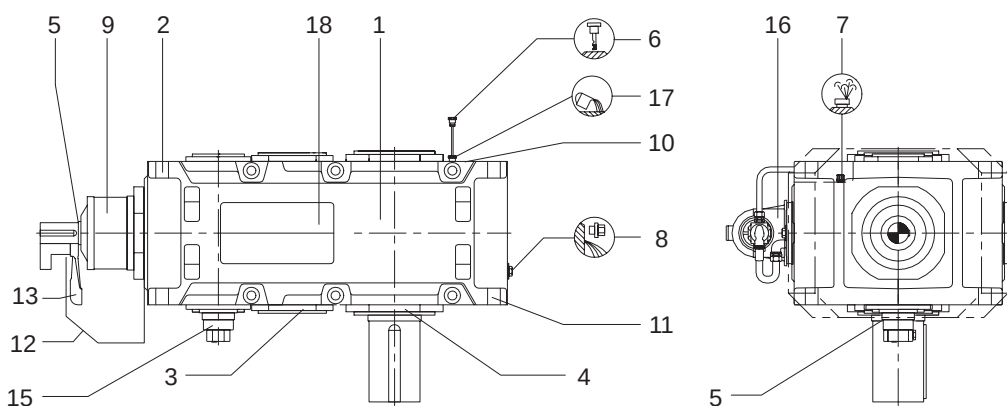
Lubrication points: red

Oil drain: white 





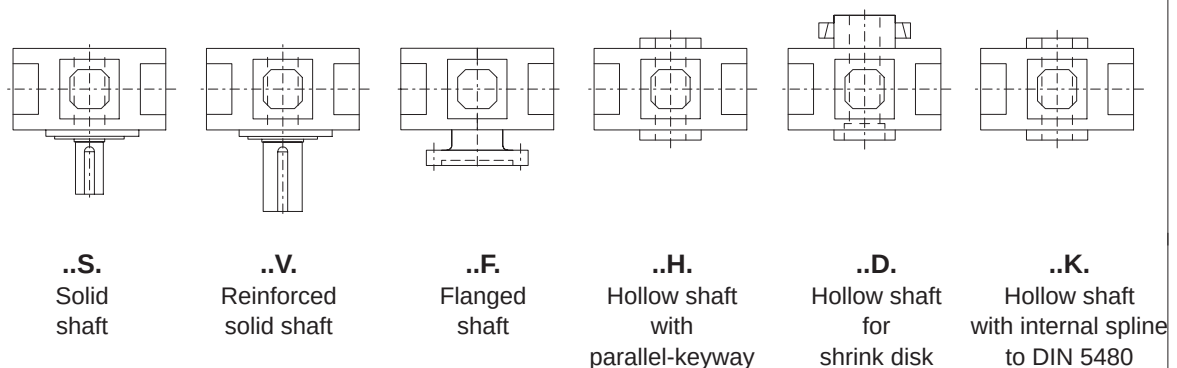
B..V ≤ 12



B..V ≥ 13

- | | |
|----------------------------|---|
| 1 Housing | 10 Rating plate |
| 2 Lifting eyes | 11 Gear-unit fastening |
| 3 Cover | 12 Fan cowl |
| 4 Cover | 13 Fan |
| 5 Shaft seals | 14 Oil-equalising tank (splash lubrication) |
| 6 Oil dipstick | 15 Flange pump |
| 7 Housing ventilation | 16 Motor pump |
| 8 Oil drain plug | 17 Oil inlet |
| 9 Cover or bearing journal | 18 Inspection or assembly cover |
- } optional

Output types



For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

5.3 Toothed components

The toothed components of the gear unit are case hardened. Helical-gear teeth are ground. Bevel helical-gear teeth are lapped, ground or gear cut. The high quality of the teeth lead to a significant noise reduction and ensure safe and reliable running.

The gear wheels are joined to the shafts by interference fitting and parallel keys. These types of joints transmit the torques generated with adequate reliability.

5.4 Lubrication

5.4.1 Splash lubrication

When installed vertically, all teeth and bearings are submerged in oil. The space required for expansion of the oil is provided by an oil-equalising tank bolted to the unit.

5.4.2 Pressure lubrication through added oil-supply unit

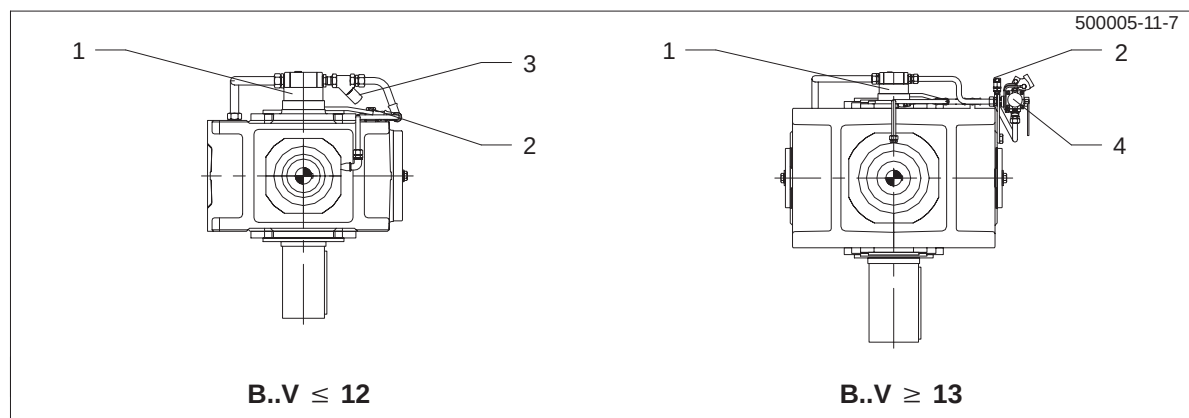
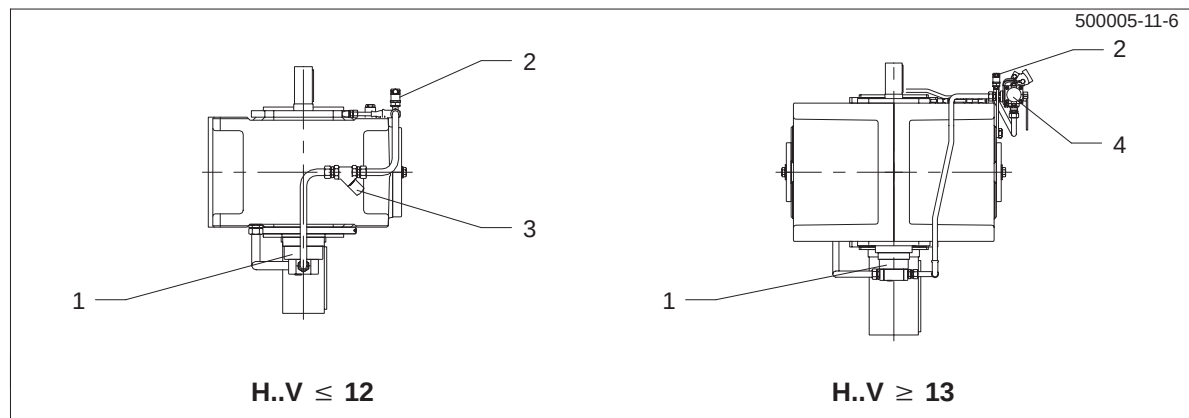
In non-vertical positions, high bearing speeds or peripheral velocities on the teeth, the splash lubrication system may be replaced or supported by a pressure lubrication system.

Gear units larger than size 13 are fitted with pressure-lubrication systems. Depending on order requirement, sizes 5 ... 12 can also be fitted with pressure-lubrication systems.

The oil-supply unit is permanently attached to the gear unit and consists of a flange or motor pump, a coarse filter, a pressure-monitoring device and pipework. In gear units of sizes 13 - 26, the coarse filter is replaced by a switchover filter.

The direction of flow from the flange pumps is independant of the direction of rotation.

Type fitted with flange pump



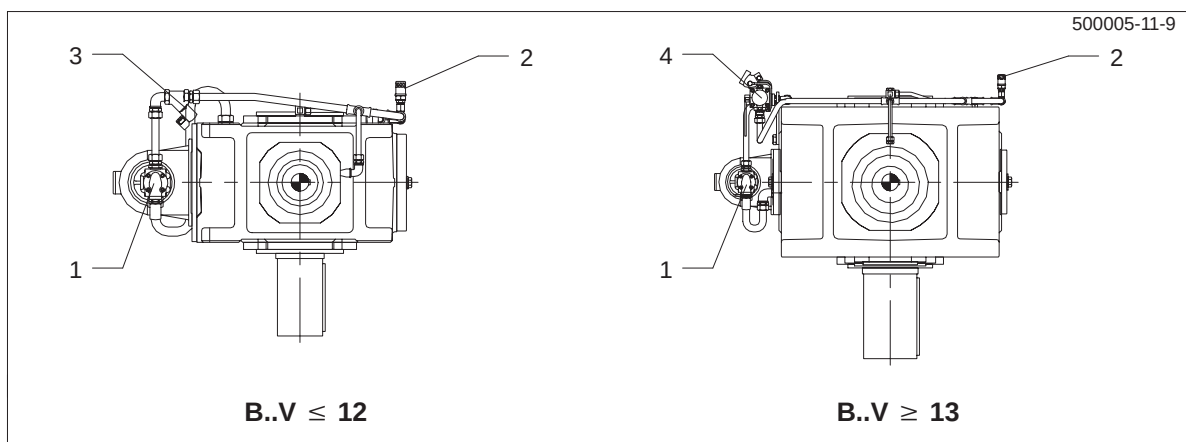
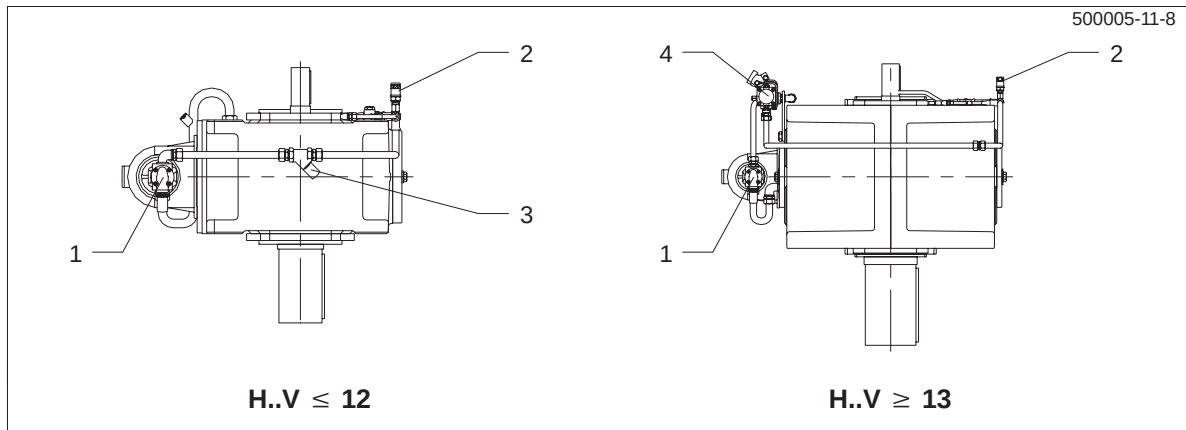
- 1 Flange pump
- 2 Pressure monitor

- 3 Coarse filter
- 4 Switchover filter

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Depending on the order specification and application, the flange pump may be replaced by a motor pump.

Type fitted with motor pump



- 1 Motor pump
- 2 Pressure monitor

- 3 Coarse filter
- 4 Switchover filter

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Caution!

With gear units with added oil-supply unit, the pressure monitor must always be connected before starting the unit up.

Note:

The operation and maintenance of the pump, pressure monitor, coarse filter or switchover filter are described in instructions included (where applicable) with the products.

For technical details, refer to the order-specific equipment list.

5.5 Bearings

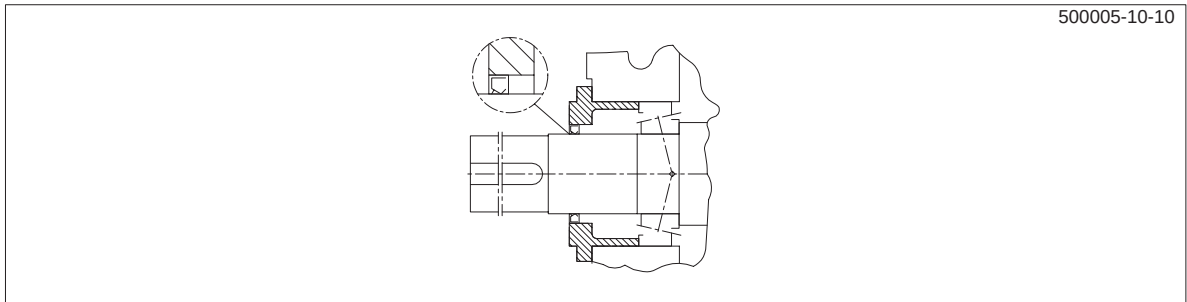
All shafts are mounted on anti-friction bearings.

5.6 Shaft seals

Depending on requirements, radial shaft sealing rings, labyrinth seals or Taconite seals are mounted at the shaft exits to prevent oil from leaking from the housing and dirt from entering it.

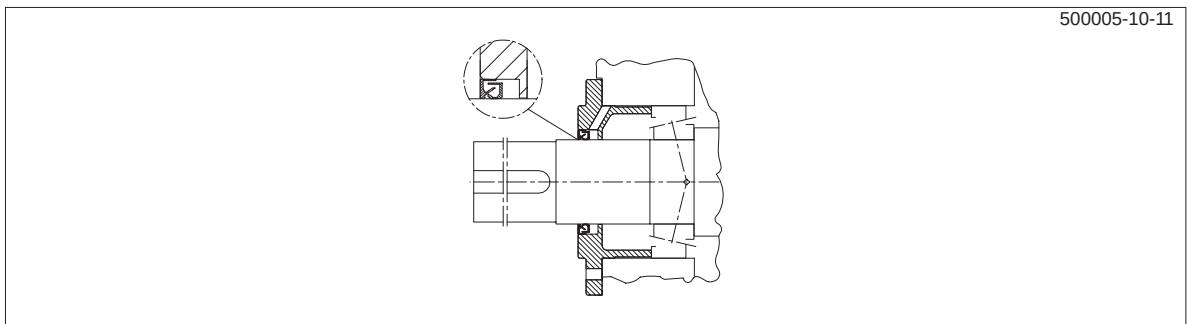
5.6.1 Radial shaft sealing rings

Radial shaft sealing rings are the standard type of seal. They are fitted with an additional dust lip to protect the actual sealing lip from external contamination.



5.6.2 Labyrinth seals

Labyrinth seals are contact-free and avoid wear to the shaft. They therefore require no maintenance and ensure favourable temperature characteristics.



Caution!

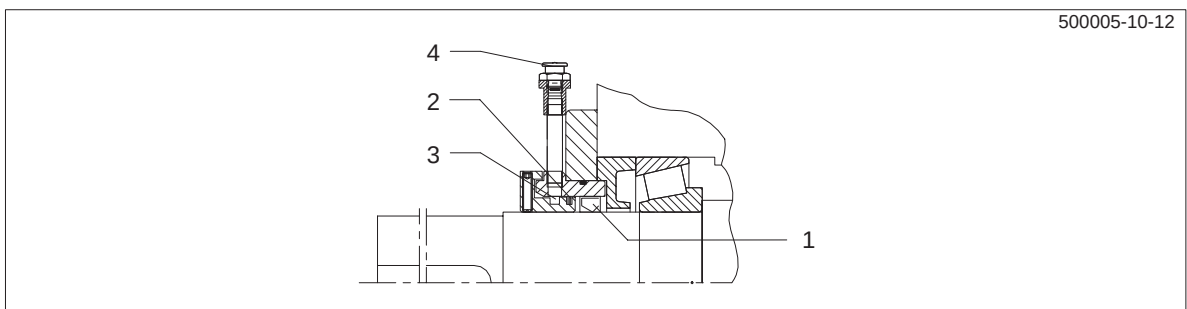
Used only for horizontal shaft ends of type B..V.

No splash lubrication.

For reliable operation, this type of seal requires horizontal positioning in a splash-free environment with very little dust.

5.6.3 Taconite seals

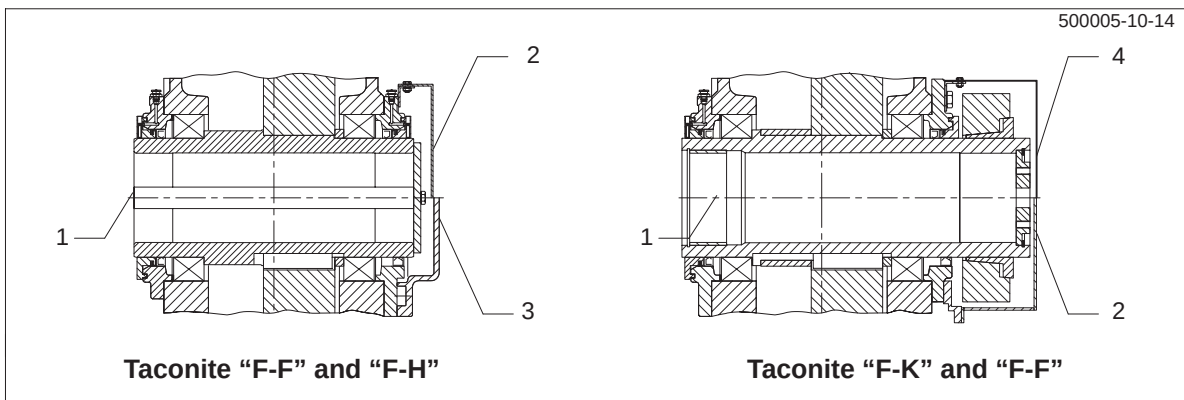
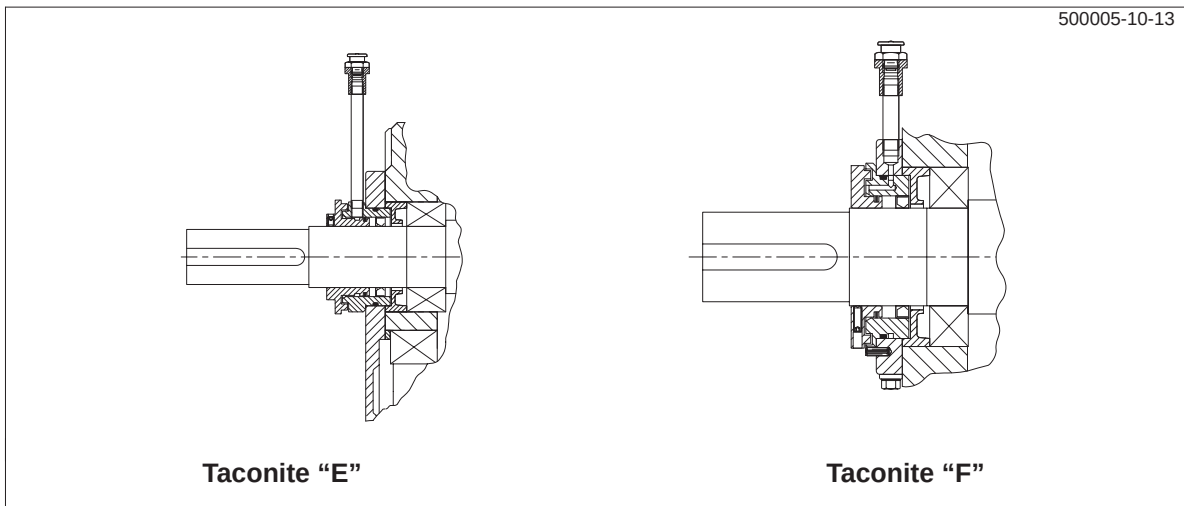
Taconite seals were specially developed for use in a dusty environment. The penetration of dust is prevented by a combination of seal elements (radial shaft sealing ring, lamellar seal and grease-charged labyrinth seal).



- 1 Radial shaft sealing ring
- 2 Lamellar seal

- 3 Grease-charged labyrinth seal (re-chargable)
- 4 Flat grease nipple AM10x1 to DIN 3404

Taconite seals are divided into the following types:



- 1 Output
- 2 Taconite "F-F"

- 3 Taconite "F-H"
- 4 Taconite "F-K"

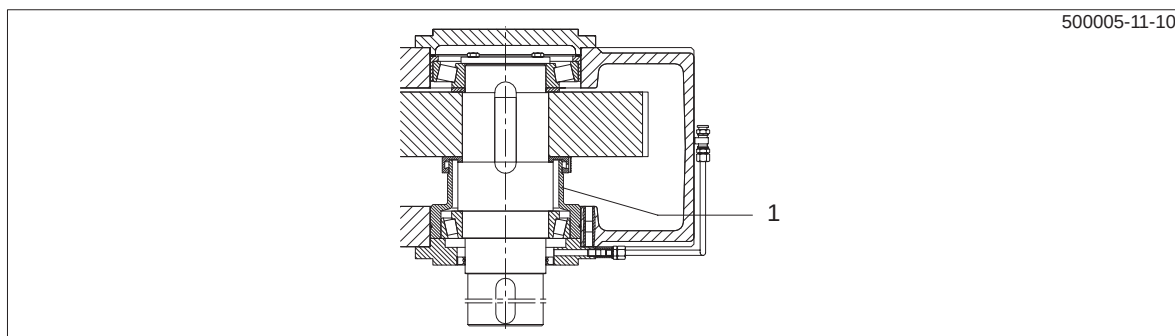
Taconite type	Application	Remarks
"E"	All input shafts with or without fan	Re-chargeable labyrinth
"F"	Output shaft Type S (solid shaft) Type F (flanged shaft)	
"F-F"	Output shaft Type H (hollow shaft with parallel-keyway) Type K (hollow shaft with internal spline to DIN 5480) Type D (hollow shaft for shrink disk)	Labyrinth re-chargeable on both sides, incl. cowl to prevent contact on gear side facing away from output
"F-H"	Output shaft Type H (hollow shaft with parallel-keyway) Type K (hollow shaft with internal spline to DIN 5480)	Re-chargeable labyrinth on output side; dustproof cowl on opposite side
"F-K"	Output shaft Type D (hollow shaft for shrink disk)	

Note: For re-charging the labyrinth seals with grease, the specified frequency must be observed (see section 10 "Maintenance and repair").

5.6.4 Oil-dam pipe

By sealing the vertically downward pointing input shaft with a contact-free and wear-free shaft seal (a so-called "oil-dam pipe") the shaft outlet is sealed absolutely tight to prevent oil escaping. This system is used only for pressure lubrication.

Note: The prescribed lubrication interval for the shaft seal must be observed at all times (see section 10. "Maintenance and repair").



1 Oil-dam pipe

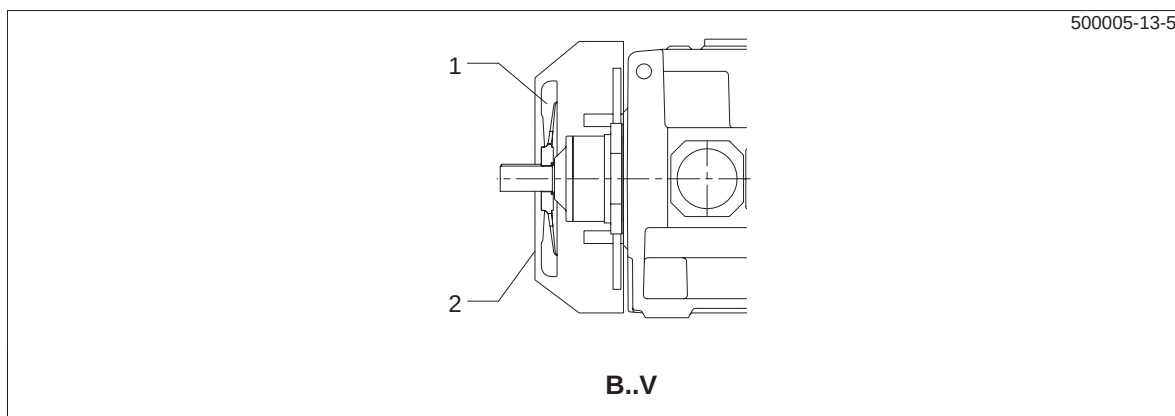
5.7 Cooling

Depending on requirement, the gear unit is fitted with a fan, a cooling coil, water or air oil-cooling systems or separate oil-supply units. In the case of separate oil-supply units, the specific instructions for these units must be observed.

5.7.1 Fan

Installation of a fan is possible only with type B..V (bevel-helical gear unit).

The fan is mounted on a high-speed shaft of the gear unit and is protected from accidental contact by a cowl. The fan sucks air through the grid on the cowl and blows it along the air ducts on the side of the gear housing, thereby dissipating a certain amount of heat from the housing.



1 Fan

2 Fan cowl

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

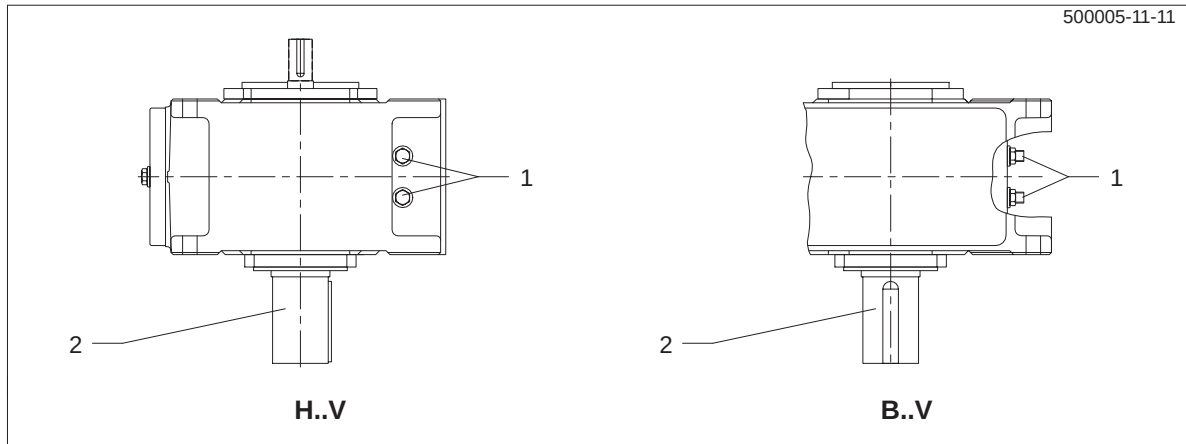
Caution!

For gear units fitted with a fan, a sufficient amount of space must be allowed for air intake when mounting the safety guards for the coupling or other components.
The correct distance is given in the dimensional drawing in the gear-unit documentation.

Note: The cooling effect is considerably reduced if the fan or the gear housing are dirty (see section 10 "Maintenance and repair").

5.7.2 Cooling coil

The cooling coil is located in the oil sump of the gear unit and is supplied with cooling water via a connection to be provided by the user. Either fresh water, sea water or brackish water can be used for cooling purposes. When water is flowing through the coil, heat is transferred from the oil to the water and thereby removed from the system.



1 Cooling-water connection

2 Output shaft

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Caution!

The water can flow through the gear unit in either direction. The pressure of the cooling water must not exceed 8 bars.

If the gear unit is being withdrawn from service for a longer period or if there is a danger of the water freezing, it must be drained off. Remove any remaining water with compressed air.

Note: To avoid excess pressure, the water inlet must be fitted with a flow-control device, e.g. a pressure reducer or stop valves.

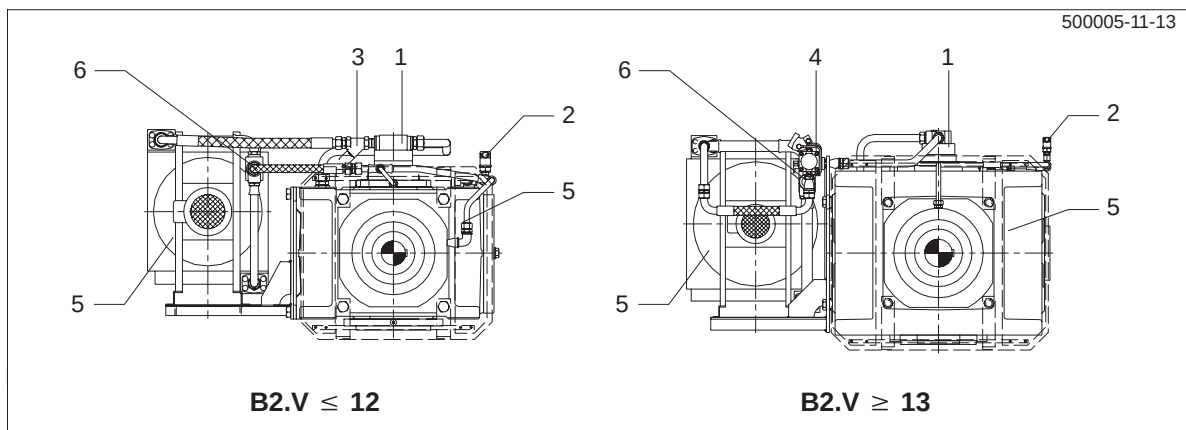
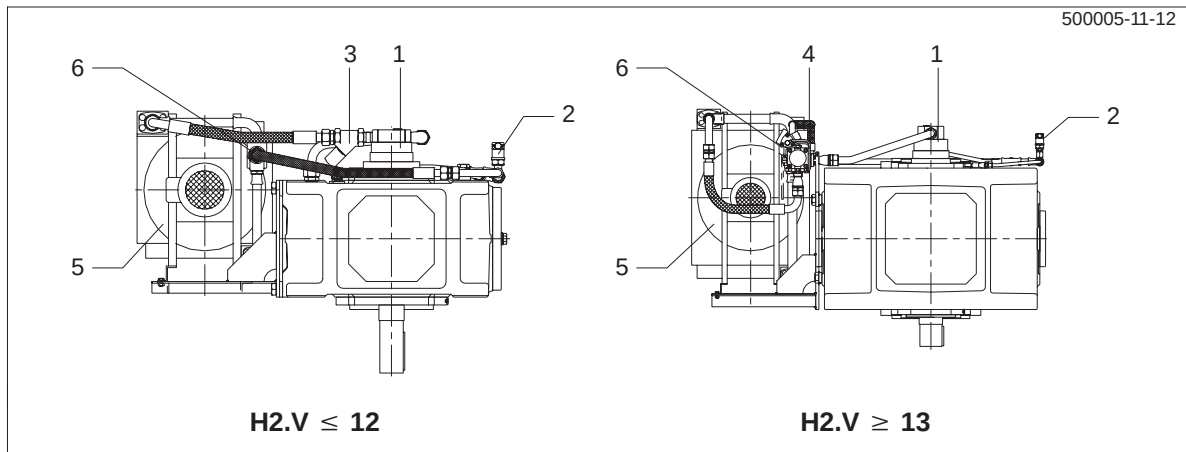
Note: For technical details, refer to the order-specific equipment list.

5.7.3 Built-on air oil-cooling units

For types H2.V and B2.V an air oil-cooling unit may be added if required. This unit is permanently mounted on the gear unit and consists of a flange pump, a coarse filter, a pressure monitor, a temperature-control valve, the actual air oil-cooling unit and the necessary pipework. For gear units of sizes 13 - 26, the coarse filter is replaced by a switchover filter.

The air oil-cooler is designed to cool the gear oil by means of air from the surrounding atmosphere. Depending on the volume flow, the oil passes through the cooler in one or several streams and through current of air blown in by the fan. For cold starts, a bypass pipe with a temperature-control valve are provided.

The direction of flow from the flange pumps is independant of the direction of rotation.



- | | |
|--------------------|-----------------------------|
| 1 Flange pump | 4 Switchover filter |
| 2 Pressure monitor | 5 Air oil-cooler |
| 3 Coarse filter | 6 Temperature control valve |

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Caution!

When installing gear units with added air oil-cooling units, it must be ensured that the air circulation is not obstructed. The required minimum distance from adjacent components, walls etc. is given in the drawings in the unit documentation.

Note: For operation and maintenance, always observe the instructions given in the order-specific appendix.
For technical details, refer to the order-specific equipment list.

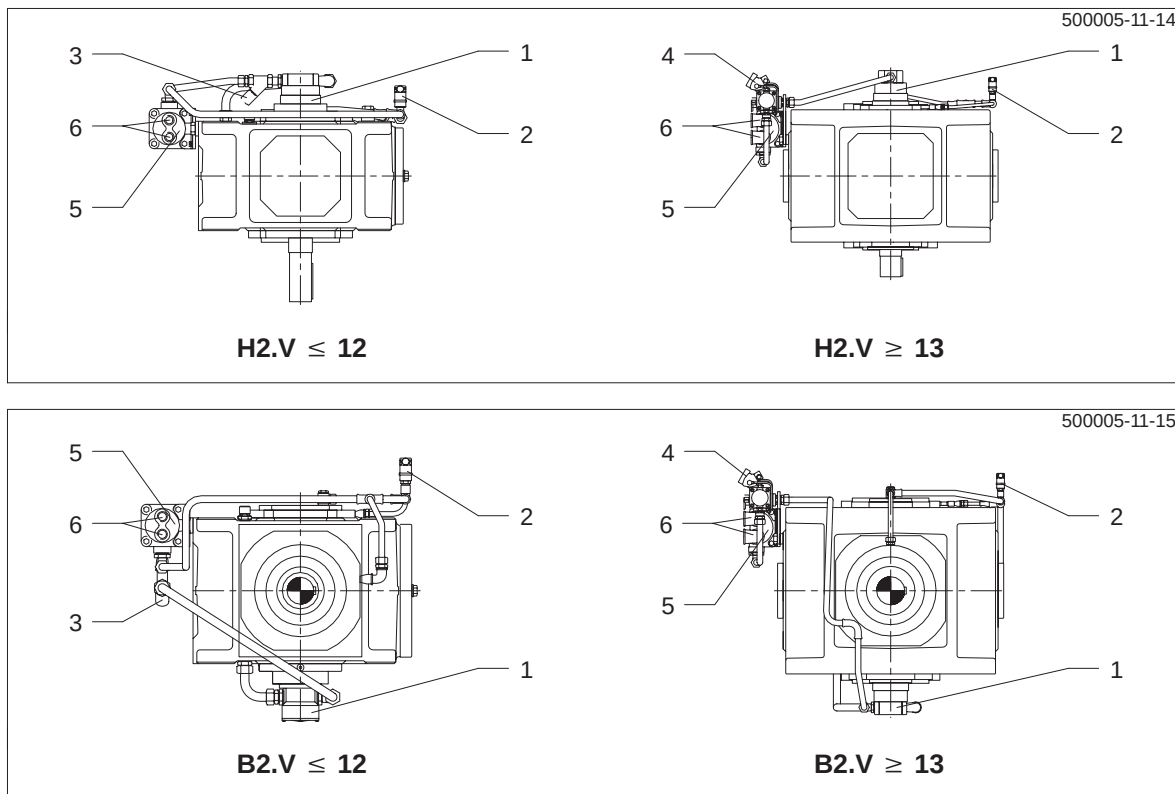
The cooling effect is considerably reduced if the cooler or the gear housing are dirty (see section 10 "Maintenance and repair").

5.7.4 Built-on water oil-cooling units

For types H2.V and B2.V a water oil-cooling unit may be added if required. This oil-cooling unit is permanently mounted on the gear unit and consists of a flange pump, a coarse filter, a pressure monitor, the actual water oil-cooling unit and the necessary pipework. For gear units of sizes 13 - 26, the coarse filter is replaced by a switchover filter.

The direction of flow from the flange pumps is independant of the direction of rotation.

The required water connection must be provided by the user.



- 1 Flange pump
- 2 Pressure monitor
- 3 Coarse filter

- 4 Switchover filter
- 5 Water oil-cooler
- 6 Cooling water inlet and outlet

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Caution!

To ensure optimum cooling performance, the specified direction of flow in the cooling unit must be observed. The cooling-water inlet and outlet must not be reversed.

The pressure of the cooling water must not exceed 8 bars.

If the gear unit is being withdrawn from service for a longer period or if there is a danger of the water freezing, it must be drained off. Remove any remaining water with compressed air.

Depending on the order specification and application, the flange pump may be replaced by a motor pump.

Note:

For operation and maintenance, always observe the instructions given in the order-specific appendix.

For technical details, refer to the order-specific equipment list.

5.8 Couplings

Generally speaking, flexible couplings or hydraulic couplings combined with flexible couplings should be provided for the input drive.

Note: For gear units with fan, the hydraulic unit of the hydraulic coupling should be located on the motor shaft to ensure that a sufficient area is available for intake of cooling air (see point 5.7.1).

For gear units fitted with solid output shafts (types ..S. and ..V.) flexible couplings are also provided in most cases.

If rigid couplings or other input or output elements are used, which create additional radial or axial forces, (e.g. gear wheels, belt pulleys, disk flywheels, hydraulic couplings), these must be agreed by contract.

Caution!

Couplings with peripheral velocities on the outer diameter of up to 30 m/s, must be statically balanced. Couplings with peripheral velocities over 30 m/s, must be dynamically balanced.

For maintenance and operation of the couplings, refer to the specific operating instructions for the coupling.

Note: When installing the drive system, the individual components must be very precisely aligned with one another in order to minimise restoring forces caused by angle and shaft offset, and avoid premature wear to flexible coupling components.

For gear units with hollow output shafts or flange output shafts, the coupling on the output side is not required. Gear units with hollow output shafts should be mounted on the shaft of the customer's machinery. Gear units with flange output shafts should be mounted on the customer's shaft via a mating flange.

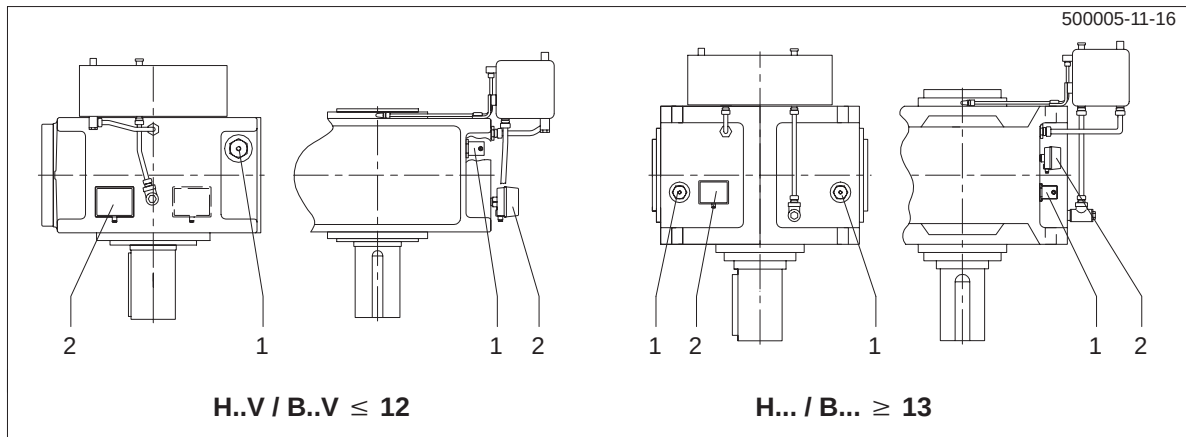
5.9 Heating element

At low temperatures, it may be necessary to heat the gear oil before switching on the drive unit, or even during operation. In such cases, one or two such heating elements may be provided if required. These convert the electrical energy into heat which is conducted to the surrounding oil. The heating elements are located in protective tubes inside the housing thus making it possible to replace them without draining off the oil.

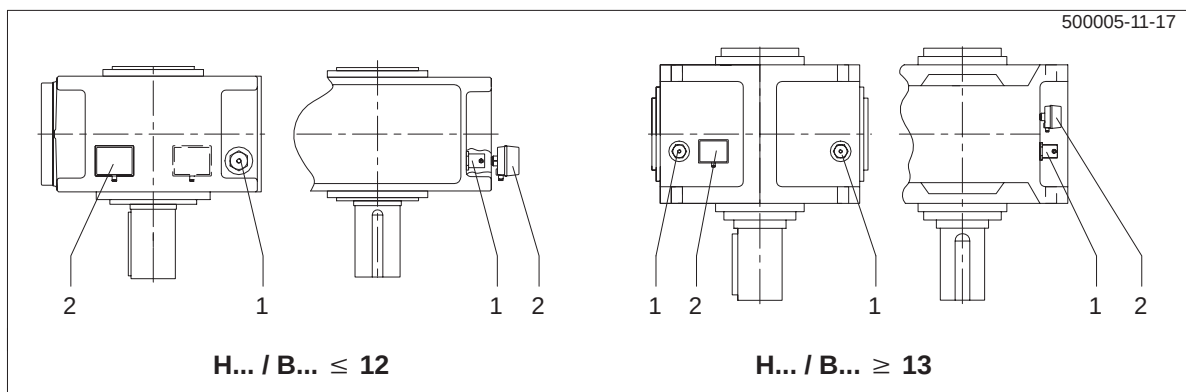
The heating elements must be fully immersed in the oil bath. Due to the higher oil level required for radial shaft seals, only seals of this type are suitable for use with heating elements.

The heating elements are controlled by a temperature monitor which provides a signal when maximum and minimum temperatures are reached. The signal requires amplification.

Type with oil-equalising tank



Type without oil-equalising tank



1 Heating element

2 Temperature monitor

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Caution!

Never switch the heating element on unless it is fully immersed in the oil bath as there is a serious risk of fire.

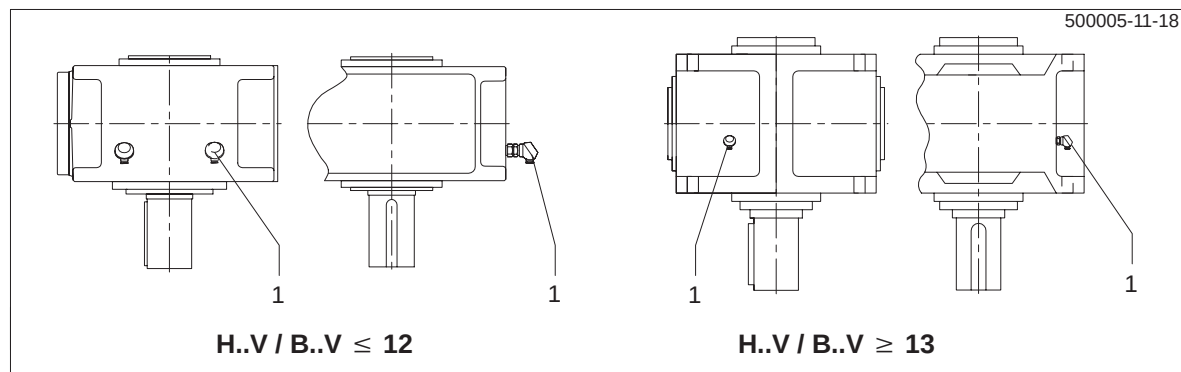
Note:

For operation and maintenance, always observe the instructions given in the order-specific appendix.
For technical details, refer to the order-specific equipment list.

5.10 Oil-temperature measurement

Depending on the order specification, the gear unit may be fitted with a resistance thermometer PT100 for monitoring the oil temperature in the sump. In order to measure the temperatures or temperature differences, the resistance thermometer PT100 should be connected to a suitable instrument provided by the customer. The thermometer has a connection head (protection type IP54) for the wiring. A two-conductor circuit is fitted by the manufacturer. However, the customer may fit his own three or four-conductor circuit if required.

Note: A certain level of oil is necessary for reliable temperature measurement. For this reason, the resistance thermometer is not suitable for use with labyrinth seals.



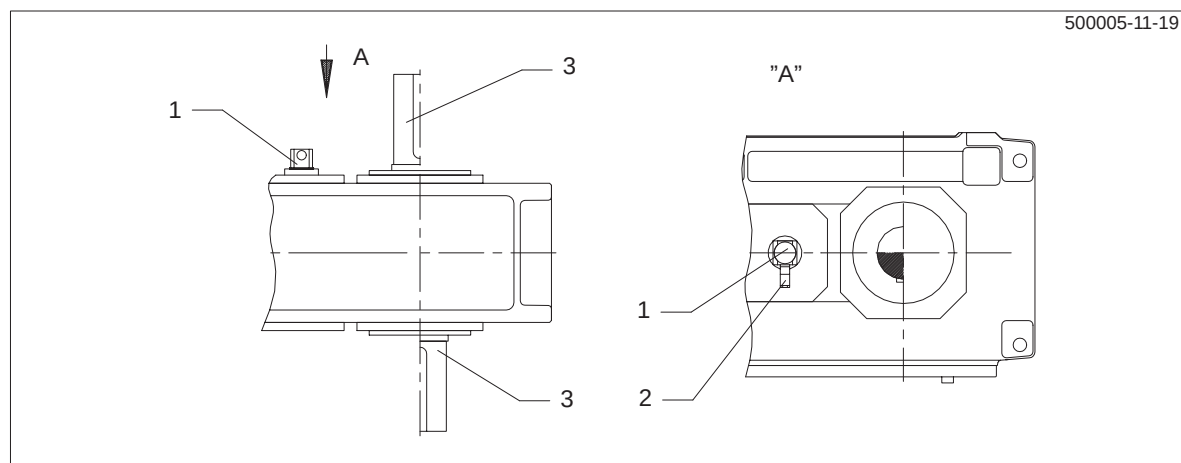
1 Resistance thermometer PT 100

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Note: For operation and maintenance, always observe the instructions given in the order-specific appendix.
For technical details, refer to the order-specific equipment list.

5.11 Speed transmitter

Depending on order specification, an incremental speed transmitter can be mounted on the second last shaft. Wiring and evaluation instrument should be provided by the customer.



1 Incremental transmitter

2 12-pole brass plug

3 Output shaft

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Note: For operation and maintenance, always observe the instructions given in the order-specific appendix.
For technical details, refer to the order-specific equipment list.

6. Assembly

Note: Observe the "Safety notes" in section 3.

6.1 General information on installation

Assembly and installation work must be done with great care by trained and qualified personnel. The manufacturer cannot be held liable for damage caused by incorrect assembly and installation.

During the planning phase, sufficient space should be allowed around the gear unit for subsequent care and maintenance work.

If the gear unit is fitted with a fan, there should be sufficient space for air intake.

Caution!

Exposure to direct sunlight may cause overheating. Protective features such as covers, roofing etc. should be provided as required.

Sufficient lifting equipment should be available before beginning the installation and assembly work.

Caution!

**No welding work should be done on the drive.
The drives must not be used as an earthing point for welding operations.
Delicate teeth and bearings may be irreparably damaged by welding.**

Caution!

All the fastening points provided by the design of the unit should be used.

To ensure proper lubrication, the installation position specified in the order must always be observed.

6.2 Installation of gear unit on housing base

6.2.1 Foundation

The foundation must be horizontal and level.

It should be designed in such a way that no resonance vibrations are created and that no vibrations are transmitted from adjacent foundations. The structure on which the unit is to be mounted, must be rigid. It must be designed according to the weight and torque, taking into account the forces acting on the gear unit.

When fastening the gear unit to concrete foundations by means of foundation blocks, suitable recesses should be made in the foundation.

Hexagon bolts and nuts should be tensioned to the specified torque. For the correct torque, refer to point 10.2.11. We recommend bolts of strength class 8.8.

Slide rails should be correctly aligned and embedded in the concrete foundation.

Note: For dimensions, space requirement, arrangement of supply connections (e.g. with separate oil-cooling units), refer to the drawings in the gear-unit documentation.

6.2.2 Description of installation work

- Remove the anti-corrosion paint on the shafts with benzine.

Caution!

Do not allow the benzine to contact the shaft sealing rings



**Ensure adequate ventilation. Do not smoke when working with benzine!
Explosion hazard!**

- Mount and secure input and output drive elements (e.g. coupling components) on shafts. These should be heated before mounting. For the correct joining temperatures, refer to the dimensional drawings in the coupling documentation.

Unless otherwise instructed, the components may be heated inductively, with a burner or in a furnace.



Take precautions to avoid burns from hot components!

Caution!

Protect shaft sealing rings from damage and heating over +100 °C. Use heat-protection screens to reduce radiant heat.

Caution!

The coupling components should be fitted using suitable equipment to avoid possible damage to the shaft bearings through axial joining forces. Always use suitable lifting equipment. When fitting the components, ensure that the shaft sealing rings and running surface are not damaged.

The components should be pushed smartly onto the shaft, up to the position specified in the order specific dimensional drawing.

Caution!

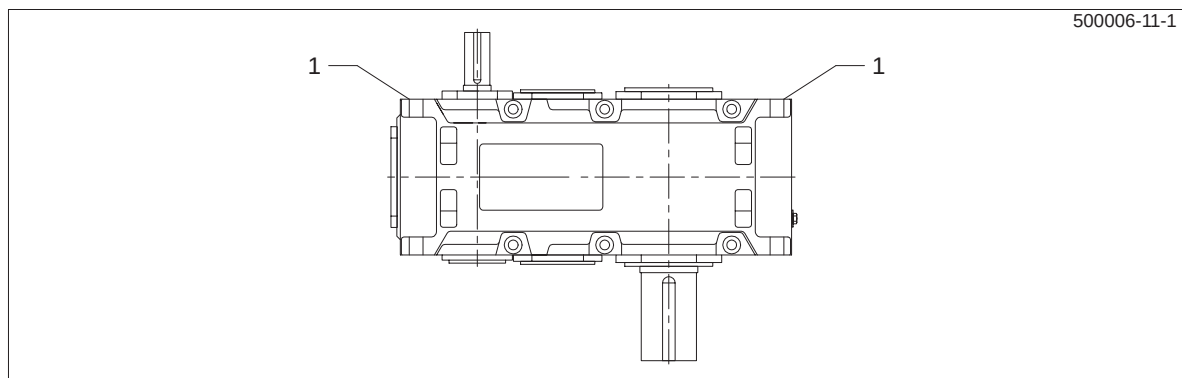
Fit the coupling using suitable fitting equipment. Never use force or knock the couplings into position, as this will cause irreparable damage to the gearwheels, anti-friction bearings, retaining rings etc.

Note:

Gear units whose weight requires the use of lifting gear must be attached at the points shown in section 4 "Handling and storage". When additional components are mounted on the gear unit, appropriate attachment points should be provided in accordance with the order-specific dimensional drawing.

6.2.2.1 Aligning surfaces

The machined surfaces on the top of the housing are intended for initial alignment of the gear unit.



1 Alignment surfaces

The final fine alignment with the assemblies on the output side must be carried out accurately to the shaft axes using:

- rulers
- spirit level
- dial gauge
- feeler gauge etc.

The gear unit can now be fixed in its position.

Note:

The accuracy of the alignment process is an important factor in determining the life span of shafts, bearings and couplings. If possible, the deviation should be zero. For the special requirements for the couplings, refer to the specific operating instructions.

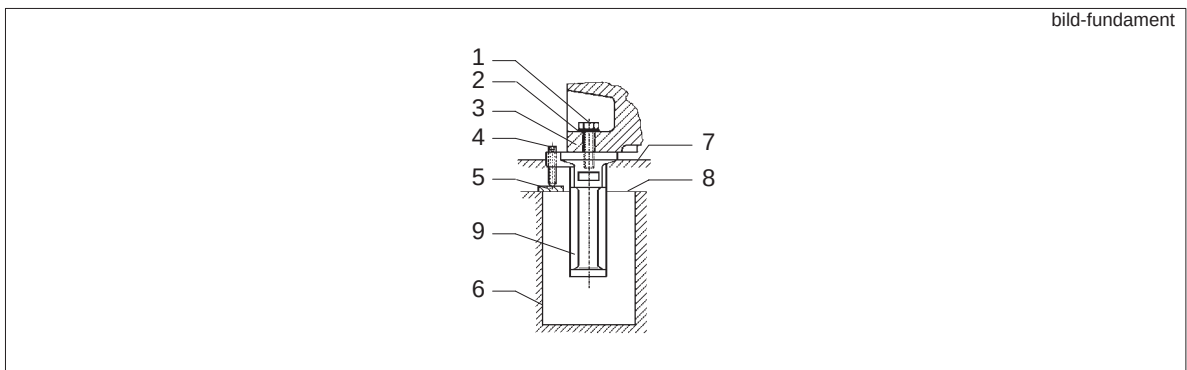
6.2.2.2 Mounting on a foundation frame

- Clean the lower surfaces of the gear-unit base.
- With suitable lifting gear, place the gear unit on the foundation frame.
- Tighten the foundation bolts to the specified torque (see point 10.2.11). If necessary use stops to prevent shifting.

Caution! The gear unit must not be twisted when tensioning the fastening bolts.

6.2.2.3 Mounting on a concrete foundation

- Clean the lower surfaces of the gear-unit base.
- Hook the foundation blocks with washers and fastening bolts into the foundation fastening points on the housing and tighten the fastening bolts (see diagram below).



- Using suitable lifting gear, place the gear unit on the concrete foundation.
- Align the gear unit horizontally by the input and output shafts using foundation blocks (and set screws, if applicable).
- If applying considerable force, use stops to prevent the unit from shifting.
- Pour concrete into the recesses for the foundation blocks.

Note: Before pouring the concrete foundation, fill up the openings in the foundation blocks with polystyrene.
When the concrete has set, tighten the fastening bolts of the foundation blocks to the **specified torque** (see point 10.2.11).

Caution! The gear unit must not be twisted when tensioning the fastening bolts.

6.3 Assembly of a shaft-mounting gear unit with hollow shaft and parallel-keyway

6.3.1 Assembly

- Remove the anti-corrosion paint from the hollow shaft and machine shaft with benzine.

Caution!

Do not allow the benzine to contact the shaft sealing rings.



**Ensure adequate ventilation. Do not smoke when working with benzine!
Danger of explosion.**

- Check the hollow and machine shafts to ensure that seat and edges are not damaged. If necessary, rework the parts with a suitable tool and clean them again.

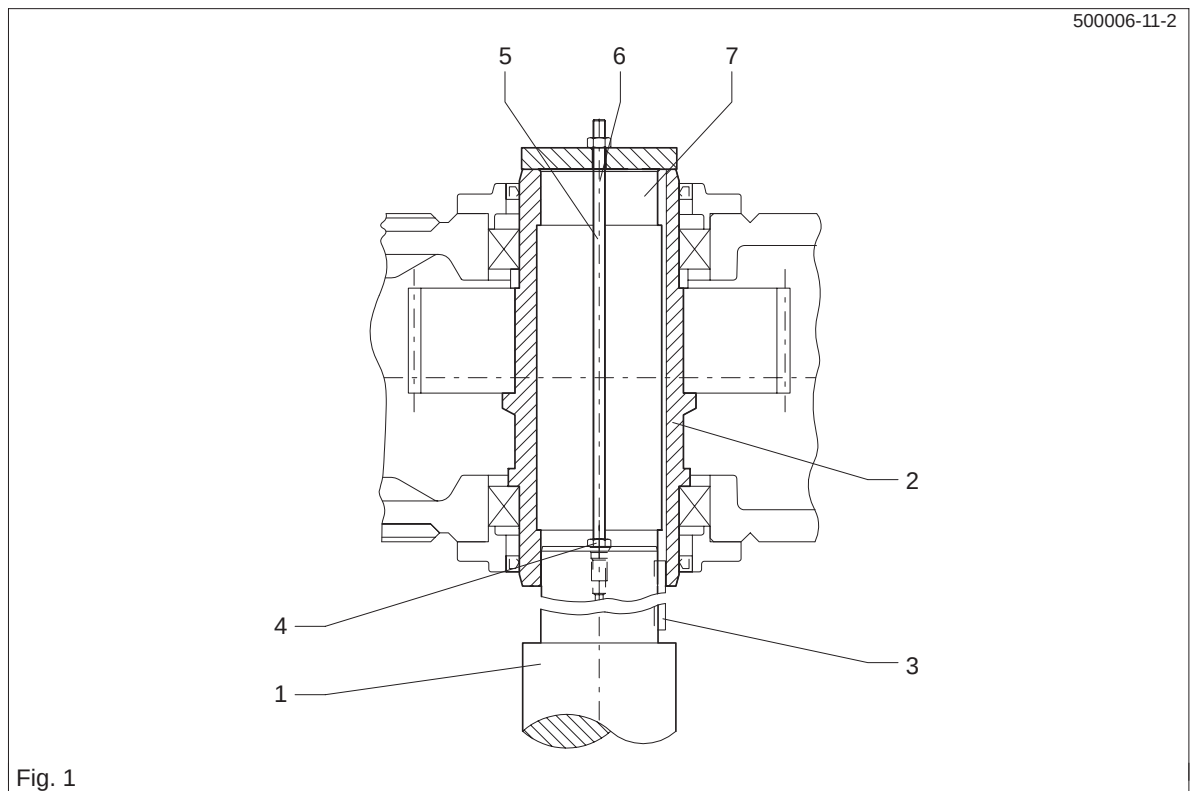
Coat with a suitable lubricant (e.g. Calypsol grease type H 443 HD88) to prevent frictional corrosion of the contact surfaces.

6.3.1.1 Fitting

- Fit the gear unit by means of nut and threaded spindle. The counter force is provided by the hollow shaft.

Caution!

The hollow shaft must be exactly aligned with the machine shaft to avoid canting.



- 1 Machine shaft
2 Hollow shaft
3 Parallel key

- 4 Nut
5 Threaded spindle
6 Nut

- 7 End plate

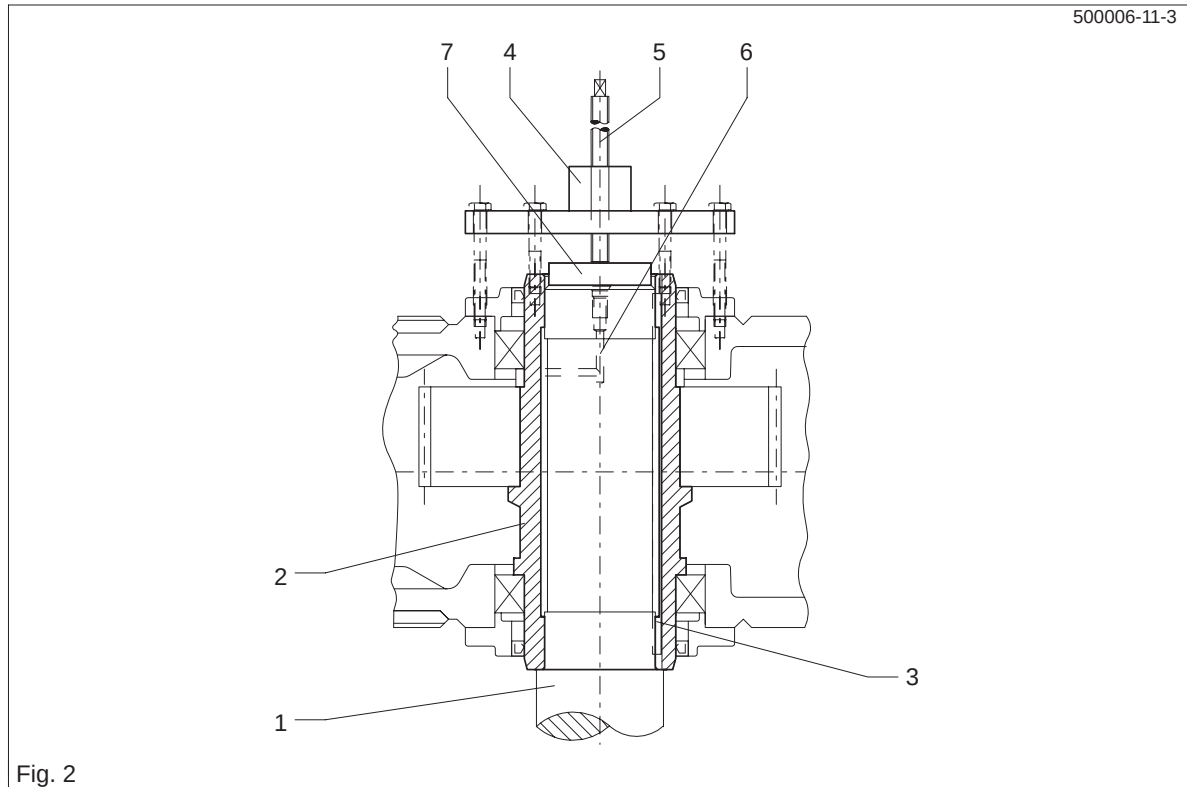
Instead of the nut and threaded spindle shown in the diagram, other types of equipment such as a hydraulic lifting equipment (type Lucas) may be used.

6.3.1.2 Axial securing device

Depending on type, secure the hollow shaft axially on the machine shaft (e.g. with circlip, end plate, set screw etc.).

6.3.2 Dismantling

- Remove the axial securing device from the hollow shaft.
- If frictional corrosion has formed on the seating surfaces, press rust-releasing or other agent (e.g. with a pump) into the bore cavity to reach the seats. The rust releaser is forced in through the bores in the machine shaft (see fig. 2) or end plate (fig. 3).
- Depending on the facilities available, the gear unit can be forced off the shaft on site using forcing screws in an end plate (see fig. 3), a central threaded spindle or a Lucas hydraulic lifting unit.
- When the rust-releasing agent has taken effect, pull the gear unit off with the device as shown in figs. 2 or 3.



- 1 Machine shaft
2 Hollow shaft
3 Parallel key
4 Hydraulic lifting unit

- 5 Threaded spindle
6 Bore for rust-releasing agent
7 Plate for forcing out

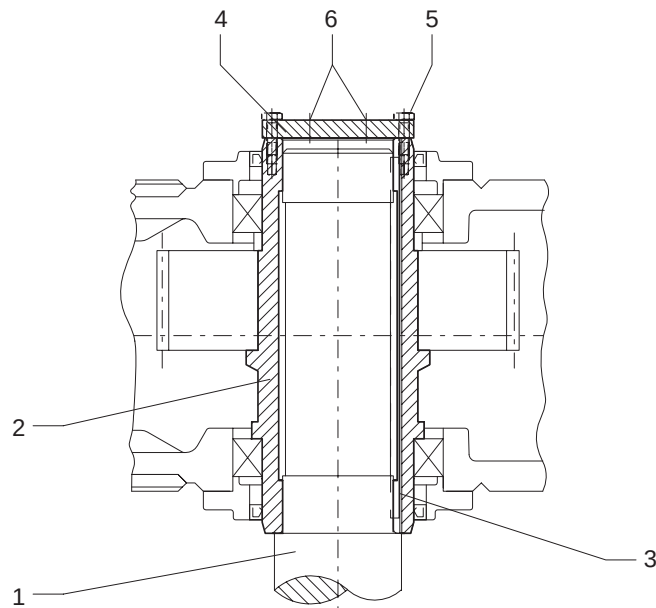


Fig. 3

- | | |
|-----------------|---------------------------------|
| 1 Machine shaft | 4 End plate |
| 2 Hollow shaft | 5 Seal |
| 3 Parallel key | 6 Bore for rust-releasing agent |

Caution!

Avoid twisting or canting when pulling the unit off.

Caution!

If the counter force is provided not only by the hollow shaft, but by the housing also, as shown in fig. 2, the forces used must not exceed the values given in the following table.

Gear-unit size	Maximum force N	Gear-unit size	Maximum force N
2	11700	11	97200
3	15200	12	113600
4	22600	13	140000
5	33000	14	160000
6	37500	15	193000
7	50000	16	215000
8	56000	17	240000
9	65000	18	266000
10	82000	19 ... 26	On request

Table 6.1: Maximum press-off forces

Caution!

If the above figures are exceeded, the hollow-shaft bearings or other gear components may be irreparably damaged. Before replacing the gear unit on the machine shaft, always check the bearings for any signs of damage.

Note:

When using forcing screws or threaded spindles, the head of the thread pushing against the working machine should be rounded and well greased to reduce the risk of seizing at this point.

6.4 Shaft-mounting gear unit with hollow shaft and internal spline to DIN 5480

6.4.1 Assembly

- Remove the anti-corrosion paint from the hollow shaft and machine shaft with benzene.

Caution!

Do not allow the benzene to contact the shaft sealing rings.



**Ensure adequate ventilation. Do not smoke when working with benzene!
Danger of explosion.**

- Check the hollow and machine shafts to ensure that seats, teeth, and edges are not damaged. If necessary, rework the parts with a suitable tool and clean them again.

Coat with a suitable lubricant (e.g. Calypsol grease type H 443 HD88) to prevent frictional corrosion of the contact surfaces.

6.4.1.1 Fitting

- Fitting with integrated DU bush
- Fit the gear unit by means of nut and threaded spindle. The counter force is provided by the hollow shaft.

Caution!

The hollow shaft must be exactly aligned with the machine shaft to avoid canting. When fitting, ensure that the position of the teeth between the machine shaft and hollow shaft is correct. The correct position can be determined by turning the input shaft or by swivelling the gear unit lightly around the hollow shaft.

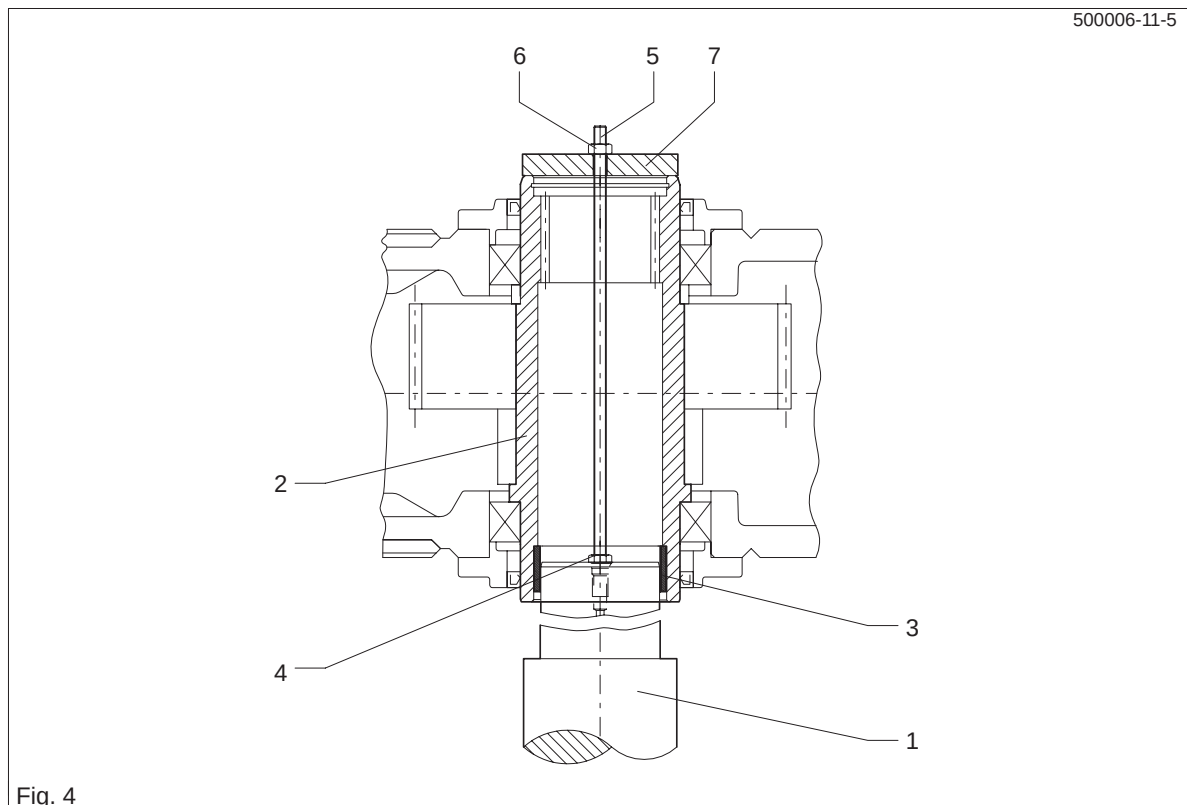


Fig. 4

- | | | |
|-----------------|--------------------|-------------|
| 1 Machine shaft | 4 Nut | 7 End plate |
| 2 Hollow shaft | 5 Threaded spindle | |
| 3 DU bush | 6 Nut | |

- Fitting with loose DU bush

The loose DU bush is pushed onto the machine shaft, fixed rigidly in position with a locating tie and then pulled into the hollow shaft along with the machine shaft (see fig. 4).

Caution!

The hollow shaft must be exactly aligned with the machine shaft to avoid canting. When fitting, ensure that the position of the teeth between the machine shaft and hollow shaft is correct. The correct position can be determined by turning the input shaft or by swivelling the gear unit lightly around the hollow shaft.

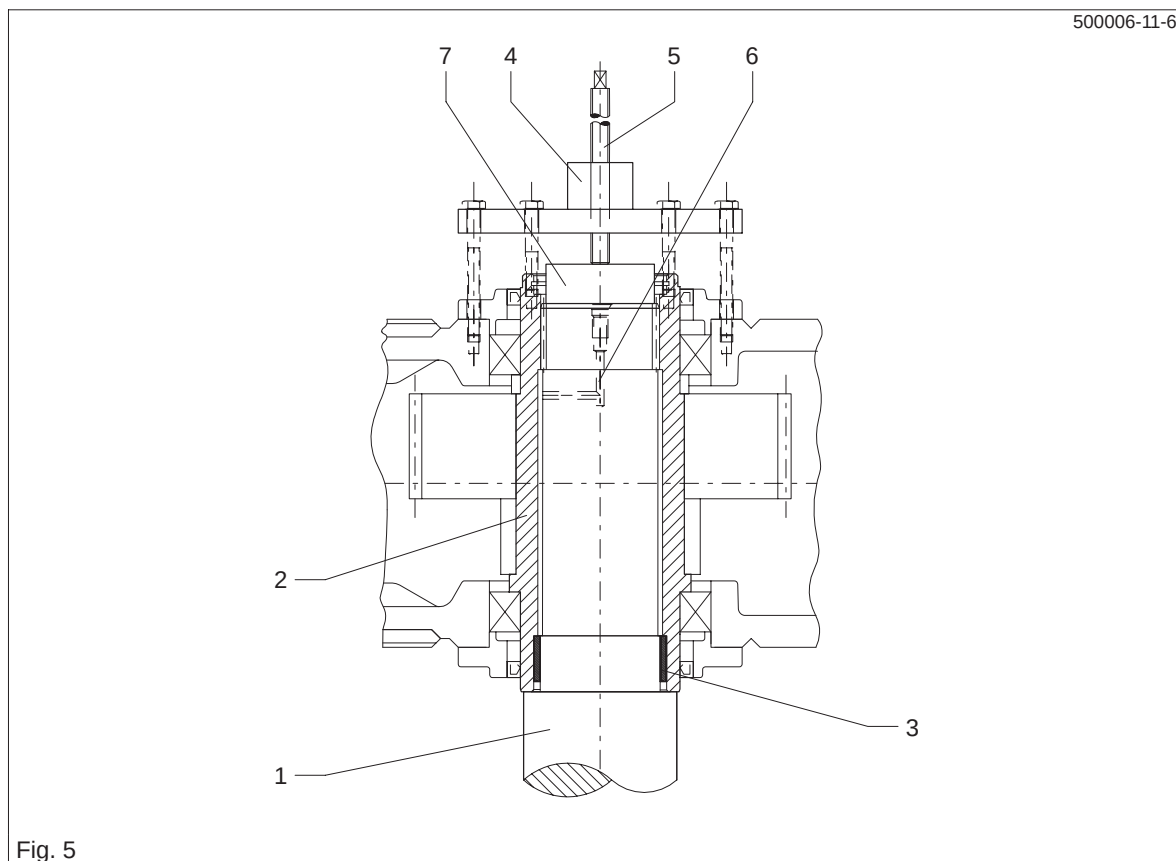
Instead of the nut and threaded spindle shown in the diagram, other types of equipment such as a hydraulic lifting equipment (type Lucas) may be used.

6.4.1.2 Axial securing device

Depending on type, secure the hollow shaft axially on the machine shaft (e.g. with circlip, end plate, set screw etc.).

6.4.2 Dismantling

- Remove the axial securing device from the hollow shaft.
- If frictional corrosion has formed on the seating surfaces, press rust-releasing or other agent (e.g. with a pump) into the bore cavity to reach the seats. The rust releaser is forced in through the bores in the machine shaft (see fig. 5) or end plate (fig. 6).
- Depending on the facilities available, the gear unit can be forced off the shaft on site using forcing screws in an end plate (see fig. 6), a central threaded spindle or a Lucas hydraulic lifting unit.
- When the rust-releasing agent has taken effect, pull the gear unit off with the device as shown in figs. 5 or 6.



- 1 Machine shaft
- 2 Hollow shaft
- 3 DU bush
- 4 Hydraulic lifting unit

- 5 Threaded spindle
- 6 Bore for rust-releasing agent
- 7 Plate for forcing out

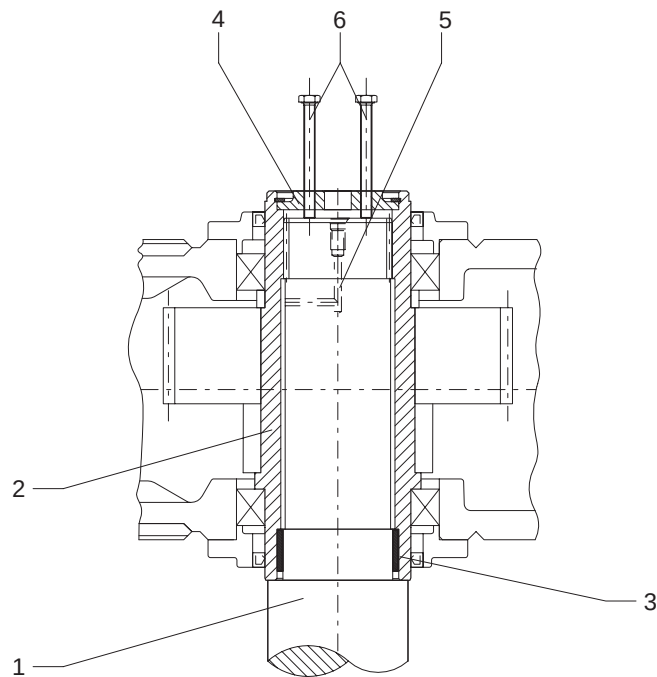


Fig. 6

- 1 Machine shaft
- 2 Hollow shaft
- 3 DU bush

- 4 End plate
- 5 Bore for rust-releasing agent
- 6 Forcing screws

Caution!

Avoid twisting or canting when pulling the unit off.

Caution!

If the counter force is provided not only by the hollow shaft, but by the housing also, as shown in fig. 5, the forces used must not exceed the values given in the following table.

Gear-unit size	Maximum force N	Gear-unit size	Maximum force N
2	11700	11	97200
3	15200	12	113600
4	22600	13	140000
5	33000	14	160000
6	37500	15	193000
7	50000	16	215000
8	56000	17	240000
9	65000	18	266000
10	82000	19 ... 26	On request

Table 6.2: Maximum press-off forces

Caution!

If the above figures are exceeded, the hollow-shaft bearings or other gear components may be irreparably damaged. Before replacing the gear unit on the machine shaft, always check the bearings for any signs of damage.

Note:

When using forcing screws or threaded spindles, the head of the thread pushing against the working machine should be rounded and well greased to reduce the risk of seizing at this point.

6.5 Shaft-mounting gear unit with hollow shaft and shrink disk

6.5.1 Assembly

- Remove the anti-corrosion paint from the hollow shaft and machine shaft with benzene.

Caution!

Do not allow the benzene to contact the shaft sealing rings.



**Ensure adequate ventilation. Do not smoke when working with benzene!
Danger of explosion.**

- Check the hollow and machine shafts to ensure that seat and edges are not damaged. If necessary, rework the parts with a suitable tool and clean them again.

Caution!

The bore of the hollow shaft and the machine shaft must be absolutely free of grease in the area of the shrink-disk seat.

This is essential for safe and reliable torque transmission.

Do not use contaminated solvents or dirty cloths for removing grease.

6.5.1.1 Fitting

- Fitting with integrated DU bush
- Fit the gear unit by means of nut and threaded spindle. The counter force is provided by the hollow shaft.

Caution!

The hollow shaft must be exactly aligned with the machine shaft to avoid canting.

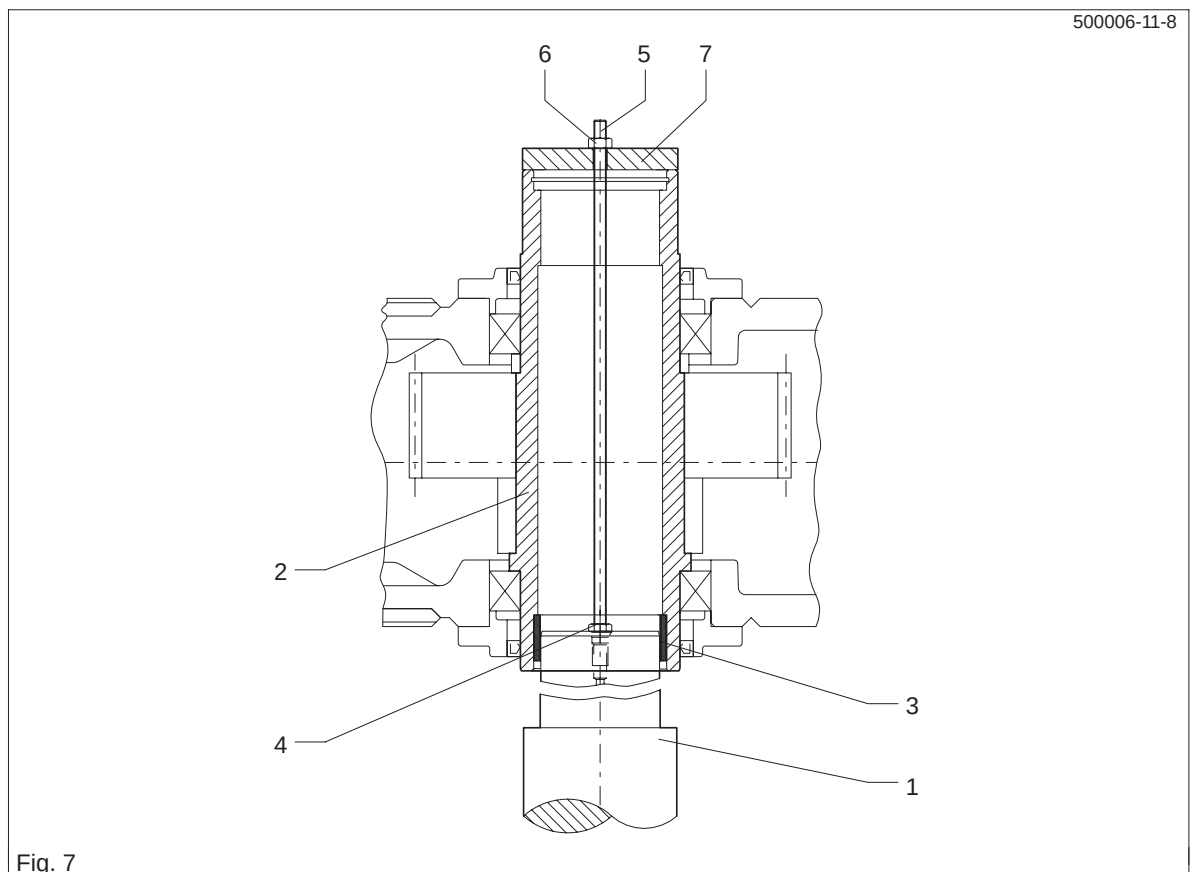


Fig. 7

- | | | |
|-----------------|--------------------|-------------|
| 1 Machine shaft | 4 Nut | 7 End plate |
| 2 Hollow shaft | 5 Threaded spindle | |
| 3 DU bush | 6 Nut | |

- Fitting with loose DU bush

The loose DU bush is pushed onto the machine shaft, fixed rigidly in position with a locating tie and then pulled into the hollow shaft along with the machine shaft (see fig. 7).

Caution!

The hollow shaft must be exactly aligned with the machine shaft to avoid canting.

Instead of the nut and threaded spindle shown in the diagram, other types of equipment such as a hydraulic lifting equipment (type Lucas) may be used.

6.5.1.2 Axial securing device

If the shrink disk is fitted according to instructions, the gear unit is fixed securely in the axial direction. Additional securing is not required.

6.5.2 Shrink disk type HSD

6.5.2.1 Assembly of shrink disk

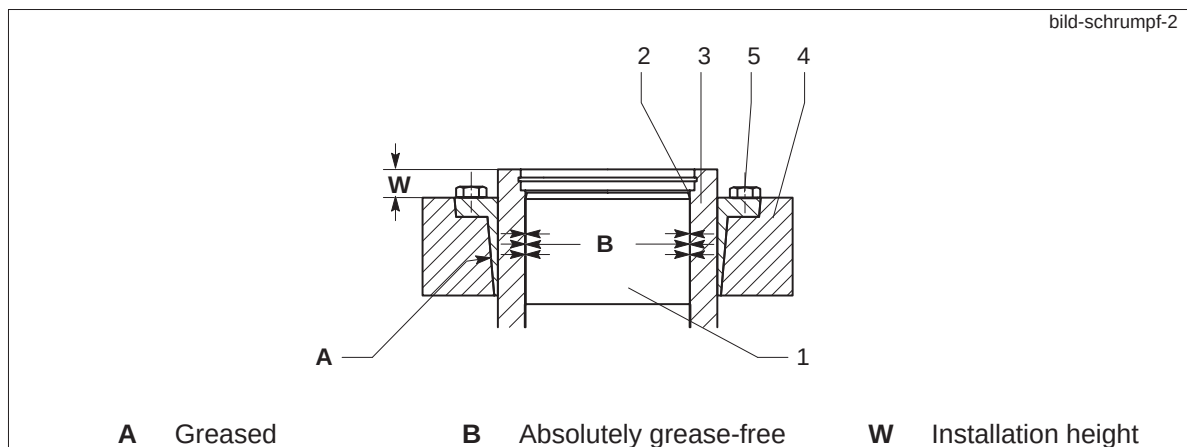
The shrink disk is delivered ready for installation.

Caution!

It must not be dismantled before tensioning for the first time.

Caution!

The bore of the hollow shaft and the machine shaft must be absolutely free of grease in the area of the shrink-disk seat. This is essential for safe and reliable torque transmission. Do not use contaminated solvents or dirty cloths for removing grease.



- 1 Machine shaft
- 2 Hollow shaft

- 3 Inner ring
- 4 Outer ring

- 5 Tensioning bolt

For the exact installation height (W) of the shrink disk, refer to the dimensional drawing (see section 1 "Technical data")

Note:

The outer surface of the hollow shaft may be greased in the area of the shrink-disk seat.

Caution!

Do not tighten the tensioning bolts until the machine shaft is installed.

Tighten all the tensioning bolts gradually one after the other, working round several times.

Caution!

Do not tighten diametrically opposite bolts one after the other.

Tighten the tensioning bolts until the front surfaces of the inner and outer rings are aligned.

Note:

This allows the tension state to be checked visually.

Caution!

To avoid overloading the individual bolts, the maximum tensioning torque (see table) must never be exceeded. The alignment of the front surfaces has priority. If they cannot be aligned by tightening, consult the manufacturer.

Tensioning-bolt thread	Max. tensioning torque per bolt (with $\mu = 0.1$)	
	Strength class 10.9 Nm	Strength class 12.9 Nm
M 6	12	14.5
M 8	29	35
M 10	58	70
M 12	100	121
M 14	160	193
M 16	240	295
M 20	470	570
M 24	820	980
M 27	1210	1450
M 30	1640	1970
M 33	2210	2650
M 36	2850	3420

Table 6.3: Maximum torques for tensioning bolts

Replace the protective cowl.

6.5.2.2 Removing the shrink disk

The loosening process is similar to that of tightening.

Loosen the tensioning bolts gradually one after the other, working round several times.

If the outer ring does not release from the inner ring, several tensioning bolts can be removed and then screwed into adjacent forcing threads.

The rings can then be released without difficulty.

Pull the shrink disk off the hollow shaft.

6.5.2.3 Cleaning and greasing the shrink disk

Shrink disks do not have to be dismantled and re-greased before being re-installed.

The shrink disk should only be dismantled and cleaned if it is dirty.

Caution!

Following cleaning, only the inner sliding surfaces of the shrink disk should be re-greased.

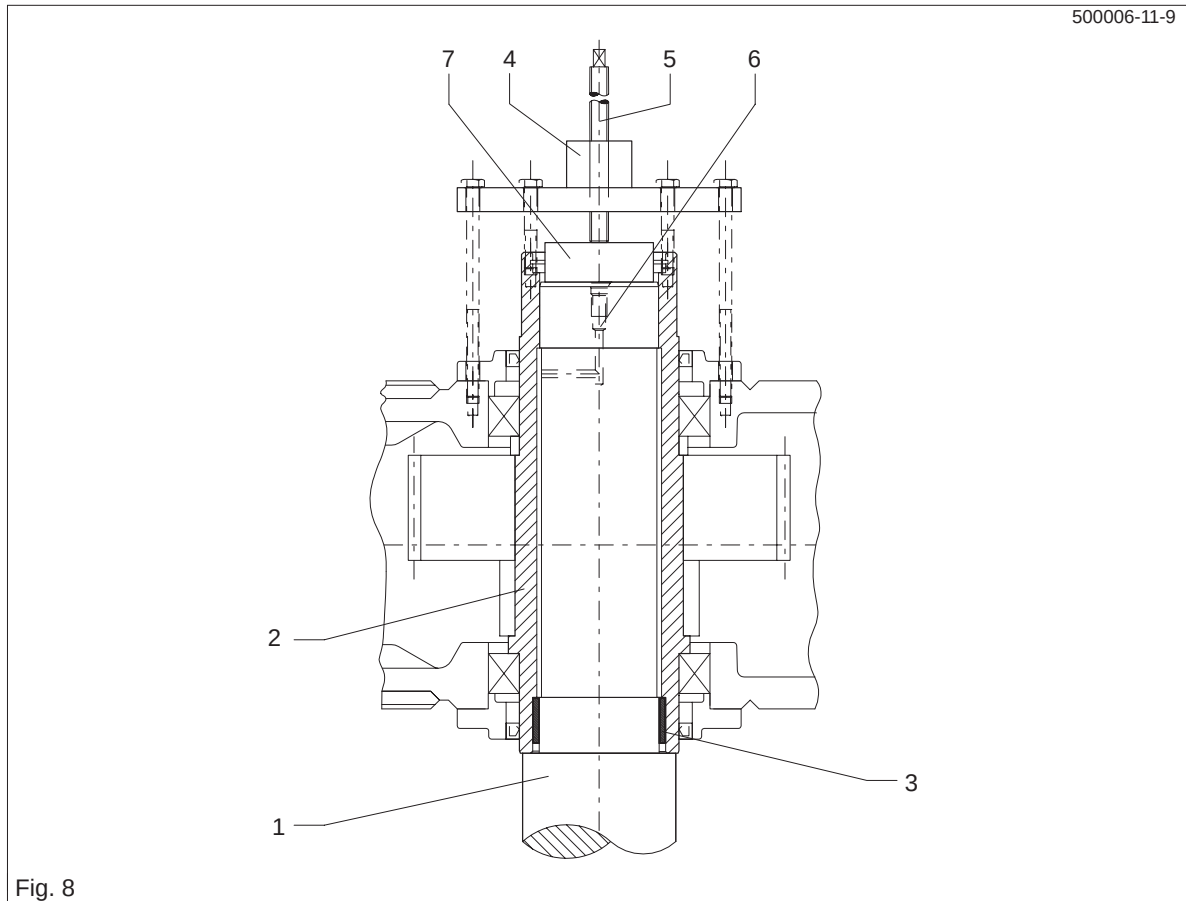
Use a solid grease with a coefficient of friction of $\mu = 0.04$ as shown in the following table.

Grease	Form	Manufacturer
Molykote 321 R (lubricating paint)	Spray	DOW Corning
Molykote Spray (powder spray)	Spray	DOW Corning
Molykote G Rapid	Spray or Paste	DOW Corning
Aemasol MO 19 P	Spray or Paste	A. C. Matthes
Molykombin UMFT 1	Spray	Klüber Lubrication
Unimoly P 5	Powder	Klüber Lubrication

Table 6.4: Lubricants for shrink disk after cleaning

6.5.3 Dismantling

- If frictional corrosion has formed on the seating surfaces, press rust-releasing or other agent (e.g. with a pump) into the bore cavity to reach the seats. The rust releaser is forced in through the bores in the machine shaft (see fig. 8) or end plate (fig. 9).
- Depending on the facilities available, the gear unit can be forced off the shaft on site using forcing screws in an end plate (see fig. 9), a central threaded spindle or a Lucas hydraulic lifting unit.
- When the rust-releasing agent has taken effect, pull the gear unit off with the device as shown in figs. 8 or 9.



- 1 Machine shaft
- 2 Hollow shaft
- 3 DU bush
- 4 Hydraulic lifting unit

- 5 Threaded spindle
- 6 Bore for rust-releasing agent
- 7 Plate for forcing out

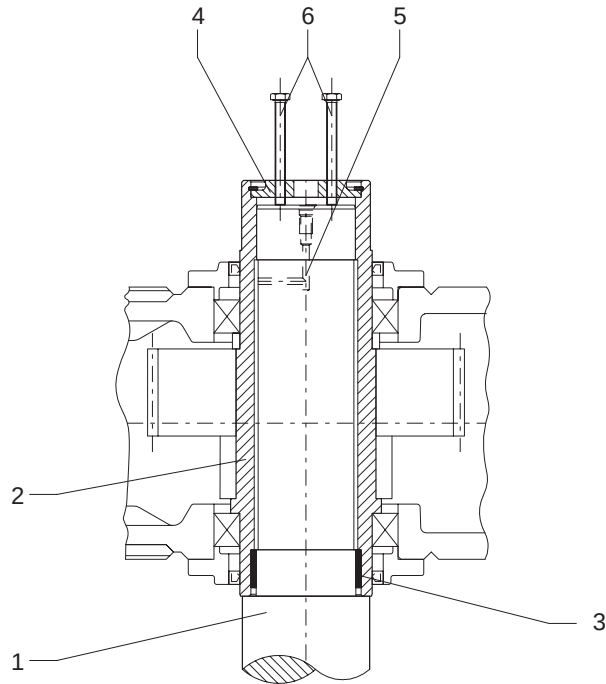


Fig. 9

- | | |
|-----------------|---------------------------------|
| 1 Machine shaft | 4 End plate |
| 2 Hollow shaft | 5 Bore for rust-releasing agent |
| 3 DU bush | 6 Forcing screws |

Caution!

Avoid twisting or canting when pulling the unit off.

Caution!

If the counter force is provided not only by the hollow shaft, but by the housing also, as shown in fig. 8, the forces used must not exceed the values given in the following table.

Gear-unit size	Maximum force N	Gear-unit size	Maximum force N
2	11700	11	97200
3	15200	12	113600
4	22600	13	140000
5	33000	14	160000
6	37500	15	193000
7	50000	16	215000
8	56000	17	240000
9	65000	18	266000
10	82000	19 ... 26	On request

Table 6.5: Maximum press-off forces

Caution!

If the above figures are exceeded, the hollow-shaft bearings or other gear components may be irreparably damaged. Before replacing the gear unit on the machine shaft, always check the bearings for any signs of damage.

Note:

When using forcing screws or threaded spindles, the head of the thread pushing against the working machine should be rounded and well greased to reduce the risk of seizing at this point.

6.6 Shaft-mounting gear unit with flanged shaft

Caution!

The front area of the flanged shaft must be absolutely free of grease. This is essential for safe and reliable torque transmission. Do not use contaminated solvents or dirty cloths for removing grease.

Caution!

Tighten diametrically opposite tensioning bolts to full torque.

- Tightening torque of flange bolts for gear units:

Gear-unit size	Strength class		Tightening torque
	Bolt DIN 931	Nut DIN 934	
5 ... 6	10.9	10	610 Nm
7 ... 10	10.9	10	1050 Nm
11 ... 16	10.9	10	2100 Nm
17 ... 20	10.9	10	3560 Nm
21 ... 26	10.9	10	5720 Nm

Table 6.6: Tightening torques for flange connections

Note:

Damaged bolts should be replaced with new ones of the same type and strength class.

6.7 Shaft mounting gear unit with block flange

Caution!

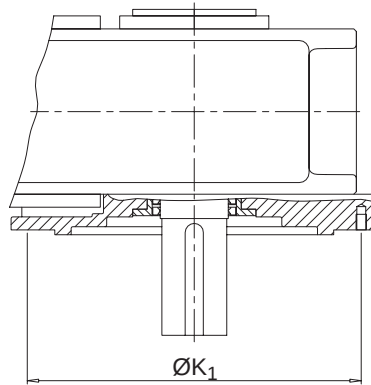
The front area of the block flange must be absolutely free of grease. This is essential for safe and reliable torque transmission. Do not use contaminated solvents or dirty cloths for removing grease.

Caution!

Tighten diametrically opposite tensioning bolts to full torque.

The joint bolts must be tightened to the prescribed torque. For the correct torque, refer to section 10.2.11. We recommend bolts of strength class 8.8. The transmittable tensioning torque is limited by the bolted joint on bolt circle K_1 .

500006-11-11



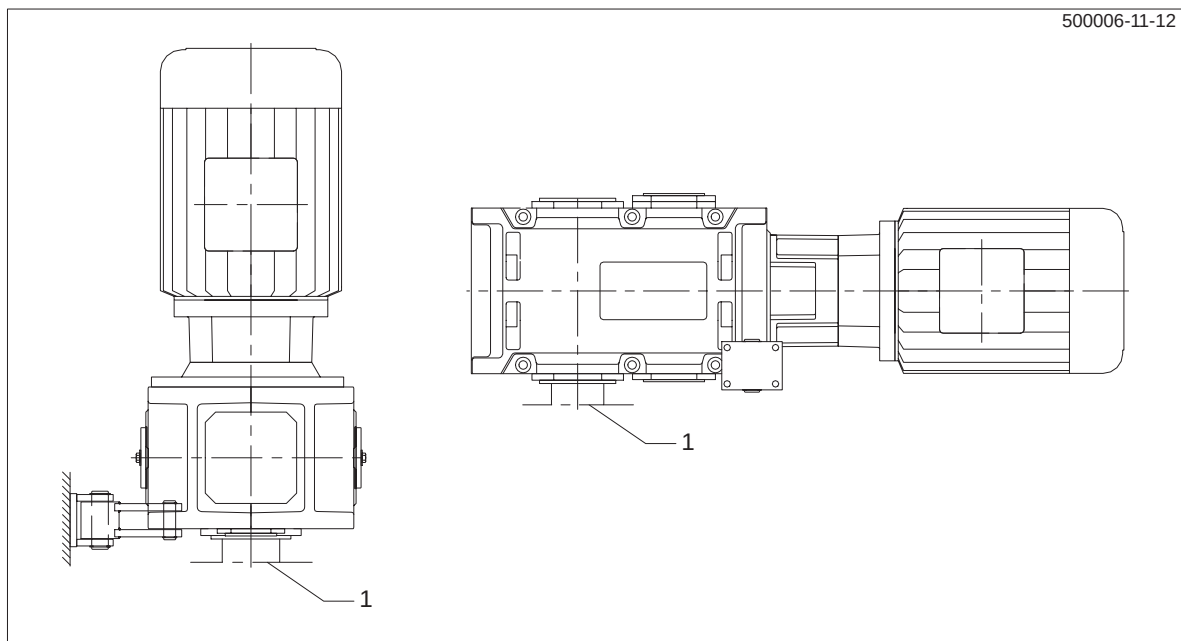
Type	Configuration 1)	
	A	B
H2.V		
H3.V		
H4.V		
B2.V		
B3.V		
B4.V		

1) Types with hollow output shaft available on request.

6.8 Mounting the torque arm for gear housing

6.8.1 Attaching the torque arm

Caution! The torque arm must be mounted stress-free on the machine side



1 Machine side

Gear-unit size	Max. standard-motor size					
	Gear type			Gear type		
	H2	H3	H4	B2	B3	B4
1	–	–	–	112	–	–
2	–	–	–	132	–	–
3	180	–	–	180	180	–
4	200	–	–	200	200	–
5 ... 6	225	225	–	225	225	160
7 ... 8	280	280	180	280	280	200
9 ... 10	280	280	225	280	280	225
11 ... 12	315M	315M	250	315M	315M	280
13 ... 14	–	355	315M	355	355	315M
15 ... 16	–	355	315	–	355	355M
17 ... 18	–	355	355M	–	355	355
19 ... 26	On request					

Table 6.7: Motor types and torque arms

Note: Larger motors should be used only with the consent of FLENDER.

- Foundation type for fastening torque arm, see point 6.2.1 "Foundation".

6.9 Gear unit with cooling coil

- Before connecting the cooling coil remove the plugs from the connection bush of the cooling coil and flush the coil well to remove any dirt.
- Install the cooling-water in- and outflow pipes (for exact position of connections, see dimensional drawing).

Note: To avoid excess pressure, the water inlet must be fitted with a flow-control device, e.g. a pressure reducer or suitable stop valve.

6.10 Gear unit with air oil-cooler

- Connect the pressure monitor and indicator for switchover filter (for gear sizes ≥ 13) electrically
- Connect the fan motor electrically

6.11 Gear unit with water oil-cooler

- Before connecting the water oil-cooler remove the plugs from the water connection and flush the cooler well to remove any dirt.
- Install the cooling-water in- and outflow pipes. The direction of flow is of no importance. (for exact position of connections, see dimensional drawing).
- Connect pressure monitor electrically

6.12 Gear units with heating element

- Connect temperature monitor electrically
- Connect heating elements electrically

6.13 Gear units with oil-temperature measurement

- Connect resistance thermometer with evaluating instrument (to be provided by customer) electrically

6.14 Gear units with speed transmitter

- Connect speed transmitter electrically

6.15 General note on add-on components

Note: For operation and maintenance of the components described in points 6.10 to 6.14, observe the instructions in the order-specific appendix.
For technical details, refer to the order-specific equipment list.

7. Start-up

Note: Observe the "Safety notes" in section 3.

7.1 Procedure before start-up

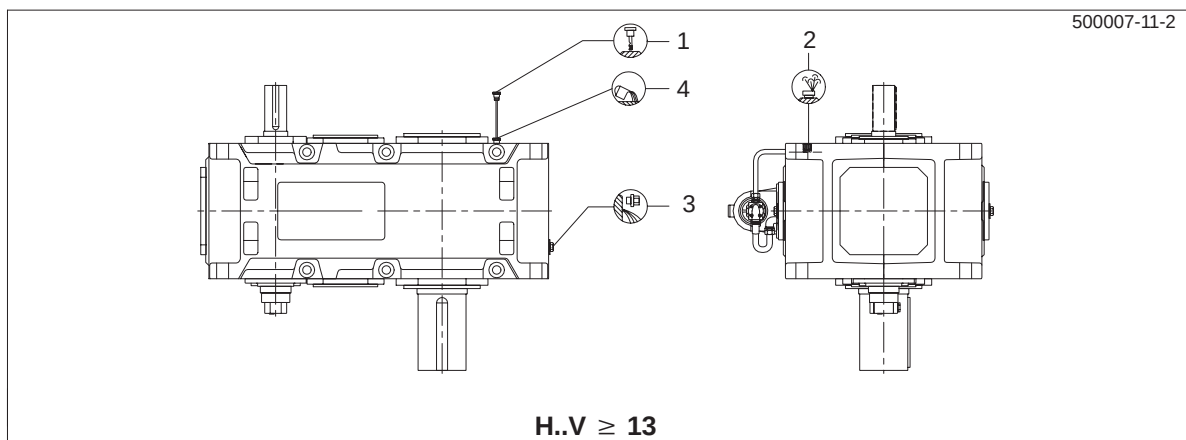
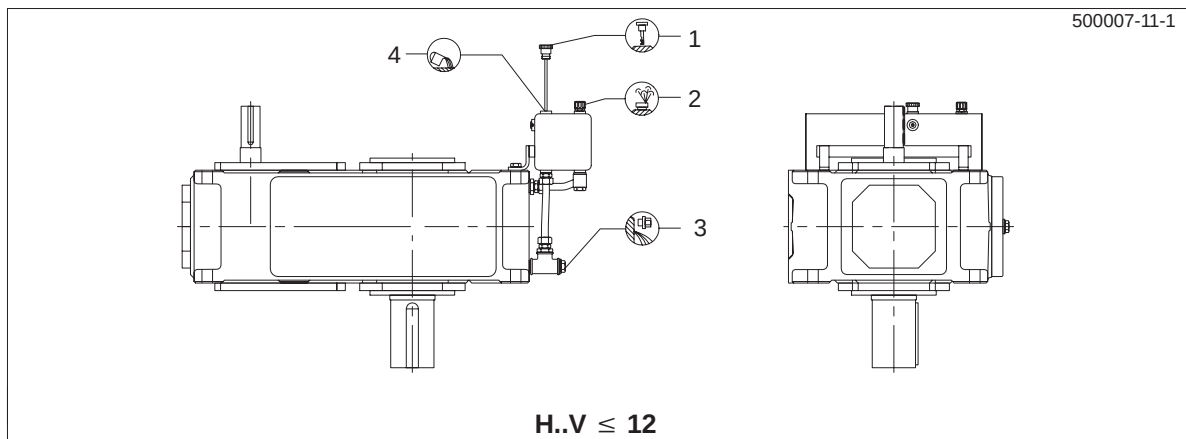
7.1.1 Removal of conservation agent

- Unscrew the oil drain plug or open the draining valve, allow the conservation agent or run-in oil to drain into a suitable container and dispose of it in accordance with regulation. Any residual-oil draining screws should also be opened e.g. in the case of gear units with backstop). The location of the oil draining point is marked by the symbol in the dimensional drawing in the unit documentation.



Remove any oil spillage immediately with an oil-binding agent.

- Replace the oil drain plug



- | | |
|-------------------------|------------------|
| 1 Oil dipstick | 3 Oil drain plug |
| 2 Venting or plug screw | 4 Oil inlet |

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Caution!

Before start-up, replace the yellow plastic plug with venting screws with cap (see notice on gear unit).

7.1.2 Charging with lubricant

- To fill with oil, screw out the oil dipstick on the gear unit or oil-equalising tank

Caution!

Fill the gear unit with fresh oil using a filter (max. mesh 25 µm).

Note:

For the correct type of oil (of various brands) to be used refer to the operating instructions BA 7300 EN supplied separately.

Information on the type, viscosity and quantity of the oil is given on the rating plate on the gear unit.

The quantity of oil stated on the rating plate is an approximation only. The markings on the oil dipstick must **always** be observed.

Note:

With gear units fitted with pressure lubrication or oil-cooling systems, the oil circuit must also be charged. To do this, briefly start up the gear unit with added pump as described in section 8.

- Check the oil level in the gear housing or in the oil-equalising tank using the dipstick

Note:

The oil level must be at the upper mark on the oil dipstick.



Remove any oil spillage immediately with an oil-binding agent.

- Replace the oil dipstick.

7.1.2.1 Oil quantities

Type	Oil quantity (approx.) in litres for size											
	1	2	3	4	5	6	7	8	9	10	11	12
H2.V	–	–	14	25	23	27	58	62	100	110	160	180
H3.V	–	–	–	–	35	37	60	72	100	110	170	190
H4.V	–	–	–	–	–	–	50	60	95	110	165	180
B2.V	7	11	16	28	41	50	75	90	115	135	190	215
B3.V	–	–	15	28	32	35	52	67	115	125	180	200
B4.V	–	–	–	–	36	40	60	70	110	130	180	195

Table 7.1: Approximate figures for required oil quantities in vertical gear units with **splash lubrication** (gear units with oil-equalising tank)

Type	Oil quantity (approx.) in litres for size																	
	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
H2.V	10	11	22	25	42	46	60	70	80	90	140	150	175	185	On request			
H3.V	13	15	25	30	40	45	66	75	115	126	180	190	190	200				
H4.V	–	–	20	25	38	45	65	75	95	110	150	160	190	200				
B2.V	20	23	35	38	53	60	86	95	100	110	145	160	210	220				
B3.V	12	13	22	28	48	52	75	85	95	110	165	190	210	240				
B4.V	15	16	30	35	60	67	75	85	130	150	200	235	215	250				

Table 7.2: Approximate figures for required oil quantities in vertical gear units with **pressure lubrication**

7.2 Start-up

- Check the oil level in the housing with the oil dipstick

Note: When the oil is cool, the level should be at the upper oil dipstick mark. When the oil is warm it may slightly exceed this mark. It must not be allowed to fall below the mark. If necessary, top up to the correct level.

Gear unit with cooling coil or water oil-cooling systems:

- Fully open the stop valves in the coolant in- and outflow pipes.

Note: For technical details, refer to the order-specific equipment list.

7.3 Removal from service

- To take the gear unit out of service, first switch off the drive assembly.



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.

- With gear units fitted with cooling coil or water oil-coolers, close the stop valve on the water inflow and outflow pipes. To prevent freezing, drain the water from the cooling coil or the water oil-cooler.
- During longer periods of disuse, start the unit up briefly at intervals of approx. 3 weeks. If it is to remain out of service for longer than six months, fill it with conservation agent (see point 7.3.1).

7.3.1 Interior conservation during longer disuse

Depending on the type of lubrication or shaft sealing, the following types of internal corrosion protection can be used.

7.3.1.1 Interior conservation with gear oil

Gear units with splash-lubrication systems and contacting shaft seals can be filled with the correct type of service oil up to a point just below the venting screw.

7.3.1.2 Interior conservation with special conservation agent

Before longer periods of storage, gear units with pressurised lubrication systems, oil-circuit cooling or non-contacting shaft seals should be allowed to idle with a charge of conservation agent.

Duration of protection	Conservation agent	Special measures
less than 24 months	Castrol Alpha SP 220 S	Close gear unit, replace venting screw or air filter with a yellow plug screw, (replace the original parts before start-up)
For storage periods longer than 24 months, renew the conservation agent. For storage periods over 36 months, FLENDER should be consulted.		

Table 7.3: Conservation procedure when using mineral oil or PAO-based synthetic oil

Duration of protection	Conservation agent	Special measures
less than 36 months	Special anti-corrosion oil TRIBOL 1390 1)	Close gear unit, replace venting screw or air filter with a yellow plug screw, (replace the original parts before start-up)
For storage periods over 36 months, FLENDER should be consulted.		

Table 7.4: Conservation procedure when using PG-based synthetic oil

- 1) Resistant to tropical conditions and sea water. Max. ambient temperature 50 °C

7.3.1.3 Interior conservation procedure

- Switch the gear unit off and drain the oil as described in section 10 "Maintenance and repair".
- Pour in the conservation agent given in Table 7.3 or 7.4 through the venting hole until it reaches the top mark on the oil dipstick.
- Close the venting hole.
- Start the gear unit and allow it to idle for a short time.
- Unscrew the oil drain plug and allow the conservation agent to drain into a suitable container and dispose of it according to regulation.



**The conservation agent emerging from the gear unit may be hot.
Wear gloves to avoid scalding.**

- Replace the oil drain plug.

Caution!

**Before re-starting the gear unit, replace the plug screw with the venting screw or air filter.
See also point 7.1.1**

7.3.2 Exterior conservation

Duration of protection	Conservation agent	Layer thickness	Remarks
less than 12 months	Tectyl 846 K19	approx. 50 µm	Long-term wax-based conservation agent, resistant to sea water and tropical conditions, soluble in benzine

Table 7.5: Exterior protection of shaft ends and other bare surfaces

7.3.2.1 Exterior conservation procedure

- Clean the surfaces
- Apply grease to shaft-sealing rings to protect the sealing lips from conservation agent
- Apply conservation agent

8. Operation

Note: Observe the "Safety notes" in section 3.

8.1 General operating data

During operation, the unit must be monitored for:

- overheating (In continuous operation, the gear unit is designed for a temperature of 90 °C with mineral oil. For higher temperatures, synthetic oils must be used. The maximum short-term operating temperature is 100 °C. See also section 10.)

- changes in running noise

- possible oil leakage at the housing and shaft seals

and

- correct oil level (see section 7 "Start-up")

Note: To check the oil level, stop operation of gear unit.
When the oil is warm it may slightly exceed the top mark of the oil dipstick. It must not be allowed to fall below the lower mark. If necessary, top up to the correct level.

Caution!

If any irregularities are noticed during operation, or if the pressure monitor in the oil-cooling system (if installed) triggers the alarm, switch the drive assembly off at once. Determine the cause of the fault using the table in section 9.

This table contains a list of possible faults, their causes and suggested remedies.

Should it not be possible to identify the cause, or the unit cannot be repaired with the facilities available, you are advised to contact one of our customer-service offices for specialist assistance. See section 11.

9. Disturbances, reasons and remedy

Note: Observe the "Safety notes" in section 3.

9.1 General information on faults and malfunctions

Note: Faults and malfunctions occurring during the guarantee period and which require repair work on the gear unit, must be carried out only by FLENDER Customer Service. In the case of faults and malfunctions occurring after the guarantee period, and whose cause cannot be precisely identified, we advise our customers to contact our Customer Service.

Caution!

FLENDER will not be bound by the terms of the guarantee or otherwise be responsible in cases of improper use of the gear unit, modifications carried out without the consent of FLENDER, or use of spare parts not supplied by FLENDER.



**To remedy faults and malfunctions, the gear unit must always be taken out of service.
Secure the drive assembly to prevent it from being started up accidentally.
Attach a warning notice to the start switch.**

9.2 Possible faults

Malfunctions	Causes	Remedy
Changes in gear noise	Damage to gear teeth	Contact Customer Service. Check teeth on all components, replace any damaged parts.
	Excessive bearing play	Contact Customer Service. Adjust bearing play.
	Defective bearing	Contact Customer Service. Replace defective bearing.
Loud noises in area of gear-unit fastening	Fastening has worked loose	Tighten bolts / nuts to prescribed torque. Replace damaged bolts / nuts.
Increased temperature at bearing points	Oil level in housing too low	Check oil level at room temperature, top up if necessary
	Oil too old	Check date of last oil change, change oil if necessary. See section 10.
	Mechanical oil pump defective	Check oil pump or replace if necessary
	Bearing defective	Contact Customer Service. Check bearings and replace if necessary
Gear unit is oily	Inadequate sealing of housing cover or joints	Seal joints
	Labyrinth seals oiled up	Check oil charge, clean labyrinth seals if necessary, incorrect transport position
Oil leakage from gear unit	Inadequate sealing of housing cover or joints	Check seals, replace if necessary, seal joints
	Radial shaft sealing rings defective	Check radial shaft sealing rings, replace if necessary

Malfunctions	Causes	Remedy
Water in oil	Oil froths in sump	Check oil state using test-tube method for water contamination. Have oil analysed by laboratory.
	Defective oil cooler or cooling coil	Repair or replace oil cooler or cooling coil. Fill with oil, find leaks and repair.
	Gear unit exposed to cold air from machine-room ventilator causing condensation	Protect gear unit with suitable insulation. Close air outlet or alter its direction.
Operating temperature too high	Oil level in housing too high	Check oil level and adjust if necessary
	Oil too old	Check date of last oil change, change oil if necessary. See section 10.
	Oil contaminated	Change oil. See section 10.
	Gear units with oil-cooling system: coolant flow too low	Fully open valves in inflow and outflow pipes. Check free flow through oil cooler.
	Coolant temperature too high	Check temperature and correct if necessary
	Oil flow through cooler too low through clogged filter	Clean oil filter. See section 10.
	Defective oil pump	Check function of oil pump, repair or replace oil pump if necessary
	Gear unit with fan: suction opening in cowl and/or housing clogged by dirt	Clean fan cowl and housing
Pressure monitor triggers alarm (gear units with pressurised lubrication, water oil-cooler, air oil-cooler)	Oil pressure < 0.5 bar	Clean or replace cooling coil. See section 10.
		Check oil level at room temperature, top up if necessary. Check oil pump, replace if necessary. Check oil filter, clean if necessary, see section 10.
Indicator on switchover filter triggers alarm	Switchover filter clogged	Switch filter over as instructed, remove clogged filter element and clean
Fault in oil-supply unit		Consult operating instructions for oil-supply unit

Table 9.1: Faults, causes and remedies

10. Maintenance and repair

Note: Observe the "Safety notes" in section 3.

10.1 General notes on maintenance

Note: Maintenance and repair work should be done with great care by trained and qualified personnel only.

Caution!

The periods stated in table 10.1 depend on the conditions under which the gear unit is operated. For this reason only average figures can be given which refer to:

a daily operating time of 24 hours,
a duty factor of 100%,
an input-drive speed of 1500 1/min,
and a maximum oil temperature of 90 °C (mineral oil only)
100 °C (synthetic oil only)

Note: Under different conditions, the frequencies stated below should be adjusted accordingly.

Measures	Periods	Remarks
Check oil temperature	Daily	
Check for unusual gear unit noise	Daily	
Check oil level	Monthly	
Check gear unit for leaks	Monthly	
Test oil for water content	After 400 service hours, once per year at least	See point 10.2.1
First oil change after start-up	After 400 service hours	See point 10.2.2
Subsequent oil changes	Every 18 months or 5000 service hours ¹⁾	See point 10.2.2
Clean oil filter	Every 3 months	See point 10.2.3
Clean venting screw	Every 3 months	See point 10.2.4
Re-grease Taconite seals	Every 3000 service hours, every 6 months at least	See point 10.2.5
Type with oil-dam pipe - recharge with grease	Every 5000 service hours, every 10 months at least	See point 10.2.5
Clean fan, fan cowl and gear unit housing	Simultaneously with oil change	See point 10.2.6
Check cooling coil for deposits	Approx. every 2 years along with oil change	See point 10.2.7
Check condition of air oil-cooler	Simultaneously with oil change	See point 10.2.8
Check condition of water oil-cooler	Simultaneously with oil change	See point 10.2.9
Check tightness of fastening bolts	After first oil change, thereafter every second oil change	See point 10.2.11
Complete inspection of gear unit	Approx. every 2 years along with oil change	See point 10.2.12

Table 10.1: Maintenance and repair work

1) when using synthetic oils the periods can be tripled.

10.2 Description of maintenance and repair work

10.2.1 Test water-content of oil

For detailed information on testing the water-content of oil, apply to the lubricant manufacturer.

10.2.2 Changing the oil

Caution!

When changing the oil, always re-charge the gear unit with the same type of oil. Never mix different types of oil or oils made by different manufacturers. Never mix synthetic oils with mineral-based oils or with other synthetic oils. When changing from mineral-based oil to synthetic oil or from one type of synthetic oil to another, flush the gear unit well with the new type of oil beforehand.

When changing the oil, the housing must be flushed with oil to remove sludge, metallic particles and oil residue. Use the same type of oil as is used for normal operation. High-viscosity oils should be heated beforehand. Ensure that all residues have been removed before filling with fresh oil.

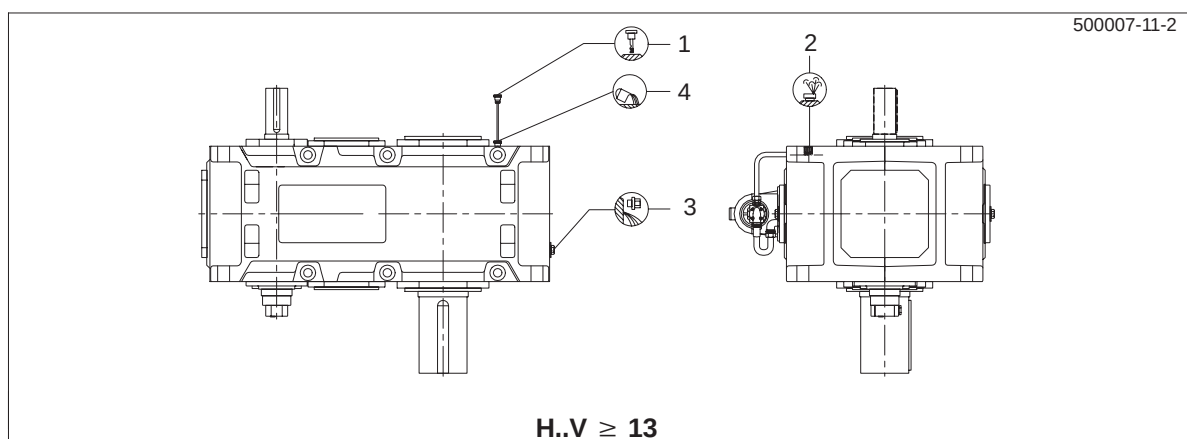
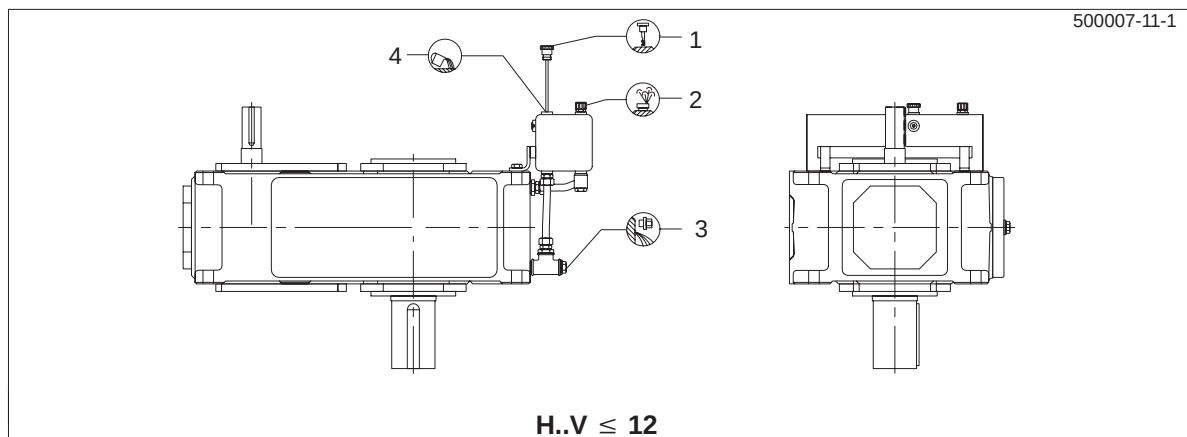
Note: Drain the oil while it is still warm, i.e. immediately after switching the machinery off.

- Stop the gear unit by switching off the drive assembly



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.

- Close the stop valves in the coolant inflow and outflow pipes (gear units with cooling coil or water oil-cooling systems)



- | | |
|-------------------------|------------------|
| 1 Oil dipstick | 3 Oil drain plug |
| 2 Venting or plug screw | 4 Oil inlet |

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

- Place a suitable container under the oil drain plug
- Remove the venting screw on the top of the housing
- Remove the oil drain plug and allow the oil to drain into the container



**The oil emerging from the housing is hot.
Wear gloves to prevent scalding.
Remove any oil spillage immediately with an oil-binding agent.**

- Clean the permanent magnet of the oil drain plug thoroughly
- Replace the oil drain plug

Note: Check the condition of the sealing ring (the sealing ring is vulcanised onto the oil drain plug). If necessary, use a new oil drain plug.

- Clean the oil filter in the oil-cooling system. See point 10.2.3 (gear units with oil-cooling system only)
- To fill with oil, screw out the oil dipstick on the gear unit or oil-equalising tank

Caution!

Fill the gear unit with fresh oil using a filter (max. mesh 25 µm).

Note: For the correct type of oil (of various brands) to be used refer to the operating instructions BA 7300 EN supplied separately.
Information on the type, quantity and viscosity of the oil is given on the rating plate on the gear unit.
The quantity of oil stated on the rating plate is an approximation only. The markings on the oil dipstick must **always** be observed.

Note: With gear units fitted with pressure lubrication or oil-cooling systems, the oil circuit must also be charged. To do this, briefly start up the gear unit with added pump as described in section 8.

- Check the oil level in the gear housing or in the oil-equalising tank using the dipstick

Note: The oil level must be at the upper mark on the oil dipstick.



Remove any oil spillage immediately with an oil-binding agent.

- Replace the oil dipstick.

10.2.3 Cleaning the oil filter

Note: For operation and maintenance, always observe the instructions given in the order-specific appendix.
For technical details, refer to the order-specific equipment list.

10.2.4 Cleaning the venting screw

Remove any dust which has collected on the venting screw. It should be cleaned at least every 3 months. To do this, the venting screw is removed, cleaned with benzine or similar agent and dried. It can also be cleaned by blowing out with compressed air.

10.2.5 For types fitted with Taconite seals or oil-dam pipe, recharge with grease

- Stop the gear unit by switching off the drive assembly



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.

- Inject approx. 30 g lithium-based bearing grease into each of the lubrication points of the Taconite seal.
- Recharge the lubrication point on the oil-dam pipe with 40 g of lithium-based grease for anti-friction bearings.
- The lubrication points are fitted with flat grease nipples type AM10x1 to DIN 3404.



Remove and dispose of any old grease emerging from the seals.

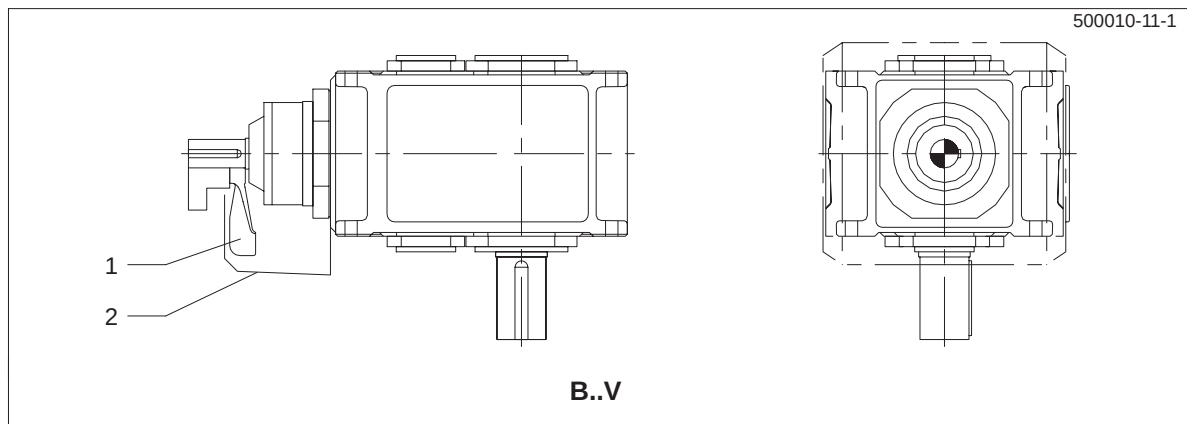
10.2.6 Cleaning the fan and gear unit

Note: A fan can be installed only on type B..V (bevel-helical gear unit).

- Stop the gear unit by switching off the drive assembly



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.



1 Fan

2 Fan cowl

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

- Remove the fan cowl
- Using a stiff brush, remove any dirt adhering to the fan wheel, cowl or safety grid.
- Remove any corrosion.
- Screw safety grid back onto the fan cowl

Caution!

Never clean the gear unit with high-pressure cleaning equipment.

10.2.7 Check cooling coil

- Stop the gear unit by switching off the drive assembly



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.

- Disconnect the water inflow and outflow pipes from the cooling coil
- Check the inside walls of the cooling coil and remove any deposits

Caution!

If the cooling coil is dirty, heat is no longer withdrawn effectively from the gear unit. Any dirt adhering to the inside of the coil should be removed by chemical cleaning or the coil replaced by a new one.

- If deposits have formed on the inside of the cooling coil, the cooling water (or the deposits themselves) should be chemically analysed. These analyses are carried out by companies specialising in chemical cleaning. They also supply the special cleaning agents required. Before using these agents, ensure that they will not damage the coil materials. Observe the manufacturer's instructions at all times.



To avoid burns when working with corrosive cleaning agents, always observe the instructions on safety and application provided by the manufacturer.

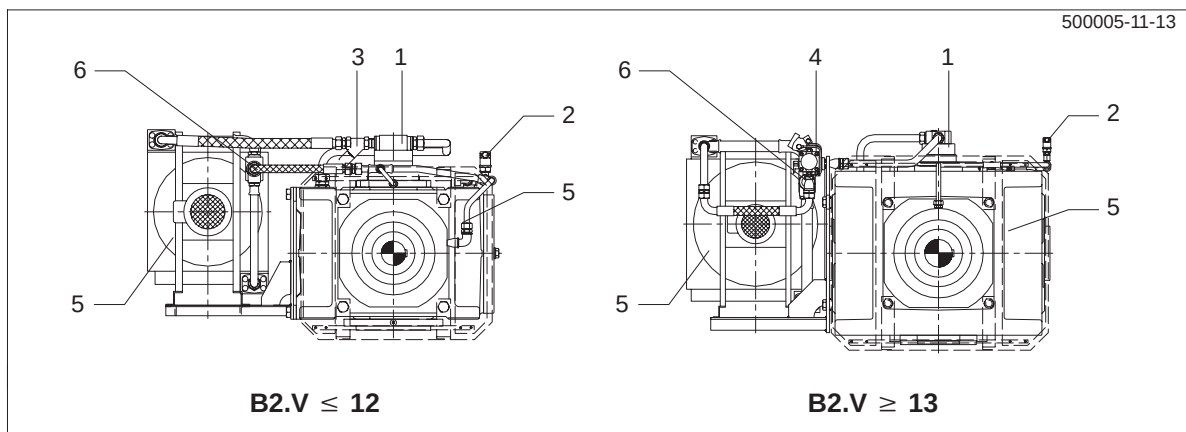
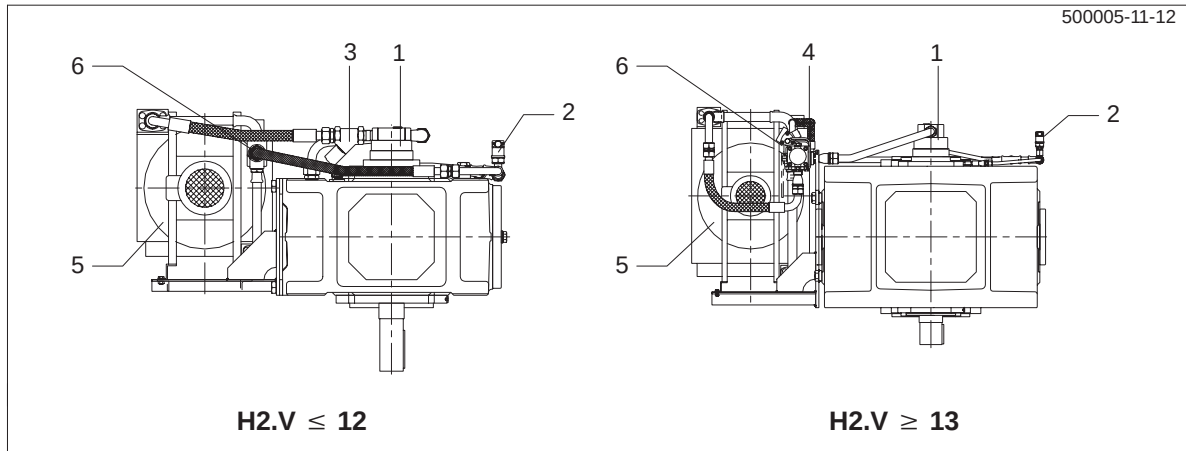
- Cooling coils which are seriously contaminated must be replaced. Consult our Customer Service.
- Re-connect the water inflow and outflow pipes.

10.2.8 Checking air oil-cooler

- Stop the gear unit by switching off the drive assembly



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.



- | | |
|---|---|
| <p>1 Flange pump</p> <p>2 Pressure monitor</p> <p>3 Coarse filter</p> | <p>4 Switchover filter</p> <p>5 Air oil-cooler</p> <p>6 Temperature control valve</p> |
|---|---|

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Note: For operation and maintenance, always observe the instructions given in the order-specific appendix.
For technical details, refer to the order-specific equipment list.

Note: Check the condition of screw connections and replace if necessary.

- Start the gear unit up briefly
- Stop the gear unit and check the oil level with the oil dipstick

Note: The oil level must be at the upper mark on the oil dipstick.

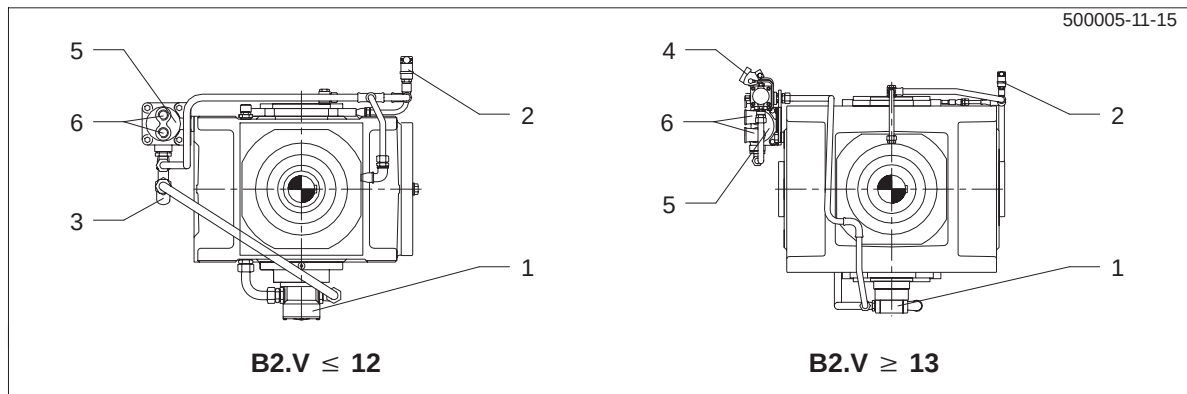
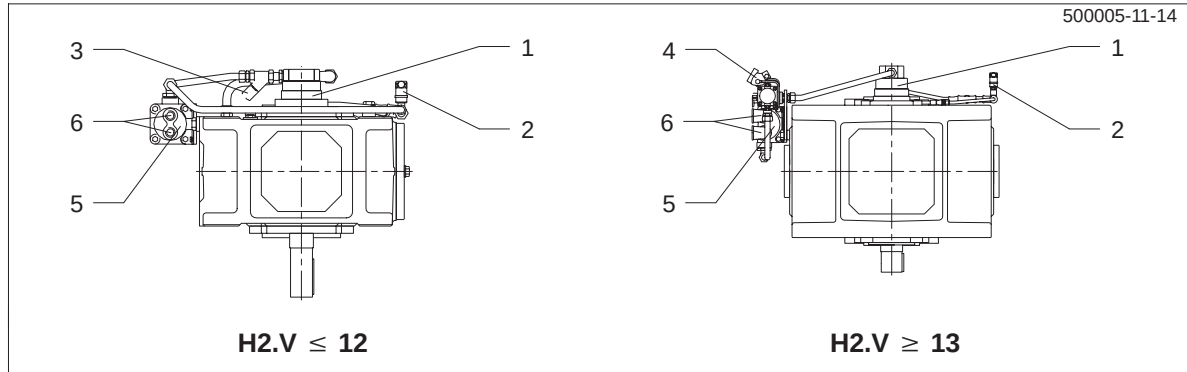
10.2.9 Checking water oil-cooler

- Stop the gear unit by switching off the drive assembly



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.

- Close the stop valves in the coolant inflow and outflow pipes



- 1 Flange pump
- 2 Pressure monitor
- 3 Coarse filter

- 4 Switchover filter
- 5 Water oil-cooler
- 6 Water inlet and outlet

For a detailed illustration of the gear unit, refer to the drawings in the unit documentation.

Note: For operation and maintenance, always observe the instructions given in the order-specific appendix.
For technical details, refer to the order-specific equipment list.

Note: Check the condition of screw connections and replace if necessary.

- Start the gear unit up briefly
- Stop the gear unit and check the oil level with the oil dipstick

Note: The oil level must be at the upper mark on the oil dipstick.

10.2.10 Topping up oil

- Stop the gear unit by switching off the drive assembly



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.

- To fill with oil, screw out the oil dipstick on the gear unit or oil-equalising tank

Caution!

Fill the gear unit with fresh oil using a filter (max. mesh 25 µm).

Note: Always top up with the same type of oil as already in the gear unit (see point 10.2.2). Information on the type, quantity and viscosity of the oil is given on the rating plate on the gear unit.

Note: With gear units fitted with pressure lubrication or oil-cooling systems, the oil circuit must also be charged. To do this, briefly start up the gear unit with added pump as described in section 8.

- Check the oil level in the gear housing or in the oil-equalising tank using the dipstick

Note: The oil level must be at the upper mark on the oil dipstick.



Remove any oil spillage immediately with an oil-binding agent.

- Replace the oil dipstick.

10.2.11 Check tightness of fastening bolts

- Stop the gear unit by switching off the drive assembly



Secure the drive assembly to prevent it from being started up accidentally. Attach a warning notice to the start switch.

- Close the stop valves in the coolant inflow and outflow pipes (gear units with cooling coil or water oil-cooling systems)
- Check tightness of all fastening bolts with a torque wrench

Thread size	Strength class	Tightening torque (with $\mu = 0.20$)
M 10	8.8	60 Nm
M 12	8.8	105 Nm
M 16	8.8	255 Nm
M 20	8.8	500 Nm
M 24	8.8	870 Nm
M 30	8.8	1750 Nm
M 36	8.8	3050 Nm
M 42	8.8	4950 Nm
M 48	8.8	7400 Nm
M 56	8.8	11700 Nm

Table 10.2: Tightening torques

Note: Damaged bolts should be replaced with new ones of the same type and strength class.

10.2.12 General inspection of gear unit

The general inspection of the gear unit should be carried out by the FLENDER Customer Service as our engineers have the experience and training necessary to identify any components requiring replacement.

10.3 Lubricants

The oil selected for use in the gear unit must be of the viscosity (VG class) stated on the rating plate. The viscosity class applies for the contractually agreed operating conditions.

FLENDER must be consulted for any change in operating conditions.

Note: For the correct type of oil (of various brands) to be used refer to the operating instructions BA 7300 EN supplied separately.

We are familiar with the composition of these lubricants and, as far as we are currently aware, they possess the properties with regard to bearing capacity, corrosion resistance, micropitting-bearing capacity and compatibility with seals and interior paint coatings which are necessary for the type of gear unit in question.

We therefore advise our customers to select one of the lubricants stated in the instructions BA 7300 EN, taking into account the VG class given on the rating plate.

Note: This recommendation is in no way intended as a guarantee for the quality of the lubricant supplied. Each lubricant manufacturer is responsible for the quality of his own product.

Should you not wish to follow our recommendation, you assume responsibility for the technical suitability of the lubricant used.

11. Spare parts, service facility addresses

11.1 Stocking spare parts

By stocking the most important spare and consumable parts on site, you can ensure that the gear unit is ready for use at any time.

To order spare parts, refer to the spare-parts list.

For further information, refer to the spare-parts drawing stated in the list.

We guarantee only the original list of spare parts supplied by us.

Caution!

Please note that spare parts and accessories not supplied by us have not been tested or approved by us. The installation or use of such products may therefore impair essential characteristics of the gear unit, thereby posing an active or passive risk to safety. FLENDER will assume no liability for damage caused by spare parts and accessories not supplied by FLENDER.

Please note that certain components often have special production and supply specifications and that we supply you with spare parts which comply fully with the current state of technical development as well as current legislation.

When ordering spare parts always state the following:

Order no. / Item	Type / Size	Part no.	Quantity
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11.2 Spare-part and service facility addresses

When ordering spare parts or the services of our specialist engineers, apply to FLENDER AG.

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12. Declaration by the manufacturer

Declaration by the manufacturer

in accordance with EC Engineering Guideline 98/37/EC, Appendix II B

We hereby declare that the

Gear Unit Types
H.SV, H.VV, H.HV, H.KV, H.DV, H.FV
B.SV, B.VV, B.HV, B.KV, B.DV, B.FV
Sizes 1 to 26

described in these Operating Instructions are intended for incorporation in a machine, and that it is prohibited to put them into service before verifying that the machine into which they are incorporated complies with the EC Guidelines (original edition 98/37/EC including any subsequent amendments thereto).

This Manufacturer's Declaration takes into account all the unified standards (inasmuch as they apply to our products) published by the European Commission in the Official Journal of the European Community.

Bocholt, 1998-08-03



Signature (person responsible for products)

Operating Instructions

BA 7300 EN 07.04

Recommended Lubricants

FLENDER

A. Friedr. Flender AG · 46393 Bocholt · Tel. 02871/92-0 · Telefax 02871/92-2596 · www.flender.com

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1. Lubricants for gear units

Caution!

This lubricant recommendation does not apply to WINERGY AG gear units.

For **axle drives, marine gearboxes and turbo gear units** also special lubricant recommendations apply.

According to manufacturer's information the gear oils listed in these operating instructions are manufactured and supplied **world-wide** in the above-mentioned quality as required by FLENDER (Exceptions: Observe footnotes in the following tables).

1.1 Required quality of gear oils

For FLENDER gear units, only CLP-quality oils containing active agents to DIN 51 517-3 for increasing corrosion protection and resistance to ageing and for reducing wear in mixed-friction areas are approved. The scuffing resistance in the FZG test to DIN 51354-2 must comply with stage 12 or higher under A/8,3/90 test conditions. In the FE-8 rolling bearing test to DIN 51 819-3 rolling element wear must be under 30 mg and cage wear under 100 mg under test conditions D-7,5/80-80.

In addition, the gear oils must meet the following quality requirements demanded by FLENDER:

- Sufficiently high grey-staining resistance in accordance with FVA 54 grey-staining test
 - a) for helical-gear, bevel-gear and planetary-gear units in table A, GS-criteria stage 10 or higher applies
 - b) for all other gear units in tables B and C, GS-criteria stage 9 or higher applies
- Low degree of foaming with less than 15 % foam formation in the FLENDER foam test
- Compatibility with elastomer materials of the shaft-sealing rings
- Compatibility with residues of corrosion-protection agent and run-in oils used by FLENDER
- Compatibility with the paints used by FLENDER in its gear-unit interiors
- Compatibility with liquid seals between bolted-joint surfaces.

For use in worm-gear units moreover:

- Low wear, high pitting resistance and high efficiency (low temperature) in the cylindrical worm-gear unit test

Note:

The use of gear oils which do not comply with the above quality requirements may invalidate our product guarantee. Please note that every oil manufacturer or supplier is responsible for the quality of his product.

Always select the gear oil in accordance with the viscosity class indicated on the rating plate on the gear unit. If a different viscosity is selected, or oil of a type other than those recommended in these instructions, the operator assumes the responsibility for its technical suitability. In order to minimise the technical risk in such a case, we advise the use of a CLP oil of the above quality which should be accompanied by a statement of suitability by the oil manufacturer.

Caution!

The oil viscosities indicated on the rating plates and the instructions in the operating instructions for the gear units must be adhered to.

Deviations are permitted only after consultation with FLENDER!

1.1.1 Oil groups

In tables A, B and C five oil groups are distinguished:

- Mineral oils (MIN oil)
- polyglycol (PG oil)
- poly- α olefin (PAO oil)
- Biologically degradable oils (BIO oil)
- Physiologically safe oils approved in accordance with USDA H1 (PHY oil).

1.1.2 Oil temperatures

The synthetic oils have a wider temperature range and a higher viscosity index (i.e. a flatter viscosity-temperature gradient) than the mineral oils. Guideline values for temperature range:

mineral oils approx. $-10\text{ }^{\circ}\text{C}$ to $+90\text{ }^{\circ}\text{C}$ (short term $+100\text{ }^{\circ}\text{C}$);
polyglycols and poly- α olefins approx. $-20\text{ }^{\circ}\text{C}$ to $+100\text{ }^{\circ}\text{C}$ (short term $+110\text{ }^{\circ}\text{C}$);
bio-degradable oils (synthetic Ester) approx. -15 to $+90\text{ }^{\circ}\text{C}$.

Note: The upper and lower operating temperatures of certain gear oils may deviate widely from the values indicated. For operating conditions outside the above mentioned temperature ranges, the flash point and pour point of the oils must be observed. For these and other data and properties of the gear oils, refer to the oil manufacturers' technical data sheets (to this purpose Flender must be consulted).

1.1.3 General oil service lives

According to the manufacturers, the following are the minimum periods during which the oils can be used without undergoing any significant change in quality. They are calculated on the basis of an average oil temperature of $80\text{ }^{\circ}\text{C}$:

- mineral oils, bio-degradable oils and physiologically safe oils: 2 years or 10 000 hours of service
- for poly- α olefins and polyglycols: 4 years or 20 000 hours of service.

Note: The actual service lives may be higher or lower (for temperatures over $80\text{ }^{\circ}\text{C}$). The general rule is that an increase in temperature of 10 K will halve the service life.

1.1.4 Oil change

The degree of purity of the oil affects the operating reliability and life span of the oil and the gear unit. It should therefore be ensured that the oil in the gear unit is clean. For the first oil change after start-up and for subsequent oil changes, the specifications in the gear-unit operating instructions must be observed. In the case of larger oil quantities, an analysis should be carried out before cleaning or changing the oil.

When changing oil of the same type, the quantity of oil remaining in the gear unit should be kept as low as possible. Generally speaking, a small remaining quantity will cause no particular problems. Gear oils of different types and manufacturers must not be mixed. If necessary, the manufacturer should confirm that the new oil is compatible with residues of the old oil. When changing to very different types of oil or oils with very different additives, especially when changing from polyglycols to another gear oil or vice versa, the gear unit must always be well flushed out with the new oil. Residues of old oil must be completely removed from the gear unit.

Caution!

Gear oils must never be mixed with other substances. Flushing with paraffin or other solvents is not permitted, as traces of these substances always remain inside the unit.

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Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant						
Mineral oils (MIN oil)	A11	VG 1000				
	A12	VG 680	CLP 680 S ECO GEAR 680 M		Degol BG 680 Plus Degol BMB 680 ¹⁾	GEAR RSX 680
	A13	VG 460	CLP 460 S ECO GEAR 460 M		Degol BG 460 Plus Degol BMB 460 ¹⁾	GEAR RSX 460
	A14	VG 320	CLP 320 S ECO GEAR 320 M		Degol BG 320 Plus Degol BMB 320 ¹⁾	GEAR RSX 320
	A15	VG 220	CLP 220 S ECO GEAR 220 M		Degol BG 220 Plus Degol BMB 220 ¹⁾	GEAR RSX 220
	A16	VG 150	CLP 150 S ECO GEAR 150 M		Degol BG 150 Plus Degol BMB 150 ¹⁾	GEAR RSX 150
	A17	VG 100	ECO GEAR 100 M		Degol BG 100 Plus Degol BMB 100 ¹⁾	GEAR RSX 100
Polyglycol (PG oil)	A21	VG 1000			Degol GS 1000	
	A22	VG 680			Degol GS 680	GEAR VSG 680
	A23	VG 460			Degol GS 460	GEAR VSG 460
	A24	VG 320			Degol GS 320	GEAR VSG 320
	A25	VG 220			Degol GS 220	GEAR VSG 220
	A26	VG 150			Degol GS 150	GEAR VSG 150
	A27	VG 100				GEAR VSG 100
Poly- α -olefin (PAO oil)	A31	VG 1000		4111 XEP 5999 XEP		SYNTOGEAR PE 1000
	A32	VG 680	ECO GEAR 680 S	4680 XEP 5680 XEP		SYNTOGEAR PE 680 EVOGEAR SX 680
	A33	VG 460	ECO GEAR 460 S	4460 XEP 5460 XEP	Degol PAS 460	SYNTOGEAR PE 460 EVOGEAR SX 460
	A34	VG 320	ECO GEAR 320 S	4320 XEP 5320 XEP	Degol PAS 320	SYNTOGEAR PE 320 EVOGEAR SX 320
	A35	VG 220	ECO GEAR 220 S	4220 XEP 5220 XEP	Degol PAS 220	SYNTOGEAR PE 220 EVOGEAR SX 220
	A36	VG 150	ECO GEAR 150 S	4150 XEP 5150 XEP	Degol PAS 150	SYNTOGEAR PE 150
	A37	VG 100	ECO GEAR 100 S	4100 XEP		SYNTOGEAR PE 100
Biologically degradable oils (BIO oil)	A41	VG 1000				
	A42	VG 680				
	A43	VG 460			Degol BAB 460	
	A44	VG 320				
	A45	VG 220			Degol BAB 220	
	A46	VG 150				
	A47	VG 100			Degol BAB 100	
Physiologically safe oils (PHY oil)	A51	VG 1000				
	A52	VG 680				
	A53	VG 460				
	A54	VG 320				
	A55	VG 220				
	A56	VG 150				
	A57	VG 100				
1) with solid content MoS ₂						

FLENDER

Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant						
Mineral oils (MIN oil)	A11	VG 1000				Beslux Gear XP 1000
	A12	VG 680	ECO GEAR 680 M		Energol GR-XF 680	Beslux Gear XP 680
	A13	VG 460	ECO GEAR 460 M		Energol GR-XF 460	Beslux Gear XP 460
	A14	VG 320	ECO GEAR 320 M		Energol GR-XF 320	Beslux Gear XP 320
	A15	VG 220	ECO GEAR 220 M		Energol GR-XF 220	Beslux Gear XP 220
	A16	VG 150	ECO GEAR 150 M		Energol GR-XF 150	
	A17	VG 100	ECO GEAR 100 M			
Polyglycol (PG oil)	A21	VG 1000				Beslux Sincart W 1000
	A22	VG 680	BERUSYNTH EP 680		Enersyn SG-XP 680	Beslux Sincart W 680
	A23	VG 460	BERUSYNTH EP 460		Enersyn SG-XP 460	Beslux Sincart W 460
	A24	VG 320	BERUSYNTH EP 320		Enersyn SG-XP 320	Beslux Sincart W 320
	A25	VG 220	BERUSYNTH EP 220		Enersyn SG-XP 220	Beslux Sincart W 220
	A26	VG 150	BERUSYNTH EP 150		Enersyn SG-XP 150	Beslux Sincart W 150
	A27	VG 100	BERUSYNTH EP 100		Enersyn SG-XP 100	
Poly- α -olefin (PAO oil)	A31	VG 1000		Bel-Ray Synth. Gear Oil 6698		Beslux Gearsint XP 1000
	A32	VG 680	ECO GEAR 680 S	Bel-Ray Synth. Gear Oil 6696		Beslux Gearsint XP 680
	A33	VG 460	ECO GEAR 460 S	Bel-Ray Synth. Gear Oil 6694	Enersyn EP-XF 460	Beslux Gearsint XP 460
	A34	VG 320	ECO GEAR 320 S	Bel-Ray Synth. Gear Oil 6692	Enersyn EP-XF 320	Beslux Gearsint XP 320
	A35	VG 220	ECO GEAR 220 S	Bel-Ray Synth. Gear Oil 6690	Enersyn EP-XF 220	Beslux Gearsint XP 220
	A36	VG 150	ECO GEAR 150 S	Bel-Ray Synth. Gear Oil 6688	Enersyn EP-XF 150	Beslux Gearsint XP 150
	A37	VG 100	ECO GEAR 100 S		Enersyn EP-XF 100	
Biologically degradable oils (BIO oil)	A41	VG 1000				
	A42	VG 680				
	A43	VG 460				
	A44	VG 320				
	A45	VG 220				
	A46	VG 150				
	A47	VG 100				
Physiologically safe oils (PHY oil)	A51	VG 1000				
	A52	VG 680				
	A53	VG 460				
	A54	VG 320				
	A55	VG 220				
	A56	VG 150				
	A57	VG 100				

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Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant			B&L	 classic	 performance	
Mineral oils (MIN oil)	A11	VG 1000		Alpha SP 1000	Optigear BM 1000 Tribol 1100 / 1000	CEPSA AEROGear 1000
	A12	VG 680		Alpha SP 680	Optigear BM 680 Tribol 1100 / 680	CEPSA AEROGear 680
	A13	VG 460		Alpha SP 460 Alpha MAX 460	Optigear BM 460 Tribol 1100 / 460	CEPSA AEROGear 460
	A14	VG 320		Alpha SP 320 Alpha MAX 320	Optigear BM 320 Tribol 1100 / 320	CEPSA AEROGear 320
	A15	VG 220		Alpha SP 220 Alpha MAX 220	Optigear BM 220 Tribol 1100 / 220	CEPSA AEROGear 220
	A16	VG 150		Alpha SP 150 Alpha MAX 150	Optigear BM 150 Tribol 1100 / 150	
	A17	VG 100		Alpha SP 100 Alpha MAX 100	Optigear BM 100 Tribol 1100 / 100	
Polyglycol (PG oil)	A21	VG 1000			Tribol 800 / 1000	
	A22	VG 680			Tribol 800 / 680	
	A23	VG 460			Tribol 800 / 460	
	A24	VG 320			Tribol 800 / 320	
	A25	VG 220			Tribol 800 / 220	
	A26	VG 150			Tribol 800 / 150	
	A27	VG 100			Tribol 800 / 100	
Poly-α-olefin (PAO oil)	A31	VG 1000				
	A32	VG 680		Alphasyn EP 680		AEROGear SYNT 680
	A33	VG 460		Optigear Synth. A 460 Alphasyn EP 460	Optigear Synth. X 460 Tribol 1710 / 460	AEROGear SYNT 460
	A34	VG 320		Optigear Synth. A 460 Alphasyn EP 320	Optigear Synth. X 320 Tribol 1710 / 320	AEROGear SYNT 320
	A35	VG 220		Optigear Synth. A 460 Alphasyn EP 220	Optigear Synth. X 220 Tribol 1710 / 220	AEROGear SYNT 220
	A36	VG 150		Alphasyn EP 150		
	A37	VG 100				
Biologically degradable oils (BIO oil)	A41	VG 1000				
	A42	VG 680	RIVOLTA S.G.L. 680			
	A43	VG 460			Tribol BioTop 1418 / 460	
	A44	VG 320	RIVOLTA S.G.L. 320		Tribol BioTop 1418 / 320	
	A45	VG 220	RIVOLTA S.G.L. 220		Tribol BioTop 1418 / 220	
	A46	VG 150			Tribol BioTop 1418 / 150	
	A47	VG 100	RIVOLTA S.G.L. 100			
Physiologically safe oils (PHY oil)	A51	VG 1000				
	A52	VG 680			Optileb GT 680	
	A53	VG 460			Optileb GT 460	
	A54	VG 320			Optileb GT 320	
	A55	VG 220			Optileb GT 220	
	A56	VG 150			Optileb GT 150	
	A57	VG 100			Optileb GT 100	

FLENDER

Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant						
Mineral oils (MIN oil)	A11	VG 1000				
	A12	VG 680		Falcon CLP 680		SPARTAN EP 680 ²⁾
	A13	VG 460		Falcon CLP 460	EP Industrial G 460	SPARTAN EP 460 ²⁾
	A14	VG 320		Falcon CLP 320	EP Industrial G 320	SPARTAN EP 320 ²⁾
	A15	VG 220		Falcon CLP 220	EP Industrial G 220	SPARTAN EP 220 ²⁾
	A16	VG 150		Falcon CLP 150		SPARTAN EP 150 ²⁾
	A17	VG 100		Falcon CLP 100		SPARTAN EP 100 ²⁾
Polyglycol (PG oil)	A21	VG 1000	Breox SL 1000	Polydea PGLP 1000		
	A22	VG 680	Breox SL 680	Polydea PGLP 680		
	A23	VG 460	Breox SL 460	Polydea PGLP 460		
	A24	VG 320	Breox SL 320	Polydea PGLP 320		
	A25	VG 220	Breox SL 220	Polydea PGLP 220		
	A26	VG 150	Breox SL 150	Polydea PGLP 150		
	A27	VG 100		Polydea PGLP 100		
Poly- α -olefin (PAO oil)	A31	VG 1000				
	A32	VG 680				
	A33	VG 460		Intor HCLP 460		
	A34	VG 320		Intor HCLP 320		
	A35	VG 220		Intor HCLP 220		
	A36	VG 150		Intor HCLP 150		
	A37	VG 100				
Biologically degradable oils (BIO oil)	A41	VG 1000				
	A42	VG 680				
	A43	VG 460				
	A44	VG 320		Ergon ELP 320		
	A45	VG 220		Ergon ELP 220		
	A46	VG 150		Ergon ELP 150		
	A47	VG 100		Ergon ELP 100		
Physiologically safe oils (PHY oil)	A51	VG 1000	Breox FGL 1000			
	A52	VG 680	Breox FGL 680			
	A53	VG 460	Breox FGL 460			
	A54	VG 320	Breox FGL 320			
	A55	VG 220	Breox FGL 220			
	A56	VG 150	Breox FGL 150			
	A57	VG 100				
2) ESSO guarantees the qualities required by FLENDER only for products from Europe						

FLENDER

Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant						
Mineral oils (MIN oil)	A11	VG 1000	RENOLIN CLP1000 PLUS	GEARMASTER CLP 1000		
	A12	VG 680	RENOLIN CLP 680 RENOLIN CLP 680 PLUS	GEARMASTER CLP 680		
	A13	VG 460	RENOLIN CLP 460 RENOLIN CLP 460 PLUS	GEARMASTER CLP 460		
	A14	VG 320	RENOLIN CLP 320 RENOLIN CLP 320 PLUS	GEARMASTER CLP 320		
	A15	VG 220	RENOLIN CLP 220 RENOLIN CLP 220 PLUS	GEARMASTER CLP 220		
	A16	VG 150	RENOLIN CLP 150 RENOLIN CLP 150 PLUS	GEARMASTER CLP 150		
	A17	VG 100	RENOLIN CLP 100 RENOLIN CLP 100 PLUS	GEARMASTER CLP 100		
Polyglycol (PG oil)	A21	VG 1000	RENOLIN PG 1000	GEARMASTER PGP 1000	Ucolub BSL-IG 1000	
	A22	VG 680	RENOLIN PG 680	GEARMASTER PGP 680	Ucolub BSL-IG 680	
	A23	VG 460	RENOLIN PG 460	GEARMASTER PGP 460	Ucolub BSL-IG 460	
	A24	VG 320	RENOLIN PG 320	GEARMASTER PGP 320	Ucolub BSL-IG 320	
	A25	VG 220	RENOLIN PG 220	GEARMASTER PGP 220	Ucolub BSL-IG 220	
	A26	VG 150	RENOLIN PG 150	GEARMASTER PGP 150	Ucolub BSL-IG 150	
	A27	VG 100	RENOLIN PG 100	GEARMASTER PGP 100	Ucolub BSL-IG 100	
Poly- α -olefin (PAO oil)	A31	VG 1000		GEARMASTER SYN 1000		
	A32	VG 680	RENOLIN UNISYN CLP 680	GEARMASTER SYN 680		
	A33	VG 460	RENOLIN UNISYN CLP 460	GEARMASTER SYN 460	Ucolub N-PA 460	Galp Transgear SMP 460
	A34	VG 320	RENOLIN UNISYN CLP 320	GEARMASTER SYN 320	Ucolub N-PA 320	Galp Transgear SMP 320
	A35	VG 220	RENOLIN UNISYN CLP 220	GEARMASTER SYN 220	Ucolub N-PA 220	Galp Transgear SMP 220
	A36	VG 150	RENOLIN UNISYN CLP 150	GEARMASTER SYN 150		Galp Transgear SMP 150
	A37	VG 100	RENOLIN UNISYN CLP 100	GEARMASTER SYN 100		
Biologically degradable oils (BIO oil)	A41	VG 1000	PLANTOGEAR 1000 S	GEARMASTER ECO 1000		
	A42	VG 680	PLANTOGEAR 680 S	GEARMASTER ECO 680		
	A43	VG 460	PLANTOGEAR 460 S	GEARMASTER ECO 460		
	A44	VG 320	PLANTOGEAR 320 S	GEARMASTER ECO 320		
	A45	VG 220	PLANTOGEAR 220 S	GEARMASTER ECO 220		
	A46	VG 150	PLANTOGEAR 150 S	GEARMASTER ECO 150		
	A47	VG 100	PLANTOGEAR 100 S	GEARMASTER ECO 100		
Physiologically safe oils (PHY oil)	A51	VG 1000			Ucolub BSL 1000	
	A52	VG 680			Ucolub BSL 680	
	A53	VG 460			Ucolub BSL 460	
	A54	VG 320			Ucolub BSL 320	
	A55	VG 220			Ucolub BSL 220	
	A56	VG 150			Ucolub BSL 150	
	A57	VG 100				

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Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant						
Mineral oils (MIN oil)	A11	VG 1000				
	A12	VG 680	Klüberoil GEM 1 - 680 N	609 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear XMP 680
	A13	VG 460	Klüberoil GEM 1 - 460 N	608 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear XMP 460
	A14	VG 320	Klüberoil GEM 1 - 320 N	605 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear XMP 320
	A15	VG 220	Klüberoil GEM 1 - 220 N	607 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear XMP 220
	A16	VG 150	Klüberoil GEM 1 - 150 N	604 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear XMP 150
	A17	VG 100	Klüberoil GEM 1 - 100 N	606 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear XMP 100
Polyglycol (PG oil)	A21	VG 1000				
	A22	VG 680				
	A23	VG 460				
	A24	VG 320				
	A25	VG 220				
	A26	VG 150				
	A27	VG 100				
Poly- α -olefin (PAO oil)	A31	VG 1000				Mobil SHC 639
	A32	VG 680	Klübersynth. GEM 4-680 N		megol Gearoil Synth 680	
	A33	VG 460	Klübersynth. GEM 4-460 N		megol Gearoil Synth 460	Mobil SHC 634
	A34	VG 320	Klübersynth. GEM 4-320 N		megol Gearoil Synth 320	Mobil SHC 632
	A35	VG 220	Klübersynth. GEM 4-220 N		megol Gearoil Synth 220	Mobil SHC 630
	A36	VG 150	Klübersynth. GEM 4-150 N			Mobil SHC 629
	A37	VG 100				
Biologically degradable oils (BIO oil)	A41	VG 1000				
	A42	VG 680				
	A43	VG 460				
	A44	VG 320	Klübersynth. GEM 2-320			
	A45	VG 220	Klübersynth. GEM 2-220			
	A46	VG 150				
	A47	VG 100				
Physiologically safe oils (PHY oil)	A51	VG 1000				
	A52	VG 680				
	A53	VG 460				
	A54	VG 320				
	A55	VG 220				
	A56	VG 150				
	A57	VG 100				

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Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant					 Petrol Ofisi	
Mineral oils (MIN oil)	A11	VG 1000	GEAR COMPOUND PLUS 1000	OMV gear HST 1000	PO Gravis MP 1000	
	A12	VG 680	GEAR COMPOUND PLUS 680	OMV gear HST 680	PO Gravis MP 680	
	A13	VG 460	GEAR COMPOUND PLUS 460	OMV gear HST 460	PO Gravis MP 460	
	A14	VG 320	GEAR COMPOUND PLUS 320	OMV gear HST 320	PO Gravis MP 320	Q8 Goya NT 320
	A15	VG 220	GEAR COMPOUND PLUS 220	OMV gear HST 220	PO Gravis MP 220	Q8 Goya NT 220
	A16	VG 150	GEAR COMPOUND PLUS 150	OMV gear HST 150	PO Gravis MP 150	Q8 Goya NT 150
	A17	VG 100	GEAR COMPOUND PLUS 100	OMV gear HST 100	PO Gravis MP 100	
Polyglycol (PG oil)	A21	VG 1000				
	A22	VG 680				
	A23	VG 460				
	A24	VG 320				
	A25	VG 220				
	A26	VG 150				
	A27	VG 100				
Poly-α-olefin (PAO oil)	A31	VG 1000	GEAR SINTEC CLP 1000		PO Gravis SP 1000	
	A32	VG 680	GEAR SINTEC CLP 680		PO Gravis SP 680	
	A33	VG 460	GEAR SINTEC CLP 460		PO Gravis SP 460	Q8 El Greco 460
	A34	VG 320	GEAR SINTEC CLP 320		PO Gravis SP 320	Q8 El Greco 320
	A35	VG 220	GEAR SINTEC CLP 220		PO Gravis SP 220	Q8 El Greco 220
	A36	VG 150	GEAR SINTEC CLP 150			Q8 El Greco 150
	A37	VG 100	GEAR SINTEC CLP 100			
Biologically degradable oils (BIO oil)	A41	VG 1000				
	A42	VG 680				
	A43	VG 460				
	A44	VG 320				
	A45	VG 220				
	A46	VG 150				
	A47	VG 100				
Physiologically safe oils (PHY oil)	A51	VG 1000				
	A52	VG 680				
	A53	VG 460				
	A54	VG 320				
	A55	VG 220				
	A56	VG 150				
	A57	VG 100				

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Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant						
Mineral oils (MIN oil)	A11	VG 1000				KASSILLA GMP 1000 CARTER EP 1000
	A12	VG 680		LoadWay EP 460		KASSILLA GMP 680 CARTER EP 680
	A13	VG 460	Shell Omala F 460	LoadWay EP 320	Meropa WM 460 Auriga EP 460	KASSILLA GMP 460 CARTER EP 460
	A14	VG 320	Shell Omala F 320	LoadWay EP 320	Meropa WM 320 Auriga EP 320	KASSILLA GMP 320 CARTER EP 320
	A15	VG 220	Shell Omala F 220	LoadWay EP 220	Meropa WM 220 Auriga EP 220	KASSILLA GMP 220 CARTER EP 220
	A16	VG 150			Meropa WM 150 Auriga EP 150	
	A17	VG 100			Meropa WM 100 Auriga EP 100	
Polyglycol (PG oil)	A21	VG 1000	Shell Tivela S 1000 Shell Cassida WG 1000			
	A22	VG 680	Shell Tivela S 680 Shell Cassida WG 680			
	A23	VG 460	Shell Tivela S 460 Shell Cassida WG 460			
	A24	VG 320	Shell Tivela S 320 Shell Cassida WG 320			
	A25	VG 220	Shell Tivela S 220 Shell Cassida WG 220			
	A26	VG 150	Shell Tivela S 150			
	A27	VG 100				
Poly- α -olefin (PAO oil)	A31	VG 1000				CARTER SH 1000
	A32	VG 680				CARTER SH 680
	A33	VG 460				CARTER SH 460
	A34	VG 320		MERETA 320		CARTER SH 320
	A35	VG 220		MERETA 220		CARTER SH 220
	A36	VG 150		MERETA 150		CARTER SH 150
	A37	VG 100				
Biologically degradable oils (BIO oil)	A41	VG 1000				
	A42	VG 680				
	A43	VG 460				
	A44	VG 320	Shell Omala EPB 320			
	A45	VG 220	Shell Omala EPB 220			
	A46	VG 150	Shell Omala EPB 150			
	A47	VG 100	Shell Omala EPB 100			
Physiologically safe oils (PHY oil)	A51	VG 1000	Shell Cassida WG 1000			
	A52	VG 680	Shell Cassida WG 680			
	A53	VG 460	Shell Cassida WG 460			Keystone Nevastane SY 460
	A54	VG 320	Shell Cassida WG 320			Keystone Nevastane SY 320
	A55	VG 220	Shell Cassida WG 220			Keystone Nevastane SY 220
	A56	VG 150				
	A57	VG 100				

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Table A	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER helical-gear, bevel helical-gear and planetary-gear units			
Lubricant						
Mineral oils (MIN oil)	A11	VG 1000		VECO MATRANOL XP 1000		
	A12	VG 680	TUNGEAR 680 ³⁾	VECO MATRANOL XP 680	COMPOUND MP 680	
	A13	VG 460	TUNGEAR 460 ³⁾	VECO MATRANOL XP 460	COMPOUND MP 460	
	A14	VG 320	TUNGEAR 320 ³⁾	VECO MATRANOL XP 320	COMPOUND MP 320	
	A15	VG 220	TUNGEAR 220 ³⁾	VECO MATRANOL XP 220	COMPOUND MP 220	
	A16	VG 150			COMPOUND MP 150	
	A17	VG 100				
Polyglycol (PG oil)	A21	VG 1000				
	A22	VG 680				
	A23	VG 460				
	A24	VG 320				
	A25	VG 220				
	A26	VG 150				
	A27	VG 100				
Poly-α-olefin (PAO oil)	A31	VG 1000				
	A32	VG 680				
	A33	VG 460				
	A34	VG 320				
	A35	VG 220				
	A36	VG 150				
	A37	VG 100				
Biologically degradable oils (BIO oil)	A41	VG 1000				
	A42	VG 680				
	A43	VG 460				
	A44	VG 320				
	A45	VG 220				
	A46	VG 150				
	A47	VG 100				
Physiologically safe oils (PHY oil)	A51	VG 1000				
	A52	VG 680				
	A53	VG 460				
	A54	VG 320				
	A55	VG 220				
	A56	VG 150				
	A57	VG 100				

3) TUNGEAR is approved for Brazil under the name of GEAROIL in VG 220, 320, 460 and 680. Distributor: TRIBOTECHNICA Lubrificantes Sinteticos Sao Paulo.
TUNGEAR is approved for India under the name of Mox-Active Gear oil in VG 220, 320 and 460. Distributor: OKS Speciality Lubricants Bombay.

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Table B	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER worm-gear units			
Lubricant						
Mineral oils (MIN oil)	B11	VG 1000				Energol GR-XP 1000
	B12	VG 680	Degol BG 680			Energol GR-XP 680
	B13	VG 460	Degol BG 460			Energol GR-XP 460
	B14	VG 320	Degol BG 320			Energol GR-XP 320
	B15	VG 220	Degol BG 220			Energol GR-XP 220
	B16	VG 150	Degol BG 150			Energol GR-XP 150
	B17	VG 100	Degol BG 100			Energol GR-XP 100
Polyglycol (PG oil)	B21	VG 1000	Degol GS 1000			
	B22	VG 680	Degol GS 680	GEAR VSG 680	BERUSYNTH EP 680	Enersyn SG-XP 680
	B23	VG 460	Degol GS 460	GEAR VSG 460	BERUSYNTH EP 460	Enersyn SG-XP 460
	B24	VG 320	Degol GS 320	GEAR VSG 320	BERUSYNTH EP 320	Enersyn SG-XP 320
	B25	VG 220	Degol GS 220	GEAR VSG 220	BERUSYNTH EP 220	Enersyn SG-XP 220
	B26	VG 150	Degol GS 150	GEAR VSG 150	BERUSYNTH EP 150	Enersyn SG-XP 150
	B27	VG 100		GEAR VSG 100	BERUSYNTH EP 100	Enersyn SG-XP 100
Poly- α -olefin (PAO oil)	B31	VG 1000				
	B32	VG 680				
	B33	VG 460				
	B34	VG 320				
	B35	VG 220				
	B36	VG 150				
	B37	VG 100				
Biologically degradable oils (BIO oil)	B41	VG 1000				
	B42	VG 680				
	B43	VG 460				
	B44	VG 320				
	B45	VG 220				
	B46	VG 150				
	B47	VG 100				
Physiologically safe oils (PHY oil)	B51	VG 1000				
	B52	VG 680				
	B53	VG 460				
	B54	VG 320				
	B55	VG 220				
	B56	VG 150				
	B57	VG 100				

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Table B	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER worm-gear units			
Lubricant						
Mineral oils (MIN oil)	B11	VG 1000	Optigear BM 1000 Tribol 1000 / 1000			
	B12	VG 680	Optigear BM 680 Tribol 1000 / 680		Falcon CLP 680	SPARTAN EP 680 ¹⁾
	B13	VG 460	Optigear BM 460 Tribol 1000 / 460		Falcon CLP 460	SPARTAN EP 460 ¹⁾
	B14	VG 320	Optigear BM 320 Tribol 1000 / 320		Falcon CLP 320	SPARTAN EP 320 ¹⁾
	B15	VG 220	Optigear BM 220 Tribol 1000 / 220		Falcon CLP 220	SPARTAN EP 220 ¹⁾
	B16	VG 150	Optigear BM 150 Tribol 1000 / 150		Falcon CLP 150	SPARTAN EP 150 ¹⁾
	B17	VG 100	Optigear BM 100 Tribol 1000 / 100		Falcon CLP 100	SPARTAN EP 100 ¹⁾
Polyglycol (PG oil)	B21	VG 1000	Optiflex A 1000 Tribol 800 / 1000	Breox SL 1000	Polydea PGLP 1000	
	B22	VG 680	Optiflex A 680 Tribol 800 / 680	Breox SL 680	Polydea PGLP 680	
	B23	VG 460	Optiflex A 460 Tribol 800 / 460	Breox SL 460	Polydea PGLP 460	GLYCOLUBE 460
	B24	VG 320	Optiflex A 320 Tribol 800 / 320	Breox SL 320	Polydea PGLP 320	
	B25	VG 220	Optiflex A 220 Tribol 800 / 220	Breox SL 220	Polydea PGLP 220	GLYCOLUBE 220
	B26	VG 150	Optiflex A 150 Tribol 800 / 150	Breox SL 150	Polydea PGLP 150	
	B27	VG 100	Optiflex A 100 Tribol 800 / 100		Polydea PGLP 100	
Poly- α -olefin (PAO oil)	B31	VG 1000				
	B32	VG 680				
	B33	VG 460				
	B34	VG 320				
	B35	VG 220				
	B36	VG 150				
	B37	VG 100				
Biologically degradable oils (BIO oil)	B41	VG 1000				
	B42	VG 680				
	B43	VG 460				
	B44	VG 320				
	B45	VG 220				
	B46	VG 150				
	B47	VG 100				
Physiologically safe oils (PHY oil)	B51	VG 1000		Breox FGL 1000		
	B52	VG 680	Tribol FoodProof 1800 / 680	Breox FGL 680		
	B53	VG 460	Tribol FoodProof 1800 / 460	Breox FGL 460		
	B54	VG 320	Tribol FoodProof 1800 / 320	Breox FGL 320		
	B55	VG 220	Tribol FoodProof 1800 / 220	Breox FGL 220		
	B56	VG 150		Breox FGL 150		
	B57	VG 100				

1) ESSO guarantees the qualities required by FLENDER only for products from Europe

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Table B		Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER worm-gear units			
Lubricant	Code no.					
Mineral oils (MIN oil)	B11	VG 1000				
	B12	VG 680			Klüberoil GEM 1 - 680	Mobilgear 636 Mobilgear XMP 680
	B13	VG 460			Klüberoil GEM 1 - 460	Mobilgear 634 Mobilgear XMP 460
	B14	VG 320			Klüberoil GEM 1 - 320	Mobilgear 632 Mobilgear XMP 320
	B15	VG 220			Klüberoil GEM 1 - 220	Mobilgear 630 Mobilgear XMP 220
	B16	VG 150			Klüberoil GEM 1 - 150	Mobilgear 629 Mobilgear XMP 150
	B17	VG 100			Klüberoil GEM 1 - 100	Mobilgear 627 Mobilgear XMP 100
Polyglycol (PG oil)	B21	VG 1000	RENOLIN PG 1000	GEARMASTER PGP 1000		
	B22	VG 680	RENOLIN PG 680	GEARMASTER PGP 680		
	B23	VG 460	RENOLIN PG 460	GEARMASTER PGP 460		
	B24	VG 320	RENOLIN PG 320	GEARMASTER PGP 320		
	B25	VG 220	RENOLIN PG 220	GEARMASTER PGP 220		
	B26	VG 150	RENOLIN PG 150	GEARMASTER PGP 150		
	B27	VG 100	RENOLIN PG 100	GEARMASTER PGP 100		
Poly-α-olefin (PAO oil)	B31	VG 1000				
	B32	VG 680				
	B33	VG 460				
	B34	VG 320				
	B35	VG 220				
	B36	VG 150				
	B37	VG 100				
Biologically degradable oils (BIO oil)	B41	VG 1000				
	B42	VG 680				
	B43	VG 460				
	B44	VG 320				
	B45	VG 220				
	B46	VG 150				
	B47	VG 100				
Physiologically safe oils (PHY oil)	B51	VG 1000				
	B52	VG 680				
	B53	VG 460				
	B54	VG 320				
	B55	VG 220				
	B56	VG 150				
	B57	VG 100				

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Table B	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER worm-gear units			
Lubricant						
Mineral oils (MIN oil)	B11	VG 1000				
	B12	VG 680		Ersolan 680		
	B13	VG 460		Ersolan 460		
	B14	VG 320		Ersolan 320		
	B15	VG 220		Ersolan 220		
	B16	VG 150		Ersolan 150		
	B17	VG 100		Ersolan 100		
Polyglycol (PG oil)	B21	VG 1000	Shell Tivela S 1000 Shell Cassida WG 1000			
	B22	VG 680	Shell Tivela S 680 Shell Cassida WG 680			
	B23	VG 460	Shell Tivela S 460 Shell Cassida WG 460			
	B24	VG 320	Shell Tivela S 320 Shell Cassida WG 320			
	B25	VG 220	Shell Tivela S 220 Shell Cassida WG 220			
	B26	VG 150	Shell Tivela S 150			
	B27	VG 100				
Poly-α-olefin (PAO oil)	B31	VG 1000				
	B32	VG 680				
	B33	VG 460				
	B34	VG 320				
	B35	VG 220				
	B36	VG 150				
	B37	VG 100				
Biologically degradable oils (BIO oil)	B41	VG 1000				
	B42	VG 680				
	B43	VG 460				
	B44	VG 320				
	B45	VG 220				
	B46	VG 150				
	B47	VG 100				
Physiologically safe oils (PHY oil)	B51	VG 1000	Shell Cassida WG 1000			
	B52	VG 680	Shell Cassida WG 680			
	B53	VG 460	Shell Cassida WG 460			
	B54	VG 320	Shell Cassida WG 320			
	B55	VG 220	Shell Cassida WG 220			
	B56	VG 150				
	B57	VG 100				

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Table C	Lubricating oils for FLENDER gear motors					
Lubricant	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)				
Mineral oils (MIN oil)	C15	VG 220	CLP 220 S ECO GEAR 220 M		Degol BG 220 Plus Degol BG 220	GEAR RSX 220
	C16	VG 150	CLP 150 S ECO GEAR 150 M		Degol BG 150 Plus Degol BG 150	GEAR RSX 150
	C17	VG 100	ECO GEAR 100 M		Degol BG 100 Plus Degol BG 100	GEAR RSX 100
Polyglycol (PG oil)	C23	VG 460			Degol GS 460	GEAR VSG 460
	C24	VG 320			Degol GS 320	GEAR VSG 320
	C25	VG 220			Degol GS 220	GEAR VSG 220
Poly- α -olefin (PAO oil)	C33	VG 460		4460 XEP 5460 XEP	Degol PAS 460	SYNTOGEAR PE 460 EVOGEAR SX 460
	C34	VG 320		4320 XEP 5320 XEP	Degol PAS 320	SYNTOGEAR PE 320 EVOGEAR SX 320
	C35	VG 220		4220 XEP 5220 XEP	Degol PAS 220	SYNTOGEAR PE 220 EVOGEAR SX 220
Biologically degradable oils (BIO oil)	C44	VG 320	ECO GEAR 320 S			
	C45	VG 220	ECO GEAR 220 S		Degol BAB 220	
	C46	VG 150	ECO GEAR 150 S			
Physiologically safe oils (PHY oil)	C53	VG 460				
	C54	VG 320				
	C55	VG 220				

Table C	Lubricating oils for FLENDER gear motors					
Lubricant	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)				
Mineral oils (MIN oil)	C15	VG 220	ECO GEAR 220 M		Energol GR-XF 220 Energol GR-XP 220	BESLUX GEAR XP 220
	C16	VG 150	ECO GEAR 150 M		Energol GR-XF 150	
	C17	VG 100	ECO GEAR 100 M		Energol GR-XF 100	
Polyglycol (PG oil)	C23	VG 460	BERUSYNTH EP 460		Enersyn SG-XP 460	
	C24	VG 320	BERUSYNTH EP 320			
	C25	VG 220	BERUSYNTH EP 220		Enersyn SG-XP 220	
Poly- α -olefin (PAO oil)	C33	VG 460	ECO GEAR 460 S	Bel-Ray Synth. Gear Oil 6694	Enersyn EP-XF 460	Beslux Gearsint XP 460
	C34	VG 320	ECO GEAR 320 S	Bel-Ray Synth. Gear Oil 6692	Enersyn EP-XF 320	Beslux Gearsint XP 320
	C35	VG 220	ECO GEAR 220 S	Bel-Ray Synth. Gear Oil 6690	Enersyn EP-XF 220	Beslux Gearsint XP 220
Biologically degradable oils (BIO oil)	C44	VG 320				
	C45	VG 220				
	C46	VG 150				
Physiologically safe oils (PHY oil)	C53	VG 460				
	C54	VG 320				
	C55	VG 220				

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Table C	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER gear motors			
Lubricant						
Mineral oils (MIN oil)	C15	VG 220		Alpha SP 220 Alpha MAX 220	Optigear BM 220 Tribol 1100 / 220	CEPSA AEROGear 220
	C16	VG 150		Alpha SP 150 Alpha MAX 150	Optigear BM 150 Tribol 1100 / 150	
	C17	VG 100		Alpha SP 100 Alpha MAX 100	Optigear BM 100 Tribol 1100 / 100	
Polyglycol (PG oil)	C23	VG 460			Tribol 800 / 460	
	C24	VG 320			Tribol 800 / 320	
	C25	VG 220			Tribol 800 / 220	
Poly- α -olefin (PAO oil)	C33	VG 460		Optigear Synth. A 460 Alphasyn EP 460	Optigear Synth. A 460 Tribol 1710 / 460	AEROGear SYNT 460
	C34	VG 320		Optigear Synth. A 460 Alphasyn EP 320	Optigear Synth. A 320 Tribol 1710 / 320	AEROGear SYNT 320
	C35	VG 220		Optigear Synth. A 460 Alphasyn EP 220	Optigear Synth. A 220 Tribol 1710 / 220	AEROGear SYNT 220
Biologically degradable oils (BIO oil)	C44	VG 320	RIVOLTA S.G.L. 320		Tribol BioTop 1418 / 320	
	C45	VG 220	RIVOLTA S.G.L. 220		Tribol BioTop 1418 / 220	
	C46	VG 150			Tribol BioTop 1418 / 150	
Physiologically safe oils (PHY oil)	C53	VG 460			Optileb GT 460 FoodProof 1800 / 460	
	C54	VG 320			Optileb GT 320 FoodProof 1800 / 320	
	C55	VG 220			Optileb GT 220 FoodProof 1800 / 220	

Table C	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER gear motors			
Lubricant						
Mineral oils (MIN oil)	C15	VG 220		Falcon CLP 220	EP Industrial G 220	SPARTAN EP 220 ¹⁾
	C16	VG 150		Falcon CLP 150		SPARTAN EP 150 ¹⁾
	C17	VG 100		Falcon CLP 100		SPARTAN EP 100 ¹⁾
Polyglycol (PG oil)	C23	VG 460	Breox SL 460	Polydea PGLP 460		GLYCOLUBE 460
	C24	VG 320	Breox SL 320	Polydea PGLP 320		
	C25	VG 220	Breox SL 220	Polydea PGLP 220		GLYCOLUBE 220
Poly- α -olefin (PAO oil)	C33	VG 460		Intor HCLP 460		
	C34	VG 320		Intor HCLP 320		
	C35	VG 220		Intor HCLP 220		
Biologically degradable oils (BIO oil)	C44	VG 320		Ergon ELP 320		
	C45	VG 220		Ergon ELP 220		
	C46	VG 150		Ergon ELP 150		
Physiologically safe oils (PHY oil)	C53	VG 460	Breox FGL 460			
	C54	VG 320	Breox FGL 320			
	C55	VG 220	Breox FGL 220			

1) ESSO guarantees the qualities required by FLENDER only for products from Europe

FLENDER

Table C	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER gear motors			
Lubricant						
Mineral oils (MIN oil)	C15	VG 220	RENOLIN CLP 220 RENOLIN CLP 220 PLUS	GEARMASTER CLP 220		
	C16	VG 150	RENOLIN CLP 150 RENOLIN CLP 150 PLUS	GEARMASTER CLP 150		
	C17	VG 100	RENOLIN CLP 100 RENOLIN CLP 100 PLUS	GEARMASTER CLP 100		
Polyglycol (PG oil)	C23	VG 460	RENOLIN PG 460	GEARMASTER PGP 460	Ucolub BSL-IG 460	
	C24	VG 320	RENOLIN PG 320	GEARMASTER PGP 320	Ucolub BSL-IG 320	
	C25	VG 220	RENOLIN PG 220	GEARMASTER PGP 220	Ucolub BSL-IG 220	
Poly- α -olefin (PAO oil)	C33	VG 460	RENOLIN UNISYN CLP 460	GEARMASTER SYN 460	Ucolub N-PA 460	Galp Transgear SMP 460
	C34	VG 320	RENOLIN UNISYN CLP 320	GEARMASTER SYN 320	Ucolub N-PA 320	Galp Transgear SMP 320
	C35	VG 220	RENOLIN UNISYN CLP 220	GEARMASTER SYN 220	Ucolub N-PA 220	Galp Transgear SMP 220
Biologically degradable oils (BIO oil)	C44	VG 320	PLANTOGear 320 S	GEARMASTER ECO 320		
	C45	VG 220	PLANTOGear 220 S	GEARMASTER ECO 220		
	C46	VG 150	PLANTOGear 150 S	GEARMASTER ECO 150		
Physiologically safe oils (PHY oil)	C53	VG 460			Ucolub BSL 460	
	C54	VG 320			Ucolub BSL 320	
	C55	VG 220			Ucolub BSL 220	

Table C	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER gear motors			
Lubricant						
Mineral oils (MIN oil)	C15	VG 220	Klüberoil GEM 1 - 220 N	607 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear 630 Mobilgear XMP 220
	C16	VG 150	Klüberoil GEM 1 - 150 N	604 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear 629 Mobilgear XMP 150
	C17	VG 100	Klüberoil GEM 1 - 100 N	606 ALMASOL Vari- Purpose Gear Lubricant		Mobilgear 627 Mobilgear XMP 100
Polyglycol (PG oil)	C23	VG 460				
	C24	VG 320				
	C25	VG 220				
Poly- α -olefin (PAO oil)	C33	VG 460	Klübersynth. GEM 4-460 N		megol Gearoil Synth 460	Mobil SHC 634
	C34	VG 320	Klübersynth. GEM 4-320 N		megol Gearoil Synth 320	Mobil SHC 632
	C35	VG 220	Klübersynth. GEM 4-220 N		megol Gearoil Synth 220	Mobil SHC 630
Biologically degradable oils (BIO oil)	C44	VG 320	Klübersynth. GEM 2-320			
	C45	VG 220	Klübersynth. GEM 2-220			
	C46	VG 150				
Physiologically safe oils (PHY oil)	C53	VG 460				
	C54	VG 320				
	C55	VG 220				

FLENDER

Table C	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER gear motors			
Lubricant						
Mineral oils (MIN oil)	C15	VG 220	GEAR COMPOUND PLUS 220	OMV gear HST 220		PO Gravis MP 220
	C16	VG 150	GEAR COMPOUND PLUS 150	OMV gear HST 150		PO Gravis MP 150
	C17	VG 100	GEAR COMPOUND PLUS 100	OMV gear HST 100		PO Gravis MP 100
Polyglycol (PG oil)	C23	VG 460				
	C24	VG 320				
	C25	VG 220				
Poly-α-olefin (PAO oil)	C33	VG 460	GEAR SINTEC CLP 460			PO Gravis SP 460
	C34	VG 320	GEAR SINTEC CLP 320			PO Gravis SP 320
	C35	VG 220	GEAR SINTEC CLP 220			PO Gravis SP 220
Biologically degradable oils (BIO oil)	C44	VG 320				
	C45	VG 220				
	C46	VG 150				
Physiologically safe oils (PHY oil)	C53	VG 460				
	C54	VG 320				
	C55	VG 220				

Table C	Code no.	Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER gear motors			
Lubricant						
Mineral oils (MIN oil)	C15	VG 220	Q8 Goya NT 220	Shell Omala F 220	Ersolan 220	LoadWay EP 220
	C16	VG 150	Q8 Goya NT 150		Ersolan 150	
	C17	VG 100			Ersolan 100	
Polyglycol (PG oil)	C23	VG 460		Shell Tivela S 460 Shell Cassida WG 460		
	C24	VG 320		Shell Tivela S 320 Shell Cassida WG 320		
	C25	VG 220		Shell Tivela S 220 Shell Cassida WG 220		
Poly-α-olefin (PAO oil)	C33	VG 460	Q8 El Greco 460			
	C34	VG 320	Q8 El Greco 320			MERETA 320
	C35	VG 220	Q8 El Greco 220			MERETA 220
Biologically degradable oils (BIO oil)	C44	VG 320		Shell Omala EPB 320		
	C45	VG 220		Shell Omala EPB 220		
	C46	VG 150		Shell Omala EPB 150		
Physiologically safe oils (PHY oil)	C53	VG 460		Shell Cassida WG 460		
	C54	VG 320		Shell Cassida WG 320		
	C55	VG 220		Shell Cassida WG 220		

FLENDER

Table C		Viscosity ISO-VG DIN 51519 at 40 °C (mm ² /s)	Lubricating oils for FLENDER gear motors			
Lubricant	Code no.		 TEXACO	 TOTAL	 SHELL	 Verkol Lubrificantes
Mineral oils (MIN oil)	C15	VG 220	Meropa WM 220 Auriga EP 220	KASSILLA GMP 220 CARTER EP 220	TUNGEAR 220 ¹⁾	COMPOUND MP 220
	C16	VG 150	Meropa WM 150 Auriga EP 150			COMPOUND MP 150
	C17	VG 100	Meropa WM 100 Auriga EP 100			
Polyglycol (PG oil)	C23	VG 460				
	C24	VG 320				
	C25	VG 220				
Poly- α -olefin (PAO oil)	C33	VG 460		CARTER SH 460		
	C34	VG 320		CARTER SH 320		
	C35	VG 220		CARTER SH 220		
Biologically degradable oils (BIO oil)	C44	VG 320				
	C45	VG 220				
	C46	VG 150				
Physiologically safe oils (PHY oil)	C53	VG 460		Kestone Nevastane SY 460		
	C54	VG 320		Kestone Nevastane SY 320		
	C55	VG 220		Kestone Nevastane SY 220		
1) TUNGEAR is approved for Brazil under the name of GEARIL in VG 220. Distributor: TRIBOTECHNICA Lubrificantes Sinteticos Sao Paulo. TUNGEAR is approved for India under the name of Mo _x -Active Gear oil in VG 220. Distributor: OKS Speciality Lubricants Bombay.						

FLENDER

1.2 Lubricating greases for gear units and rolling bearings

For special gear unit applications, grease lubrication of the gears and/or bearings may be necessary.

Note: Greases may be used only if their use is specified in the operating instructions for the gear unit. The relubrication periods must be adhered to.

Gear greases are suitable for the lubrication of gears and bearings with closed gear units (e.g. small gear units) and open drives with low peripheral speeds (e.g. for girth gears, racks).

As well as lubrication, rolling bearing greases are used for the special sealing of bearing points, e.g. with vertical gear-unit connection shafts or against environmental action such as dust or water spray.

Note: In closed gear units with internal oil lubrication the gear oils must not be allowed to mix with bearing greases.

Table D Lubricating greases for FLENDER gear units and rolling bearings						
Lubricant	Code no.	Consistency NLGI class DIN 51818				
Gear greases (MIN-GF) Mineral oil base	G13	3				Viscogen ¹⁾
	G14	2				
	G15	1				
	G16	0				
	G17	00	Aralub FDP 00		CLS Grease	Tribol 5000 Optitemp OG 0
	G18	000				Longtime PD 00 MA Mehrzweckfett 00
Rolling bearing greases (MIN-WF) Mineral oil base Lithium saponification	H12	4				
	H13	3	Aralub HL3	Energrease LS 3	Spheerol AP 3	
	H14	2	Aralub HL2	Energrease LS 2	Spheerol AP 2 Spheerol EPL 2	Tribol 4020 / 220-2 Longtime PD 2
	H15	1				Tribol 3785 ²⁾
¹⁾ with solid content MoS ₂ ²⁾ Mixture of mineral oil and PAO						

Table D Lubricating greases for FLENDER gear units and rolling bearings						
Lubricant	Code no.	Consistency NLGI class DIN 51818				
Gear greases (MIN-GF) Mineral oil base	G13	3				
	G14	2				
	G15	1				
	G16	0	Orona FG EP 0			
	G17	00	Orona 00	Mobilux EP 004		
	G18	000				
Rolling bearing greases (MIN-WF) Mineral oil base Lithium saponification	H12	4				
	H13	3	Glissando 30	BEACON 3		Renolit FWA 220 Renolit H 443-HD 88
	H14	2	Glissando 20		Arcanol L 186 V ²⁾	Renolit H 443-HD 88 Renolit FWA 160
	H15	1				
³⁾ Li, Ca saponification						

FLENDER

Table D	Code no.	Consistency NLGI class DIN 51818	Lubricating greases for FLENDER gear units and rolling bearings			
Lubricant						
Gear greases (MIN-GF) Mineral oil base	G13	3				
	G14	2				
	G15	1				
	G16	0			Mobilplex 44	
	G17	00		MICROLUBE GB 00		Alvania GL 00
	G18	000				
Rolling bearing greases (MIN-WF) Mineral oil base Lithium saponification	H12	4				
	H13	3			Mobilux 3	Alvania RL 3 Alvania EP / LF 3
	H14	2	LAGERMEISTER EP 2	CENTOPLEX GLP 402	Mobilux 2	Alvania RL 2 Alvania EP / LF 2
	H15	1				

Table D	Code no.	Consistency NLGI class DIN 51818	Lubricating greases for FLENDER gear units and rolling bearings			
Lubricant						
Gear greases (MIN-GF) Mineral oil base	G13	3		Glissando FG 30 EP		
	G14	2				
	G15	1				
	G16	0	Wiolub GFW		Multis EP 0	
	G17	00		Marfak 00	Multis EP 00	
	G18	000				
Rolling bearing greases (MIN-WF) Mineral oil base Lithium saponification	H12	4				
	H13	3			Multis 3 Multis EP 3	
	H14	2	Wiolub LFK 2	Multifak 2 Multifak 20	Multis 2 Multis EP 2	
	H15	1			Multis 1 Multis EP 1	

2. Lubricants for ZAPEX couplings

Note: For **FLUDEX couplings**, special lubricant recommendations apply.

According to manufacturer's information the oils and greases listed in these operating instructions are manufactured and supplied world-wide in the above-mentioned quality as required by FLENDER (Exceptions: Observe footnotes in the tables).

2.1 Required oil and grease quality

For FLENDER ZAPEX couplings, only oils and greases containing active agents to DIN 51 517-3 for increasing corrosion protection and resistance to ageing and for reducing wear in mixed-friction areas are approved. The scuffing resistance of the oils in the FZG test to DIN 51 354-2 must comply with stage 12 or higher under A/8,3/90 test conditions.

In addition, the oils and greases must meet the following quality requirements specified by FLENDER:

- Compatibility with elastomer materials of the shaft-sealing rings
- Compatibility with liquid seals between bolted-joint surfaces.

Note: The use of lubricants which do not comply with the above quality requirements may invalidate our product guarantee. Please note that every lubricant manufacturer or supplier is responsible for the quality of his product.

Which lubricant to select always depends on the viscosity or NLGI class indicated in these operating instructions. If a different viscosity or NLGI class or a lubricant of a type other than those recommended in these instructions is used, the operator is responsible for its technical suitability. To minimise the technical risk in such cases, we recommend using a lubricant of the above indicated quality, which should be accompanied by a statement of suitability by the lubricant manufacturer.

Caution!

The instructions given in the operating instructions for the couplings must be observed!

Deviations are permitted only after consultation with FLENDER!

2.1.1 Oil and grease types

Note: The upper and lower limits for the operating temperatures (flash point, pour point) of individual oils and greases may vary considerably. For these and other data and properties of oils and greases, refer to the lubricant manufacturers' technical data sheets.

According to the manufacturers, the following are the minimum periods during which the oils or greases can be used without undergoing any significant change in quality:

- with use at up to 70 °C after approx. 8 000 operating hours, max. 2 years,
- with use at over 70 °C after approx. 3 000 operating hours, max. 1 year.

2.1.2 Oil and grease change

The degree of purity of the lubricant affects the operating reliability and life span of the lubricant and the coupling. Care must always be taken that the oil or grease in the coupling is clean. For the first lubricant change after start-up of the coupling and for subsequent lubricant changes, the specifications in the operating instructions for the coupling must be observed. With larger lubricant quantities, an analysis should be carried out to find out whether cleaning or changing is required.

When changing lubricant of the same type, the quantity of lubricant remaining in the coupling should be kept as low as possible. Generally speaking, a small remaining quantity will cause no particular problems. Lubricants of different types and manufacturers must not be mixed together. If necessary, confirmation that the new lubricant is compatible with residues of the old lubricant should be obtained from the manufacturer.

Caution!

Lubricants must never be mixed with other substances.

FLENDER

Table E	Code no.	Oil: Viscosity in cSt at 40 °C DIN 51519 Grease: Consistency DIN 51818		Lubricating oils and lubricating greases for FLENDER ZAPEX couplings			
Lubricant							
Mineral oils (MIN oil)	I11	ISO class	VG 1000				
	I12		VG 680	CLP 680 S ECO GEAR 680 M	Degol BG 680 Plus	GEAR RSX 680	ECO GEAR 680 M
	I13		VG 460	CLP 460 S ECO GEAR 460 M	Degol BG 460 Plus	GEAR RSX 460	ECO GEAR 460 M
	I14		VG 320				
Liquefied greases (MIN-FLF) Mineral oil base	J15	NLGI class	1				
	J16		0		Aralub Fließfett AN 0		
	J17		00				
	J18		000				

Table E	Code no.	Oil: Viscosity in cSt at 40 °C DIN 51519 Grease: Consistency DIN 51818		Lubricating oils and lubricating greases for FLENDER ZAPEX couplings			
Lubricant							
Mineral oils (MIN oil)	I11	ISO class	VG 1000				
	I12		VG 680	Energol GR-XF 680	BESLUX GEAR XP 680	Alpha SP 680	Tribol 1100 / 680 Optigear BM 680
	I13		VG 460	Energol GR-XF 460	BESLUX GEAR XP 460	Alpha SP 460 Alpha MAX 460	Tribol 1100 / 460 Optigear BM 460
	I14		VG 320				
Liquefied greases (MIN-FLF) Mineral oil base	J15	NLGI class	1				
	J16		0				
	J17		00	Energrease LS-EP 00		CLS Grease	Tribol 3020/1000-00 Longtime PD 00
	J18		000				

Table E	Code no.	Oil: Viscosity in cSt at 40 °C DIN 51519 Grease: Consistency DIN 51818		Lubricating oils and lubricating greases for FLENDER ZAPEX couplings			
Lubricant							
Mineral oils (MIN oil)	I11	ISO class	VG 1000	CEPSA AEROGear 1000			
	I12		VG 680	CEPSA AEROGear 680	Falcon CLP 680		SPARTAN EP 680 ¹⁾
	I13		VG 460	CEPSA AEROGear 460	Falcon CLP 460	EP Industrial 460	SPARTAN EP 460 ¹⁾
	I14		VG 320				
Liquefied greases (MIN-FLF) Mineral oil base	J15	NLGI class	1				
	J16		0		Orona FG EP 0		FIBRAX EP 370
	J17		00				
	J18		000				

1) ESSO guarantees the qualities required by FLENDER only for products from Europe

FLENDER

Table E	Code no.	Oil: Viscosity in cSt at 40 °C DIN 51519 Grease: Consistency DIN 51818	Lubricating oils and lubricating greases for FLENDER ZAPEX couplings			
Lubricant			FLENDER	FUCHS	FUCHS LUBRITECH	KLÖBER LUBRICATION
Mineral oils (MIN oil)	I11	ISO class	VG 1000			
	I12		VG 680	Renolin EPX 680 Renolin CLP 680 PLUS	GEARMASTER CLP 680	STRUCTOVIS BHD-MF
	I13		VG 460	Renolin EPX 460 Renolin CLP 460 PLUS	GEARMASTER CLP 460	
	I14		VG 320			
Liquefied greases (MIN-FLF) Mineral oil base	J15	NLGI class	1			
	J16		0			GRAFLOSKON C-SG 500 Plus
	J17		00	FLENDER Hochleistungsfett RENOLIT SO-D 6024		
	J18		000			

Table E	Code no.	Oil: Viscosity in cSt at 40 °C DIN 51519 Grease: Consistency DIN 51818	Lubricating oils and lubricating greases for FLENDER ZAPEX couplings			
Lubricant			LE	Mobil	MOTOREX	OMV
Mineral oils (MIN oil)	I11	ISO class	VG 1000			
	I12		VG 680 609 ALMASOL Vari-Purpose Gear Lubricant	Mobilgear XMP 680 Mobilgear 636	GEAR COMPOUND PLUS 680	OMV gear HST 680
	I13		VG 460 608 ALMASOL Vari-Purpose Gear Lubricant	Mobilgear XMP 460 Mobilgear 634	GEAR COMPOUND PLUS 460	OMV gear HST 460
	I14		VG 320			
Liquefied greases (MIN-FLF) Mineral oil base	J15	NLGI class	1			
	J16		0			
	J17		00	Mobilux EP 004		
	J18		000			

Table E	Code no.	Oil: Viscosity in cSt at 40 °C DIN 51519 Grease: Consistency DIN 51818	Lubricating oils and lubricating greases for FLENDER ZAPEX couplings			
Lubricant			PO Petrol Ofisi	Shell	STATOIL	TEXACO
Mineral oils (MIN oil)	I11	ISO class	VG 1000 PO Gravis MP 1000			
	I12		VG 680 PO Gravis MP 680	Shell Omala 680		
	I13		VG 460 PO Gravis MP 460	Shell Omala 460 Shell Omala F 460		Meropa WM 460 Auriga EP 460
	I14		VG 320 PO Gravis MP 320		LoadWay EP 320	
Liquefied greases (MIN-FLF) Mineral oil base	J15	NLGI class	1			
	J16		0			
	J17		00	Alvina GL 00		
	J18		000			

FLENDER

Table E	Code no.	Oil: Viscosity in cSt at 40 °C DIN 51519 Grease: Consistency DIN 51818		Lubricating oils and lubricating greases for FLENDER ZAPEX couplings			
Lubricant							
Mineral oils (MIN oil)	I11	ISO class	VG 1000		VECO MATRANOL XP 1000		
	I12		VG 680	TUNGEAR 680 2)	VECO MATRANOL XP 680		
	I13		VG 460	TUNGEAR 460 2)	VECO MATRANOL XP 460		
	I14		VG 320			COMPOUND MP 320	
Liquefied greases (MIN-FLF) Mineral oil base	J15	NLGI class	1				
	J16		0				
	J17		00				
	J18		000				

2) TUNGEAR is approved for Brazil under the name of GEAROIL in VG 460 and 680. Distributor: TRIBOTECHNICA Lubrificantes Sinteticos Sao Paulo. TUNGEAR is approved for India under the name of Mox-Active Gear oil in VG 460. Distributor: OKS Speciality Lubricants Bombay.

FLENDER**List of equipment**

Code: NARAYANGONJ

Type **B2SV**Size **007**Page
1 of 1

Please quote in correspondence

GL 412 3221-040 EN

Part No.	Qty.	Description	Manufacturer
774	1	Filter Type B 5911 Mesh width 250 µm Operating pressure up to 25 bar Rate of flow: approx. 12 l/min	FLENDER
700	1	Gear pump Type SF 2/8 Delivery approx. 12 l/min Startup temperature > 10 °C with pressure relief valve adjusted at p > 8 bar coupled directly to: Three-phase motor, P = 0.75kW, N = 1395 1/min, 400V, 50Hz, IP55, Iso F	STEIMEL
773	1	Pressure switch Type 604 G1 Pressure adjustment range: with falling pressure 0.02 - 1.2 bar Reconnect value approx. 0.03 bar Switching capacity: 15 A/125 - 480 V AC 0.2 A/125 V DC, 5 A/28 V DC Degree of protection: IP 65	CCS

A. Friedr. Flender AG, D 46393 Bocholt,		Datum	Name: Bdr/Die	ZGE
Tel. 02871/92-0, Telefax 02871/922596, http://www.flender.com		2004-06-15	Rev.:)	

These operating instructions are binding for coarse mesh filters according to FLENDER works standard W 5911.

Configuration and principle of operation

Coarse mesh filters protect load-side systems, units, measurement and control devices from impurities.

The coarse mesh filters consist of a housing with threaded connections (1), a strainer (2) and a drain plug (4). The mesh insert itself consists of a coarse-mesh screen with an additional fine-mesh inner screen. The medium flows through the housing, the entrained dirt particles in the line being retained by the mesh insert and collected in the filter bowl.

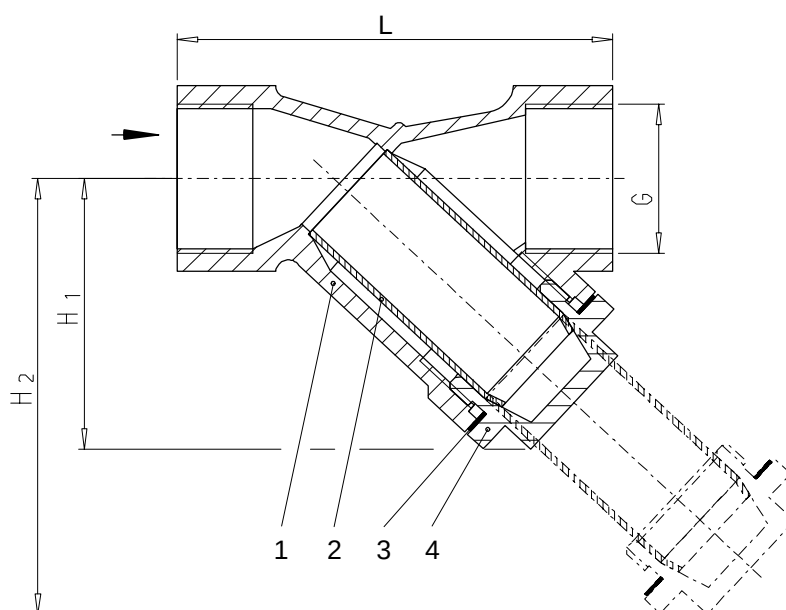
Installation

The coarse filters should be installed so that the direction of flow corresponds with the arrow cast on the housing. The filter bowl must be underneath at all times. Sufficient space should be provided for filter cleaning and filter change (see dimension H₂). For vertically running pipes with an upwards flow direction, the coarse mesh filter is exceptionally to be installed with the drain plug pointing upward.

Maintenance

Check from time to time, whether large quantities of dirt particles have collected which are adversely affecting the volume flow. For this purpose, the drain plug (4) is loosened and the strainer (2) is pulled out after the respective part of the plant has been put out of service. Damaged mesh screens (2) and seal rings (3) must be replaced.

Type 1 NI



- | | |
|--|--|
| 1. Housing | material: brass |
| 2. Mesh insert (coarse and fine-mesh screens) | material: CrNiMo (1.4401) |
| 3. Sealing ring | material: graphite with metallic support |
| 4. Drain plug | material: brass |

Connection thread size	G 1/2	G 3/4	G 1	G 1 1/2	G 2
Installation length L	64	74	90	120	150
Overall height H ₁	40	45	84	84	108
Overall height H ₂ (mesh insert withdrawn)	61	75	134	134	158
Mesh size (µm)	250				
Oil rate of flow K _{VS} (m ³ /h)	5.1	9.1	14.3	36.6	57
Screen discharge area (cm ²)	2.5 x cross section of pipe				
Weight (kg)	0.2	0.3	0.47	1.35	1.9
Nominal pressure PN (bar)	25				
Max. medium temperature (°C)	200				



OPERATING MANUAL

Steimel Gear Pumps and Pump Units

Series B, R, SF, T

Type designation :

Serial No. :

**Gebr. Steimel GmbH & Co
Maschinenfabrik**

Postfach 1565
D-53762 Hennef

Telephone (02242) 88 09 - 0 Fax (02242) 88 09 160

1	General	6	Starting up / shutting down
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1.2.1	Type designation and works serial No.	6.3	Placing back into operation
1.2.2	Technical data sheet	6.4	Shutting down
1.2.3	Permitted use		
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2.2	Qualification and training of operating personnel	7.2	Shaft seal
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2.4	Compliance with regulations pertaining to safety at work	7.2.2	Axial face seal / rotary shaft seal
2.5	Safety instructions relevant for operation	7.2.3	Twin seals
2.6	Safety instructions relevant for maintenance, inspection and assembly work	7.2.4	Pressure relief valve
2.7	Unauthorised alterations and production of spare parts	7.3	Dismantling and re-assembly
2.8	Unauthorised modes of operation	7.3.1	Basic instructions
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		7.3.3	Dismantling the pump
		7.3.4	Assembling the pump / unit
		7.4	Spare parts
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3.1	Transport	9	Faults; causes and remedies
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4.3.5	Drive		
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5.5.6	Pressure test and leakage test		
5.6	Protection against foreign bodies and contamination		

1 General

1.1 Intended application

Steimel gear pumps are suitable for pumping lube oils, fuel oils and other lubricating fluids.

The fluids pumped may not contain any abrasive components and may not chemically attack the materials of which the pump is made.

The field of application of these pumps extends from oil hydraulics through lubrication systems to pure pumping applications.

Above all, the pumped medium must always be clean and have good lubricating properties in order to ensure that the pump operates correctly, in order to ensure a high level of operating reliability and safety and in order to ensure a long service life.

1.2 Product information

1.2.1 Type designation and works serial No.

The full type designation is specified in the technical data sheet or the order confirmation. The works serial No. comprises the contract No. and a consecutive serial No. (e.g. 94 1234-2). It is imprinted directly on the rating plate or on the pump.

1.2.2 Technical data sheet

The technical data sheet is a part of our sales order confirmation and should contain all important technical details agreed between the purchaser and manufacturer, such as

- designation, type, size
- item No.
- project drawing / dimension sheet
- spare parts list for parts subject to wear
- related Operating Manuals
- suction and delivery connection
- materials
- shaft seal
- pumped medium / field of application
- direction of rotation
- accessories
- hydraulic parameters such as delivery rate, operating pressure, speed, viscosity range and temperature range

If required, also:

- pressure relief valve / safety facilities
- special installation position
- suction characteristics

- operating or working range / scheduled or permitted use
- sound pressure level
- power requirement
- internal preservation
- surface protection / paint finish
- weight
- references to tests, certificates and acceptance tests
- date of issue

in the case of pump units, also:

- drive type / motor / transmission
- motor and transmission data
- coupling / belt drive
- sub-plate, intermediate bracket / pump mount
- accessories / safety facilities
- references to required certificates
- miscellaneous.

1.2.3 Permitted use

The pump / pump unit is designed only for the field of application listed in the technical data sheet of our order confirmation. Differing operating conditions will necessitate new contractual agreements.

2 Safety

This Operating Manual gives basic instructions which are to be observed during installation, operation and maintenance of the pump. It is therefore imperative that this Manual be read by the responsible personnel / operator prior to assembly and commissioning. It is always to be kept available at the installation site.

It is not only the general safety instructions contained under this main heading "Safety" that are to be observed but also the specific information provided under the other main headings.

2.1 Identification of safety instructions in the Operating Manual

Safety instructions given in this Manual, non-compliance with which would affect safety are identified by the following symbol:



See DIN 4844-W 9

or where electrical safety is involved, with



see DIN 4844-W 8.

The word

CAUTION

indicates safety information which may result in risks to the machine and its function if the information is not observed.

It is imperative that signs affixed to the machine, e.g.

- arrow indicating the direction of rotation
- symbols indicating fluid connections

be observed and kept legible.

2.2 Qualification and training of operating personnel

The personnel responsible for operation, maintenance, inspection and assembly must be adequately qualified. Scope of responsibility and supervision of the personnel must be exactly defined by the plant operator. If the staff do not have the necessary knowledge, they must be trained and instructed, which may be performed by the machine manufacturer or supplier on behalf of the plant operator. Moreover, the plant operator is to make sure that the contents of the Operating Manual are fully understood by the personnel.

2.3 Hazards in the event of non-compliance with the safety instructions

Non-compliance with the safety instructions may produce a risk to the personnel as well as to the environment and the machine and results in a loss of any right to claim damages.

For example, non-compliance may involve the following hazards:

- Failure of important functions of the machine/plant
- Failure of specified procedures of maintenance and repair
- Exposure of people to electrical, mechanical and chemical hazards
- Endangering the environment owing to hazardous substances being released.

2.4 Compliance with regulations pertaining to safety at work

When operating the pump, the safety instructions contained in this Manual, the relevant national accident prevention regulations and any other service and safety instructions issued by the plant operator are to be observed.

2.5 Safety instructions relevant for the plant operator

- If hot or cold machine components involve hazards, they must be guarded against accidental contact on site.
- Guards for moving parts (e.g. coupling) must not be removed from the machine while in operation.
- Any leakage of hazardous (e.g. explosive, toxic, hot) fluids (e.g. from the shaft seal) must be drained away so as to prevent any risk to persons or the environment. Statutory regulations are to be complied with.
- Hazards resulting from electricity are to be prevented (see for example, the VDE Specifications and the bye-laws of the local power supply utilities).

2.6 Safety instructions relevant for maintenance, inspection and assembly work

It shall be the plant operator's responsibility to ensure that all maintenance, inspection and assembly work is performed by authorised and qualified personnel who have adequately familiarised themselves with the subject matter by studying this Manual in detail.

Any work on the machine shall only be performed when it is at a standstill, it being imperative that the procedure for shutting down of the machine described in this Manual be followed. Pumps and pump units which convey hazardous media must be decontaminated.

On completion of work all safety and protective facilities must be re-installed and made operative again.

Prior to restarting the machine, the instructions listed under "Initial commissioning" are to be observed.

2.7 Unauthorised alterations and production of spare parts

Any modification may be made to the machine only after consultation with the manufacturer. Using spare parts and accessories authorised by the manufacturer is in the interest of safety. Use of other parts may exempt the manufacturer from any liability.

2.8 Unauthorised modes of operation

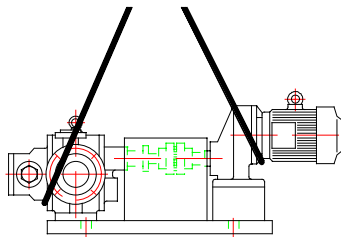
The reliability of the machine delivered will be only guaranteed if it is used in the manner intended, in accordance with Clause 1 of this Manual. The limit values specified in the data sheet must under no circumstances be exceeded.

3 Transport and intermediate storage

3.1 Transport

The pump / the pump unit is delivered in assembled condition. Lube oils for transmissions, confining and sealing fluid and valves / fittings are delivered in separate packagings if necessary.

The pump / the pump unit is packaged differently, depending on the transport route and size. The packaging complies with our general packaging guidelines unless otherwise contractually agreed. When transporting the complete unit, the cables must be attached, as shown, to pump and motor (not to the eyebolt of the motor and pump).



The motor suspension eyebolts are suitable only for the weight of the motor. Likewise, the pump eyebolt is suitable only for the weight of the pump. Consequently, additional loads attached to the motor and pump may not be raised with these eyebolts.



Please avoid jerky movements, shock and heavy vibration when raising and lowering during transportation !



3.2 Intermediate storage

The pump / the pump unit must be stored on a timber support at a protective location in the position of normal use and must be covered.



It is not permitted to stack heavy pumps / pump units one above the other.

The pump is not provided with internal preservation. In some cases, a film of test medium originating from the test bench run may still adhere to the inside faces. A mineral oil is used as the test medium unless otherwise agreed.

Should you intend to store the unit for long periods, we advise you to contractually agree on a special internal preservation.

The condition of the pump and its preservation must be checked at least once per year!

If stored in the open air, please carefully cover the pump / pump unit and ensure that it is not exposed either to moisture or foreign bodies.

4 Technical description

4.1 General description

The gear pump, being the most popular type of rotary, positive-displacement pump, owes its great popularity primarily to the sophisticated involute gearing which permits low-cost series production with high precision, besides its simple construction. Most gear pumps have straight-toothed spur gears. Certain types are fitted with helical-tooth spur gears or with double-helical (herring-bone) gears in an effort to achieve greater running smoothness.

4.2 Construction and mode of operation

In line with the diverse fields of application of the gear pumps, we have developed a variety of different models.

Very low-pressure gear pumps for pumping moderately fluid to viscous media generally have a large displacement volume and low speed of rotation. It is more economical to pump low-viscosity media with small, high-speed gear pumps. The mode of operation of the external gear pump results from the engaged teeth. This is where the actual separation process between suction and pressure chamber occurs.

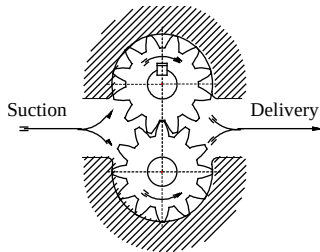
The drive gear wheel / drive pinion drives the mating gear wheel / mating pinion, thus causing the intermeshing teeth to release the tooth spaces abruptly.

When the gear wheels, which are surrounded by a casing or adapter on all sides, rotate, the tooth spaces fill with the medium to be pumped as the result of the partial vacuum produced and convey it outwards and around to the delivery end. Combing of the gear wheels which are turning in opposite directions forces the medium into the delivery line. The delivery pressure produced by a pump builds up in the delivery line depending on the resistances opposing the delivery flow. These resistances result from pipe friction and the various installed fittings, fixtures and devices.

The power demand increases with increasing delivery pressure!

CAUTION

The pump may never be operated with the suction or delivery line closed!



4.3 Mechanical construction

4.3.1 Pump casing

The pump casing encloses the gear wheels and is generally equipped with a suction and delivery connection for the pipes.

Depending on the particular type, it also contains a pressure relief valve and one side of the drive and counter-shaft bearing assembly.

In the case of the disk or plate design, the gear wheels are enclosed by a so-called adapter which has the shape of a plate, the thickness of which corresponds to approximately the length of the gear wheel.

The drive and counter-cover serves to accommodate the shaft bearing.

The shaft seal is accommodated in the drive cover or in a casing.

4.3.2 Gear wheels

Gear wheels have an involute gearing. Straight-tooth spur gears are generally used in feed pumps. Owing to the greater running smoothness, helical-tooth spur gears are given preference in high-speed lube oil pumps in the low pressure range. The gear wheel bodies are generally connected to the shaft by interference fits and, in some cases, also by feather keys. In many cases, shafts and

gear wheels are designed as pinions. Materials, surface hardnesses, wear-resistant coatings and gearing quality will depend on the customer's sales order.

4.3.3 Shafts, bearings and lubrication

The pump shafts are mounted in plain bearing bushes. The bearing points must be adequately lubricated by the medium!

CAUTION

The pump must always receive fluid and may never be allowed to run dry!

The running play on the sliding surfaces is rated so as to constantly guarantee an adequate lubrication film.

The lower the viscosity of a medium to be pumped, the closer the running play must be kept in order to achieve a favourable volumetric efficiency of the pump. However, this also necessitates a purer medium to be pumped in order to achieve high operating reliability and as long a pump service life as possible.

4.3.4 Shaft seal

On versions with stuffing box packing (P), this packing must drip slightly during operation in order to dissipate the frictional heat.

Ensure that the leakage is safely collected or discharged in controlled fashion.

On versions with axial face seal (G), this seal has only slight or invisible (vapour) leakage losses during operation. It requires no maintenance.

No maintenance is required either on versions with rotary shaft seal (R).

In the case of twin seals, such as GG or GR versions, a quench fluid recipient serves to lubricate and cool the downstream seal. These seal combinations serve to exclude air in the case of media which react in an undesirable manner when coming into contact with atmospheric oxygen.

4.3.5 Drive

CAUTION

The pump must be driven free of vibration via a flexible coupling and the drive shaft may be subjected

neither to radial forces nor axial forces!

Any radial loading of the shaft journal resulting from V belts, chains, drive gear wheels or so forth is cushioned by an additional radial bearing which is accommodated inside or outside the pump. Please state the relevant parameters for design of this bearing in your purchase order.

4.4 Accessories

4.4.1 Pressure relief valve

The pump can be delivered with a built-in or mounted pressure relief valve or circulation valve which returns the pumped medium from the delivery end to the suction end in order to protect against overloading of the pump. This valve is suitable for occasional, brief response, not owing to regular operation. Rapid heating of the medium occurs should the valve respond for a longer period, and this may result in undesirable changes and may also disturb proper operation of the pump. We recommend inserting bypass-relief valves in a bypass with a return line to the supply tank for high pressures and long response times.



The setting of the pressure relief valve may be changed only after procuring the prior consent of the pump manufacturer!

The final pressure of the valve lies 20-30 % above the opening pressure.

4.4.2 Heating and cooling units

Types TM have double-walled casings or corresponding ducts and can be heated or cooled with a liquid or gaseous medium. Temperature sensors in the pump serve to monitor and safeguard the pump against inadmissible switch-on of the drive motor.

Types TE are heated with electrical heating cartridges. The temperature is controlled steplessly by means of a system regulator.

5 Mounting / installation

5.1 Installation position

Unless otherwise agreed, the pump should be mounted / installed so that the drive shaft and counter-shaft are vertically adjacent. The fact that the line connections are then horizontal is the best method of guaranteeing that as large a fluid quantity as possible remains in the pump when the pump is switched off. It then normally inducts this fluid "reservoir" easily again under normal conditions when restarted.

5.2 Mounting for the first time

Base-mounted pumps may be mounted only on precisely level supports, in order to avoid twisting.

Flange pumps must be mounted in a precisely machined flange centring ring. If machines are to be coupled, please ensure that the machines are

carefully aligned. Wherever possible, please use torsion-resistant sub-plates or frames.

CAUTION

On pump units with a sub-plate made of grey cast iron, please ensure that the deposit surface is absolutely flat

and level!

On uneven deposit surfaces, the sub-plates may also be aligned with the aid of shims, but these shims must be well-set (snugly fitting).

5.3 Drive elements

Please also refer to 4.3.5 Drive.

- On a pump with free shaft extension, remove the shaft protection (if applicable, corrosion protection, e.g. with solvent, but not with emery paper).
- Fit and lock the drive element onto the shaft.

CAUTION

Fit the coupling using a fitting tool. On no account may you force the coupling on by knocking or tapping

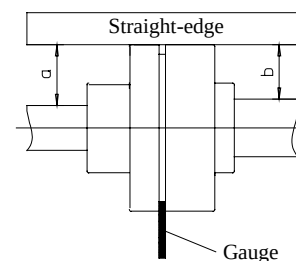
since this could damage the shaft, gear wheels, bearing assembly, thrust rings and retaining rings for instance.

5.3.1 Aligning flexible couplings

The unit is correctly aligned if a straight-edge placed axially over both coupling halves is the same distance from the relevant shaft at all points on the circumference, whereby it must be noted that the measurement point must be turned as well by hand.

Moreover, both coupling halves must be the same distance from each other at all points on the circumference. This can be verified with a probe or gauge.

Example: Aligning a flexible coupling without intermediate sleeve or gear rim.



The deviation between the two coupling halves should not exceed 0.3 mm either radially or axially. This must also be ensured at operating temperature and with supply pressure applied.

5.3.2 Coupling guard

In accordance with the accident prevention regulations, the pump may be operated only with a coupling guard. If this coupling guard is not supplied by us, at the express request of the orderer, the operator must provide such a guard.

5.4 Electrical connection

The electrical connection must be made by a qualified electrician in accordance with the applicable safety regulations. Please refer to the Installation Instructions and Operating Manual provided by the motor manufacturer. Please refer to the Installation Instructions and Operating Manual of the equipment manufacturer for safety facilities and control equipment.

5.5 Pipes

5.5.1 Purging

The suction and delivery connection of the pump are sealed with plastic caps. The pump interiors still contain residual quantities of the testing oil or, possibly, a preservation agent. If contractually agreed, pumps are also cleaned by us after the trial run. These pumps are then entirely free of such residue.

Should the pump be purged before initial commissioning, we recommend using either the medium to be pumped or a neutral agent for this purpose. Please do not use water or fluids containing water since this could entail the risk of corrosion, apart from on pumps made of high-grade steel or stainless steel.

CAUTION

Before the pump is installed in a new installation, the tanks, pipes and connections must be cleaned thoroughly, flushed through and blown through with compressed air. Welding beads, scale and other contamination frequently only become detached after a certain period. In order to prevent this, please take appropriate measures, such as sandblasting, before installation.

CAUTION

Foreign bodies will endanger the operational reliability and safety of the pump and may lead to blockage or total failure.

5.5.2 Connecting pipes

On no account may the pump be used as a fixed support on long pipes. Long pipes must be supported directly ahead of the pump and must be connected free of strain. On no account may their weight be borne by the pump.

Elongation of the pipes resulting from temperature variations must be compensated for by suitable measures.

5.5.3 Additional connections

The dimensions and position of the additional connections required for the pump, e.g. for sealing, purging or leakage fluid, are specified in the installation and piping plan.

5.5.4 Suction line

In the case of short suction lines, the nominal diameter should be at least that of the pump connection. On longer pipes, determine the most economical nominal diameter in each individual case. Adapters for increasing to larger nominal diameters should be designed with as gentle an expansion angle as possible in order to reduce pressure losses.

The suction line should always be as short and straight as possible. The total suction-end resistance may not exceed 0.4 bar in operating condition. In start-up condition, it may be max. 0.65 bar for a period of up to approx. 30 min (until operating temperature is reached).

The total resistance comprises the geodetic suction head and the pipe resistance, including the resistance of any built-in fixtures and fittings. It is measured as a partial vacuum using a vacuum meter directly at the pump suction connection. The flow velocity is a further characteristic parameter for the suction line. Always endeavour to achieve a value of 1...1.5 m/s for the flow velocity in the case of oil.

A partial vacuum greater than 0.4 bar impairs the pump's volumetric efficiency, thus reducing the delivery capacity and resulting in cavitation. Cavitation causes increasing noise and greater wear in the long term.

Installing a vacuum meter at the pump suction connection is particularly advisable for checking the partial vacuum in the suction line and assessing the degree of contamination of any suction filter installed.

CAUTION

The suction line must be absolutely tight !

5.5.5 Delivery line

The operating or delivery pressure must be checked with a pressure gauge fitted directly at the pump delivery connection. The delivery pressure or total resistance comprises the individual resistances of the actuators, the geodetic pressure head and pipe resistances.

CAUTION Line and actuator resistances depend on viscosity. It is advisable to fit a pressure relief valve in order to avoid overloading the pump as the result of inadmissibly high pressure.

The flow velocity in the delivery line should not exceed 3 m/s in the case of oil.

Should there be air in the suction line, this may cause induction problems when starting the pump, e.g.:

- if the pump is to pump against a closed system (loaded non-return valve),
- if two pumps operating in parallel are mutually safeguarded by means of non-return valves.

The situation can be remedied by bleeding the delivery line or by a small bypass line (DN 4-6) upstream of the non-return valve. The bypass is then connected back to the tank.

On forced-circulation lubrication systems such as those conventional in gear construction, cleaning filters must be fitted in the delivery line.

The overall circulating volume of the lube oil must not be selected too low so that it does not overheat as the result of frequent pumping through the circuit and inadequate cooling. A cooler or heat exchanger must be provided if necessary.

5.5.6 Pressure test and leakage test

These tests must be conducted statically **before** the pump is installed. Special-purpose boiler and pipe test pumps must be used for this purpose. On no account may the feed pump be used !

Test pressures generally lie

CAUTION substantially above the pump design pressure so that the pump could be damaged. (see also 5.5.1 Purging)

5.6 Protection against foreign bodies and contamination

A distinction is made between two types of contamination:

1. Primary contamination

- Filling and feed pumps are exposed to foreign bodies, dirt particles or mineral pigments which are located in the medium to be pumped.
- Pipe elements, in particular upstream of the pump, may be contaminated with fine dirt particles such as metal swarf, welding or soldering beads or abrasives etc. originating from production or installation. (see also 5.5.1 Purging)

2. Secondary contamination

Circulation pumps, e.g. for oil supply systems, are subjected to contamination which, after commissioning, forms continually in the sliding surfaces of the gear components as abrasion or abraded wear in the form of metallic particles and which can be entrained by the lube oil.

Gear pumps can be protected against primary contamination by coarse filters or plate-type filters fitted in the suction line.

Depending on the filter rating, these suction filters retain foreign bodies of a specific size.

CAUTION The filter clogs up gradually as the result of contamination and silting-up, thus impeding induction of the feed stream by the pump. This will be noticeable since the pump will become noisier.

(see also 8.4 Faults)

A filter rating of 0.05 to 0.13 mm is appropriate, depending on pump design. The larger the filter area, the longer will be the maintenance intervals. Gear pumps are best protected against secondary contamination by means of changeover filters or double self-bypassing filters fitted in the delivery line and designed for continuous filtration. Whilst one filter chamber is in operation, the other is at standby. The filters can be equipped with visual differential pressure indicators or electrical contacts for remote monitoring in order to permit the increasing contamination of the filter to be recognised.

One of the other aspects which must be borne in mind when selecting the filter rating and size is the viscosity range of the lube oil. Consequently, the filter manufacturer should be consulted. Filter ratings of 25-60 µm are conventional for lube oils.

CAUTION Foreign bodies endanger the operational safety and reliability of the pump and may lead to blockage or total failure. Larger quantities of ultra-fine contamination will cause abrasion wear on the sliding surfaces and will reduce the service life of the pump.

6 Starting up / shutting down

6.1 Preparations for starting up

Before starting up, inspect the pump installation in order to establish that it is in proper working order as specified in the technical documentation.

In the case of electric motor-driven pumps, the electrical connections may be made only by a qualified electrician.

CAUTION Facilities for personal safety (e.g. contact guard on clutches and belts) must be fitted.

6.2 Initial commissioning

Ensure that there is adequate medium in the pump before initial commissioning.

Gear pumps are always self-priming pumps. Under normal circumstances, they may induct the air present in the suction line and convey it into the delivery line. It is advisable to also fill the suction line with the medium to be pumped before starting, particularly in the case of extremely unfavourable suction conditions, in order to prevent the pump running dry.

CAUTION Fill the pump and, wherever possible, also the suction line with the fluid to be pumped and then bleed.

Always avoid the pump running dry!

Should there be shut-off elements in the suction or delivery line, these elements must be opened fully. Fully open all additional connections provided, e.g. for sealing or purging fluid etc., and check flow.

Heavy pumps generally have an eyebolt for attaching load suspension devices at the topmost point on the casing. There may be a screw plug at this point on small pumps. Depending on the particular design, the eyebolt is located either in a tapped blind hole or in a through-hole with thread. The latter offers the option of filling or bleeding the pump.

CAUTION Check the direction of rotation. Each pump features a direction-of-rotation arrow. If the pump is operated in the wrong direction, this could entail the risk of overloading the shaft seal and causing it to leak. Pumps suitable for clockwise and counter-clockwise rotation are exceptions to this rule. The direction of rotation can even be checked with the pump full and with the shut-off elements open by briefly switching the drive on and off.

CAUTION Gear pumps are positive-displacement pumps.

If the pump pumps into a closed delivery line, it will be overloaded and fail.

Pressure relief valves protect pumps against damage and overload (see also 4.4.1 Pressure relief valve).

Proper functioning of the pump can be checked by listening to the noise and by checking the pressure gauge. The time taken from switching on to pumping should not exceed 30 seconds.

6.3 Placing back into operation

In principle, the same procedure as that described in Point 6.2 Initial commissioning must be noted in this case.

If the pump has been placed out of operation for a long period, please check that there is still adequate fluid in the pump in order to ensure reliable re-induction.

CAUTION Check the mobility of the internal components of the pump in the case of media which could set (e.g. by turning the drive shaft).

CAUTION On no account may a cold pump be started up suddenly with a hot medium. The temperature of the pump must be increased slowly in order to avoid seizing of the pump as the result of thermal shock.

6.4 Shutting down

Depending on the particular system, the pump should be allowed to idle for an adequate period - *with the heating source switched off* - until the pumped fluid's temperature has dropped to such an extent as to avoid heat build-up in the pump. Close the shut-off element in the supply line if the pump has been out of operation for a long period. Close off additional connections.

The shaft seal must be supplied with sealing fluid even when at standstill on pumps in which the pumped fluid is supplied under vacuum.

Media which set should be drained off. The pump can be used at most only briefly and at low purging pressure in order to clean the pipe system.

CAUTION The intention to use the pump as a purging pump must be known when designing the pump in order to ensure material compatibility.

High purging pressures and long operating times should be avoided in the case of purging or cleaning fluids with poor lubrication properties owing to the risk of jamming or seizing on the sliding surfaces of the pump.

7 Maintenance / repair

7.1 General information

Steimel gear pumps largely require no maintenance under normal operating conditions if they are operated with media having good lubrication properties.

It is advisable to regularly check the performance data. Integration of corresponding monitoring devices, such as operating hours counters, pressure gauges, pressure and temperature monitors etc., will depend on the requirements and the relevant regulations.

Please note the following points when carrying out maintenance work on the overall system:

- Check that all securing screws, bolts and connections are tight and retighten if necessary.
- Check alignment of the coupling. Examine elastomers for wear.
- Check the pump for leaks, in particular the shaft seals.

7.2 Shaft seal

7.2.1 Stuffing box packing

The stuffing box gland nut may be tightened only lightly so that the packing allows slight leakage during operation, thus dissipating the frictional heat.

If the leakage rate is too high after a long operating period, tighten the nuts on the stuffing box gland uniformly by 1/6 of a turn. Then observe the leakage. If it is no longer possible to tighten the stuffing box gland nut, you must fit an extra packing ring. It is generally not necessary to replace the entire packing assembly.

Should it be necessary to renew the packing assembly nevertheless, also assess the condition of the shaft bearing assembly and the seal seat. If there are signs of major scoring on the shaft, the shaft must be exchanged since, otherwise, this will impair sealing with new packing rings.

The joints of the packing rings or cut ring sections must be mutually offset by approx. 90°. The joint should be at the top on the last ring at the gland end. The packing rings must be carefully bent open both axially and radially only sufficiently to permit them to just be slid over the shaft.

7.2.2 Axial face seal / rotary shaft seal

Maintenance of the axial face seal is restricted to occasionally checking for leaks.

Very slight leakage of the axial face seal is permitted.

Should the leakage be higher than this, the axial face seal must be renewed. When doing this, also inspect the pump bearing assembly.

The rotary shaft seal requires no maintenance either. Should it leak, it must be renewed.

7.2.3 Twin seals

On twin seals and combined seals, the quench fluid level in the recipient must be checked constantly and topped up if necessary.

7.2.4 Pressure relief valve

Please follow the steps below should it be necessary to regulate / adjust the opening pressure of the pressure relieve valve:

- Unscrew the pressure cap and undo the lock nut.
- The required opening pressure can be set by turning the threaded bushing / regulating screw which is then accessible.
Turning clockwise: *increases the opening pressure.*
Turning counter-clockwise: *reduces the opening pressure.*
Recommended values for pressure setting: approx. 10 % above the operating pressure / working pressure of the system.
- Screw the pressure cap back on (do not forget the sealant).

7.3 Dismantling and re-assembly

7.3.1 Basic instructions

The unit must be safeguarded so that it cannot be switched on before starting dismantling work. The shut-off elements in the supply and delivery line must be closed. The pump must have reached ambient temperature.

7.3.2 Dismantling the unit

1. Disconnect the electrical power supply
2. Remove any fitted additional connections
- 3.: *Unit on sub-plate:*
 - 3.1. Remove the coupling guard
 - 3.2. Disconnect the motor terminals
Detach the motor from the sub-plate
 - 3.3. Disengage the pump and motor
 - 3.4. by moving the motor's position
 - 3.5. Detach the suction and delivery connection
 - 3.6. Detach the pump from the sub-plate
- 4.: *Unit with pump support / intermediate bracket:*
 - 4.1. Disconnect the motor terminals
 - 4.2. Detach the motor
 - 4.3. Detach the suction and delivery connection
 - 4.4. Detach the pump from the pump support / intermediate bracket

7.3.3 Dismantling the pump

Please refer to the various exploded views (if available).

Identify the position of the pump components and number them consecutively!

1. Undo the casing bolts
2. Release the counter-cover from the casing using 2 forcing threads and hexagon bolts
3. Pull the counter-shaft with gear wheel out of the pump casing
4. Release the safety catch on the coupling half and detach it with an extractor tool and then detach the feather key
5. *Version with packing:*
- 5.1. Undo the gland stuffing box
6. *Direction of rotation-dependent axial face seal:*
- 6 1. Pull out the axial face seal
7. *Direction of rotation-independent axial face seal:*
(Please refer to the special Installation Instructions and Operating Manual)
8. *Version with single thrust bearing:*
- 8.1. Detach the bearing catch
9. *Version with rotary shaft seal:*
- 9.1. Detach the seal locating cover
10. Pull the drive shaft with gear wheel out of the pump casing
11. On pumps with bearing components made of carbon / ceramic material, always ensure that you take great care when dismantling!
Use absolutely no force since these components are fragile and could fracture.

7.3.4 Assembling the pump / unit

Follow the generally valid rules applicable in the field of mechanical engineering and machine construction when re-assembling.

O-rings must be inspected for damage and must be replaced by new O-rings if necessary.

Flat gaskets must always be renewed. Ensure that the new gasket has precisely the same thickness as the old gasket.

Hylomar SQ 32/M or Eppl 33 for instance may be used as permanently elastic universal seals.

All seal and gasket residue must be removed completely. When re-assembling, follow the reverse procedure to that followed when dismantling.

- Stuffing box packing, see Point 7.2.1
- Rotary shaft seal
 - Always use new rotary shaft seals.
 - Always ensure that they are fitted in the correct position.
 - The sealing edge of the new ring should not be

at the position of the sealing edge on the old ring.

This can be achieved by:

- exchanging the shaft or shaft bushing
- fitting with a spacer ring
- pressing in to different depths in the location hole.
- Use a press-in tool and a suitable press-in die when pressing in.
- *Make sure that you do not damage the sealing lip at sharp edges when sliding into the shaft!*
- On smooth shafts, round off the keyways and turned grooves. Fitted shafts should have a conical bevel. If this is not possible, an installation sleeve must be used to assist in assembly.
- Axial face seal
please refer to the special Installation Instructions and Operating Manual
- Pressure relief valve
please refer to Point 7.2.4

7.4 Spare parts

Keeping a stock of the most important spare parts and parts subject to wear at the installation site is an important precondition for ensuring that the pump is always ready for operation.

On the other hand, we recommend that you keep in stock a reserve pump and return the removed pump to our works for repair if necessary.

Please refer to our spare parts list with cutaway view when ordering spare parts.

We are able to assume warranty only for original spare parts supplied by us.

It must be expressly pointed out that spare parts and accessories not supplied by us are not tested by us and are not approved by us. Installation and/or use of such products may thus impair design characteristics of the pump and thus endanger active and/or passive safety. We are unable to accept liability and accept claims under warranty for damage and loss occurring as the result of using non-original spare parts and accessories.

Please state the following information when submitting orders for spare parts:

- Contract No. / Serial No.
- Part No.
- Quantity

8 Related documents

Please refer to the technical data sheet or the sales order confirmation for the related documents.

Faults								Cause	Remedy
a	b	c	d	e	f	g	h		
Pump not inducing	↑							Incorrect direction of rotation	1 Check direction of rotation as shown by arrow on pump; reverse polarity of motor if direction of rotation is incorrect
Pump delivery rate too low	↑			↑			↑	Pump not filled Fluid seal inadequate	2 Fill and bleed pump and, if necessary, suction line
Irregular pumping	↑			↑				- Suction line leaking - Shaft seal leaking - Screw plugs on pump leaking - Pressure cap on pressure relief valve missing or leaking	3 - Seal in suction line again - Renew shaft seal - Tighten screw plugs or seal them in again (possibly eyebolt) - Screw on and tighten pressure cap with seal
Pump overheating	↑			↑				- Suction head excessive / inflow inadequate; Suction filter clogged or too small - Internal parts worn	4 - Check suction head / resistance and, if necessary, increase the line cross section or shorten the length of the line - if necessary, clean suction filter or fit a larger suction filter - Renew worn parts
Pump pressure level too high								No bleed facility in the delivery line Inducting against non-return valve	5 Install small bypass line (DN 4) with return line into the oil tank upstream of the nonreturn line. Alternatively: attach bleed cock
Valve cone chattering	↑							- Partial flow via pressure relief valve, valve cone jamming - Suction and delivery line clogged	6 - Remove valve cone and remachine it or renew it - Clear clogging or deposits
Motor overheating				↑				- Valve spring operating in resonance range - Valve cone beating against seat	7 - Fit new valve spring - Adjust the opening pressure
Pump seizes up								Pressure relief valve opening too early	8 Set pressure relief valve to a higher value (10% above operating pressure) Renew valve spring if it is slack
				↑				- Gearwheels running unstably - Construction/mounting not noise-optimised	9 - Load pump at 1-2 bar in the case of unpressurised pumping of thin-bodied media - Isolate structure-borne noise by vibration dampers, provide acoustic hoods
				↑				Air inclusions in circulation system Inadequate bleeding Pumped medium frothing	10 Return lines must enter the tank below the oil level, lay suction and return connections further apart in the tank, increase bleed area / oil level area
	↑	↑						Volumetric efficiency of pump too low / viscosity too high	11 Reduce the speed of rotation when pumping highly viscous media

Faults							Cause	Remedy
Pump not inducing	a						- Delivery pressure too high - Speed of rotation too high - Defective bearing - Plain bearings overloaded owing to inadequate lubrication ¹⁾ - Inadequate viscosity of pumped medium - Unit poorly aligned - Stuffing box packing overtightened - Power consumption too high	- Reduce pressure / increase size of delivery line / reduce resistances - Reduce speed of rotation - Inspect bearing assembly and repair it - Improve bearing lubrication ¹⁾ - Reduce temperature - Align unit - Slacken stuffing box gland nut
Pump delivery rate too low	b	↑					- Running on 2 phases - Inadequate cooling - Excessive viscosity of pump medium - Motor bearing assembly defective	- Compare voltage and frequency with motor rating plate, install higher-power motor, reduce speed of rotation - Repair motor or renew defective fuse - Improve cooling air supply - Heat medium to be pumped / install additional heater - Repair bearing assembly
Irregular pumping	c						Foreign bodies in pump	Dismantle pump and remove any foreign bodies, smooth damaged points with oilstone / install a filter
Pump overheating	d	↑					Seized bearing assembly	Check whether the material has lost its lubricating properties owing to excessively high temperature Dismantle pump and repair it; return pump to works if necessary
Sound pressure level too high	e							
Valve cone chattering	f							
Motor overheating	g							
Pump seizes up	h							

1) Please enquire with us

Information on safety and commissioning for low voltage asynchronous motors

(in accordance with L. V. directive 73/23/EEC)

Types: 5AP, 6AP,C,F,16BA,7BA,7AA

Shaft height: 56 to 315 mm

1 General information

Electric motors have dangerous voltage-carrying and rotating components as well as surfaces that may become hot. All work involved in the transport, connection, commissioning and regular maintenance must be carried out by **qualified, responsible specialists** (note VDE 0105; IEC 364). Improper behaviour may result in serious **injury** and **damage to property**. The applicable **national, local and works regulations** and **requirements** must be complied with.

2 Intended use

These motors are intended for commercial installations. They comply with the harmonized standards of the **EN60034 (VDE 0530)** series. Utilization in areas subject to **explosion hazard is not permitted**, unless **expressly intended** for this purpose (see additional notes). In certain special cases, for example, on use in non-commercial installations, if requirements are more strict (e.g. protection against contact with children's fingers), it is the responsibility of the customer to ensure compliance on installing the equipment. The motors are rated for ambient temperatures of **-20°C to +40°C** and site altitudes **≤ 1000 m** above sea level. Any contradictory information on the rating plate **must** be observed. The conditions on site **must** correspond to all rating plate specifications.

Low voltage motors are **components** for installation in machinery in terms of the Machine directive 89/392/EEC. **Commissioning** must not take place until it has been proved that the end product conforms with this guideline (please note EN 60204-1).

3 Transport and storage

Any **damage** detected after dispatch should be reported immediately to the transport company and **commissioning** must be postponed. Tighten the eyebolts. They are designed for the weight of the motor only therefore do not attach any additional loads. If necessary, use suitable, adequately dimensioned transporting equipment (e.g. rope guides).

Remove existing **shipping braces** before commissioning; and reuse for subsequent transport. If motors are stored, a **dry, dust-free** and **low vibration** ($v_{rms} \leq 0.2 \text{ mm/s}$) environment is important (to avoid bearing standstill damage). On long-term storage, the regrease interval of the bearings is reduced.

Before commissioning, measure the impedance of the insulation. If values $\leq 1 \text{ k}\Omega$ per volt of rated voltage are measured, the windings must be dried out.

When motors with roller bearings for increased cantilever force are operated the value of cantilever force must be minimal 30% of permissible cantilever force (see catalogue). Operating with smaller cantilever force is the cause of bearing faults.

4 Installation

Ensure an even underlying surface, good foot or flange fixing and precise alignment for direct coupling. It is important to ensure that the mounting conditions do not cause resonance with the rotational frequency and the doubled supply frequency. Turn the rotor **by hand** and listen for any unusual grinding noises. **Check** the **direction of rotation** in the decoupled state (note section 5).

Only mount or remove drive components (belt pulley, coupling, etc.) using suitable tools (heat up), and cover to shield against contact. Avoid unpermissible belt tensions (see catalogue and technical data). The **balancing type** is specified on the shaft end face or rating plate (**H** = half- and **F** = full-key balancing). On mounting the drive, note the balancing type! In the case of half key balancing, the **protruding, visible part** of the half-featherkey must be removed.

A canopy is recommended for designs with the shaft end pointing downwards, and with the shaft end pointing upwards a cover must be **provided by the customer** to prevent foreign bodies from falling into the fan.

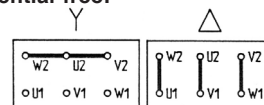
Do not obstruct ventilation! Discharged air, also from neighbouring equipment, must not be sucked in again immediately.

Checking of bearing grease must be carry out when motors are longer than 12 months stored. When storage conditions are the cause of grease depreciation (presence of condensate, consistency change) the grease must be exchanged. Grease exchangr must be carry out no later than in three year interval.

5 Electrical connection

Work is only permitted to be carried out by **qualified specialists** on the **stationary motor**, while **disconnected** and **prevented from being switched on again**. This also applies for the auxiliary power circuits (e.g. Anti-condensation heaters).

Check that the equipment is potential-free!



If the **tolerance limits** are exceeded that are specified in **EN 60034, part 1 / IEC 34-1** (voltage $\pm 5\%$, frequency $\pm 2\%$, shape of curve, symmetry) the heating effect is increased and the electromagnetic compatibility is affected. Please note the specifications on the rating plate and the connection diagram in the terminal box.

Connections must be made in such a way as to ensure that a **permanently** safe electrical connection is maintained (no protruding wire ends); use the corresponding cable end pieces. Create a **safe earth continuity connection**.

Tightening torques for terminal board connections

Thread Ø	M4	M5	M6	M8	M10	M12	M16
	0,8..1,2	1,8..2,5	2,7..4	5,5..8	9..13	16..20	36..40

Clearances in air between bare live parts themselves and between bare live parts and earth must be $\geq 5,5 \text{ mm}$ ($U_{rated} \leq 690 \text{ V}$).

It must be ensured that the terminal box does not contain **foreign** bodies, dirt or humidity. Seal any unused cable entry openings against **dust** and **water**.

Secure the featherkey on test operation without drive components. For motors with brakes, check that the brakes are operating perfectly before commissioning.

6 Operation

Vibration levels of $v_{rms} \leq 3.5 \text{ mm/s}$ ($P_N \leq 15 \text{ kW}$) or $v_{rms} \leq 4.5 \text{ mm/s}$ ($P_N > 15 \text{ kW}$) are quite acceptable in the coupled state.

If deviations from normal operation occur - e.g. **increased temperatures, noises, vibration** - the motor should be **switched off** in the event of doubt. Determine the causes and contact the manufacturer if necessary. Do not disconnect protective equipment, even under test operation.

Under dirty operating conditions, clean the air channels regularly. Open any closed **condensate water** holes from time to time!

For motors **without regreasing facilities**, bearing or grease replacement must be carried out in accordance with the manufacturer's instructions, or after 3 years, whichever is sooner.

Bearings **with regreasing facilities** must be regreased when the motor is running.

In the case of **motors with separate ventilation**, the separately-driven fan must be switched on throughout motor operation.

7 Further information

Information provided about any additional equipment must be noted!

These notes on safety and commissioning must be retained for future reference!

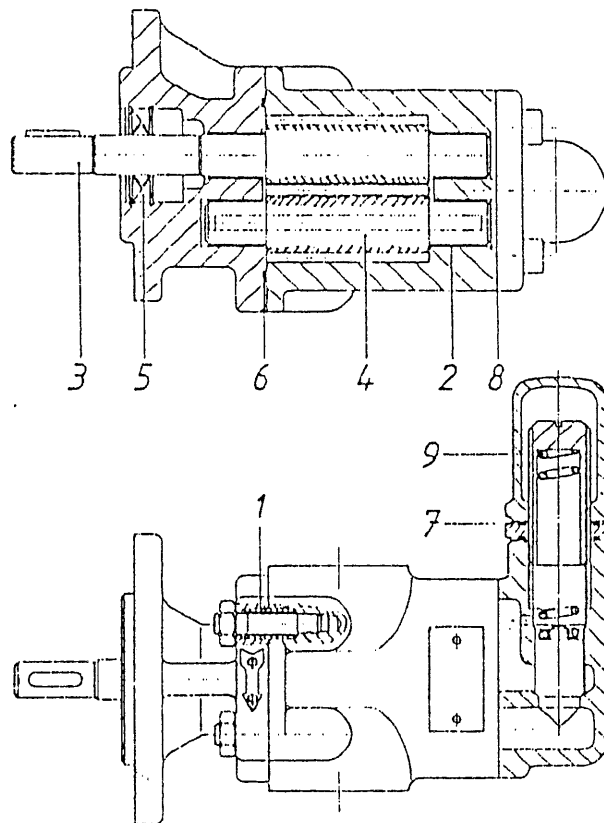
Gebr. Steimel GmbH & Co.
Maschinenfabrik
Telefon: 0 22 42 / 88 09-0
Telefax: 0 22 42 / 88 09+60
Telex : 889 503
Postfach: 15 65
5202 Hennef 1

Zahnradpumpe

Pompa ad ingranaggi SF 2/8 RD

Bomba de ruedas dentadas

Steimel



Bei Ersatzteilbestellung
bitte Fabrikations-Nr.
angeben!

In caso di ordinazione di
pezzi di ricambio indicare
per favore il n° di fabbri-
cazione!

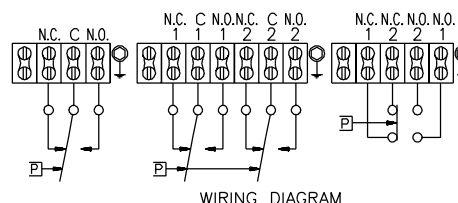
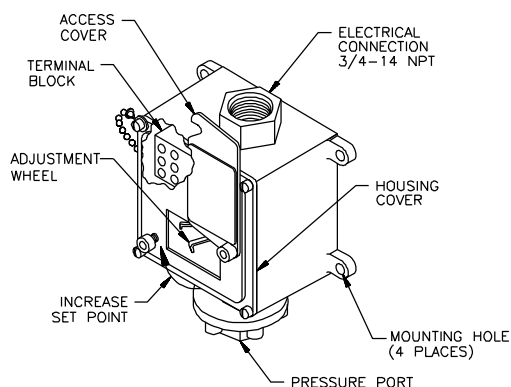
Al pedir piezas de repuesto
indicar el No.de fabricación!

V=FKM
P=NR

1	Ventilfeder molla per valvola muelle de válvula	1- 8 bar 9-18 bar 19-25 bar	EVF92213x076FA01 EVF92212x076FA01 EVF92212x076FA02	9	0,020 0,025 0,030
1	Dichtung guarnizione empaquetadura		EDI042---100IA02	8	0,005
2	O-Ring anello torico anillo redondo	IDP020x2V IDP028x2P	7	0,001	
1	O-Ring anello torico anillo redondo	IDP060x2V IDP060x2P	6	0,001	
1	Radialwellendichtring guarnizione radiale per alberi empaquetadura de eje radial	IDI015x035x06V IDI015x035x06P	5	0,009	
1	Gegenritzel contropignone piñón de contra	EGR052008025VB50	4	0,308	
1	Antriebsritzel pignone conduttore piñón de accionamiento	EAR052008025VB50	3	0,430	
4	Lagerbuchse guscio di cuscinetto buje de rodamiento	ELD---161820M001	2	0,008	
4	Zentrierbuchse boccola di contratura buje de centrado	EZD052---011BC01	1	0,004	
Stück pezzo pieza	Artikelbezeichnung denominazione dell'articolo designación del artículo	Artikel-Nr. articolo n° No. del artículo	Position posizione posición	Gewicht kg/Stück. peso kg/pezzo peso kg/pieza	Preis DM/Stück. prezzoDM/pezzo precioDM/pieza
Ausgabe : Edizione : Edición :	9.89	Ersatzteile / pezzi di ricambio / piezas de repuesto		DET 052 008 RD-- ¹ / ₀ 50	

CUSTOM CONTROL SENSORS, INC.

604 INSTALLATION INSTRUCTION



General:

- Observe all Warnings and Cautions contained on the attached Warning Sticker, the Warning Declaration and the CCS Catalog.
- Follow all applicable Governmental or other Codes or Regulations such as, but not restricted to NEC, ASME, CSA and OSHA.

Mounting:

- Units may be mounted in any position. On a flat surface using (4) mounting holes provided and 1/4" screws at least 1/2" long.

Warning: Release all pressure prior to installation or removal of this device. Do not exceed nameplate listed proof pressures.

- Units are supplied with female NPT ports suitable for connection to NPT pipe nipples or NPT taper threaded pipe. CCS recommends use of either a union or bendable tubing. This will eliminate excessive strain on the housing.
- Pipe joint sealing compound is preferred over Teflon tape. However, either method of sealing is acceptable with proper care to avoid introducing sealing material into pressure switch or plumbing.

Electrical Connections:

WARNING: Do not make electrical connections while electrical power is ON.

- Remove housing cover to allow access to the terminal block per illustration.

- The electrical connection is provided with a 3/4" NPT female conduit connection. Flexible conduit is recommended. Connect conduit to housing.

- Make electrical connections in accordance with the NEC and all other appropriate codes and regulations, see wiring diagrams.

Note: Circuitry coding is provided on instruction label inside housing.

- When making wire connections disconnect electrical power. Check for multiple circuits.

- Grounding: Proper grounding must be accomplished by use of green terminal screw provided.

Adjustment: (Not applicable if units are factory set) Pressure settings are externally adjustable.

- If the electrical circuit cannot be used for testing disconnect electrical power and use a suitable circuit tester or light box.

- Loosen access cover. Adjust wheel as illustrated.

- Clockwise to increase settings.
- Counterclockwise to decrease settings.

- Re-tighten access cover.

- Reconnect electrical circuit.

- Return power to unit.

WARNING DECLARATION

If this switch is used to protect equipment or personnel from unsafe pressure or to guard against the hazardous release of contained substances it must be installed and operated in accordance with applicable codes, regulations and standards. This switch must be used in conjunction with system design(s) or procedure(s) necessary to mitigate any hazard resulting from its failure. Conform to installation instructions accompanying this switch. Individuals who ignore this warning may suffer serious or fatal injury and do so at their own risk. Custom Control Sensors is not liable for any misuse, abuse, suitability or adequacy of user's application of the switch. Service by qualified personnel only.

**FLENDER
BOCHOLT****N-EUPEX**
Ersatzteilliste / Maßzeichnung
Spare parts list / Dimension drawingBauart
Type **BDS**
Größe
Size **245**Ausf.
Assembly

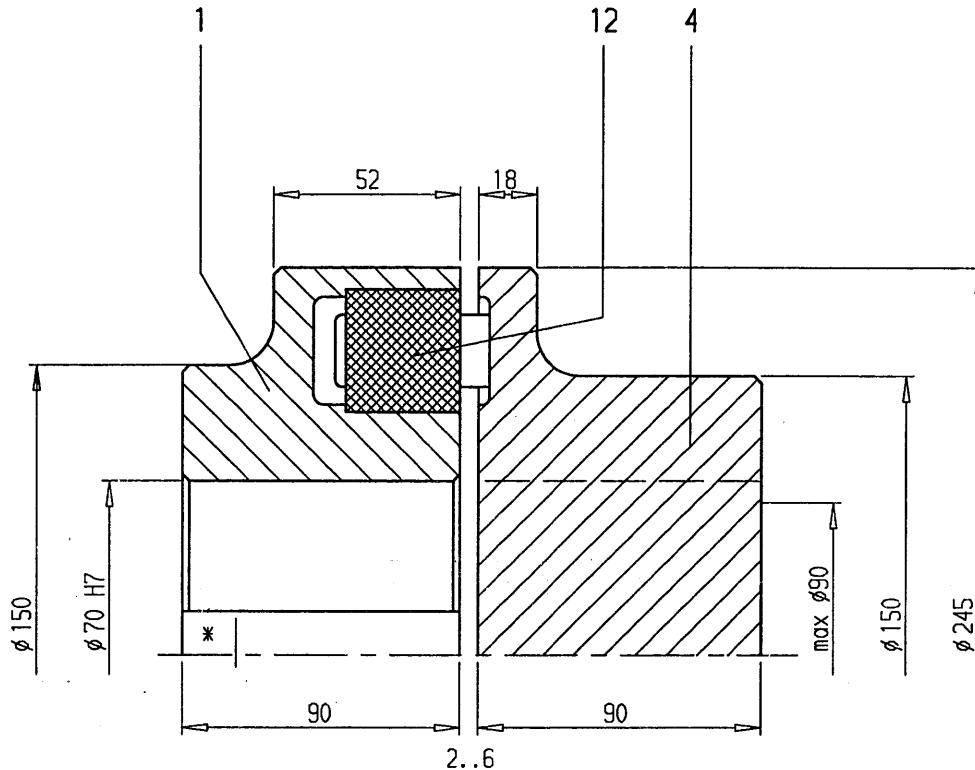
Bei Korrespondenz bitte angeben / Please quote in correspondence

NO. 4123221-20

Ident: NARAYANGONJ

BNR. 45017726

Code: MC1M10.020.50.09.01



- * Nut nach DIN 6885/1
- * Keyway acc. to DIN 6885/1

8	PAKET FLEXIBLES	245 DS-2K	12	8 N 6014825	1223188
1	TEIL PART	4 245	4	7 N 5242860	501071
1	TEIL PART	1 245	1	7 N 5242294	502240
Menge No. off	Benennung Description	Kenndaten Data	Teil-Nr. Part No.	Zeichnungs-Nr. Drawing No.	Waren-Nr. Ident No.

UmLaufende Teile müssen vom Käufer gegen unbeabsichtigtes Berühren gesichert werden! (Gesetz über techn. Arbeitsmittel)
The purchaser is responsible for the provision of safety guards.

A. Friedr. Flender AG, D46393 Bocholt,
Tel. 02871/92-0, Tx 0813841, Telefax 02871/922596

Datum/ Date
2004-06-03

Name van Gelder KUE
Z-Nr. 4123221-20

Diese technische Unterlage hat gesetzlichen Schutz (DIN 34)

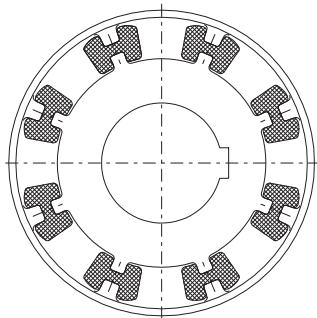
/B0

KUE5 2003-10-21
E-LISTE BDS 245 EN

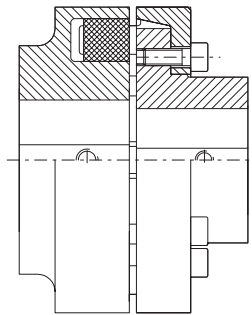
Operating Instructions

BA 3100 EN 07.03

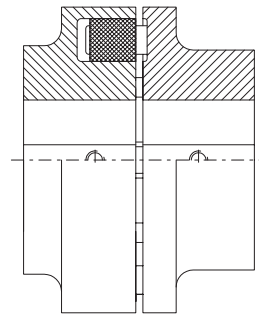
Flexible **N-EUPEX** and **N-EUPEX-DS** couplings
Types **A**, **B** and **ADS**, **BDS**



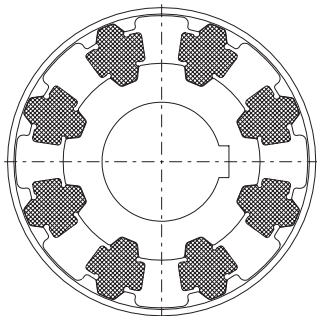
N-EUPEX



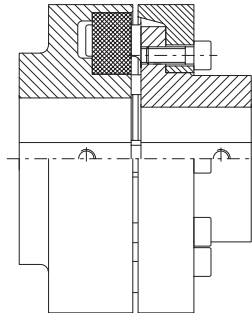
A



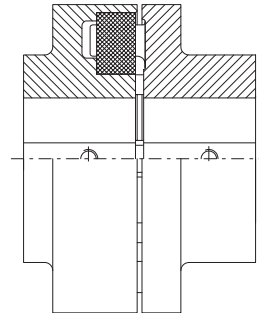
B



N-EUPEX-DS



ADS



BDS

FLENDER

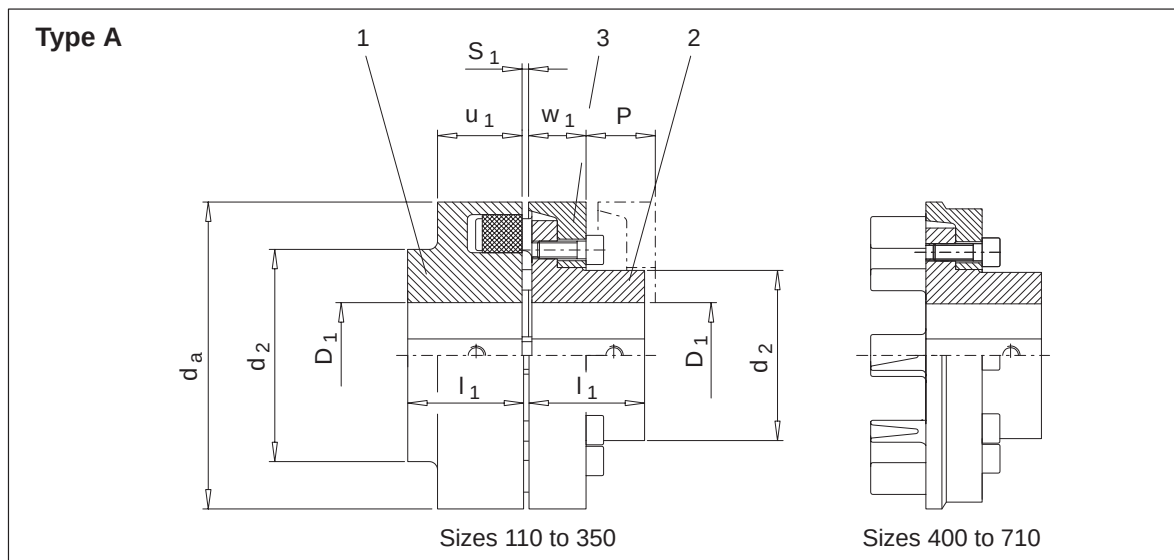
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1. Technical data

1.1 N-EUPEX coupling, Types A and B

1.1.1 Geometric data



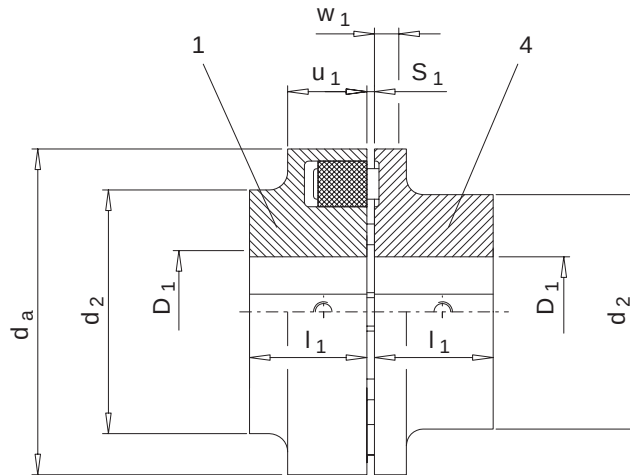
Size	Bore D ₁				d _a mm	d ₂		w ₁ mm	l ₁ mm	u ₁ mm	P mm	S ₁ mm	Weight		Mass moment of inertia	
	Part 1		Part 2			Part 1)							Part 1)			
	from mm	to mm	from mm	to mm		1 mm	2 mm						1 kg	2+3 kg	1 kgm ²	2+3 kgm ²
110		48		38	110	86	62	20	40	34	33	2.... 4	1.9	1.6	0.0027	0.002
125		55		45	125	100	75	23	50	36	38	2.... 4	2.9	2.7	0.005	0.0045
140		60		50	140	100	82	28	55	34	43	2.... 4	3.3	3.7	0.007	0.008
160		65		58	160	108	95	28	60	39	47	2.... 6	4.7	5.1	0.013	0.015
180		75		65	180	125	108	30	70	42	50	2.... 6	6.9	7.3	0.023	0.026
200		85		75	200	140	122	32	80	47	53	2.... 6	9.5	10.3	0.04	0.045
225		90		85	225	150	138	38	90	52	61	2.... 6	13	14	0.07	0.08
250	46	100	32	95	250	165	155	42	100	60	69	3.... 8	17.5	19.5	0.12	0.13
280	49	110	54	105	280	180	172	42	110	65	73	3.... 8	24	24	0.2	0.2
315	49 90	100 120	46 90	100 120	315	165 200	165 200	47	125	70	78	3.... 8	31 32	32 34	0.31 0.34	0.33 0.37
350	61 90	110 140	61 90	110 140	350	180 230	180 230	51	140	74	83	3.... 8	43 45	43 47	0.54 0.60	0.54 0.63
400	66 100	120 150	66 100	120 150	400	200 250	200 250	56	160	78	88	3.... 8	63 66	59 64	1 1.2	0.9 1
440	80 120	130 160	80 120	130 160	440	215 265	215 265	64	180	86	99	5...10	79 82	80 85	1.5 1.7	1.5 1.7
480	90 136	145 180	90 136	145 180	480	240 300	240 300	65	190	90	104	5...10	100 105	100 110	2.3 2.6	2.3 2.6
520	100 140	150 190	100 140	150 190	520	250 315	250 315	68	210	102	115	5...10	130 140	120 135	3.5 3.8	3.2 3.6
560	120	200	120	200	560	320	320	80	220	115	125	6...12	180	185	5.9	6
610	130	220	130	220	610	352	352	88	240	121	135	6...12	225	240	8.6	9.3
660	140	240	140	240	660	384	384	96	260	132	145	6...12	290	320	13	14
710	140	260	140	260	710	416	416	102	290	138	155	6...12	370	400	18.5	20

Table 1.1.1 a : Dimensions, weights and mass moments of inertia of Type A

1) Weights and mass moments of inertia apply to mean bores

FLENDER

Type B



Size	Bore D ₁				d _a	d ₂		w ₁	l ₁	u ₁	S ₁	Weight 1)		Mass moment of inertia 1)	
	Part 1		Part 4			1	4					Part		Part	
	from mm	to mm	from mm	to mm								1 mm	4 mm	1 kg	4 kg
58		19		24	58	–	40	8	20	20	2...4	0.22	0.23	0.0001	0.0001
68		24		28	68	–	50	8	20	20	2...4	0.31	0.32	0.0002	0.0001
80		30		38	80	–	68	10	30	30	2...4	0.79	0.72	0.0006	0.0006
95		42		42	95	76	76	12	35	30	2...4	1.2	1.4	0.0013	0.0014
110		48		48	110	86	86	14	40	34	2...4	1.9	2.0	0.0027	0.0028
125		55		55	125	100	100	18	50	36	2...4	2.9	3.3	0.005	0.0057
140		60		60	140	100	100	20	55	34	2...4	3.3	3.6	0.007	0.007
160		65		65	160	108	108	20	60	39	2...6	4.7	4.7	0.013	0.012
180		75		75	180	125	125	20	70	42	2...6	6.9	7.1	0.023	0.022
200		85		85	200	140	140	24	80	47	2...6	9.5	10.5	0.04	0.04
225		90		90	225	150	150	18	90	52	2...6	11.5	13	0.07	0.065
250	46	100	46	100	250	165	165	18	100	60	3...8	17.5	16.5	0.12	0.11
280	49	110	54	110	280	180	180	20	110	65	3...8	24	21	0.2	0.17

Table 1.1.1 b : Dimensions, weights and mass moments of inertia of Type B

1) Weights and mass moments of inertia apply to mean bores

1.1.2 Performance data

Note: For identification marking of the individual flexible elements, refer to section 5.

Flexible elements: 80 Shore A									
Size	Rated torque	Maximum torque	Fatigue torque	Speed	dynamic torsional stiffness				
	T_{KN}	T_{Kmax}	T_{KW}	n_{max}	$C_{T\ dyn}$				
	Nm	Nm	Nm	1/min	$1 \times T_{KN}$	$0.75 \times T_{KN}$	$0.5 \times T_{KN}$	$0.25 \times T_{KN}$	$0 \times T_{KN}$
58	19	57	7.6	5000	1200	850	600	430	300
68	34	102	13.6	5000	1300	930	670	480	350
80	60	180	24	5000	2750	1950	1400	980	700
95	100	300	40	5000	4200	3100	2300	1700	1280
110	160	480	64	5000	5700	4200	3100	2250	1670
125	240	720	96	5000	16000	10000	6200	3800	2400
140	360	1080	144	4900	24000	15000	9600	6200	4000
160	560	1680	224	4250	49000	34000	23000	16000	11000
180	880	2640	352	3800	78000	51000	33000	21500	14000
200	1340	4020	536	3400	127000	80000	51000	32000	20500
225	2000	6000	800	3000	210000	136000	87000	56000	36000
250	2800	8400	1120	2750	290000	176000	107000	65000	40000
280	3900	11700	1560	2450	365000	233000	149000	94000	60000
315	5500	16500	2200	2150	840000	540000	340000	215000	138000
350	7700	23100	3080	1950	920000	590000	380000	245000	160000
400	10300	30900	4120	1700	1350000	840000	530000	335000	210000
440	13500	40500	5400	1550	1830000	1180000	760000	490000	315000
480	16600	49800	6640	1400	2000000	1300000	830000	530000	340000
520	21200	63600	8480	1300	2700000	1770000	1150000	740000	480000
560	29000	87000	11600	1200	3600000	2300000	1500000	960000	620000
610	38000	114000	15200	1100	5000000	3200000	2070000	1330000	850000
660	49000	147000	19600	1000	6800000	4350000	2800000	1800000	1150000
710	62000	186000	24800	950	9300000	6000000	3900000	2500000	1600000

damping coefficient $\Psi = 1.1$

Flexible elements: 60 Shore A									
Size	Rated torque	Maximum torque	Fatigue torque	Speed	dynamic torsional stiffness				
	T_{KN} Nm	T_{Kmax} Nm	T_{KW} Nm	n_{max} 1/min	$1 \times T_{KN}$ Nm/rad	$0.75 \times T_{KN}$ Nm/rad	$0.5 \times T_{KN}$ Nm/rad	$0.25 \times T_{KN}$ Nm/rad	$0 \times T_{KN}$ Nm/rad
58	11	35	4	5000	360	290	235	190	150
68	21	64	8	5000	400	325	260	210	175
80	37	113	15	5000	830	670	540	430	350
95	63	190	25	5000	1340	1110	920	760	640
110	100	300	40	5000	1800	1500	1200	1000	830
125	150	450	60	5000	4000	3000	2150	1600	1200
140	230	680	90	4900	6000	4600	3500	2600	2000
160	350	1060	140	4250	14000	11000	8800	7000	5500
180	550	1660	220	3800	20700	15700	12000	9200	7000
200	850	2530	337	3400	32200	24300	18000	13400	10200
225	1260	3780	504	3000	55000	41400	31500	24000	18000
250	1760	5300	705	2750	69000	50600	37000	27000	20000
280	2460	7400	980	2450	94000	71000	53000	39500	30000
315	3500	10500	1400	2150	216500	161500	121000	91000	69000
350	4850	14500	1940	1950	239000	181000	137000	104000	80000
400	6500	19500	2600	1700	336000	252000	189000	141000	105000
440	8500	25500	3400	1550	478000	362000	275000	208000	158000
480	10500	31400	4200	1400	525000	395000	298000	225000	170000
520	13300	40000	5300	1300	720000	548000	415000	314000	240000
560	18300	54800	7300	1200	936000	715000	541000	404000	310000
610	24000	71800	9600	1100	1297000	987000	747000	562000	425000
660	30900	92600	12350	1000	1759000	1334000	1010000	763000	575000
710	39000	117000	15600	950	2440000	1860000	1400000	1050000	800000

damping coefficient $\Psi = 1.1$

The performance data for Types A and B are valid for:

- max. 25 starts per hour
- daily operating cycle of up to 24 h
- operation within the specified alignment
- Operation in the temperature range -30 °C to +80 °C in the immediate vicinity of the coupling

Caution!

For sustained faultfree operation the coupling must be designed with a service factor f_1 in accordance with item 1.3 and appropriate to the application. In the event of a change in operating conditions (e.g. output, speed, starting frequency, changes to the prime mover and driven machine) the design must always be checked (see item 1.1.3).

1.1.3 Checking the selected coupling size

The following must apply to the coupling:

$$T_{KN} \geq T_N \times f_1$$

T_{KN}	= rated coupling torque
T_N	= rated system torque - rated drive torque acting on the coupling
f_1	= service factor in accordance with item 1.3

During starting or operation torque impulses up to 25 times per hour are permissible. The following applies:

$$T_{Kmax} \geq T_{max}$$

T_{Kmax}	= maximum coupling torque
T_{max}	= maximum system torque - peak drive torque acting on the coupling

The following must apply to the alternating torques occurring during operation:

$$T_{KW} \geq T_W \times S_f \times f_1$$

T_{KW}	= fatigue torque load on the coupling
T_W	= alternating torque load on the coupling
f_1	= service factor in accordance with item 1.3

$$S_f = \sqrt{\frac{f_{Err}}{10\text{Hz}}} \quad \text{for } f_{Err} > 10 \text{ Hz}$$
$$S_f = 1.0 \quad \text{for } f_{Err} \leq 10 \text{ Hz}$$
$$f_{Err} = \text{excitation frequency of the alternating torque load in Hz}$$

Caution!

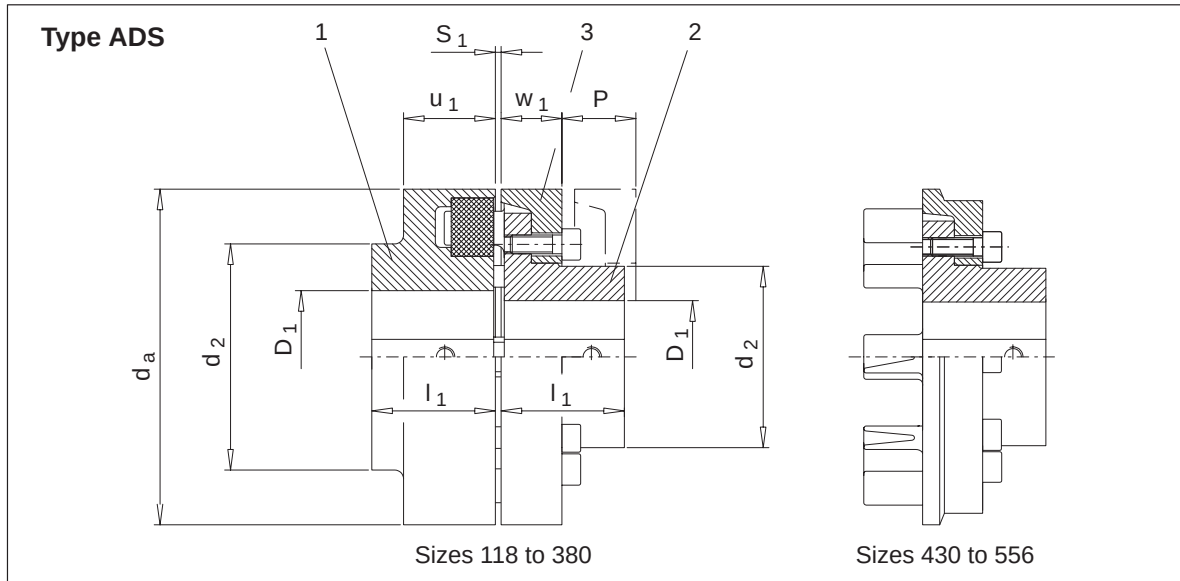
When selecting the coupling, the permissible maximum speed and the permissible maximum bore must also be taken into consideration. Selection of bore fit in accordance with section 6. item 6.1.1.

Caution!

The shaft displacement values specified in section 6, item 6.5.4, must not be exceeded.

1.2 N-EUPEX-DS coupling, Types ADS and BDS

1.2.1 Geometric data

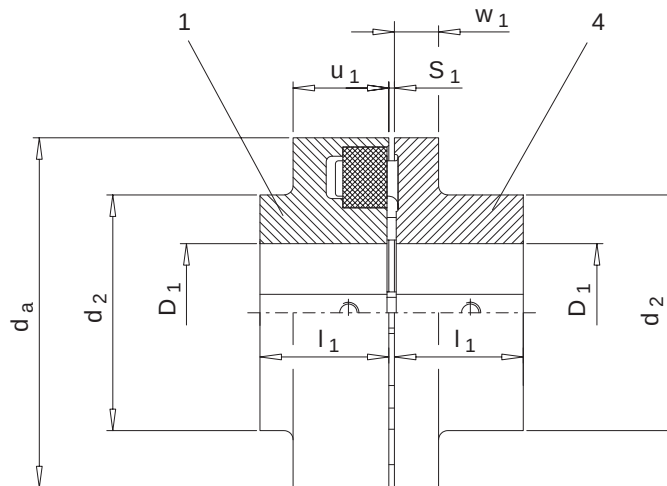


Size	Bore D ₁				d _a	d ₂		w ₁	l ₁	u ₁	P	S ₁	Weight		Mass moment of inertia	
	Part 1		Part 2			Part 1	Part 2						1) Part		1) Part	
	from mm	to mm	from mm	to mm									1 kg	2+3 kg	1 kgm ²	2+3 kgm ²
118		48		38	118	86	62	20	40	34	33	2.... 4	1.9	1.94	0.003	0.003
135		55		45	135	100	75	23	50	36	38	2.... 4	3.1	3.1	0.006	0.006
152		60		50	152	108	82	28	55	36	43	2.... 4	4.2	4.5	0.011	0.012
172		65		58	172	118	95	28	60	41	47	2.... 6	5.8	6	0.019	0.020
194		75		65	194	135	108	30	70	44	50	2.... 6	8.8	8.5	0.037	0.035
218		85		75	218	150	122	32	80	47	53	2.... 6	12	12	0.062	0.062
245		90		85	245	150	138	38	90	52	61	2.... 6	14.5	17.7	0.09	0.115
272	46	100	32	95	272	165	155	42	100	60	69	3.... 8	20	24.7	0.16	0.2
305	49	110	54	105	305	180	172	42	110	65	73	3.... 8	27	29.1	0.26	0.3
340	49	120	46 90	100 120	340	200	165 200	47	125	70	78	3.... 8	38	39.3 40.3	0.41 0.44	0.49 0.53
380	61	140	61 90	110 140	380	230	180 230	51	140	74	83	3.... 8	54	53.5 57.5	0.71 0.77	0.84 0.93
430	66	150	66 100	120 150	430	250	200 250	56	160	78	88	3.... 8	76	69 74	1.2 1.4	1.26 1.4
472	80	160	80 120	130 160	472	265	215 265	64	180	86	99	5...10	95	91 97	1.9 2.1	2 2.1
514	90	180	90 136	145 180	514	300	240 300	65	190	90	104	5...10	119	115 122	2.8 3.1	3 3.3
556	100	190	100 140	150 190	556	315	250 315	68	210	102	115	5...10	159	138 152	4.4 4.7	4.1 4.6

Table 1.2.1 a : Dimensions, weights and mass moments of inertia of Type ADS

1) Weights and mass moments of inertia apply to mean bores

Type BDS



Size	Bore D ₁				d _a	d ₂		w ₁	l ₁	u ₁	S ₁	Weight 1)		Mass moment of inertia 1)	
	Part 1		Part 4			1	4					Part		Part	
	from mm	to mm	from mm	to mm								1 kg	4 kg	1 kgm ²	4 kgm ²
66		19		24	66		40	8	20	20	2.... 4	0.24	0.31	0.0001	0.0002
76		24		28	76		50	8	20	20	2.... 4	0.33	0.42	0.0002	0.0003
88		30		38	88		68	10	30	30	2.... 4	1	0.92	0.0007	0.0006
103		42		42	103	76	76	12	35	30	2.... 4	1.6	1.5	0.0015	0.0014
118		48		48	118	86	86	14	40	34	2.... 4	1.9	2.1	0.003	0.0031
135		55		55	135	100	100	18	50	36	2.... 4	3.1	3.5	0.006	0.007
152		60		60	152	108	100	20	55	36	2.... 4	4.2	4.4	0.011	0.011
172		65		65	172	118	108	20	60	41	2.... 6	5.8	5.7	0.019	0.018
194		75		75	194	135	125	20	70	44	2.... 6	8.8	8.2	0.037	0.032
218		85		85	218	150	140	24	80	47	2.... 6	12	12.1	0.062	0.059
245		90		90	245	150	150	18	90	52	2.... 6	14.5	14.6	0.09	0.082
272	46	100	46	100	272	165	165	18	100	60	3.... 8	20	19.1	0.16	0.132
305	49	110	54	110	305	180	180	20	110	65	3.... 8	27	24.3	0.26	0.208

Table 1.2.1 b : Dimensions, weights and mass moments of inertia of Type BDS

1) Weights and mass moments of inertia apply to mean bores

1.2.2 Performance data

Size	Rated torque T_{KN} Nm	Maximum torque T_{Kmax} Nm	Fatigue torque T_{KW} Nm	Speed n_{max} 1/min	dynamic torsional stiffness $C_{T dyn} \quad 1)$ Nm/rad
66	19	57	7.6	5000	–
76	34	102	13.6	5000	–
88	60	180	24	5000	5600
103	100	300	40	5000	9350
118	160	480	64	5000	15000
135	240	720	96	5000	22450
152	360	1080	144	4900	33650
172	560	1680	224	4250	52350
194	880	2640	352	3800	82250
218	1340	4020	536	3400	125250
245	2000	6000	800	3000	187000
272	2800	8400	1120	2750	114000
305	3900	11700	1560	2450	165000
340	5500	16500	2200	2150	239000
380	7700	23100	3080	1950	340000
430	10300	30900	4120	1700	460000
472	13500	40500	5400	1550	607000
514	16600	49800	6640	1400	750000
556	21200	63600	8480	1300	961000

damping coefficient $\Psi = 1.1$

1) The dynamic torsional stiffness applies at an ambient temperature of -30 °C to +40 °C

The performance data for the Types ADS and BDS are valid for:

- max. 25 starts per hour
- daily operating cycle of up to 24 h
- operation within the specified alignment
- Operation in the temperature range -30 °C to +80 °C in the immediate vicinity of the coupling

Caution!

For sustained faultfree operation the coupling must be designed with a service factor f_1 in accordance with item 1.3 and a temperature factor S_θ appropriate to the application. In the event of a change in operating conditions (e.g. output, speed, starting frequency, changes to the prime mover and driven machine) the design must always be checked (see item 1.2.3).

1.2.3 Checking the selected coupling size

The following must apply to the coupling:

$$T_{KN} \geq T_N \times f_1 \times S_{\vartheta}$$

T_{KN} = rated coupling torque
 T_N = rated system torque -
 rated drive torque acting on the coupling
 f_1 = service factor in accordance with item 1.3
 S_{ϑ} = temperature factor

The highest temperature in the immediate vicinity of the coupling must be applied

T_U	from -30 °C to +40 °C	from +40 °C to +60 °C	from +60 °C to +80 °C
S_{ϑ}	1	1.4	1.8

Table 1.2.3: Temperature factor S_{ϑ}

During starting or operation torque impulses up to 25 times per hour are permissible. The following applies:

$$T_{Kmax} \geq T_{max} \times S_{\vartheta}$$

T_{Kmax} = maximum coupling torque
 T_{max} = maximum system torque -
 peak drive torque acting on the coupling
 S_{ϑ} = temperature factor

The following must apply to the alternating torques occurring during operation:

$$T_{KW} \geq T_W \times S_f \times S_{\vartheta} \times f_1$$

T_{KW} = fatigue torque load on the coupling
 T_W = alternating torque load on the coupling
 S_{ϑ} = temperature factor
 f_1 = service factor in accordance with item 1.3

$$S_f = \sqrt{\frac{f_{Err}}{10\text{Hz}}} \quad \text{for } f_{Err} > 10 \text{ Hz}$$

$$S_f = 1.0 \quad \text{for } f_{Err} \leq 10 \text{ Hz}$$

f_{Err} = excitation frequency of the alternating torque load in Hz

Caution!

When selecting the coupling, the permissible maximum speed and the permissible maximum bore must also be taken into consideration. Selection of bore fit in accordance with section 6. item 6.1.1.

Caution!

The shaft displacement values specified in section 6, item 6.5.4, must not be exceeded.

1.3 Determining the service factor

The service factors taken as basis are based on empirical values which generally estimate the output of in- and output combinations in service.

Service factor f_1 (daily operating cycle of up to 24 h)			
Prime mover	Load characteristic of driven machine		
	G	M	S
Electric motors, Turbines, Hydraulic motors	1	1.25	1.75
Piston engines 4 - 6 cylinders Coefficient of cyclic variation up to 1 : 100 to 1 : 200	1.25	1.5	2
Piston engines 1 - 3 cylinders Coefficient of cyclic variation up to 1 : 100	1.5	2	2.5

Load characteristics of driven machines listed by area of application		
Dredgers S Bucket-chain conveyors S Travelling gear (caterpillar) M Travelling gear (rails) M Manoeuvring winches M Lift pumps S Bucket wheels S Cutter heads M Slewing gear Building machinery M Hoists M Concrete mixers M Road construction machinery Chemical industry M Cooling drums M Mixers G Agitators (light liquids) M Agitators (semi-liquid material) M Drying drums G Centrifuges (light) M Centrifuges (heavy) Mineral oil extraction M Pipeline pumps S Rotary drilling equipment Conveyor systems M Hauling winches S Hoists M Link conveyors M Belt conveyors (bulk material) S Belt conveyors (piece goods) M Band pocket conveyors M Endless chain transporters M Rotary conveyors M Goods lifts G Bucket-type flour conveyors M Passenger lifts M Apron conveyors M Screw conveyors M Ballast elevators S Inclined hoists M Steel belt conveyors M Trough chain conveyors Blowers, Ventilators G Rotary piston blowers $T_N \leq 75 \text{ Nm}$ M Rotary piston blowers $T_N \leq 750 \text{ Nm}$ S Rotary piston blowers $T_N > 750 \text{ Nm}$ G Blowers (axial/radial) $T_N \leq 75 \text{ Nm}$ M Blowers (axial/radial) $T_N \leq 750 \text{ Nm}$ S Blowers (axial/radial) $T_N > 750 \text{ Nm}$ G Cooling tower fans $T_N \leq 75 \text{ Nm}$ M Cooling tower fans $T_N \leq 750 \text{ Nm}$ S Cooling tower fans $T_N > 750 \text{ Nm}$ G Induced draught fans $T_N \leq 75 \text{ Nm}$ M Induced draught fans $T_N \leq 750 \text{ Nm}$ S Induced draught fans $T_N > 750 \text{ Nm}$ G Turbo blowers $T_N \leq 75 \text{ Nm}$ M Turbo blowers $T_N \leq 750 \text{ Nm}$ S Turbo blowers $T_N > 750 \text{ Nm}$	Generators, transformers S Frequency transformers S Generators S Welding generators Rubber processing machines S Extruders M Calenders S Pug mills M Mixers S Rolling mills Wood working machines S Barkers M Planing machines G Wood working machines S Saw frames Cranes G Luffing gear S Travelling gear S Hoisting gear M Slewing gear M Derricking jib gear Plastics processing machines M Extruders M Calenders M Mixers M Crushers Metal working machines M Sheet bending machines S Sheet straightening machines S Hammers S Planing machines S Presses M Shears S Forging presses S Punch presses G Countershafts, shaft trains M Machine tools, main drives G Machine tools, auxiliary drives Food processing machines G Bottling and container filling machines M Kneading machines M Mash tubs, crystallizers G Packaging machines M Cane crushers M Cane knives S Cane mills M Sugar beet cutters M Sugar beet washing machines Paper processing machines S Couches S Glazing cylinders S Pulpers S Pulp grinders S Calenders S Wet presses S Willows S Suction presses	S Suction rolls S Drying cylinders Pumps S Piston pumps G Centrifugal pumps (light liquids) M Centrifugal pumps (heavy liquids) S Plunger pumps S Pressure pumps Stone and clay working machines S Crushers S Rotary kilns S Hammer mills S Ball mills S Tube mills S Beater mills S Brick presses Textile machines M Batches M Printing and dyeing machines M Tanning vats M Willows M Looms Compressors S Piston compressors M Turbo compressors Rolling mills S Sheet shears M Sheet tilters S Ingot pushers S Blooming and slabbing mills S Ingot conveying systems M Wire drawing benches S Descaling machines S Thin sheet mills S Heavy sheet mills M Winding machines (strip and wire) S Cold rolling mills M Chain transfers S Billet shears M Cooling beds M Cross transfers M Roller tables (light) S Roller tables (heavy) M Roller straighteners S Tube welding machines M Trimming shears S Cropping shears S Continuous casting plant M Roller adjustment drives S Shifting devices Laundry machines M Tumble driers M Washing machines Water treatment M Rotary aerators G Screw pumps

G = uniform load

M = medium load

S = heavy load

2. General notes

2.1 Introduction

These Operating Instructions (BA) are an integral part of the coupling delivery and must be kept in its vicinity for reference at all times.

Caution!

All persons involved in the installation, operation, maintenance and repair of the coupling must have read and understood these Operating Instructions and must comply with them at all times. We accept no responsibility for damage or disruption caused by disregard of these Instructions.

The **"Coupling"** described in these operating instructions has been developed for stationary use in general engineering applications. The coupling serves to transmit power and torque between two shafts or flanges connected by this coupling.

The coupling is designed only for the application described in section 1. "Technical data". Other operating conditions must be contractually agreed.

The coupling described in these Instructions reflects the state of technical development at the time these Instructions went to print.

In the interest of technical progress we reserve the right to make changes to the individual assemblies and accessories which we regard as necessary to preserve their essential characteristics and improve their efficiency and safety.

2.2 Copyright

The copyright to these Operating Instructions is held by **FLENDER AG**.

These Operating Instructions must not be wholly or partly reproduced for competitive purposes, used in any unauthorised way or made available to third parties without our agreement.

Technical enquiries should be addressed to the following works

FLENDER AG
D 46393 Bocholt

Telefon: 02871/92-2868
Telefax: 02871/92-2579

or to one of our customer-service addresses. A list of our customer-service addresses is given in section 11. "Spare parts, customer-service addresses".

3. Safety notes

3.1 Proper use

- The coupling has been manufactured in accordance with the state of the art and is delivered in a condition for safe and reliable use. Any changes on the part of the user which may affect safety and reliability are prohibited. This applies equally to safety features designed to prevent accidental contact.
- The coupling must be used and operated strictly in accordance with the conditions laid down in the contract governing performance and supply.

3.2 Obligations of the user

- The operator must ensure that all persons involved in installation, operation, maintenance and repair have read and understood these Operating Instructions and comply with them at all times in order to:

- avoid injury or damage,
- ensure the safety and reliability of the coupling,

and

- avoid disruptions and environmental damage through incorrect use.
- During transport, assembly, installation, dismantling, operation and maintenance of the unit, the relevant safety and environmental regulations must be complied with at all times.
- The coupling must be operated, maintained or repaired only by authorised, duly trained and qualified personnel.
- All work must be carried out with great care and with due regard to safety.
- All work on the coupling must be carried out only when it is at a standstill.
The drive unit must be secured against being switched on accidentally (e.g. by locking the key switch or removing the fuses from the power supply). A notice should be attached to the ON switch stating clearly that work is in progress.
- The coupling must be fitted with suitable safeguards to prevent accidental contact. The operation of the coupling must not be impaired by the safeguard.
- The drive unit must be shut down as soon as changes to the coupling are detected during operation.
- If the coupling is intended for installation in plant or equipment, the manufacturer of such plant or equipment must ensure that the contents of the present Operating Instructions are incorporated in his own instructions.
- All spare parts must be obtained from FLENDER.

3.3 Warnings and symbols used in these Instructions



This symbol indicates safety measures which must be observed to avoid **personal injury**.

Caution!

This symbol indicates safety measures which must be observed to avoid **damaging the coupling**.

Note:

This symbol indicates general **operating instructions** which are of particular importance.

4. Handling and storage

4.1 Scope of supply

The products supplied are listed in the despatch papers. Check immediately on receipt to ensure that all the products listed have actually been delivered. Parts damaged during transport or missing parts must be reported in writing immediately.

The parts must be provided with explosion protection marking in accordance with section 5.

4.2 Handling

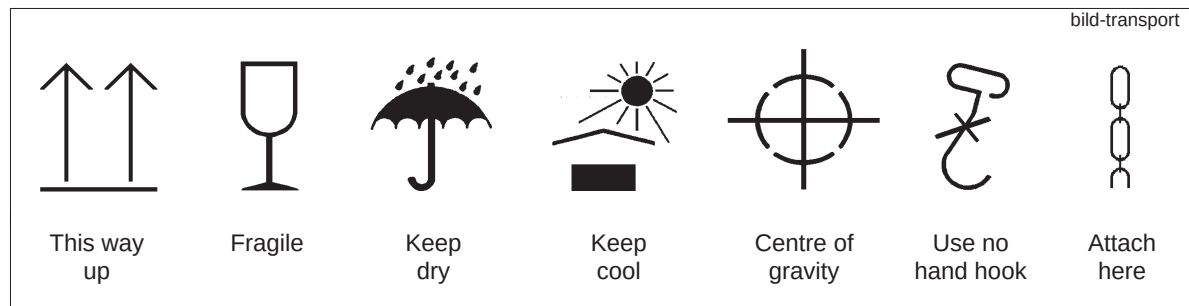


When handling FLENDER products, use only lifting and handling equipment of sufficient load-bearing capacity!

Note: The coupling must be transported using suitable transport equipment only.

Different forms of packaging may be used depending on the size of the coupling and method of transport. Unless otherwise agreed, the packaging complies with the **HPE Packaging Guidelines**.

The symbols marked on the packaging must be observed at all times. These have the following meanings:



4.3 Storage of the coupling

4.3.1 Storage of the coupling parts

Unless otherwise expressly agreed, the coupling is delivered in a preserved condition and can be stored in a covered, dry place for up to 3 months. If the coupling is to be stored for a protracted period, it should be treated with a long-term preservative agent (FLENDER must be consulted).

Caution!

Before cleaning the coupling parts and applying the long-term preservative agent, the flexible elements (12) must be removed.

4.3.2 Storing the flexible elements

4.3.2.1 General

Correctly stored flexible elements (12) retain their properties unchanged for up to five years. Unfavourable storage conditions and improper treatment will negatively affect the physical properties of the flexible elements (12). Such negative effects may be caused by e.g. the action of ozone, extreme temperatures, light, moisture, or solvents.

4.3.2.2 Storage area

The storage area must be dry and free from dust. The flexible elements (12) must not be stored with chemicals, solvents, motor fuels, acids, etc. Furthermore, they should be protected against light, in particular direct sunlight and bright artificial light with a high ultraviolet content.

Caution!

The storage areas must not contain any ozone-generating equipment, e.g. fluorescent light sources, mercury vapour lamps, high-voltage electrical equipment. Damp storage areas are unsuitable. Ensure that no condensation occurs. The most favourable atmospheric humidity is below 65 %.

5. Technical description

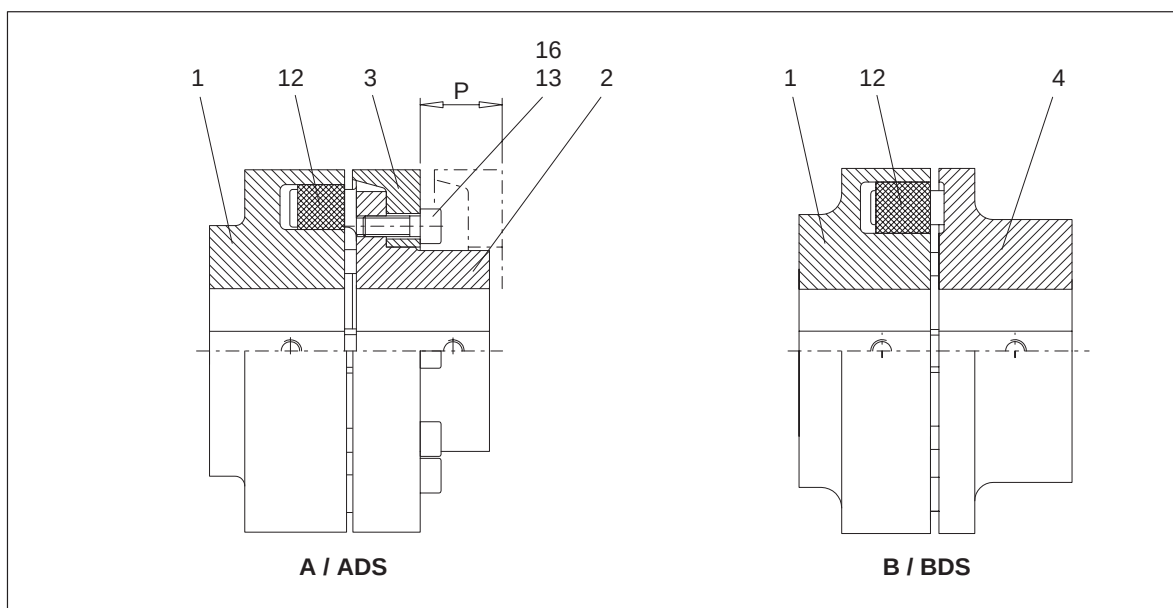
5.1 General description

N-EUPEX couplings are torsionally flexible claw couplings. They are suitable for linking machines and can compensate for small shaft misalignment caused by manufacturing inaccuracies, heat expansion, and the like.

The N-EUPEX coupling Type A / ADS comprises the coupling part 1 with the inserted flexible elements (12), the cam part 3 and the coupling part 2 which is bolted to part 3. If dimension P in section 1 is taken into consideration, the screw connection of part 2/3 enables the machines to be connected to be disconnected without shifting them axially.

On Type A size 560 to 710 Part 2 and part 3 are additionally fastened with 2 parallel pins (16).

The N-EUPEX coupling Type B / BDS comprises the coupling part 1 with the inserted flexible elements (12) and the cam part 4.



Because of the form-fitting design of the metal parts, types A and B enable an "emergency operation" even after the flexible elements have been irreparably damaged.

The flexible elements (12) are subjected primarily to pressure loads, so that the wear on the flexible elements (12) is relatively low with infrequent, substantial overload moments.



Where overload moments are excessively high, the result may be breakage of the coupling or irreparable damage to the connected machine.

On Types ADS and BDS there is no metal contact when the flexible elements (12) are irreparably damaged, and the metal parts are not of a form-fitting design. These coupling types have no "emergency running capability" in the above described sense. The flexible elements (12) are subjected to shear and pressure loads, so that with substantial overload the flexible elements (12) are irreparably damaged and the torque transmission interrupted.

5.2 Flexible elements

As well as in standard hardness 80 Shore A, the H-shaped flexible elements (12) of Types A and B are also available in the softer 60 Shore A version.

This enables to move critical speeds of the drive train out of normal operating conditions.

When using these flexible elements (12), the reduction of the transmissible torque must be noted (see section 1. "Technical Data").

For reversing operation and drives with very high masses to accelerate and strong shock loads N-EUPEX couplings, Types A and B, may be fitted with higher flexible elements packs (12) with reduced torsional backlash.

The flexible elements (12) of Type ADS and BDS are available in 90 Shore A and 95 Shore A hardnesses.

The different flexible elements (12) are distinguished as follows:

Type	Size	Material	Hardness	Configuration	Identification marking
A, B	all sizes	Perbunan	80 Shore A	normal	blue stripe
	225 ... 480	Perbunan	60 Shore A	normal	green strip
	58 ... 200	Perbunan	80 Shore A	oversized	yellow stripe
	58 ... 200	Perbunan	60 Shore A	oversized	white stripe
ADS, BDS	all sizes	Polyurethan	90 Shore A	normal	blue flexible elements
	all sizes	Polyurethan	95 Shore A	normal	white flexible elements
	66 ... 272	Perbunan 2K	80/92 Shore A	normal	black flexible elements

Caution!

Only identical flexible elements (12) may be used in one coupling.

6. Assembly

At the customer's request FLENDER also delivers unbored or prebored coupling parts.

The necessary refinishing must be carried out in strict compliance with the following specifications and with particular care!

Caution!

Responsibility for carrying out the refinishing is borne by the orderer. FLENDER can accept no guarantee claims arising from unsatisfactory refinishing!

6.1 Instructions for machining the finished bore, parallel keyway, axial retaining means, set screws and balancing

6.1.1 Finish bore

- Remove flexible elements
- Depreserve and, if necessary, clean coupling parts



Note manufacturer's instructions for handling solvent.

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When machining the finished bore the parts must be carefully aligned. For the permissible radial and axial runout errors and the permissible cylindricity tolerances, refer to DIN ISO 286. The parts must be mounted on the marked faces ().

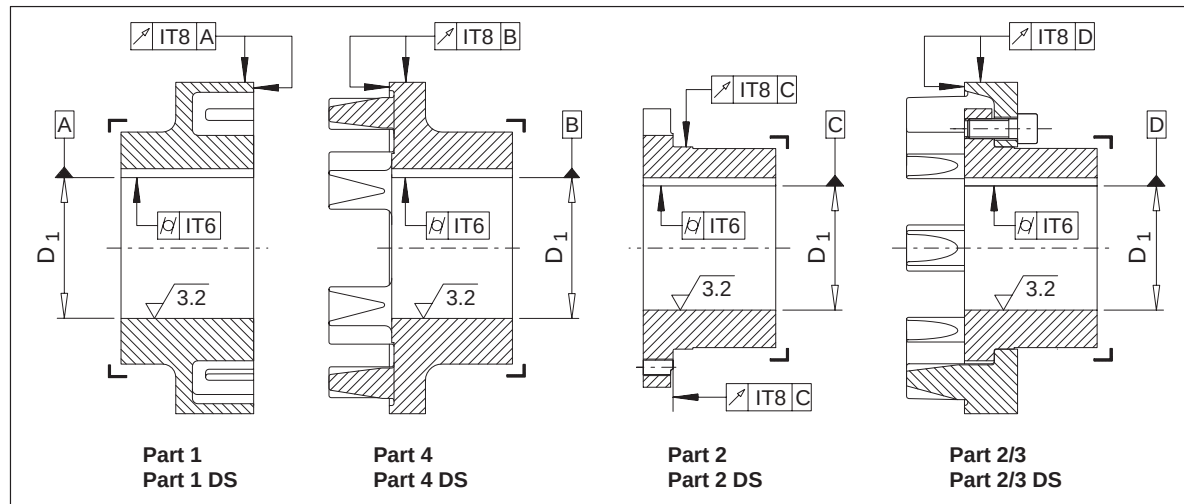


In the case of part 2/3 and part 4 great caution is necessary owing to the rotating cams.

Caution!

The maximum permissible bore diameters (see section 1.) are designed for drive-type fastenings without taper action to DIN 6885/1 and must not under any circumstances be exceeded. The finish-machined bores must be 100% checked with suitable measuring equipment.

If other shaft - hub connections (e.g. taper or stepped bore, etc.) are to be used instead of the flanged sleeve connections provided for, FLENDER must be consulted. Flanged sleeve connections with taper action are not permissible.



For drive by means of parallel keys the following fit pairs are prescribed for the bores:

Selection of fit	Bore D_1		Shaft tolerances	Bore tolerances
	over mm	to mm		
Shaft tolerances to FLENDER standard		25	k6	H7
	25	100	m6	
	100		n6	
Shaft tolerances to DIN 748/1		50	k6	H7
	50		m6	
System standard shaft		50	h6	K7
	50			M7
	all		h8	N7

Table 6.1.1: Fit pairs

Caution!

The assigned fits must be adhered to in order, on the one hand, to keep the play in the shaft-hub connection as low as possible, depending on utilisation of the tolerance zones, or, on the other, to keep the hub tension arising from the oversize within the permissible load limit. Failure to adhere to the fits may impair the shaft-hub connection. If the tolerance values of the shafts deviate from those in table 6.1.1 above, FLENDER must be consulted.



Failure to observe these instructions may result in breakage of the coupling. Danger from flying fragments!

6.1.2 Parallel keyway

The parallel keyways must be designed in accordance with DIN 6885/1. If the keyway geometry deviates, FLENDER must be consulted. Taper keys or nose keys (gib headed keys) are not permissible.

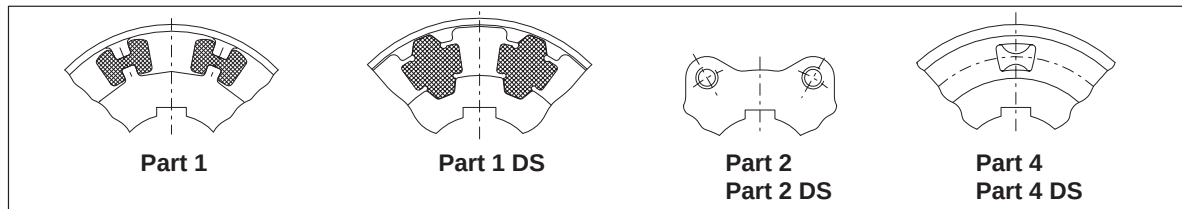
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The parallel keyways must be designed to suit the available parallel keys. For parallel keyways the tolerance zone of the hub keyway width **ISO JS 9** must be adhered to.

For **more difficult operating conditions** of the kind arising e.g. with reversing operation or operation with impulses the hub keyway tolerance zone **ISO P9** is specified.

Caution!

On part 1 the parallel keyway must be applied midway between the lands or pockets of the flexible elements, on part 2 midway between the through-holes and on part 4 below a cam.



6.1.3 Axial securing device

A set screw or end plate must be provided to secure the coupling parts axially. If end plates are used, FLENDER must be consulted with regard to machining the recesses in the coupling parts.

If the coupling part mounted on the shaft does not lie up against the shaft shoulder, we recommend using grooved spacer rings.

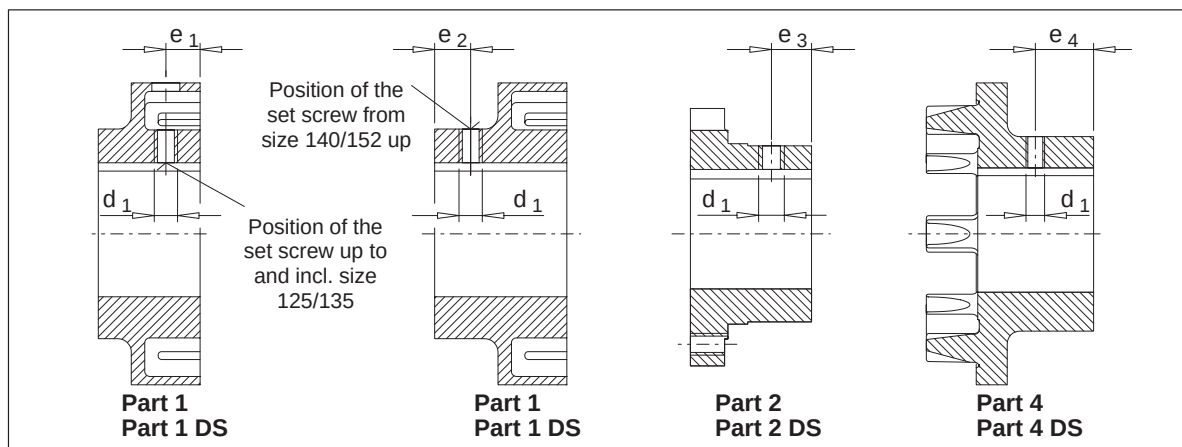
6.1.4 Set screws

Hexagon socket set screws with cup points to DIN 916 must be used for set screws.

The following guidelines must be observed!



The length of the set screw must be selected so that it fills the tapped hole, but does not project from the hub ($L_{\min} = d_1 \times 1.2$).



Size	58	68	80	95	110	125	140	160	180	200	225	250	280	315	350	400	440	480	520	560	610	660	710
	66	76	88	103	118	135	152	172	194	218	245	272	305	340	380	430	472	514	556	-	-	-	-
d ₁	M5	M6	M6	M6	M6	M8	M8	M10	M12	M12	M12	M16	M16	M16	M20	M20	M24	M24	M24	M24	M24	M24	M24
e ₁	*10	*10	*11	*15	18	20	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
e ₂	-	-	-	-	-	-	13	13	16	20	22	24	28	35	40	50	60	70	80	75	85	100	115
e ₃	-	-	-	-	*9	12	15	20	30	30	35	40	45	50	60	70	80	90	100	100	110	130	140
e ₄	*8	*8	12	15	18	20	22	25	32	40	40	45	45	-	-	-	-	-	-	-	-	-	-
1)	3	4	4	4	4	8	8	15	25	25	25	70	70	70	130	130	230	230	230	230	230	230	230

Table 6.1.4: Set screw assignment and tightening torques of the set screws

1) Tightening torques of the set screws in Nm

*) Note following position of the set screw!

Caution!

The set screws must always be positioned on the keyway. An exception are the following coupling parts:

Part 1: Size 58 / 66 :	Bore $D_1 \geq 15$ mm set screw displaced by 180° relative to the keyway.
Size 68 / 76 :	Bore $D_1 \geq 20$ mm set screw displaced by 144° relative to the keyway.
Size 80 / 88 :	Bore $D_1 \geq 25$ mm set screw displaced by 180° relative to the keyway.
Size 95 / 103 :	Bore $D_1 \geq 38$ mm set screw displaced by 180° relative to the keyway.
Part 2: Size 110 / 118 :	Bore $D_1 \geq 30$ mm set screw displaced by 180° relative to the keyway.
Part 4: Size 58 / 66 :	Bore $D_1 \geq 18$ mm set screw displaced by 180° relative to the keyway.
Size 68 / 76 :	Bore $D_1 \geq 20$ mm set screw displaced by 180° relative to the keyway.

6.1.5 Balancing

Prebored couplings or prebored coupling parts are delivered unbalanced. It is recommended that these parts are balanced to suit the application after finish-boring (see DIN ISO 1940 and DIN 740/2), but to min. balancing quality G16.

Balancing is normally done by drilling material away.

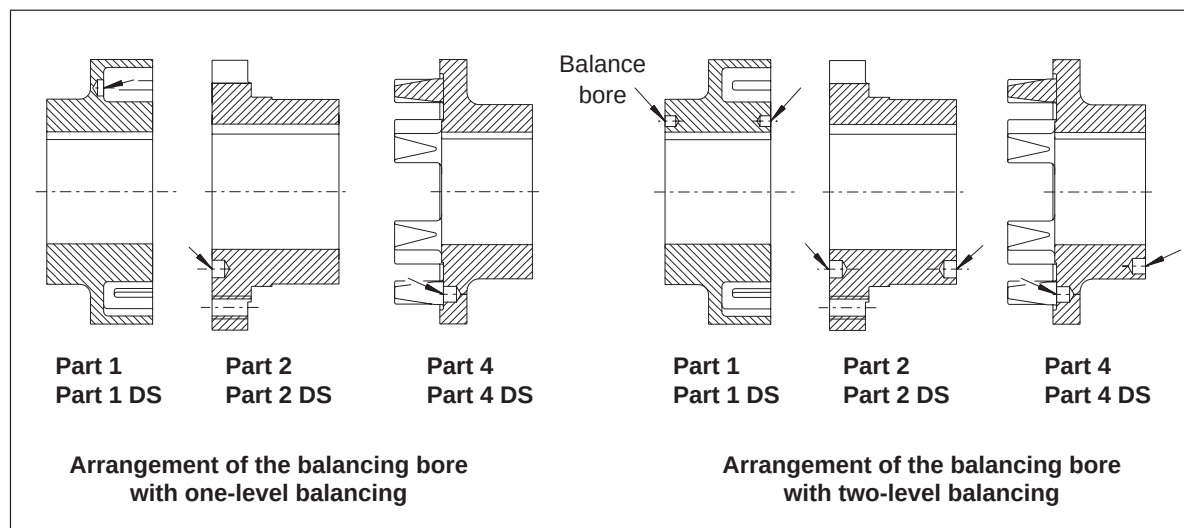
Caution!

On part 1 the material must be removed between the lands or pockets of the flexible elements without drilling right through the base and without damaging the pockets of the flexible elements.

On part 4 the material must be removed from the end face between the cams. Sufficient space must be left between the balancing hole and the cam to avoid weakening the cam connection.

Since the coupling part 3, the cam part is always in a balanced condition, the coupling part 2 can be balanced individually or together with the fitted part 3.

Finish-bored couplings or coupling parts are balanced according to the customer's specifications.



6.2 General information on installation

During assembly, Section 3. "Safety Instructions" must be observed.

Assembly and installation work must be done with great care by trained and qualified personnel.

As early as during the planning phase it must be ensured that sufficient space is available for installation and subsequent care and maintenance work.

Adequate lifting equipment must be available before beginning the installation and assembly work.

6.3 Mounting the coupling parts

Before beginning installation, the shaft ends and the coupling parts must be carefully cleaned. Before cleaning the coupling parts with solvent the flexible elements (12) must be removed.



Note manufacturer's instructions for handling solvent.

Before fitting coupling part 2 the cam part 3 must be located on the shaft.

If necessary, heating the coupling parts (to max. +150 °C) will facilitate fitting. With temperatures over +80 °C the flexible elements (12) must be removed from the coupling parts before heating.



Take precautions to avoid burns from hot components!

Caution!

**The coupling parts must be fitted with the aid of suitable equipment to avoid damaging the shaft bearings through axial joining forces.
Always use suitable lifting equipment.**

The shaft ends must not project from the inner sides of the hub. Axial securing is effected by means of the set screw or end plate.

Caution!

Tightening the set screws to a tightening torque in accordance with item 6.1.4.



**Failure to observe these instructions may result in breakage of the coupling.
Danger from flying fragments!**

After fitting the coupling parts, the flexible elements (12), if previously removed, must be fitted. Previously heated coupling parts must have cooled down again to a temperature below +80 °C. It must be ensured that the flexible elements (12) are of identical size and have identical markings.

Move together the machines to be coupled.



Danger of squeezing!

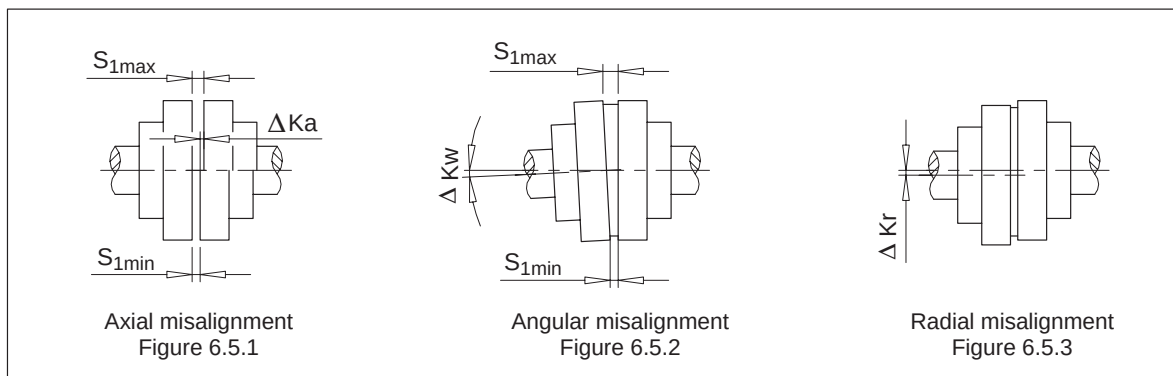
Dimension S_1 must be adhered to. On Types A and ADS the connection between parts 2 and 3 must be remade and the tightening torque of the screw connection to parts 2 and 3 checked before alignment (for tightening torques and gap dimension S_1 , see item 6.6 and section 1.).

6.4 Alignment

The couplings pick up positional errors in the shaft ends to be connected up to the data shown in item 6.5.

When aligning, the radial and angular misalignment of the shaft ends must be kept as small as possible, because, other conditions being equal, this increases the service life of the flexible elements.

6.5 Possible misalignments



Misalignments of the coupling parts in relation to each other can be caused by inaccurate alignment during assembly, but also by actual operation of the equipment (expansion due to heat, shaft deflection, insufficiently rigid machine frames, etc.).

Caution!

The following maximum permissible misalignments must by no means be exceeded during operation.

6.5.1 Axial misalignment

Axial misalignment ΔK_a (Fig. 6.5.1) of the coupling parts relative to one another is possible within the "permissible error" for dimension S_1 (see section 1.).

6.5.2 Angular misalignment

The angular misalignment ΔK_w (Fig. 6.5.2) can usefully be measured as the difference in the gap dimension ($\Delta S_1 = S_{1max} - S_{1min}$). For the permissible values for the difference in the gap dimension, refer to item 6.5.4.

If required, the permissible angular misalignment ΔK_w can be calculated as follows:

$$\Delta K_{w_{perm}} \text{ in Rad} = \frac{\Delta S_{1_{perm}}}{d_a}$$

$$\Delta K_{w_{perm}} \text{ in Degrees} = \frac{180}{\pi} \times \frac{\Delta S_{1_{perm}}}{d_a}$$

$\Delta S_{1_{perm}}$ see item 6.5.4
for d_a , see section 1., item 1.1.1 or item 1.2.1

6.5.3 Radial misalignment

For the permissible radial misalignment $\Delta K r_{perm.}$ (Fig. 6.5.3), - depending upon the operating speed -, refer to item 6.5.4.

6.5.4 Permissible shaft misalignment values for radial misalignment $\Delta K r_{perm.}$ and difference in gap dimension ΔS_{1perm}

Values given in mm, rounded off

Type / Size		Coupling speed in 1/min								
A, B	ADS, BDS	250	500	750	1000	1500	2000	3000	4000	5000
58	66	0.4	0.3	0.25	0.2	0.2	0.15	0.15	0.1	0.1
68	76	0.4	0.3	0.25	0.2	0.2	0.15	0.15	0.1	0.1
80	88	0.4	0.3	0.25	0.2	0.2	0.15	0.15	0.1	0.1
95	103	0.5	0.35	0.25	0.25	0.2	0.2	0.15	0.1	0.1
110	118	0.5	0.35	0.3	0.25	0.2	0.2	0.15	0.1	0.1
125	135	0.5	0.4	0.3	0.25	0.25	0.2	0.15	0.15	0.1
140	152	0.6	0.4	0.35	0.3	0.25	0.2	0.2	0.15	
160	172	0.6	0.5	0.4	0.35	0.3	0.25	0.2	0.15	
180	194	0.6	0.5	0.4	0.35	0.3	0.25	0.2		
200	218	0.8	0.55	0.45	0.4	0.3	0.3	0.2		
225	245	0.8	0.55	0.5	0.4	0.35	0.3	0.25		
250	272	0.8	0.6	0.5	0.4	0.35	0.3			
280	305	1	0.7	0.6	0.5	0.4	0.35			
315	340	1	0.7	0.6	0.5	0.4	0.35			
350	380	1	0.8	0.6	0.6	0.5				
400	430	1.2	0.9	0.7	0.6	0.5				
440	472	1.3	1	0.7	0.7	0.6				
480	514	1.4	1	0.8	0.7					
520	556	1.5	1.1	0.9	0.8					
560		1.6	1.2	1	0.8					
610		1.8	1.3	1	0.9					
660		1.9	1.4	1.1	1					
710		2	1.5	1.2						

The numerical values of the table can be calculated as follows:

Coupling speed n in 1/min	
Coupling size designation d_a in mm (see section 1, item 1.1.1 or item 1.2.1)	
Radial misalignment $K r_{perm}$ in mm	
$\Delta K r_{perm} = \Delta S_{1perm} = \left(0.1 + \frac{d_a}{1000} \right) \times \frac{40}{\sqrt{n}}$	

Caution!

Angular and radial misalignment may occur simultaneously.

6.6 Tightening torques

N-EUPEX Coupling	N-EUPEX-DS Coupling	Tightening torque T_A and spanner size S_w for hexagon socket screws to DIN EN ISO 4762	
Size	Size	T_A Nm	S_w mm
110	118	14	6
125	135	17.5	6
140	152	29	8
160	172	35	8
180	194	44	8

200	218	67.5	10
225	245	86	10
250	272	145	14
280	305	185	14
315	340	200	14
350	380	260	17
400	430	340	17
440	472	410	17
480	514	550	19
520	556	670	19
560		710	19
610		1450	22
660		1450	22
710		1450	22

Table 6.6: Tightening torques for part 13 of Types A and ADS

Note: Tightening torques apply to screws with untreated surfaces which are not or only lightly oiled (coefficient of friction $\mu = 0.14$). The use of lubricant paint or the like, which affects the coefficient of friction μ , is not permitted.

Note: The tightening torques of the set screws are specified in item 6.1.4.

7. Start-up

7.1 Procedure before start-up

Before starting up check the flexible elements (12) for correct seating, i.e. the flexible elements (12) must sit flush with the end face of the hub, and the set screws for tightness, check and, if necessary, adjust the alignment and the gap dimension S_1 and check all screw connections for the specified tightening torques (see section 6.).



Then fit the coupling guard to prevent unintentional contact.

8. Operation

8.1 General operating data

During operation of the coupling watch for:

- changes in running noise
- sudden shocks

Caution!

If any irregularities are noticed during operation, switch the drive assembly off at once. Determine the cause of the fault, using the table in section 9.

This table contains a list of possible faults, their causes and suggested remedies.

If the cause cannot be identified or the unit repaired with the facilities available, you are advised to contact one of our customer-service offices for specialist assistance (see section 11.).

9. Faults, causes and remedy

9.1 General

The following irregularities can serve as a guide for fault tracing.

Where the system is a complex one, all the other component units must be included when tracing faults.

The coupling must run with little noise and without vibration in all operating phases. Irregular behaviour must be treated as a fault requiring immediate remedy.

Caution!

FLENDER will not be bound by the terms of the guarantee or otherwise be responsible in cases of improper use of the coupling, modifications carried out without FLENDER's agreement, or use of spare parts not supplied by FLENDER.



When remedying faults and malfunctions, the coupling must always be taken out of service.

Secure the drive unit to prevent it from being started up unintentionally. Attach a warning notice to the start switch.

9.2 Possible faults

Malfunctions	Causes	Remedy
Sudden changes in the noise level and/or sudden vibrations	Change of alignment	take the system out of service if necessary, rectify causes of alignment change (e.g. tighten loose foundation bolts) Check and, if necessary, adjust alignment (see section 6). Wear check, procedure as described in section 10
	Flexible elements (12) worn	take the system out of service Demount coupling and remove remains of flexible elements (12) Check and replace damaged coupling parts Flexible elements (12) must be changed in sets; use only identical N-EUPEX flexible elements (12) Assembly of coupling according to section 6. and section 7.

Table 9.2: Possible faults

9.3 Incorrect use

Experience has shown that the following faults can result in incorrect use of the N-EUPEX coupling. In addition to observing the other instructions in this BA, care must therefore be taken to avoid these faults.



Failure to observe these instructions may result in breakage of the coupling. Danger from flying fragments!

Caution!

Incorrect use of the N-EUPEX coupling may result in damage to the coupling.

Caution!

Coupling damage may result in stoppage of the drive and the entire system.

9.3.1 Possible faults when selecting the coupling or coupling size

- Important information for describing the drive and the environment will not be communicated to others
- System torque too high
- System speed too high
- Application factor not correctly selected
- Chemically aggressive environment not taken into consideration
- The ambient temperature is not permissible. See also section 1.
- Finished bore with impermissible diameter (see section 1) and/or impermissible fit classification (see section 6.)
- The transmission capacity of the shaft-hub connection is not appropriate to the operating conditions

9.3.2 Possible faults when installing the coupling

- Components with transport or other damage are being fitted
- When mounting coupling parts in a heated condition, already mounted N-EUPEX flexible elements (12) are being excessively heated
- The shaft diameter is outside the specified tolerance range
- Coupling parts are being interchanged, i.e. their assignment to the specified shaft is incorrect
- Prescribed tightening torques are not being adhered to
- Alignment or shaft misalignment values do not match the operating instructions
- The coupled machines are not correctly fastened to the foundation, so a shifting of the machines e.g. through loosening of the foundation screw connection is causing excessive displacement of the coupling parts
- N-EUPEX flexible elements (12) are being omitted or incorrectly positioned
- Operating instructions are being changed without authorisation

9.3.3 Possible faults in maintenance

- Maintenance intervals are not being adhered to
- Original FLENDER N-EUPEX flexible elements (12) are not being used
- Old or damaged N-EUPEX flexible elements (12) are being used
- Different N-EUPEX flexible elements (12) are being used (see section 5.)
- Leakage in the vicinity of the coupling is not being identified and as a result chemically aggressive media are damaging the coupling

10. Maintenance and repair



All work on the coupling must be carried out only when it is at a standstill. The drive unit must be secured against being switched on accidentally (e.g. by locking the key switch or removing the fuses from the power supply). A notice should be attached to the ON switch stating clearly that work is in progress.

10.1 Maintenance interval



**On Types A and B the torsional backlash between the two coupling parts must be checked after three months, then at least once a year.
In the case of types ADS and BDS checking the torsional backlash at regular intervals is recommended in terms of preventive maintenance.**

If an increased coupling backlash does not impair the operation of the coupling, the flexible elements (12) can continue to be used up to a specified wear limit before being replaced. To assess wear, the permitted circumferential backlash, converted to the chord dimension ΔS_V on the outer coupling diameter, is shown in table 10.1a. and table 10.1b. To obtain the dimension ΔS_V , one coupling part is rotated without torque as far as the stop and a mark applied to a coupling part (see Fig. 10.1). If the coupling part is rotated in the opposite direction of rotation as far as the stop, the marks move apart. The distance between the marks is the chord dimension ΔS_V . If the dimension ΔS_V exceeds the value in table 10.1a or table 10.1b, the flexible elements (12) must be replaced.

Caution!

**The flexible elements (12) must be replaced in sets.
Only identically marked flexible elements (12) must be used.**

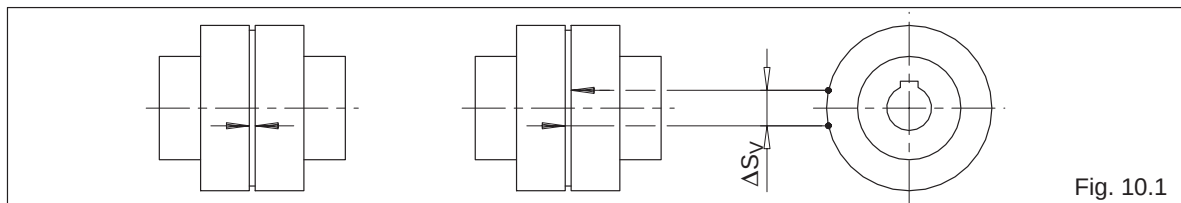


Fig. 10.1

Size	58	68	80	95	110	125	140	160	180	200	225	250	280	315	350	400	440	480	520	560	610	660	710
Wear mark ΔS_V (mm)	5.5	5.5	5.0	6.0	7.0	8.0	8.0	8.0	8.0	8.5	9.0	10.0	11.5	10.5	11.5	13.0	14.0	15.5	17.5	17.5	19.5	21.0	22.5

Table 10.1 a: Wear mark of N-EUPEX coupling

Size	66	76	88	103	118	135	152	172	194	218	245	272	305	340	380	430	472	514	556
Wear mark ΔS_V (mm)	6.0	7.0	5.0	7.0	9.0	10.5	11.5	9.0	8.0	7.0	6.5	7.0	8.0	6.5	7.0	10.0	12.0	14.0	16.0

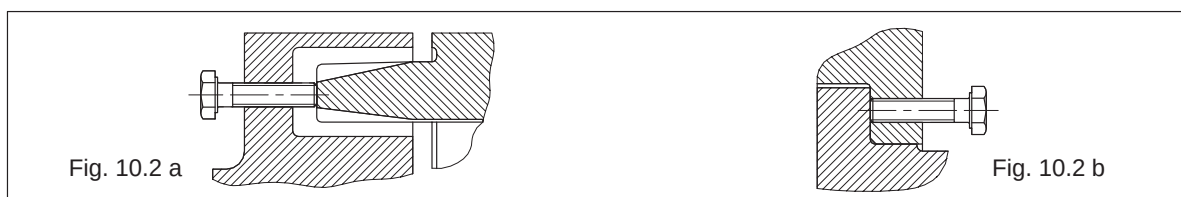
Table 10.1 b : Wear mark of the N-EUPEX-DS coupling

10.2 Replacement of wearing parts

Only **original N-EUPEX flexible elements** must be used for replacement to guarantee troublefree torque transmission and faultfree operation.

Note: Replacement of the flexible elements (12) is possible without shifting the coupled machines only on Types A and ADS.

After releasing the connection of parts 2 and 3, part 3 is shifted axially. The flexible elements (12) will then be made freely accessible by rotating part 2. To facilitate the release of part 3, on sizes 225 - 430 forcing-off threads are provided in part 1. From size 440 up the forcing-off threads are located in part 3 (see Figures 10.2 a and 10.2 b).



For re-assembly, the instructions in section 6. "Assembly" and section 7. "Start-up" must be carefully observed.

11. Spare parts, customer-service addresses

By stocking the most important spare and wearing parts on site, you can ensure that the coupling is ready for use at any time.

When ordering spare parts, always state the following:

- Original order no.
- Part no. (see item 11.1)
- Specification / size (the size designation corresponds to the outside diameter d_a in mm)
- Quantity

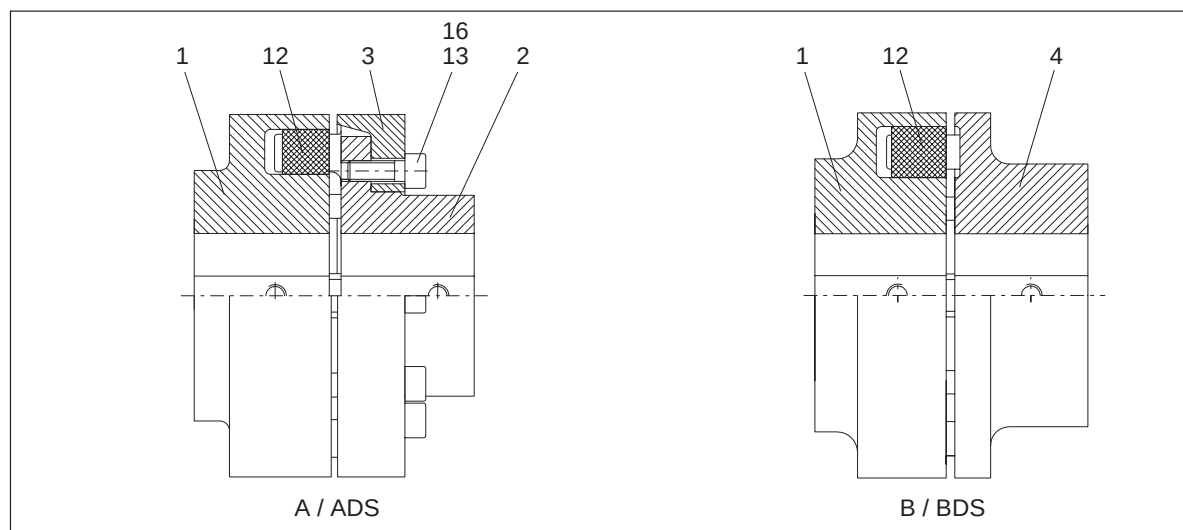
We guarantee only the original spare parts supplied by us.

Caution!

Please note that spare parts and accessories not supplied by us have not been tested or approved by us. The installation or use of such products may therefore impair essential characteristics of the coupling under certain circumstances and so pose an active or passive hazard. FLENDER will assume no liability or guarantee for damage caused by spare parts and accessories not supplied by FLENDER.

Please note that certain components often have special production and supply specifications and that we supply you with spare parts which comply fully with the current state of technical development as well as current legislation.

11.1 Spare parts list



Spare parts Type A, ADS		Spare parts Type B, BDS	
Part no.	Description	Part no.	Description
1	Part 1	1	Part 1
2	Part 2	4	Part 4
3	Part 3	12	Flexible elements
12	Flexible elements		
13	Cheese head screw		
16	Parallel pin only on Type A Size 560 to 710		

Table 11.1: Spare parts list, Types A, ADS, B and BDS

11.2 Spare-part and customer service addresses

When ordering spare parts or requesting the services of our specialist engineers, please apply first to FLENDER AG.

FLENDER Germany

A. FRIEDR. FLENDER AG

46393 Bocholt - Tel.: (0 28 71) 92-0 - Fax: (0 28 71) 92 25 96
E-mail: contact@flender.com • www.flender.com
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12. Declaration by the manufacturer

Declaration by the manufacturer

in accordance with EC Engineering Guideline 98/37/EC, Appendix II B

We hereby declare that the

**Flexible N-EUPEX and N-EUPEX-DS couplings
Types A, B and ADS, BDS**

described in these Operating Instructions are intended for incorporation in a machine, and that it is prohibited to put them into service before verifying that the machine into which they are incorporated complies with the EC Guidelines (original edition 98/37/EC including any subsequent amendments thereto).

This Manufacturer's Declaration takes into account all the unified standards (inasmuch as they apply to our products) published by the European Commission in the Official Journal of the European Community.



Bocholt, 2003-07-10

Signature (person responsible for products)

FLENDER BOCHOLT	RUPEX Ersatzteilliste / Maßzeichnung Spare parts list / Dimension drawing	Bauart Type	RWN	Ausf. Assembly
		Größe Size	450	

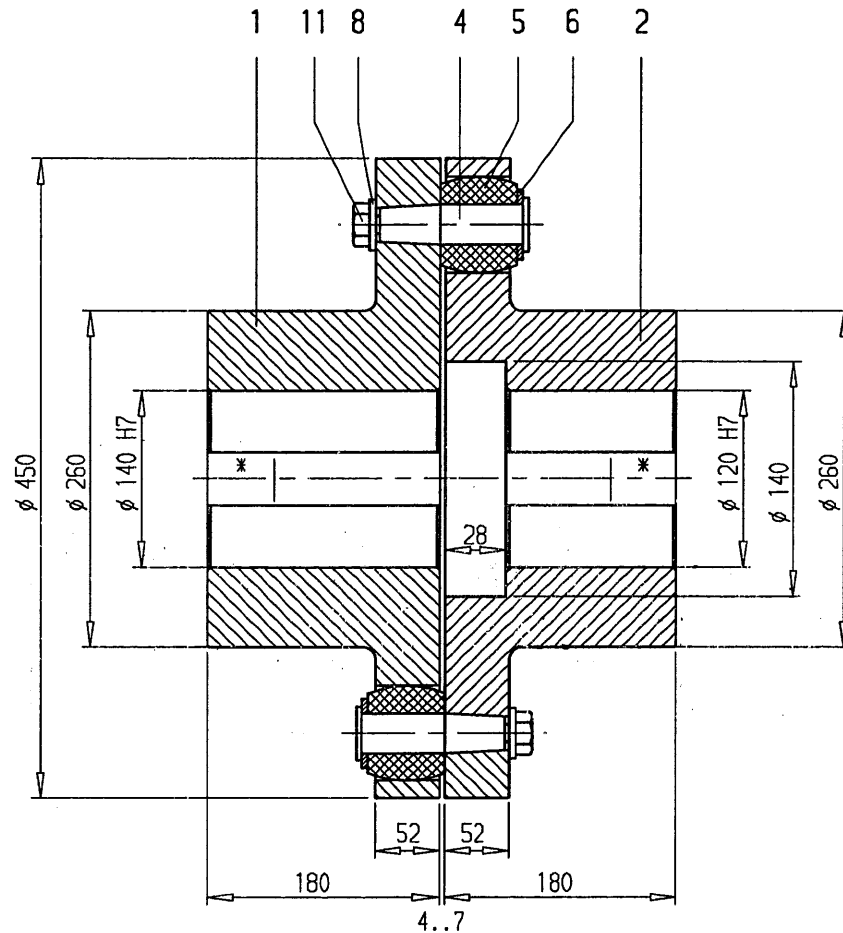
Bei Korrespondenz bitte angeben / Please quote in correspondence

NO. 4123221-60

Ident: NARAYANGONJ

BNR. 45017726

Code: MC1M10.020.50.09.02



* Nut nach DIN 6885/1

* Keyway acc. to DIN 6885/1

12	SCHRB-SKT HEXAGON HEAD SCREW	M16X30	11	D 933	321374
12	SCHB-END END PLATE	.	8	9 N 5285040	500383
12	SCHEIBE DISC	32	6	D 1440	342034
12	PUFFER BUFFER	450	5	4 T 9003343	500084
12	BLZN BOLT	.	4	9 N 5285046	500376
1	TEIL PART	1/2 450	2	2 N 5213965	500028
1	TEIL PART	1/2 450	1	2 N 5213965	500028
Menge No. off	Benennung Description	Kenndaten Data	Teil-Nr. Part No.	Zeichnungs-Nr. Drawing No.	Waren-Nr. Ident No.

Umlaufende Teile müssen vom Käufer gegen unbeabsichtigtes Berühren gesichert werden! (Gesetz über techn. Arbeitsmittel)

The purchaser is responsible for the provision of safety guards.

A. Friedr. Flender AG, D 46393 Bocholt,
Tel. 02871/92-0, Tx 0813841, Telefax 02871/922596

Datum/ Date
2004-06-03

Name van Gelder KUE
Z-Nr. 4123221-60

Diese technische Unterlage hat gesetzlichen Schutz (DIN 34)

/B0

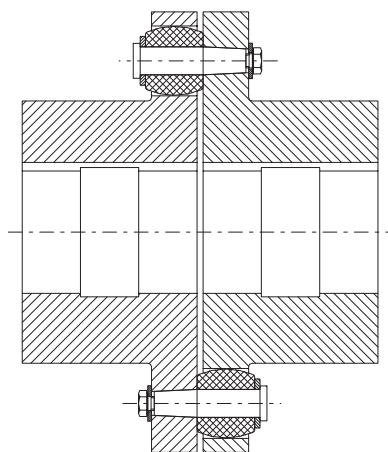
PEK

E-LISTE RWN 450 EN

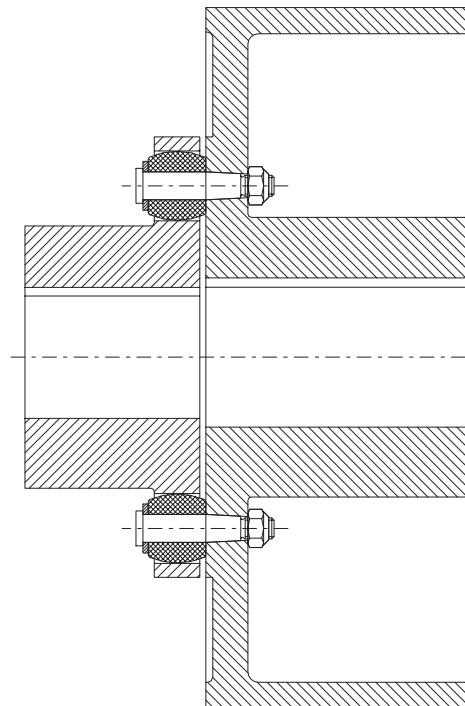
Operating Instructions

BA 3600 EN 08.98

Flexible **RUPEX** couplings,
type **RWN**, **RWS** and **RWB**, **RBS**



RWN
RWS



RWB
RBS

FLENDER

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1. Technical data

1.1 Validity of the nominal torques T_{KN}

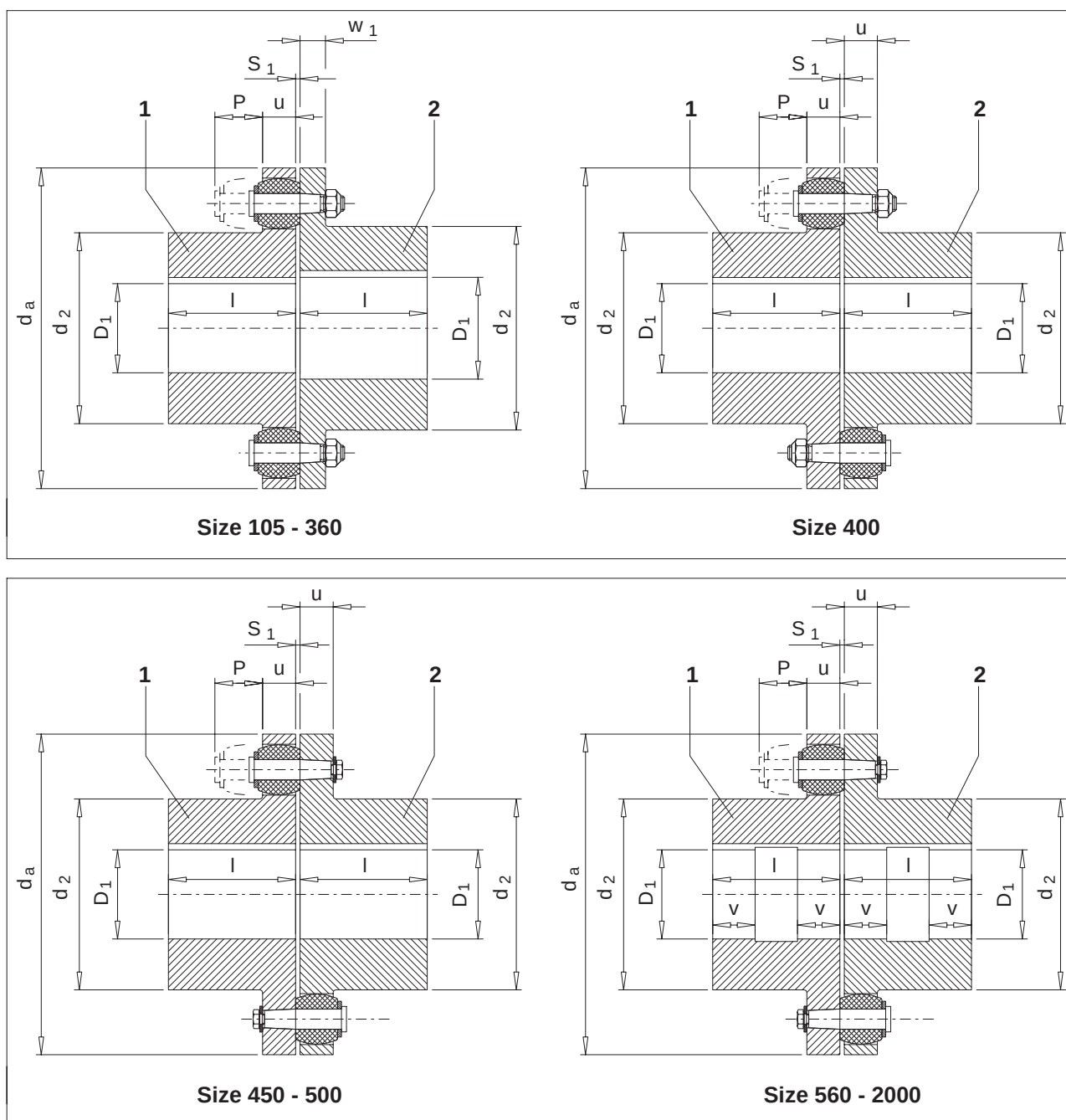
Validity of nominal torques T_{KN} (with original RUPEX buffer of buna N, hardness 80 Shore **only**):

- daily operating cycle up to 24 h
- during the starting process or during operation, torque impulses up to the triple nominal torque are allowed up to 25 times per hour.
- Operation within the prescribed alignment
- Operation in the temperature range from $-30\text{ }^{\circ}\text{C}$ to $+80\text{ }^{\circ}\text{C}$ (ambient temperature or temperature of the shaft ends).

Caution!

For permanent trouble-free operation, the coupling has to be designed with a service factor appropriate to the respective application. When changing the operating conditions (performance, speed, changes on power engine and machine), a check of the design is absolutely necessary.

1.2 Types RWN and RWS



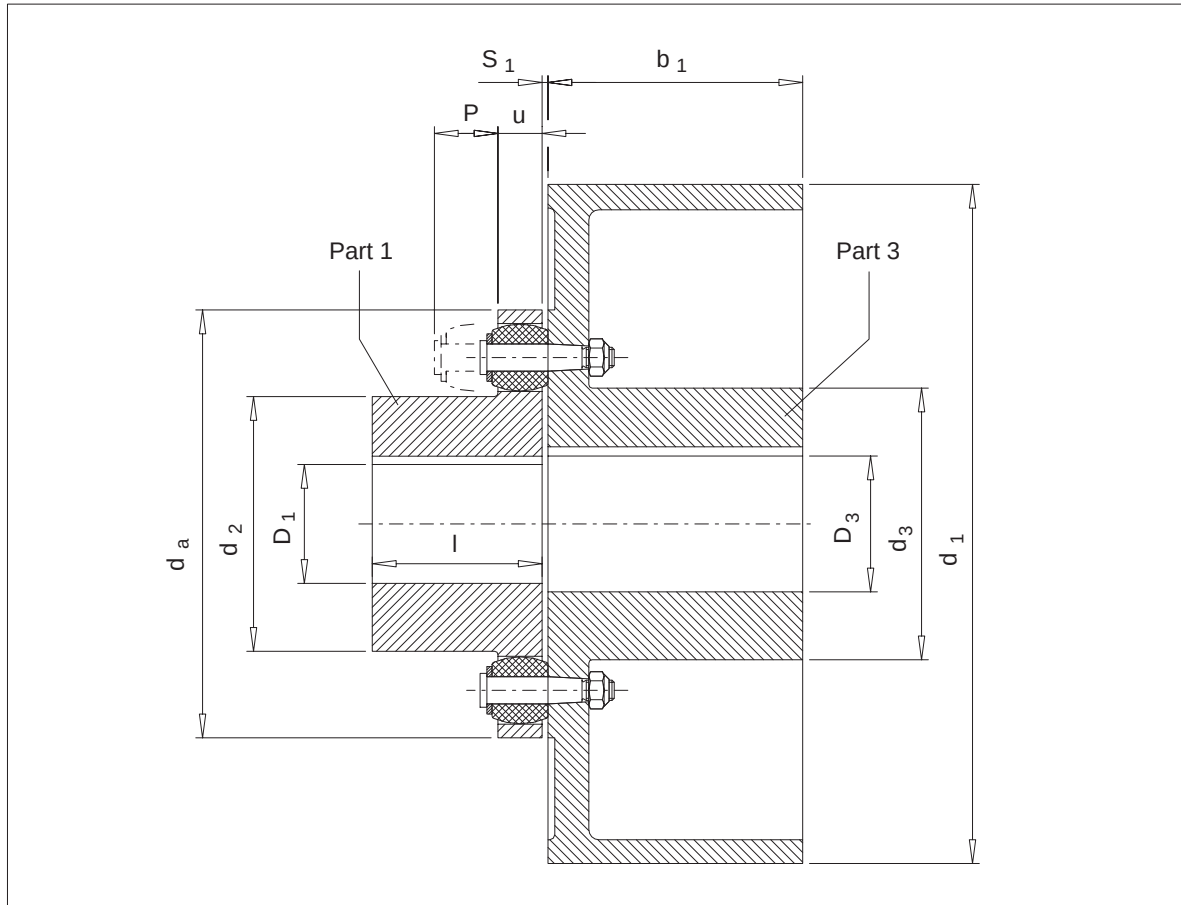
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Size	Nominal torque T _{KN} 1) Nm	Speed n _{max}		Bore						d _a mm			l mm	v mm	P mm	S ₁ mm	w ₁ mm	u mm	Weight 2)			
		RWN 1/min	RWS 1/min	from		to		RWN			RWS								RWN		RWS	
				D _{1/2} mm	D ₁ mm	D ₂ mm	D ₁ mm	D ₂ mm	D ₁ mm		D ₂ mm	Part 1 kg							Part 2 kg	Part 1 kg	Part 2 kg	
105	200	5000	5000	10	32	38	32	38	105	53	59	45		30	2... 4	12	13	0.8	1	0.9	1.1	
125	350	5000	5000	14	40	48	40	48	125	65	68	50		35	2... 4	15	16	1.4	1.5	1.5	1.6	
144	500	4900	5000	18	45	55	50	60	144	76	84	55		35	2... 4	15	16	1.9	2.6	2.1	2.8	
162	750	4300	5000	22	50	60	55	65	162	85	92	60		40	2... 5	18	20	2.8	3.9	3	4.2	
178	950	3800	4900	24	60	70	780	75	178	102	108	70		40	2... 5	18	20	4.1	5.6	4.4	6.1	
198	1300	3400	4600	28	70	80	80	85	198	120	128	80		40	2... 5	18	20	5.9	7	6.4	7.6	
228	2200	3000	4400	28	80	90	85	95	228	129	140	90		50	2... 5	24	26	7.3	11.5	7.9	12.5	
252	2750	2700	4200	38	90	100	100	110	252	150	160	100		50	2... 5	24	26	10.5	15.5	11.5	16.5	
285	4300	2400	3900	48	100	110	110	120	285	164	175	110		60	3... 6	30	32	16	23	17.5	25	
320	5500	2100	3500	55	110	120	125	130	320	180	192	125		60	3... 6	30	32	23	30	25	32	
360	7800	1900	3100	65	120	130	135	140	360	200	210	140		75	3... 6	42	42	32	46	35	50	
400	12500	1700	2800	75	140	140	150	150	400	230	230	160		75	3... 6		42	53	53	57	57	
450	18500	1500	2500	85	160	160	170	170	450	260	260	180		90	4... 7		52	78	78	84	84	
500	25000	1350	2200	95	180	180	190	190	500	290	290	200		90	4... 7		52	99	99	105	105	
560	39000	1200	2000	> 100	140	140	165	165	560	250	250	220	70	120	4... 8		68	140	140	150	150	
				140	180	200	200	300		300	145							145	155	155		
				> 180	200	210	210	320		320	145							145	155	155		
630	52000	1050	1800	> 100	140	140	165	165	630	250	250	240	80	120	4... 8		68	175	175	190	190	
				140	180	200	200	300		300	185							185	200	200		
				> 180	220	235	246	355		355	200							200	215	215		
710	84000	950	1600	> 110	160	160	190	190	710	290	290	260	80	140	5... 9		80	255	255	275	275	
				160	200	200	220	220		330	330							260	260	280	280	
				> 200	240	240	250	250		385	385							270	270	290	290	
800	110000	850	1400	> 125	180	180	210	210	800	320	320	290	90	140	5... 9		80	330	330	360	360	
				180	220	220	240	240		360	360							350	350	380	380	
				> 220	260	260	280	280		420	420							360	360	390	390	
900	150000	750	1250	> 140	180	180	210	210	900	320	320	320	100	160	5...10		90	450	450	490	490	
				180	220	220	240	240		360	360							450	450	490	490	
				> 220	260	260	280	280		425	425							480	480	520	520	
				> 260	290	290	310	310		465	465							510	510	550	550	
1000	195000	680	1100	> 150	200	200	230	230	1000	355	355	350	110	160	5...10		90	560	560	610	610	
				200	240	240	260	260		395	395							580	580	630	630	
				> 240	280	280	300	300		460	460							630	630	680	680	
				> 280	320	320	340	340		515	515							660	660	710	710	
1120	270000	600	1000	> 160	200	200	240	240	1120	360	360	380	120	180	6...11		100	730	730	790	790	
				200	250	250	270	270		410	410							750	750	810	810	
				> 250	300	300	330	330		495	495							800	800	870	870	
				> 300	350	350	370	370		560	560							840	840	910	910	
1250	345000	550	900	> 180	230	230	270	270	1250	410	410	420	130	180	6...11		100	920	920	1000	1000	
				230	280	280	300	300		460	460							950	950	1050	1050	
				> 280	330	330	360	360		540	540							1000	1000	1050	1050	
				> 330	380	380	400	400		610	610							1100	1100	1150	1150	
1400	530000	490	800	> 200	260	260	310	310	1400	465	465	480	145	210	6...12		120	1400	1400	1500	1500	
				260	320	320	350	350		525	525							1450	1450	1550	1550	
				> 320	380	380	410	410		620	620							1550	1550	1650	1650	
				> 380	440	440	460	460		700	700							1650	1650	1750	1750	
1600	750000	430	700	> 260	320	320	370	370	1600	565	565	540	165	210	6...12		120	1900	1900	2050	2050	
				320	380	380	410	410		625	625							1950	1950	2100	2100	
				> 380	440	440	480	480		720	720							2050	2050	2200	2200	
				> 440	480	480	510	510		770	770							2150	2150	2300	2300	
1800	975000	380	600	> 320	380	380	440	440	1800	660	660	600	185	240	8...16		140	2750	2750	3000	3000	
				380	440	440	480	480		720	720							2800	2800	3050	3050	
				> 440	500	500	540	540		820	820							3000	3000	3250	3250	
				> 500	540	540	580	580		870	870							3150	3150	3400	3400	
2000	1300000	340	550	> 380	440	440	500	500	2000	760	760	660	200	240	8...16		140	3400	3400	3700	3700	
				440	500	500	540	540		820	820							3500	3500	3800	3800	
				> 500	560	560	610	610		920	920							3700	3700	4000	4000	
				> 560	600	600	640	640		960	960							3800	3800	4100	4100	

Table 1.1: Torques T_{KN} , speeds n_{max} , weights and sizes of type RWN and RWS

- 1) Observe the validity of the nominal torques T_{KN} according to 1.1!
- 2) Weights are valid for max. bores.

1.3 Types RWB and RBS



Size	Nominal torque	Speed	Bore					d _a	d ₂	d ₃	l	u	P	S ₁	d ₁	b ₁	Weight 2)						
	T _{KN} 1)	n _{max}	from	to			RWB										RBS		RWB		RBS		
				D _{1/3}	D ₁	D ₃	D ₁										D ₃	Part 1	Part 3	Part 1	Part 3		
																						mm	mm
Nm	1/min	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	kg	kg	kg	kg				
144	500	3400	18	45	55	50	60	144	76	84	55	16	35	2... 4	200	75	1.9	6.6	2.1	7.1			
162	750	2750	22	50	60	55	65	162	85	92	60	20	40	2... 5	250	95	2.8	13	3	14			
178	950	2750	24	60	70	70	75	178	102	108	70	20	40	2... 5			4.1	13.5	4.4	14.5			
198	1300	2750	28	70	80	80	85	198	120	128	80	20	40	2... 5			5.9	15	6.4	16			
178	950	2150	24	60	70	70	75	178	102	108	70	20	40	2... 5	315	118	4.1	21	4.4	22.5			
198	1300	2150	28	70	80	80	85	198	120	128	80	20	40	2... 5			5.9	22.5	6.4	24.5			
228	2200	1700	28	80	90	85	95	228	129	140	90	26	50	2... 5	400	150	7.3	41	7.9	44			
252	2750	1700	38	90	100	100	110	252	150	160	100	26	50	2... 5			10.5	46	11.5	50			
252	2750	1400	38	90	100	100	110	252	150	160	100	26	50	2... 5	500	190	10.5	74	11.5	80			
285	4300	1400	48	100	110	110	120	285	164	175	110	32	60	3... 6			16	79	17.5	86			
285	4300	1100	48	100	110	110	120	285	164	175	110	32	60	3... 6	630	236	16	130	17.5	140			
320	5500	1100	55	110	120	125	130	320	180	192	125	32	60	3... 6			23	135	25	145			
360	7800	1100	65	120	130	135	140	360	200	210	140	42	75	3... 6			32	145	35	155			
320	5500	950	55	110	120	125	130	320	180	192	125	32	60	3... 6	710	265	23	180	25	195			

Table 1.1: Torques T_{KN} , speeds n_{max} , weights and sizes of type RWB and RBS

- 1) Observe the validity of the nominal torques T_{KN} according to 1.1!
- 2) Weights are valid for max. bores.

2. General notes

2.1 General

These Operating Instructions constitute part of the coupling shipment and should be kept in the immediate vicinity of the coupling at all times.

Only a precise knowledge of the Operating Instructions will ensure trouble-free operation of the coupling. It is therefore in the interest of our customer that the operating instructions are read, understood and observed in all respects by the persons responsible for handling, installation and operation.

Note: We accept no liability for any damage or malfunction resulting from non-observance of the operating instructions.

The "**coupling**" dealt with in these operating instructions was developed for stationary use in general engineering.

The coupling is only designed for the field of application as specified in Section 1 "Technical data". Operating conditions which differ from the stated will require fresh contractual agreements.

The coupling described here is in accordance with the state of the art at the time of these operating instructions go into print.

In the interest of further development, we reserve the right to introduce modifications which, while retaining the essential features, can be regarded as desirable to increase its efficiency and safety.

The copyright of these Operating Instructions remains the property of **FLENDER AG**.

These operating instructions may not be duplicated in part or whole, utilized for the purpose of publicity or communicated to third parties without our expressed consent.

Please contact our works listed below in respect of all technical queries.

FLENDER AG
D 46393 Bocholt

Telephone: 02871/92-2800
Telefax: 02871/92-2801

or one of our service branches which are listed in Section 11 "Stocking spare parts, service facility addresses".

3. Safety notes

3.1 Safety notes

- The coupling is constructed in accordance with the state of the art and is reliable in the condition as shipped. Unauthorized modifications which impair its reliability are not permissible. This also applies to guards which are fitted as protection against accidental contact.
- The coupling may only be used and operated within the scope of the condition specified in the contract of performance and supply.
- The customer should ensure that the persons entrusted with installation, operation, care and maintenance as well as repair have read and understood the operating instructions and observe them in all respects in order to:
 - prevent hazard to life and limb of the user and third parties
 - ensure the reliability of the coupling

and

- prevent failure and environmental pollution due to incorrect handling.
- The relevant regulations concerning industrial safety and pollution control should be observed during handling, installation, operation as well as care and maintenance.
- The coupling may only be operated, serviced and repaired by authorized, trained and properly instructed personnel.
- All work should be carried out with care with the safety aspect in mind.
- All work on the coupling may only be carried out when it is stationary.
The coupling must be secured to prevent accidental start up (e.g. by locking the key switch or by removing the fuses and the power supply). A notice should be displayed at the switch-on point stating that work is in progress on the coupling.
- The drive unit should be shut off at once if changes in the coupling are detected during operation, such as e.g. changed running noises.
- The coupling must be protected by means of suitable guards to prevent accidental contact.
- During installation of the coupling in units or systems, the manufacturer of the units or systems is obliged to incorporate the requirements, notes and descriptions contained in these operating instructions in his own operating instructions.

3.1.1 Notes and symbols in the operating instructions

Instructions in the operating instructions which concern operating safety are emphasized as follows:



This symbol draws attention to safety measures which must be observed to prevent **personal injury**.

Caution!

This symbol draws attention to safety measures which must be observed to prevent **damage to the coupling**.

Note:

This note draws attention to general **operating notes** which should be especially observed.

4. Handling and storage

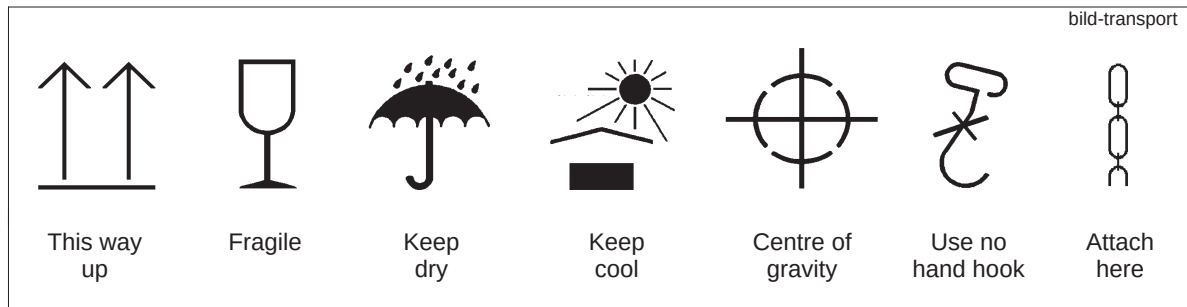
4.1 Scope of supply

The scope of the shipment is listed in the shipping documents. They should be checked for completeness on receipt. Any shipping damage and/or missing parts should be reported in writing at once. After consulting FLENDER an expert is to be called in.

4.2 Handling

The packing of the coupling will differ depending on method of shipment and size. The packing, unless otherwise agreed contractually, complies with **HPE Packing Guidelines**.

The symbols shown on the packing should be noted. Their meaning is as follows:



Caution!

Make sure that suitable hoists are used.

4.3 Storage

4.3.1 Storage of the coupling parts

The coupling is delivered in a preserved state and can be stored at a covered dry place up to 6 months. If the coupling shall be stored for a longer period of time, an appropriate long-term preservation is necessary (consultation with FLENDER required).

Caution!

Before cleaning the coupling parts and applying the long-term preservation, the buffers are to be removed.

4.3.2 Storage of the buffers

4.3.2.1 General

Properly stored buffers keep their characteristics for up to 5 years. Unfavourable storage conditions and improper treatment of the buffers result in a negative change of the physical characteristics. These changes can be caused by the effects of e.g. ozone, extreme temperatures, light, moisture or solvents.

4.3.2.2 Storage room

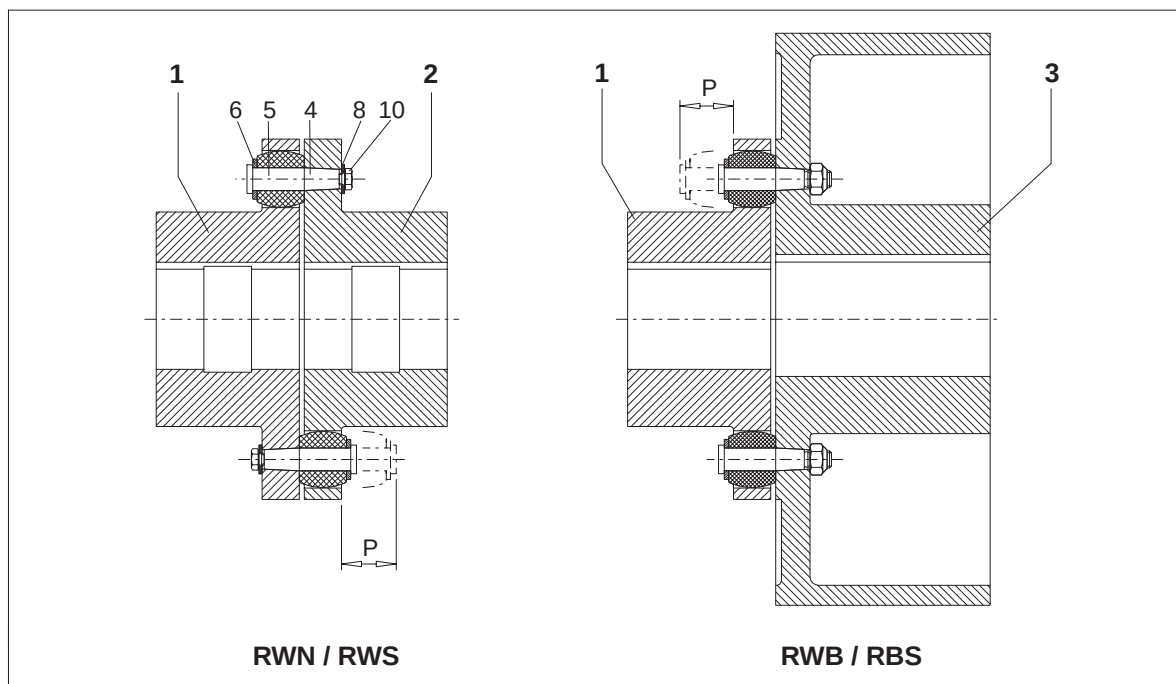
The storage room should be dry and dust-free. The buffers must not be stored together with chemicals, solvents, fuels, acids, etc. Furthermore, they should be protected against light, especially against direct sun light and strong artificial light with a high ultra-violet percentage.

Caution!

The storage rooms must not contain any ozone-producing devices like e.g. fluorescent light sources, mercury-vapour lamps, electric high-voltage devices. Damp storage rooms are unsuitable. Make sure that no condensation develops. The relative humidity of air is most favourable below 65 %.

5. Technical description

5.1 General description



The RUPEX coupling type RWN and RWS consists of two coupling parts and the pins with the flexible plastic buffers required for torque transmission. In the RWN version, the hub parts (1 and 2) are of grey cast iron. In the RWS version, they are of steel.

The RUPEX coupling type RWB and RBS consists of a coupling part and a brake drum, with the pins and flexible plastic buffers required for torque transmission. In the RWB version, the hub part (1) and the brake drum (3) are of grey cast iron. In the RBS version, they are of steel.

Up to size 360, the ground steel pins are attached to the buffers only in the coupling part (2) (RWN and RWS) or the brake drum (3) (RWB and RBS). From size 400 they are attached alternately in the coupling parts (1 and 2). In the assembled state, the buffers engage the buffer bores of the counterpart.

The flexible plastic buffers of buna N have a hardness of 80 Shore. Buffers of differing material quality and/or hardness are available within limits. With regard to the availability of these special buffers and the extent to which the properties of the coupling are affected, please consult FLENDER.

6. Assembly

6.1 Notes on fitting the finished bore, the axial securing, the set screws, the balancing

6.1.1 Finished bore

- Remove buffers.
- Remove the anti-corrosion agent from coupling parts



Observe manufacturer's instructions on handling solvents.

When fitting the finished bore, align the parts carefully. For the permissible radial and axial excentricity see DIN ISO 286 degree of fundamental tolerance IT8. The location of the parts () is to be carried out on the marked surfaces.

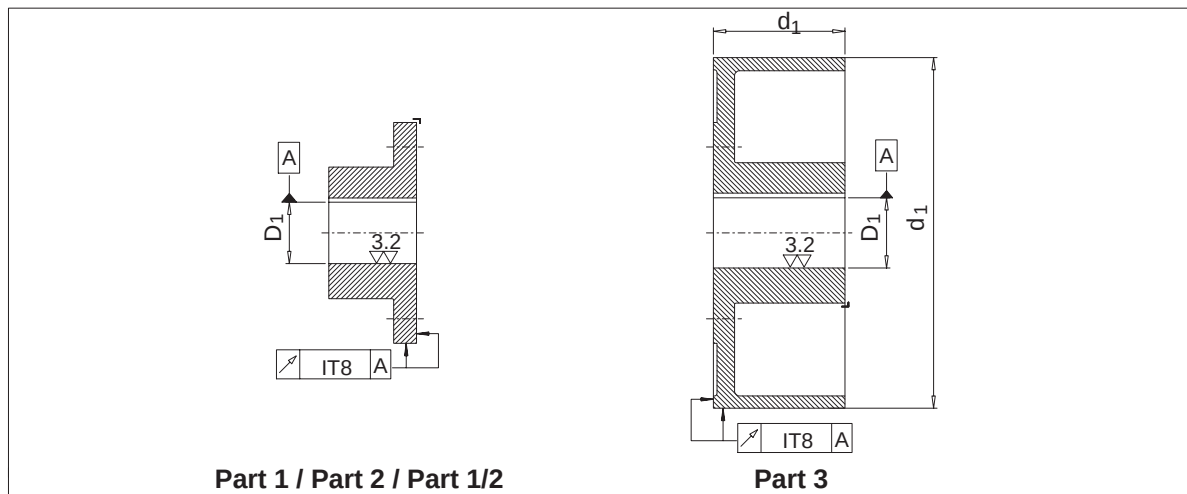
Caution!

The maximum permissible boring diameters (see Section 1.) are designed for driving connections without tightening according to DIN 6885/1 and must not be exceeded in any case.

If other shaft hub connections (e.g. splined hub profile, tapered or graded bore, driving connection with tightening, etc.) shall be fitted instead of the intended driving connections, FLENDER is to be consulted.



**Non-observance of these notes may lead to the drifting of coupling.
There is a danger to life due to broken pieces flying around!**



In case of drive by means of parallel keys, the following fit pairings are prescribed for the bores:

Fit selection	Bore		Shaft tolerances	Boring tolerances
	above mm	to mm		
Shaft tolerance to FLENDER standard		25	k6	H7
	25	100	m6	
	100		n6	
Shaft tolerance to DIN 748/1		50	k6	H7
	50		m6	
Unified system shaft		50	h6	K7
	50			M7
		all	h8	N7

Table 6.1: Fit pairings

Caution!

Observing the fit correspondance is absolutely necessary, on the one hand in order to keep low the backlash in the shaft hub connection or, on the other hand, to keep the hub tension caused by the overdimension within the permissible load depending on the use of the tolerance fields. It cannot be excluded, that the shaft hub connection is endangered when the fit correspondance is not observed.



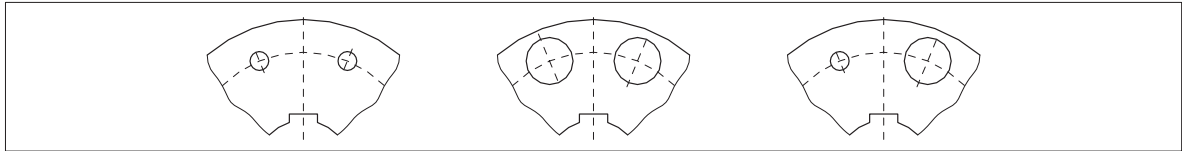
**Non-observance of these notes may lead to the bursting of the coupling.
There is a danger to life due to broken pieces flying around!**

6.1.1.1 Keyway

The keyways have to be designed according to the existing parallel keys. For keyways, the tolerance field of the hub keyway width **ISO JS 9** is to be observed.

For more difficult operating conditions, as it is the case for e.g. reversing operation of operation with impulses, the tolerance field of the hub keyway width **ISO P9** is prescribed.

Caution! The keyways are to be set and centered between the buffer bores.



6.1.2 Axial securing

For axially securing the coupling parts, a set screw or an end plate has to be provided for. When using end plates, FLENDER is to be consulted with regard to the insertion of recess in the coupling parts.

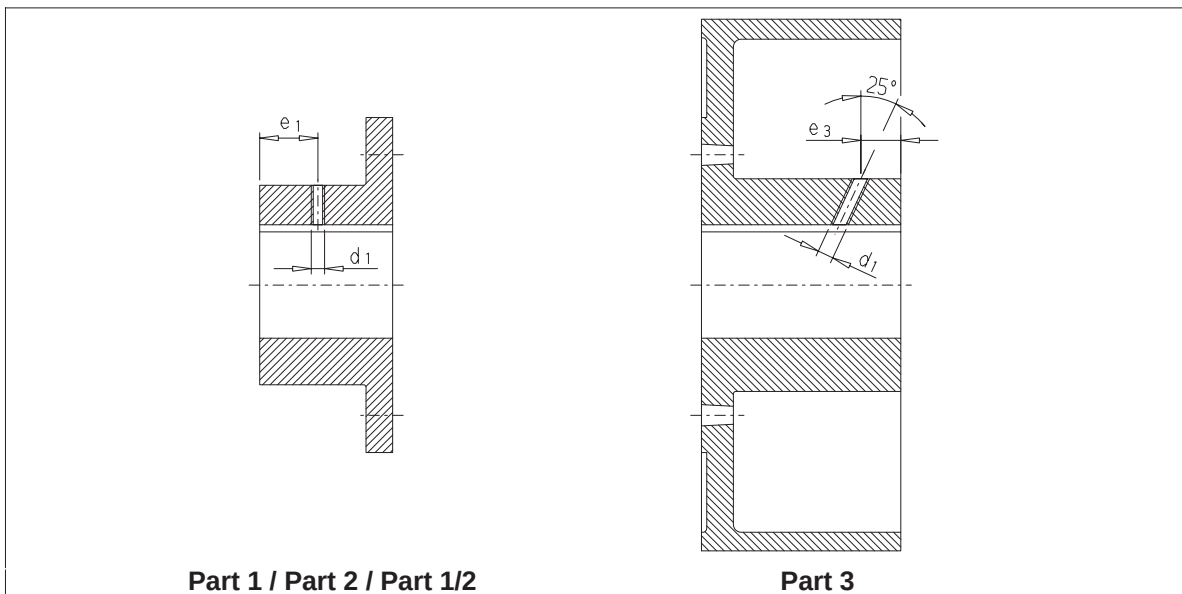
6.1.3 Set screws

Headless pins with notched cut point according to DIN 916 are to be used as set screws.

It is absolutely necessary to observe the following guidelines!



The length of the set screw is to be chosen so that it completely fills the cut hole but that it does not protect over the hub ($L_{\min} = d_1 \times 1.2$).



Type RWN			Type RWS		
Bore range		Set screw size d_1 mm	Bore range		Set screw size d_1 mm
above mm	to mm		above mm	to mm	
8	30	M 6	8	30	M 6
30	38	M 8	30	75	M 8
38	65	M10	75	95	M12
65	95	M12	95	110	M16
95	110	M16	110	150	M20
110	150	M20	150	230	M24
150	230	M24	230	600	M30
230	600	M30			

Table 6.2: Set screw assignment

Size	105	125	144	162	178	198	228	252	285	320	360	400	450
Distance dimension e_1	15	20	25	25	35	40	40	50	55	60	70	80	80
Size	500	560	630	710	800	900	1000	1120	1250	1400	1600	1800	2000
Distance dimension e_1	90	100	110	130	115	160	175	160	200	240	250	300	330

Break drum	200 x 75	250 x 95	315 x 118	400 x 150	500 x 190	630 x 236	710 x 265
Distance dimension e_3	12	12	25	30	50	75	90

Tabelle 6.3: Distance dimensions of the set screws

Caution!

Generally, the set screws are to be arranged on the keyway. This does not apply to coupling parts 1+2 of sizes 105 and 125 where the set screw should be arranged by 180° shifted to the keyway.

6.1.4 Balancing

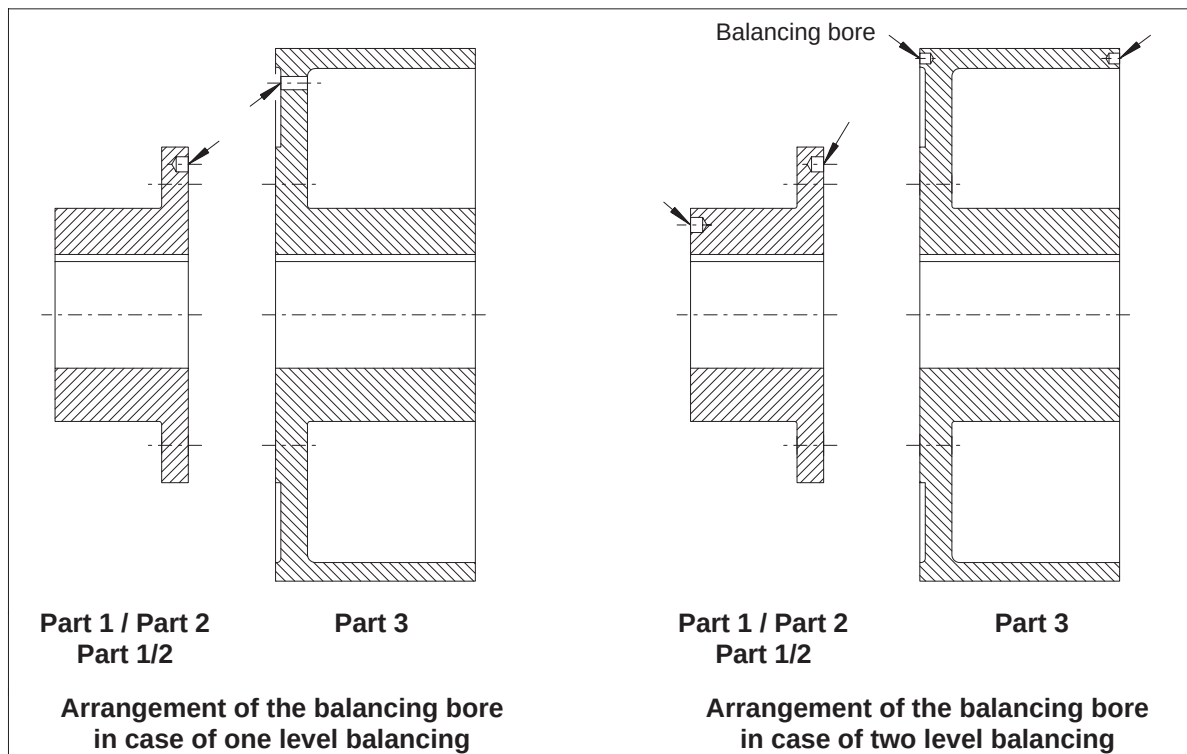
Prebored couplings resp. prebored coupling parts are shipped unbalanced. For these parts it is recommended to balance them depending on the application case after finish boring (see also DIN 740, VDI guideline 2060).

Balancing is usually carried out by material cutting through boring. In order to restrict the material quantity to be cut to a minimum, the biggest possible balancing radius is to be selected.

Caution!

For parts 1/2, cutting has to be carried out between the bores without through-boring of the bottom. For part 32, the break surface must not be damaged under any circumstances.

Finished bored couplings or coupling parts are balanced according to the instructions of the ordering party.



6.2 General installation notes

For the installation the safety notes in Section 3. are to be observed.

The installation has to be carried out with utmost care by trained personnel.

Already during the planning phase it is to be observed that sufficient room is to be provided for the installation and later inspection and maintenance work.

Before starting the installation a sufficient number of hoists must be provided for.

6.3 Mounting the coupling parts

Before starting the assembly, clean the shaft ends, the brake drum and the coupling parts carefully. Before cleaning the coupling parts with solvents, remove the buffers.



Observe the manufacturer's instructions on handling the solvents.

Caution!

The coupling parts and the brake drum are to be fitted by means of suitable devices in order to prevent damage to the shaft bearing arrangement by the axial fitting force.

Make sure that suitable hoists are used.

If it shall be possible to exchange the buffers or to move the coupled machines, the dimension P should be taken into consideration during assembly according to 6.6.

Caution!

Tightening of the set screws only by means of a hexagon socket head wrench according to DIN 911, without an extension pipe.



Non-observance of these notes may lead to the bursting of the coupling. There is danger to life due to broken pieces flying around!

The assembly process is made easier by heating the coupling parts and the brake drum (3) (to max. +150 °C). If using temperatures greater than +80 °C, the buffers/bolts must be removed from the coupling parts before heating them.



Protect yourself against burns by hot parts!

After mounting the coupling parts or the brake drum, replace the buffers (if removed beforehand). Tighten the nuts (7) and bolts (11) using a torque wrench (for correct torques, see section 6.6). Secure bolts (11) with a few spots of adhesive (e.g. Loctite, type 242). Coupling parts which have been previously heated must be allowed to cool to a temperature of less than 80 °C.

Loop together the machines to be coupled.



Danger of squeezing!

Observe the dimension S_1 (see Section 1.).

6.4 Aligning

The couplings compensate for positional variations of the shaft ends to be connected up to the data shown under item 6.5.

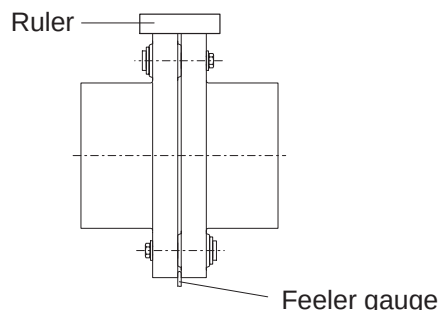
When aligning, keep the radial and angular misalignment of the shaft ends as small as possible because hereby the service life of the flexible is increased under otherwise the same operating conditions.

The alignment should be realised in the order:

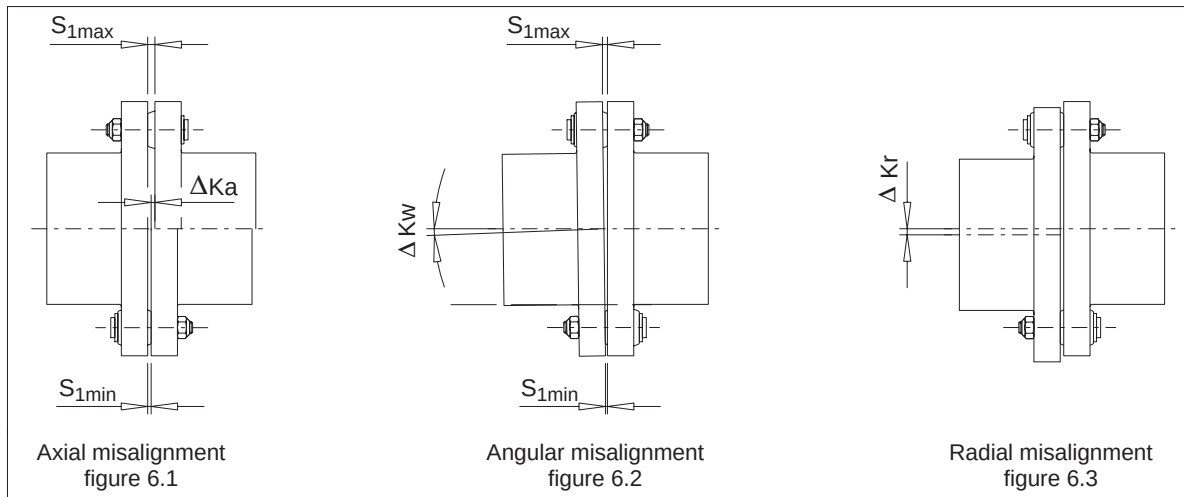
1. angular alignment
2. radial alignment

and should be carried out in two axial planes vertical to each other. This is possible by means of a feeler gauge (angular misalignment) and a ruler (radial misalignment). The distance dimension S_1 is to be kept (see Section 1.).

By using a dial gauge, the alignment precision can be increased.



6.5 Possible misalignments



Misalignments of the coupling parts may result from an inexact alignment during the assembly but also from the operation of the plant (expansion due to heat, bending of the shaft, machine frame too soft, etc.).

Caution!

The following max. permissible misalignments must not be exceeded during operation under any circumstances.

6.5.1 Axial misalignment

Axial misalignment ΔK_a (figure 6.1) of the coupling parts to each other is permissible within the "permissible variation" for the dimension S_1 (see Section 1.).

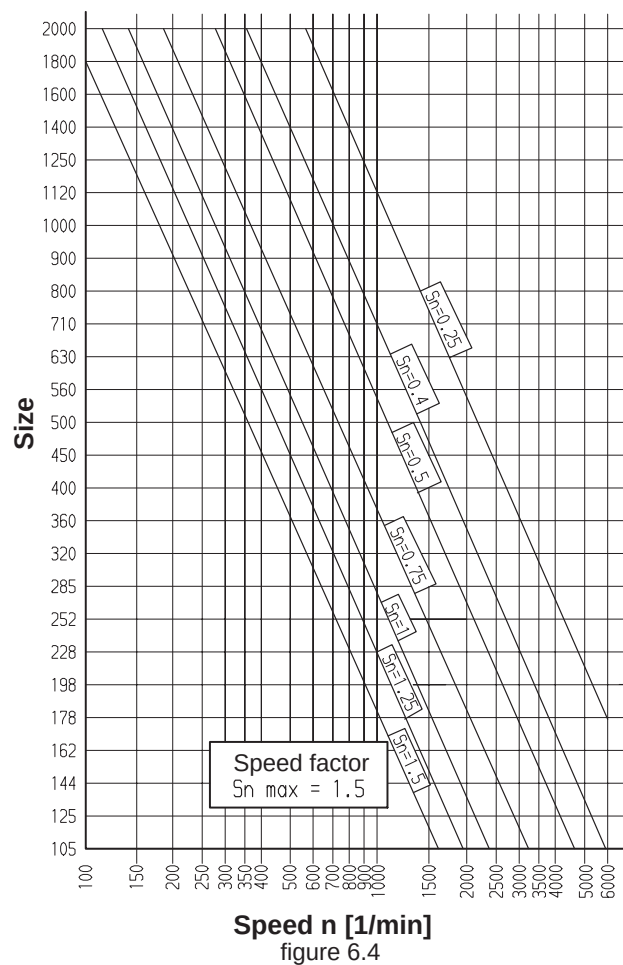
6.5.2 Angular misalignment

The permissible angular misalignment ΔK_w (figure 6.2) has to be determined taking into consideration the speed factor S_n of figure 6.4.
 $\Delta K_{wperm.} = S_{1max} - S_{1min} = 0.00175 \times da \times S_n$
 da = Coupling size

6.5.3 Radial misalignment

The permissible radial misalignment ΔK_r (figure 6.3) has to be determined taking into consideration the speed factor S_n of figure 6.4.

$\Delta K_{rperm.} = 0.00175 \times da \times S_n$
 da = Coupling size



Caution!

Angular and radial misalignments may occur at the same time. The sum of both misalignments must not exceed the max. permissible value of the angular or radial misalignment.

$$(K_w + K_r)_{existing} \leq \Delta K_w \times S_n \text{ or } \Delta K_r \times S_n$$

6.6 Assignments of the tightening torques

Size	P mm	Tightening torque T_A Nm	Wrench width S_w mm
105	30	12	10
125	35	30	13
144	35	30	13
162	40	60	17
178	40	60	17
198	40	60	17
228	50	65	19
252	50	65	19
285	60	150	24
320	60	150	24
360	75	220	27
400	75	220	27
450	90	180	24
500	90	180	24
560	120	340	30
630	120	340	30
710	140	580	36
800	140	580	36
900	160	600	36
1000	160	600	36
1120	180	1150	46
1250	180	1150	46
1400	210	1150	46
1600	210	1150	46
1800	240	2000	55
2000	240	2000	55

Table 6.4: Assignments of the tightening torques

7. Startup

7.1 Measures before startup

Before startup, check the tightening of the set screws, the alignment and the distance dimension S_1 and correct if necessary and check all screw connections for the prescribed tightening torques (see Section 1. and Section 6.). Finally, the coupling guard to prevent accidental contact is to be fixed.

8. Operation

8.1 General operating data

During operation, the coupling should be checked for:

- any changes in running noises
- sudden shaking.

Caution!

If irregularities are detected during operation, the drive assembly should be set off immediately. The cause of the malfunction should be determined with the aid of the Troubleshooting Table (Section 9.).

The Troubleshooting Table lists possible malfunctions, their causes and suggestions for remedying them.

If the cause cannot be determined or there is no facility for repair with suitable equipment, we recommend calling in one of our service fitters (see Section 11.).

9. Disturbances, reasons and remedy

9.1 General

The malfunctions listed below can only be hints for a troubleshooting.

In case of a complex plant, all other components have always to be included in the troubleshooting.

The coupling has to run with low noise and without shaking in all operating phases. Any deviating behaviour is to be regarded as malfunction and should be repaired immediately.



Before maintenance work, repairs or other work, the operator should make sure that the whole drive train is stationary. Especially the drive motors are to be secured against accidental start up.

Futhermore, we refer to the relevant regulations on the prevention of accidents at the place of installation.

9.2 Possible malfunctions

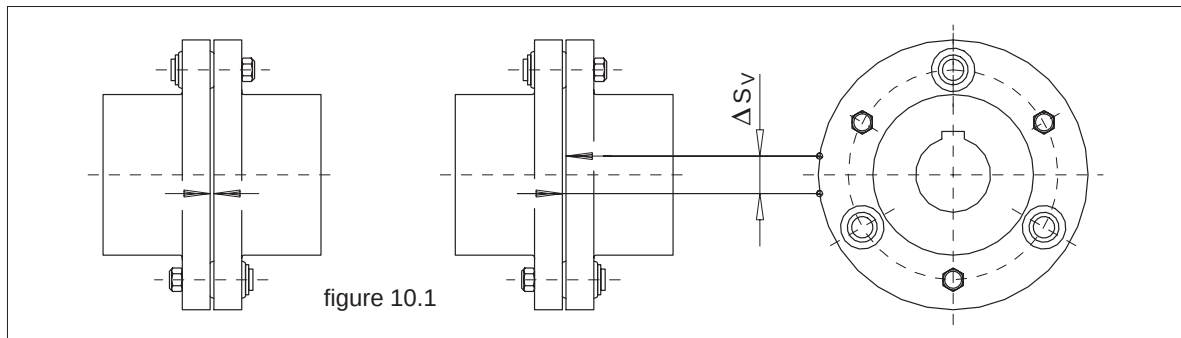
Malfunctions	Causes	Remedy
Sudden change of the noise level and/or sudden shaking	Change of alignment	Shut down the plant possibly repair the reason for the change of alignment (e.g. fasten loose foundation bolts)
	Buffers worn, no dampening	Shut down the plant Disassemble coupling and remove the rests of the buffers Check coupling parts and exchange damaged coupling parts Buffers are to be changed in sets Check alignment and correct if necessary, see Section 6. Assembly of coupling acc. to Section 6. "Assembly" and Section 7. "Start up".

10. Maintenance and repair

10.1 General

The circumferential backlash between the two coupling parts should be checked according to the maintenance intervals, however, at least once a year. As far as an increased backlash of the coupling is not detrimental to the operation of the coupling, the elastic buffers may remain in use until a defined wear limit is reached before they should be changed. For evaluating the wear, in Table 10.1, the permissible circumferential backlash is listed, which is converted to the chord dimension ΔS_V on the outer coupling diameter. In order to determine the dimension ΔS_V , a coupling part is turned until stop without torque and a mark is made on the coupling part (see figure 10.1). By turning the coupling part in the opposite direction of rotation until stop, the marks move apart. The distance between the marks is the chord dimension ΔS_V . If the dimension ΔS_V exceeds the value listed in Table 10.1, the buffers should be changed.

Caution! The buffers should be changed in sets.



Size	105	125	162	228	285	360	450	560	710	900	1120	1400	1800
		144	198	252	320	400	500	630	800	1000	1250	1600	2000
ΔS_V	3.0	3.5	4.0	4.5	6.0	7.0	8.5	10.0	12.0	13.5	15.0	18.0	20.0

Table 10.1: Wear mark

10.2 Change of wear parts

Only use original **RUPEX** buffers as spare buffers, in order to guarantee a perfect transmission of the torque and a trouble-free function.

Note: Changing the buffers is possible without moving the coupled machines.

Up to coupling size 400, the bolts with the plastic buffers can be removed through the buffer holes after releasing and removing the nuts, and up to coupling size 450, after releasing and removing the hexagon bolts and end plates.

After renewing the plastic buffers, assemble the unit in the opposite sequence. Secure the nuts or bolts with adhesive (e.g. Loctite 242).

Carefully observe the instructions of Section 6. "Installation" and Section 7. "Start up" for the re-installation.

11. Spare parts stock, service facility addresses

Maintaining a stock of the most essential spare and wear parts is an important prerequisite for the permanent service ability of the coupling.

When ordering spare parts, the following data should be stated:

- Part No. (see Section 5.)
- Description / Size (the size designation corresponds to the outer diameter d_a in mm)
- Quantity

We assume warranty only for original spare parts supplied by us.

Caution!

We would expressly draw attention to the fact that spare parts and accessories not supplied by us have not been tested or approved by us either. Fitting and/or use of such products can therefore under certain circumstances adversely affect structurally specified properties of the coupling and will thus impair active and/or passive safety. No form of reliability or warranty will be assumed by FLENDER for damage occasioned by use of non-original spare parts and accessories.

Please note that production and supply specifications frequently exist for components and we will always offer spare parts in accordance with the state of the art and in accordance with the latest legal requirements.

11.1 Service facility addresses

When ordering spare parts or requesting a service fitter, please contact FLENDER AG first of all.

FLENDER Germany

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E-mail: contact@flender.com • www.flender.com
Shipping address: Alfred - Flender - Strasse 77 - 46395 Bocholt

A. FRIEDR. FLENDER AG - Kupplungswerk Mussum

Industriepark Bocholt - Schlavenhorst 100 - 46395 Bocholt - Tel.: (0 28 71) 92 28 68 - Fax: (0 28 71) 92 25 79
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A. FRIEDR. FLENDER AG - Werk Friedrichsfeld

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Thierbacher Strasse 24 - 09322 Penig - Tel.: (03 73 81) 60 - Fax: (03 73 81) 8 02 86
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12. Declaration by the manufacturer

Declaration by the manufacturer

in accordance with EC Engineering Guideline 98/37/EC, Appendix II B

We hereby declare that the

**Flexible RUPEX couplings,
type RWN, RWS and RWB, RBS**

described in these Operating Instructions are intended for incorporation in a machine, and that it is prohibited to put them into service before verifying that the machine into which they are incorporated complies with the EC Guidelines (original edition 98/37/EC including any subsequent amendments thereto).

This Manufacturer's Declaration takes into account all the unified standards (inasmuch as they apply to our products) published by the European Commission in the Official Journal of the European Community.



Bocholt, 1998-08-24

Signature (person responsible for products)